

Final Year Design Project Report

Project Nexus: The Unified Intelligent Campus Platform



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Project Nexus: The Unified Intelligent Campus Platform

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CERTIFICATE OF APPROVAL

It is to certify that the final year design project (FYDP) of BSIT “Project Nexus: The Unified Intelligent Campus Platform” was developed by Muhammad Asad(BIT22031) ,Muhammad Saad(BIT22034) and Muhammad Hanzla(BIT22002) under the supervision of “Dr. Ghulam Mustafa” in my opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Information Technology.

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Project Nexus: The Unified Intelligent Campus Platform

Executive Summery

Modern educational institutions operate in a digital environment that is deep fragmented, typically relying on disconnected systems for administrative records (SIS) and academic learning (LMS). This separation creates critical "data silos," where valuable insights about student performance are trapped in isolation, preventing the institution from providing timely, data-driven support. Project Nexus addresses this by establishing a "Unified Intelligent Campus Platform" that integrates these distinct domains into a single, cohesive ecosystem. By bringing the office side and the classroom side together, the system turns the university from a simple record-keeper into an active partner that uses data to help students succeed.

Technologically, the platform is built upon a **Polyglot Microservices Architecture**, containerized via **Docker Swarm** for scalability and using **Apache Kafka** as a durable event bus to ensure "Zero Data Loss" for critical operations. The system optimizes data management by employing a specialized persistence strategy: **PostgreSQL** for financial integrity, **MongoDB** for flexible content, **Redis** for high-speed caching, and **ChromaDB** for vector-based AI memory. Key innovations include a **Hybrid AI Chatbot** (combining CAG and RAG) for context-aware support, **Spoof-Proof Biometric Attendance** with liveness detection, and a proactive data science layer that automatically predicts and assists "at-risk" students in real-time.

Keywords: Microservices Architecture, Unified Campus Platform, Retrieval-Augmented Generation (RAG), Biometric Attendance, Predictive Analytics.

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Abbreviations

The following table lists all the abbreviations and acronyms used throughout this document, along with their respective full forms and brief descriptions.

Abbreviation	Full Form / Description
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BM25	Best Match 25 (ranking function)
CAG	Cache-Augmented Generation
CGPA	Cumulative Grade Point Average
CNIC	Computerized National Identity Card
CP-SAT	Constraint Programming - Satisfiability
CPU	Central Processing Unit
CRUD	Create, Read, Update, Delete
CV	Computer Vision
DB	Database
DNS	Domain Name System
Docker	Docker Container Platform
EAR	Eye Aspect Ratio
FAC	Faculty Advisory Committee
FastAPI	Fast Application Programming Interface (Python web framework)
FCIT	Faculty of Computing and Information Technology
FR	Functional Requirement
FYDP	Final Year Design Project
GPA	Grade Point Average
GPS	Global Positioning System
GUI	Graphical User Interface
HNSW	Hierarchical Navigable Small World
HOD	Head of Department
HR	Human Resources
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token
KPI	Key Performance Indicator
LLM	Large Language Model
LMS	Learning Management System
MCQ	Multiple Choice Question
ML	Machine Learning
MUI	Material User Interface
NFR	Non-Functional Requirement
NLP	Natural Language Processing

OOP	Object-Oriented Programming
ORM	Object-Relational Mapping
OR-Tools	Operations Research Tools (Google)
OTP	One-Time Password
P2P	Peer-to-Peer
PDF	Portable Document Format
PII	Personally Identifiable Information
PWA	Progressive Web Application
QR	Quick Response (code)
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
RRF	Reciprocal Rank Fusion
SDS	Software Design Specification
SGPA	Semester Grade Point Average
SIS	Student Information System
SLA	Service Level Agreement
SPA	Single Page Application
SQL	Structured Query Language
SRS	Software Requirements Specification
SSL	Secure Sockets Layer
TLS	Transport Layer Security
TTL	Time-To-Live
UC	Use Case
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
UX	User Experience
VM	Virtual Machine
WSS	WebSocket Secure

Table of Contents

Chapter No.1 Introduction	1
1. Introduction.....	2
1.1 Problem Statement	2
1.2 Problem Solution	2
1.3 Objectives	2
1.4 Scope	4
1.4.1 Inclusions (In Scope)	4
1.4.2 Exclusions (Out of Scope)	4
1.4.3 Optional (If Time Permits)	5
1.5 System Components.....	5
1.5.1 Client Web App (Frontend)	5
1.5.2 Administrative & Finance Services	5
1.5.3 Academic & Communication Services	5
1.5.4 Intelligence & Biometric Services	5
1.6 Related System Analysis/Literature Review	6
1.7 Vision Statement	6
1.8 System Limitations and Constraints	7
1.9 Tools, Libraries & Technologies With Reasoning	7
1.9.1 Technology Stack Overview.....	7
1.9.2 Hardware Requirements	8
1.9.3 Tools	9
1.9.4 Libraries	10
1.9.5 Technologies.....	11
1.9.6 Analysis of Alternatives.....	12
1.10 Project Deliverables	14
1.11 Project Planning.....	14
1.11.1 Project Timeline	15
1.12 Summary	16
Chapter No. 2 Requirements Analysis	17
2. Requirements Analysis	18
2.1 User classes and characteristics	18
2.2 Requirement Identifying Technique	19
2.2.1 Gap Analysis	19
2.2.2 Use Case Analysis.....	19
2.2.3 Storyboarding	19
2.3 Functional Requirements	20
2.3.1 FR-01: User Registration and Account Creation	20
2.3.2 FR-02: User Login and Authentication.....	20
2.3.3 FR-03: Secure Session Management	20
2.3.4 FR-04: Role Based Access Control	21
2.3.5 FR-05: Student Profile Management	21
2.3.6 FR-06: Semester Course Registration	22
2.3.7 FR-07: Faculty Grade Submission & Sync.....	22
2.3.8 FR-08: Digital Transcript Generation	22

2.3.9	FR-09: Automated Fee Voucher Generation	23
2.3.10	FR-10: Online Payment and Real Time Ledger.....	23
2.3.11	FR-11: Course Content Upload and Organization.....	23
2.3.12	FR-12: Assignment Creation and Deadline Management.....	24
2.3.13	FR-13: Student Assignment Submission	24
2.3.14	FR-14: Hybrid AI Chatbot	25
2.3.15	FR-15: Context Aware Responses	25
2.3.16	FR-16: Biometric Liveness Detection	25
2.3.17	FR-17: GPS Geofencing Verification	26
2.3.18	FR-18: Facial Recognition and Auto Marking.....	26
2.3.19	FR-19: Event Driven Data Synchronization.....	27
2.3.20	FR-20: Real Time Push Notifications	27
2.3.21	FR-21: Peer to Peer Chat Initiation	27
2.3.22	FR-22: Real Time Messaging & Persistence	28
2.3.23	FR-23: Group Chat Creation	28
2.3.24	FR-24: Timetable Constraints Definition	28
2.3.25	FR-25: Automated Conflict-Free Scheduling	29
2.3.26	FR-26: Digital Leave Application	29
2.3.27	FR-27: Employee Profile & Salary Management	30
2.3.28	FR-28: Alumni Portal & Networking	30
2.3.29	FR-29: Digital Library Catalogue Search	30
2.3.30	FR-30: QR-Based Book Issuance	31
2.3.31	FR-31: Automated Library Fine Calculation	31
2.3.32	FR-32: Faculty “Student Success” Dashboard	31
2.3.33	FR-33: “At-Risk” Student Prediction	32
2.3.34	FR-34: Smart Grievance Submission & Sentiment Analysis	32
2.3.35	FR-36: Grievance Status Tracking & SLA.....	33
2.3.36	FR-36: Secure System Audit Logging.....	33
2.3.37	FR-37: Online Quiz Creation.....	33
2.3.38	FR-38: Timed Quiz Attempt	34
2.3.39	FR-39: Automated Grading and Feedback	34
2.3.40	FR-40: View Current Semester Details	35
2.3.41	FR-41: Executive Analytics Dashboard.....	35
2.3.42	FR-42: Global Announcements System.....	35
2.3.43	FR-43: Student Leave Application & Academic Exemption	36
2.4	Non-Functional Requirements.....	37
2.4.1	Reliability.....	37
2.4.1.1	NFR-01 (System Uptime)	37
2.4.1.2	NFR-02 (Self-Healing)	37
2.4.1.3	NFR-03 (Zero Data Loss).....	37
2.4.1.4	NFR-04 (Data Backup).....	37
2.4.2	Usability	37
2.4.2.1	NFR-05 (Mobile Responsiveness)	37
2.4.2.2	NFR-06 (Efficiency).....	37
2.4.2.3	NFR-07 (AI Transparency).....	37
2.4.2.4	NFR-08 (Ease of Learning)	38
2.4.3	Performance	38
2.4.3.1	NFR-09 (General Speed).....	38
2.4.3.2	NFR-10 (AI Response Time)	38
2.4.3.3	NFR-11 (Real-Time Chat).....	38
2.4.3.4	NFR-12 (Concurrency)	38

2.4.4	Security	38
2.4.4.1	NFR-13 (Encryption)	38
2.4.4.2	NFR-14 (Role-Based Access)	38
2.4.4.3	NFR-15 (Biometric Privacy).....	38
2.4.4.4	NFR-16 (Session Security).....	39
2.5	External Interface Requirements	40
2.5.1	User Interfaces Requirements.....	40
2.5.1.1	General GUI Standards	40
2.5.1.2	Standard Screen Elements	40
2.5.2	Software Interfaces	41
2.5.2.1	Database Interfaces	41
2.5.2.2	External APIs	41
2.5.3	Hardware Interfaces	41
2.5.3.1	Client-Side Hardware	42
2.5.3.2	Server-Side Hardware.....	42
2.5.4	Communications Interfaces	42
2.5.4.1	Communication Protocols	42
2.5.4.2	Message Queuing (Internal).....	42
2.5.4.3	Data Formats.....	43
3.	Use Case Analysis.....	44
3.1	Use Case #1: User Registration (UC-01)	44
3.2	Use Case #2: User Login (UC-02)	44
3.3	Use Case #3: Manage Student Profile (UC-03)	45
3.4	Use Case #4: Register for Courses (UC-04)	45
3.5	Use Case #5: Download Digital Transcript (UC-05)	46
3.6	Use Case #6: Generate Fee Voucher (UC-06)	46
3.7	Use Case #7: Pay Fees Online (UC-07)	47
3.8	Use Case #8: Submit Final Grades (UC-08)	47
3.9	Use Case #9: Sync Grades via Kafka (UC-09)	48
3.10	Use Case #10: Upload Course Material (UC-10)	49
3.11	Use Case #11: Create Assignment (UC-11)	49
3.12	Use Case #12: Submit Assignment (UC-12)	50
3.13	Use Case #13: Query AI Chatbot (UC-13)	50
3.14	Use Case #14: Mark Attendance (UC-14)	51
3.15	Use Case #15: Receive Notification (UC-15)	51
3.16	Use Case #16: Initiate P2P Chat (UC-16).....	52
3.17	Use Case #17: Send Real-Time Message (UC-17)	52
3.18	Use Case #18: Create Group Chat (UC-18)	53
3.19	Use Case #19: Define Timetable Constraints (UC-19)	53
3.20	Use Case #20: Generate Master Timetable (UC-20).....	54
3.21	Use Case #21: Submit Leave Application (UC-21)	55
3.22	Use Case #22: Approve/Reject Leave (UC-22).....	55
3.23	Use Case #23: View Leave Balance (UC-23).....	56
3.24	Use Case #24: Manage Employee Profile (UC-24).....	56
3.25	Use Case #25: Register as Alumni (UC-25).....	57
3.26	Use Case #26: Post Job Opportunity (UC-26)	57
3.27	Use Case #27: Issue Book via QR (UC-27).....	58
3.28	Use Case #28: View At-Risk Analytics (UC-28)	58
3.29	Use Case #29: Submit Grievance (UC-29).....	59
3.30	Use Case #30: Track Grievance Status (UC-30)	59

3.31	Use Case #31: Create Online Quiz (UC-31)	60
3.32	Use Case #32: Attempt Online Quiz (UC-32)	60
3.33	Use Case #33: View Semester Details (UC-33)	61
3.34	Use Case #34: Resolve Grievance (UC-34)	61
3.35	Use Case #35: View Executive Dashboard (UC-35)	62
3.36	Use Case #36: Post Global Announcement (UC-36)	62
3.37	Use Case #37 Process Student Leave (UC-37)	63
3.38	Use Case Diagrams	64
3.38.1	Use Case Diagram 1	64
3.38.2	Use Case Diagram 2	65
3.38.3	Use Case Diagram 3	66
3.38.4	Use Case Diagram 4	67
3.38.5	Use Case Diagram 5	68
3.38.6	Use Case Diagram 6	69
3.38.7	Use Case Diagram 7	70
4.	Storyboards	71
4.1	AI Chatbot (Context-Aware RAG)	71
4.2	AI Analytics & Prediction (Data Science)	72
4.3	Smart Attendance (Geo-Fence + Liveness)	73
4.4	Online Quiz Attempt (Auto-Grading)	74
4.5	Automated Timetable Generation	75
5.	Summary	76
Chapter No. 3 Design and Architecture		77
6.	System Design	78
6.1	Product Perspective	78
6.2	Dependencies	78
6.3	Design Constraints	79
6.3.1	Hardware Constraints	79
6.3.2	Performance Constraints	79
6.3.3	Security & Privacy Constraints	79
6.4	External Integrations	79
7.	Design Considerations	81
7.1	Assumptions and Dependencies	81
7.1.1	Assumptions	81
7.1.2	Dependencies	81
7.2	General Constraints	82
7.2.1	Hardware & Resource Constraints	82
7.2.2	Performance Constraints	82
7.2.3	Security & Privacy Constraints	82
7.3	Goals and Guidelines	83
7.4	Development Methods	83
8.	Design Models	84
8.1	Architectural Design	84
8.1.1	High Level Architecture	85
8.1.2	System Component Structure	89
8.1.3	Deployment Architecture	93
8.1.4	UML Class Diagrams	95

8.1.4.1	High-Level System Map	95
8.1.4.2	User Hierarchy & Security Module	96
8.1.4.3	Academic Sub-Systems (SIS & LMS).....	98
8.1.5	Behavioral Model	102
8.1.5.1	Sequence Diagrams	102
8.1.5.2	State Transition Diagrams	108
8.2	Data Design	114
8.2.1	Data Dictionary	114
8.2.2	Database Design	129
8.2.2.1	Relational Database (PostgreSQL)	129
8.2.2.2	NoSQL Database (MongoDB)	143
8.2.2.3	REDIS CACHE	148
8.2.2.4	VECTOR STORE (ChromaDB).....	150
8.3	User Interface Design	153
8.3.1	Screen Images	153
8.3.2	Screen Objects and Actions	158
8.3.2.1	Login & Account Recovery	158
8.3.2.2	Fee Payment & Financial Ledger	158
8.3.2.3	Academic Course & Transcript View	159
8.3.2.4	AI Chatbot Interaction	159
9.	Design Decisions	160
9.1	Architectural Pattern: Microservices vs. Monolithic	160
9.2	Database Strategy: Polyglot Persistence vs. Single Database	160
9.3	Communication Style: Event-Driven vs. Synchronous REST	160
9.4	AI Implementation: RAG Pipeline vs. Fine-Tuning.....	161
9.5	Biometric Privacy: Vector Embeddings vs. Image Storage	161
10.	Summery.....	162
Chapter No. 4	Implementation	163
11.	Implementation	164
11.1	Algorithm	164
11.1.1	Algorithm 1: Hybrid AI Chatbot Pipeline (CAG + RAG)	164
11.1.2	Algorithm 2: At-Risk Student Prediction (RandomForest Classifier).....	166
11.1.3	Algorithm 3: Spoof-Proof Biometric Attendance (3-Step Verification)	167
11.1.4	Algorithm 4: Conflict-Free Automated Timetable Generation (CP-SAT).....	169
11.2	External APIs/SDKs	170
11.3	Code Repository	171
11.3.1	Metrics of the Git Repository	171
11.4	Summary	171
Chapter No. 5	Testing and Evaluation	172
12.	Introduction	173
12.1	Testing Strategy	173
12.2	Unit Testing (UT)	173
12.3	Functional Testing (FT).....	174
12.4	Integration Testing (IT)	176
12.5	Performance Testing (PT)	177
12.6	Security / Validation Checks	177
12.7	Summary	178

Chapter No. 6 System Conversion 179

13. Introduction 180

13.1 Conversion Method 180

13.2 Deployment..... 180

13.3 Data Conversion 181

13.4 Training (User Manual)..... 182

13.5 Post Deployment Testing 182

13.6 Challenges 183

13.7 Summary 183

Chapter No. 7 Conclusion 184

14. Introduction 185

14.1 Evaluation 185

14.2 Traceability Matrix 185

14.3 Conclusion 188

14.4 Future Work..... 188

References 190

Appendix-A 193

Appendix-B 195

List of Tables

Table 1 Analysis	6
Table 2 Hardware Requirements	8
Table 3 Tools	9
Table 4 Libraries	10
Table 5 Technologies.....	11
Table 6 Technology Justification Matrix.....	12
Table 7 User Classes & Characteristics	18
Table 8 User Registration & Account Creation [FR-01].....	20
Table 9 User Login & Authentication [FR-02]	20
Table 10 Secure Session Management (JWT) [FR-03].....	20
Table 11 RBAC [FR-04]	21
Table 12 Student Profile Management [FR-05].....	21
Table 13 Semester Course Registration [FR-06]	22
Table 14 Faculty Grade Submission & Sync [FR-07]	22
Table 15 Digital Transcript Generation [FR-08].....	22
Table 16 Automated Fee Voucher Generation [FR-09].....	23
Table 17 Online Payment & Real-Time Ledger [FR-10].....	23
Table 18 Course Content Upload & Organization [FR-11]	23
Table 19 Assignment Creation & Deadline Management [FR-12].....	24
Table 20 Student Assignment Submission [FR-13]	24
Table 21 Hybrid AI Chatbot [FR-14].....	25
Table 22 Context Aware Response [FR-15].....	25
Table 23 Biometric Liveness Detection [FR-16]	25
Table 24 GPS Geofencing Verification [FR-17]	26
Table 25 Facial Recognition and Auto Marking [FR-18]	26
Table 26 Event-Driven Data Synchronization [FR-19]	27
Table 27 Real-Time Push Notification [FR-20].....	27
Table 28 P2P-Chat Initiation [FR-21]	27
Table 29 Real Time Message [FR-22]	28
Table 30 Group Chat Creation [FR-23]	28
Table 31 Timetable Constraints [FR-24].....	28
Table 32 Conflict Free Scheduling [FR-23].....	29
Table 33 Leave Application [FR-26].....	29
Table 34 Employee Profile & Salary Management [FR-27].....	30
Table 35 Alumni Portal & Networking [FR-28]	30
Table 36 Digital Library Catalogue Search [FR-29].....	30
Table 37 Book Issuance [FR-30].....	31
Table 38 Library Fine [FR-31].....	31
Table 39 Student Success Dashboard [FR-32].....	31
Table 40 Student Prediction [FR-33]	32
Table 41 Grievance Submission [FR-34].....	32
Table 42 Grievance Status [FR-35].....	33
Table 43 System Audit Logging [FR-36].....	33
Table 44 Online Quiz Creation [FR-37].....	33

Table 45 Timed Quiz Attempt [FR-38]	34
Table 46 Automated Grading & Feedback [FR-39]	34
Table 47 Semester Details [FR-40]	35
Table 48 Executive Analytics Dashboard [FR-41]	35
Table 49 Global Announcement System [FR-42]	35
Table 50 Student Leave [FR-43]	36
Table 51 UC-1	44
Table 52 UC-2	44
Table 53 UC-3	45
Table 54 UC-4	45
Table 55 UC-5	46
Table 56 UC-6	46
Table 57 UC-7	47
Table 58 UC-8	47
Table 59 UC-9	48
Table 60 UC-10	49
Table 61 UC-11	49
Table 62 UC-12	50
Table 63 UC-13	50
Table 64 UC-14	51
Table 65 UC-15	51
Table 66 UC-16	52
Table 67 UC-17	52
Table 68 UC-18	53
Table 69 UC-19	53
Table 70 UC-20	54
Table 71 UC-21	55
Table 72 UC-22	55
Table 73 UC-23	56
Table 74 UC-24	56
Table 75 UC-25	57
Table 76 UC-26	57
Table 77 UC-27	58
Table 78 UC-28	58
Table 79 UC-29	59
Table 80 UC-30	59
Table 81 UC-31	60
Table 82 UC-32	60
Table 83 UC-33	61
Table 84 UC-34	61
Table 85 UC-35	62
Table 86 UC-36	62
Table 87 UC-37	63
Table 88 Storyboard-1	71
Table 89 Storyboard-2	72

Table 90 Storyboard-3	73
Table 91 Storyboard-4	74
Table 92 Storyboard-5	75
Table 93 Data Dictionary	114
Table 94 auth_users	132
Table 95 auth_roles	132
Table 96 auth_user_roles	132
Table 97 auth_permissions	132
Table 98 auth_api_keys	133
Table 99 sis_departments	133
Table 100 sis_programs	133
Table 101 sis_students	133
Table 102 sis_faculty	134
Table 103 sis_semesters	134
Table 104 sis_transcripts	134
Table 105 sis_classrooms	135
Table 106 sis_classrooms	135
Table 107 lms_sections	135
Table 108 sis_enrollments	135
Table 109 lms_course_materials	136
Table 110 lms_assignments	136
Table 111 lms_submissions	136
Table 112 lms_quizzes	137
Table 113 lms_questions	137
Table 114 lms_answers	137
Table 115 lms_attendance	137
Table 116 lms_timetable_slots	138
Table 117 fin_fee_heads	138
Table 118 fin_invoices	138
Table 119 fin_invoice_items	139
Table 120 fin_transactions	139
Table 121 fin_fines	139
Table 122 lib_books	139
Table 123 lib_issues	140
Table 124 ops_leaves	140
Table 125 ops_grievances	140
Table 126 alumni_registry	141
Table 127 alumni_jobs	141
Table 128 chat_sessions	141
Table 129 sys_notifications	141
Table 130 chat_messages	144
Table 131 audit_trails	144
Table 132 media_assets	145
Table 133 notifications	145
Table 134 analytics_events	146

Table 135 content_announcements	146
Table 136 system_logs	147
Table 137 feedback_surveys	147
Table 138 User Sessions.....	148
Table 139 Attendance Status	148
Table 140 Course Grades Cache	148
Table 141 API Rate Limiting	149
Table 142 CGPA Leaderboard.....	149
Table 143 Notification Queue	149
Table 144 OTP Codes	149
Table 145 Cached SQL Queries	149
Table 146 Feature Flags	149
Table 147 vectors_face_biometrics.....	150
Table 148 vectors_student_transcripts	150
Table 149 vectors_course_materials	151
Table 150 vectors_university_policies.....	151
Table 151 vectors_faq_knowledge.....	151
Table 152 vectors_assignment_submissions.....	151
Table 153 Login & Account Recovery (User Authentication).....	158
Table 154 Fee Payment & Financial Ledger.....	158
Table 155 Academic Course & Transcript View	159
Table 156 AI Chatbot Interaction.....	159
Table 157 AI-Service Algorithm	164
Table 158 AT-Risk Student Algorithm	166
Table 159 Attendance Algorithm.....	167
Table 160 Timetable Generation	169
Table 161 External API/SDK	170
Table 162 UT-01 — JWT Token Generation & Hashing.....	173
Table 163 UT-02 — EAR Blink Liveness Detection.....	174
Table 164 FT-01 — Course Registration Workflow	174
Table 165 FT-02 — Fee Payment Online.....	174
Table 166 FT-03 — Query AI Chatbot.....	175
Table 167 FT-04 — Master Timetable Generation	175
Table 168 FT-06 Quiz Creation and Automated Grading.....	176
Table 169 IT-01 — Sync Grades via Kafka	176
Table 170 PT-01 — AI Chatbot Response Latency	177
Table 171 PT-02 — Real-Time Chat Delivery.....	177
Table 172 Project Objectives Evaluation	185
Table 173 Requirements Traceability Matrix.....	185

List of Figures

Figure 1 Gantt Chart	15
Figure 2 Student Academic System	64
Figure 3 LMS & Analytics Module	65
Figure 4 Financial System.....	66
Figure 5 AI & Chat Service.....	67
Figure 6 HR & Admin System.....	68
Figure 7 Library & Career Portal.....	69
Figure 8 System Interaction Context	70
Figure 9 AI-Chat	71
Figure 10 AI-Analysis & Prediction	72
Figure 11 Smart Attendance.....	73
Figure 12 Online Quiz Attempt.....	74
Figure 13 Automated Timetable.....	75
Figure 14 High Level Architecture Diagram	85
Figure 15 UML Component Diagram [High-Level Layered Architecture]	89
Figure 16 UML Component Diagram [External System Integrations].....	90
Figure 17 UML Component Diagram [Microservices Component Decomposition]	91
Figure 18 UML Component Diagram [Polyglot Data Persistence Layer].....	92
Figure 19 UML Deployment Diagram.....	93
Figure 20 UML Class Diagram [High Level System Map].....	95
Figure 21 UML Class Diagram [User Inheritance Hierarchy].....	96
Figure 22 UML Class Diagram [Communication & Notification Entities].....	97
Figure 23 UML Class Diagram [Faculty Operational Model.].....	98
Figure 24 UML Class Diagram [LMS Content Structure].....	99
Figure 25 UML Class Diagram [Student Academic Relations].....	100
Figure 26 UML Class Diagram [Financial Transaction Model]	101
Figure 27 Sequence Diagram [Asynchronous Grade Synchronization]	102
Figure 28 Sequence Diagram [Marking Attendance]	103
Figure 29 Sequence Diagram [Hybrid AI Chatbot (RAG Pipeline)]	104
Figure 30 Sequence Diagram [Fee Payment]	105
Figure 31 Sequence Diagram [At- Risk Student Predication]	106
Figure 32 Sequence Diagram [Real-Time Chat & Offline Push Notification]	107
Figure 33 State Transition Diagram [Fee Submission]	108
Figure 34 State Transition Diagram [Students Complaints]	109
Figure 35 State Transition Diagram [Leave Request].....	110
Figure 36 State Transition Diagram [Assignment Submission].....	111
Figure 37 State Transition Diagram [AI Chatbot Session Lifecycle]	112
Figure 38 State Transition Diagram [Library Book Status Lifecycle].....	113
Figure 39 Identity & SIS Schema	129
Figure 40 LMS & Assessment Schema.....	130
Figure 41 Financial Billing Schema.....	131
Figure 42 Operations & Support Schema	131
Figure 43 MongoDB Collections.....	143
Figure 44 Redis Cache Patterns	148

Figure 45 Chroma DB Vector Store	150
Figure 46 Unified Login Page.....	153
Figure 47 Forget Password.....	154
Figure 48 Fee Payment	154
Figure 49 Payment Processing.....	155
Figure 50 Payment Successful	155
Figure 51 Course Page	156
Figure 52 Transcript Page	156
Figure 53 AI-Powered Chat Assistant	157
Figure 54-Admin Dashboard	195
Figure 55-Department& Course Management.....	195
Figure 56-Grievance Management.....	196
Figure 57-My Assignments	196
Figure 58- Nexus-AI	197
Figure 59-Fee Receipt.....	197

Chapter No.1

Introduction

1. Introduction

This chapter provides a comprehensive overview of Project Nexus. The primary purpose of this chapter is to clearly define the fundamental motivations behind the project, outline the specific problem domain, and present the proposed technical solution. By establishing quantifiable objectives, defining the precise scope and system components, and analyzing related literature, this chapter lays the foundational groundwork necessary to achieve the project's goal of delivering a unified, intelligent campus platform.

1.1 Problem Statement

Modern educational institutions currently operate in a highly fragmented digital environment. They typically rely on entirely disconnected systems to manage administrative records and academic learning. The administrative side utilizes Student Information Systems (SIS) for enrollments, fee management, and official transcripts. Conversely, the academic side relies on Learning Management Systems (LMS) for course materials, assignments, and grading. Because these two domains do not natively communicate, they create critical, isolated "data silos." Valuable insights regarding a student's daily engagement remain trapped within the LMS, while financial records are locked within the SIS. This severe separation forces faculty to rely on tedious, manual data transfers (glue work), wasting thousands of institutional man-hours and introducing human error. Most importantly, this fragmented infrastructure enforces a strictly reactive model of student support—creating an "opportunity lag" where universities only identify struggling students after they have already failed.

1.2 Problem Solution

Project Nexus resolves this fragmentation by establishing a Unified Intelligent Campus Platform. The proposed software serves as a cohesive ecosystem that natively merges both administrative and academic operations into a single interface. By employing a Polyglot Microservices Architecture and an Apache Kafka event bus, the system ensures absolute zero data loss for critical operations. This deep integration allows the system to establish a powerful "data flywheel" effect: real-time academic data from the LMS immediately informs the administrative and intelligence layers. To modernize support, the system incorporates a Hybrid AI Chatbot (CAG+RAG) that securely queries live records to provide context-aware answers. Furthermore, the solution replaces manual roll-calls with a Spoof-Proof Biometric Attendance system utilizing computer vision, and integrates an analytics layer to proactively predict and flag at-risk students before academic failure occurs.

1.3 Objectives

The main objectives of Project Nexus are to:

1. **Establish a Microservices Architecture**

Build the system as independent services linked by an event bus to ensure stability and zero data loss.

- **Containerization:** Use Docker to run SIS, LMS, and AI as separate apps so one crash doesn't stop the system.
- **Event Automation:** Use **Apache Kafka** to instantly sync grades and payments across modules without manual entry.
- **Scalability:** Design the backend to easily handle thousands of students simultaneously.

2. Develop a Unified Administrative & Financial Core

Digitize and automate the daily business operations in one central portal.

- **Student Hub (SIS):** Manage profiles, course registration, and digital transcripts.
- **Smart Finance:** Auto-generate fee vouchers and process online payments instantly.
- **Smart Grievance:** Use AI to prioritize urgent complaints and route them to the right department.

3. Build a Comprehensive Academic Ecosystem

Create a seamless digital environment for teaching, learning, and scheduling.

- **Next-Gen LMS:** Upload content, manage assignments, and block late submissions automatically.
- **Intelligent Timetabling:** Auto-generate conflict-free class schedules based on room capacity.
- **Digital Library:** Provide a searchable catalog of e-books and manage issuance.
- **Faculty Portal:** Give teachers a single dashboard for grading, attendance, and advising.

4. Deploy a Context-Aware AI Intelligence Layer

Integrate **Google Gemini** and a **Vector Database** to provide personalized, 24/7 assistance.

- **Hybrid Chatbot:** Answer queries like "What is my GPA?" using live database records.
- **Personal Tutor:** Recommend study videos and other material.
- **Instant FAQs:** Use caching to answer general policy questions in milliseconds.

5. Implement Proactive Student Success & Career Analytics

Move from reactive record-keeping to proactive student support using Machine Learning.

- **Risk Prediction:** Flag students with low attendance or grades automatically.
- **Early Intervention:** Send alerts to faculty before a student fails the course.

6. Ensure Secure & Real-Time Campus Operations

Modernize campus security and communication using biometric and real-time technologies.

- **Spoof-Proof Attendance:** Verify presence using Face Recognition[7], [8], Liveness Detection, and GPS.
- **P2P Communication:** Enable secure, instant messaging between students and teachers.
- **Real-Time Alerts:** Send instant notifications for grade updates, fee deadlines, and campus news.

1.4 Scope

The scope of this project is to design, develop, and deploy a comprehensive, production-ready software ecosystem.

1.4.1 Inclusions (In Scope)

- **Full-Stack Development:** The project will deliver a fully functional, web-based intelligent campus platform unifying SIS, LMS, Finance, and AI support.
- **Microservices Deployment:** Complete containerization of all services using **Docker** to ensure the system is portable and scalable.
- **Unified Web Platform:** Creation of a single, mobile-responsive web interface (React.js) that adapts to Students, Faculty, and Admin roles.
- **Proof of Concept:** Demonstration of the system using a robust dataset of 50-100 test users.
- **Deliverables:** 14 containerized backend microservices, a React-based Single Page Application (SPA), deployed databases, and complete system documentation.
- **Feature Priorities:** Priority 1 is the core SIS/LMS integration and Kafka event bus. Priority 2 is the AI Chatbot and Biometric Attendance. Priority 3 includes secondary modules like the Alumni portal and Library management.
- **Milestones:** Divided into four phases: **Frontend Prototype**, **Core Backend Integration**, **Intelligence Layer Activation**, and **Final Deployment/Validation**.

1.4.2 Exclusions (Out of Scope)

- **Legacy Data Migration:** We will not migrate millions of old paper records from the university's legacy systems. We will use fresh data for the demonstration.
- **Native Mobile Apps:** We are building a Progressive Web App (PWA) accessible via mobile browsers. We will not develop separate installable apps (.apk or .ipa) for the Play Store or App Store.

- **Hardware Integration:** The attendance system relies entirely on software (Computer Vision). We will not integrate with physical biometric thumb scanners or turnstiles.

1.4.3 Optional (If Time Permits)

- **Voice-Enabled Chatbot:** Adding a "Speech-to-Text" feature so students can talk to the AI Assistant instead of typing.
- **Blockchain Verification:** Implementing a basic blockchain ledger to make digital transcripts tamper-proof.
- **Multi-Language Support:** Adding an Urdu translation layer for the student portal interface.

1.5 System Components

The platform is organized by system types to efficiently serve all user classes through the unified web portal.

1.5.1 Client Web App (Frontend)

- Built with React.js, Vite, and Material UI (MUI).
- Provides 60+ role-based routes and dashboards for Students, Teachers, Admins, Librarians, and Alumni.

1.5.2 Administrative & Finance Services

- **Auth Service:** Secure JWT-based Role Authentication and session management.
- **SIS Service:** Core Student Information System for enrollment and transcripts.
- **Finance Service:** Automated fee voucher generation and Stripe online payment.
- **HR Service:** Encrypted salary management and leave applications.
- **Operations & Alumni Services:** Smart grievance routing and alumni network management.

1.5.3 Academic & Communication Services

- **LMS Service:** Course materials, assignments, and automated quiz grading.
- **Scheduler Service:** Constraint-based academic scheduling (OR-Tools).
- **Chat & Notification Services:** Real-time WebSocket P2P messaging and push alerts.
- **Library Service:** QR-based book issuance and fine calculation.

1.5.4 Intelligence & Biometric Services

- **AI Service:** Hybrid CAG+RAG AI Chatbot acting as a 24/7 personal campus assistant.
- **Analytics Service:** Predictive Dashboard utilizing a RandomForest classifier.

- **Attendance Service:** Biometric attendance utilizing computer vision for facial recognition.

1.6 Related System Analysis/Literature Review

Existing software solutions in the educational technology market suffer from distinct feature gaps, generally operating as either purely administrative tools or purely academic tools without intelligent integration.

Table 1 Analysis

Application Name	Weakness	Proposed Project Solution
OpenEduCat & ERPNext	Strictly focused on administration. They lack native pedagogical (LMS) tools and cannot track daily student classroom engagement, missing data needed for student success prediction.	Provides a unified ecosystem where the SIS and LMS share the exact same database and event bus, allowing administrative and academic data to merge seamlessly for a holistic view.
Moodle & Google Classroom	Heavily focused on academics. They lack native financial ledgers, automated fee vouchers, and HR capabilities, operating as isolated data silos.	Utilizes a microservices architecture to natively tie LMS assignment and grading data directly into the financial and SIS core via Kafka, guaranteeing zero manual data entry.
Traditional NLU (e.g., Rasa)	Rely on strict keyword matching and heavy manual training. They cannot accurately query live student records to answer dynamic, personal questions.	Implements a Hybrid CAG+RAG system using the Google Gemini API to securely "read" personal student records, delivering accurate, context-aware responses instantly.

1.7 Vision Statement

For universities and higher education institutions in Pakistan and the broader developing world, **Who** are burdened by disconnected administrative and academic systems that create data silos, waste institutional resources, and prevent timely support for struggling students, **The Project Nexus: Unified Intelligent Campus Platform** **Is a** cloud-ready, AI-powered university management ecosystem **That** integrates student information management, learning management, financial operations, HR workflows, biometric attendance, intelligent advising, and alumni networking into a single unified platform. **Unlike** traditional fragmented solutions such as standalone SIS tools and separate LMS platforms that require manual data transfer between systems, **Our product** eliminates the data silo problem entirely by building a natively unified ecosystem where administrative data and academic data share a single intelligent backbone transforming the university from a passive record-keeper into an active, proactive partner in every student's success.

1.8 System Limitations and Constraints

- **Legacy Data Migration:** We will not migrate millions of old paper records from the university's legacy systems. We will use fresh data for the demonstration.
- **Native Mobile Apps:** We are building a Progressive Web App (PWA) accessible via mobile browsers. We will not develop separate installable apps (.apk or .ipa) for the Play Store or App Store.
- **Hardware Integration:** The attendance system relies entirely on software (Computer Vision). We will not integrate with physical biometric thumb scanners or turnstiles.
- **Server Resources:** The backend infrastructure requires a minimum of **16GB RAM** and a **Multi-Core Processor (i5/i7)** to simultaneously host resource-intensive microservices, particularly the Apache Kafka event bus and AI processing containers.
- **System Latency:** Standard user interactions, such as loading the Dashboard or viewing a Profile, must complete in under **200 milliseconds** to ensure a smooth user experience.
- **Real-Time Delivery:** Chat messages and critical push notifications (e.g., "Grade Posted") must be delivered to the recipient's screen in under **1 second**.
- **AI Response Time:** Simple, cached AI queries (CAG) must load in under **50 milliseconds**. Complex, database-driven queries (RAG) must begin streaming results to the user within **3-4 seconds**.
- **Biometric Privacy:** To protect user privacy, the system is strictly forbidden from storing raw photographs of students. It must only store **128-dimensional mathematical vectors** (numerical representations of a face), which cannot be reverse-engineered into an image.
- **Encryption:** All data transmission must occur over secure **HTTPS (TLS 1.2+)** channels. Sensitive fields at rest, such as Faculty Salary, must be encrypted in the database.
- **Session Management:** To prevent unauthorized access on public computers, user login sessions must automatically expire after **30 minutes of inactivity**.

1.9 Tools, Libraries & Technologies With Reasoning

1.9.1 Technology Stack Overview

The technical architecture of Project Nexus is built on a modern, industry-standard stack designed for high scalability and fault tolerance. We have adopted a **Microservices-First** approach, utilizing **Docker Swarm** for orchestration and **Apache Kafka** for durable, event-driven communication. The backend leverages the speed of **FastAPI** (Python) for AI and

operational tasks, while the frontend is built on **React.js** to ensure a responsive, mobile-first user experience. The data layer employs a **Polyglot Persistence** strategy, using the best database for each specific data type (Relational, Document, and Vector).

1.9.2 Hardware Requirements

Given the complexity of running microservices (including AI models, Kafka, and Databases), hardware resource management is critical. We will utilize **Docker Swarm** to cluster our development laptops, pooling resources to run the full system.

Table 2 Hardware Requirements

Component	Minimum Requirements	Recommended Specifications
Development Node	• Processor: Intel Core i5 (3rd Gen) or AMD Ryzen 5 • RAM: 8GB - 12GB • Storage: 256GB SSD/Hard drive	• Processor: Intel Core i7 • RAM: 16GB - 32GB • Storage: 512GB NVMe SSD
Production Server	• vCPU: 4 Cores • RAM: 16GB • Network: 1 Gbps	• vCPU: 8+ Cores • RAM: 32GB (To handle concurrent AI & Kafka) • Storage: 1TB SSD
Client Device	Any smartphone or computer with a modern web browser (Chrome, Edge, Safari) with Internet.	4G/5G or Wi-Fi connection for real-time WebSocket features (Chat/Notifications).

1.9.3 Tools

We selected these industry-standard tools to ensure efficient collaboration and a consistent development environment across our team.

Table 3 Tools

Tools	Version	Rationale
Visual Studio Code	Latest	(Core Editor) A lightweight, powerful, and extensible code editor. Its rich ecosystem of extensions for Python, React.js, Docker, and database management makes it the ideal all-in-one IDE for managing this project's diverse services.
Git / GitHub	Latest	(Version Control) The industry standard for source code management. It is critical for a multi-person team, allowing for parallel development, branch management, and a complete, auditable history of all code changes.
Docker (with Swarm & Compose)	Latest	(Containerization & Orchestration) Essential for a microservices architecture. We will use Docker Swarm [9] mode to create a development cluster across our team's laptops. This allows us to distribute the heavy resource load of 10+ microservices (AI, Kafka, DBs) instead of overwhelming a single machine.

1.9.4 Libraries

These modern libraries accelerate development by providing the essential components for our frontend interface, AI orchestration, and real-time backend processing.

Table 4 Libraries

Libraries	Version	Rationale
FastAPI	Latest	(API Framework) A modern, high-performance Python framework. Chosen over Django for its raw speed (comparable to Node.js) and native support for asynchronous programming , which is essential for our AI and real-time chat microservices[10].
React.js	Latest	(UI Library) Chosen for its vast ecosystem and proven scalability for large, data-intensive, enterprise-grade applications. Its component-based architecture is perfect for building the many complex and reusable dashboards required for our various user roles.
LangChain	Latest	(AI Orchestration) This is the "brain" that connects our Google Gemini model to our data. It manages the logic to decide when to use CAG for static, cached data (fast responses) and when to use RAG via the Vector Database for dynamic answers.
Scikit-learn & Pandas	Latest	(Predictive Analytics) The industry-standard Python libraries for machine learning. We will use Pandas for data analysis and Scikit-learn to train and evaluate the classification models needed for the "Student Success Prediction" module[11].
OpenCV & face_recognition	Latest	(Computer Vision) These are the Python libraries for our CV-based attendance system. OpenCV is used for processing the real-time video feed and performing "liveness detection." The face_recognition library (built on dlib) is a highly accurate model for matching the student's selfie to their profile photo[8].

1.9.5 Technologies

Our core stack supports a distributed microservices architecture, utilizing high-performance languages and specialized databases for optimal scalability.

Table 5 Technologies

Technology	Version	Rationale
Python	3.10+	(Core Language) Chosen as the primary backend language because it is the undisputed standard for AI and Data Science . It gives us direct, native access to all required AI/CV and data analysis libraries
WebSockets	(Protocol)	(Real-time Communication) This protocol is required for the Peer-to-Peer Chat module. Unlike traditional HTTP, it provides a persistent, bidirectional communication channel for instant message delivery[1].
PostgreSQL	15+	(Relational Database) Our "single source of truth" for all structured, transactional data. Chosen for its strict ACID compliance and data integrity, which is non-negotiable for handling financial records, student grades, and official enrollments
MongoDB	6+	(NoSQL Database) Part of our "polyglot persistence" strategy. Chosen for its flexible schema, which is ideal for storing the unstructured data from our LMS content modules and the millions of messages from the P2P chat logs [12], [13].
Redis	7+	(In-Memory Cache / Session Store) A high-speed, in-memory key-value store used for Caching (to reduce database load and increase speed) and Session Management (to store user login tokens across all microservices)[14].
ChromaDB	Latest	(Vector Database) The specific database used for our AI's "Long-Term Memory." It stores the vector embeddings of student handbooks and policies locally within our Docker cluster. We chose ChromaDB because it is open-source, lightweight, and integrates natively with LangChain, unlike cloud-only options like Pinecone.
Google Gemini API	(API)	(Generative AI Model) The state-of-the-art model for our AI Chatbot Assistant. We chose this over traditional NLU frameworks (like Rasa) because it provides far more powerful conversational abilities and natively supports the caching features needed for our Hybrid CAG+RAG model[15].
Apache Kafka	Latest	(Durable Event Bus) The "central nervous system" for our automation. Chosen over lighter alternatives (like Redis Pub/Sub) for its durability . It stores critical events (like grades and payments) on disk, <i>guaranteeing</i> zero data loss even if a service is temporarily offline[16].

1.9.6 Analysis of Alternatives

This matrix explains our strategic choices, highlighting the specific performance and scalability advantages of our selected stack over common alternatives.

Table 6 Technology Justification Matrix

Component / Decision	Chosen Technology	Alternatives Considered	Justification for Choice
Backend Framework	FastAPI	Django, Node.js (Express)	Performance & Concurrency: FastAPI is built on ASGI (asynchronous) principles, making it significantly faster than Django (WSGI) and ideal for handling concurrent AI and WebSocket connections. Its speed is comparable to Node.js, but it allows us to stay within the Python ecosystem, which is essential for our AI libraries.
Frontend Framework	React.js	Vue.js, Angular	Scalability & Ecosystem: For a large, enterprise-grade application with complex state and numerous dashboards, React.js is the most strategic choice. Its vast ecosystem, community support, and proven scalability make it a more robust selection than Vue.js for a project of this magnitude.
Relational Database	PostgreSQL	MySQL	Data Integrity & Features: While MySQL is common, PostgreSQL is widely regarded as superior for data integrity (full ACID compliance), concurrent performance, and its rich set of advanced data types and indexing capabilities, which are beneficial for complex queries in our SIS[28].
NoSQL Database	MongoDB	PostgreSQL (JSONB)	Schema-less Flexibility: While PostgreSQL's JSONB is capable, MongoDB is a purpose-built document database. It is optimized for the flexible, unstructured, and high-

			volume data generated by our LMS content modules and, most importantly, our P2P chat logs.
Caching / Session Store	Redis	Memcached	Data Structures & Versatility: Memcached is a simple memory cache. Redis, however, is a multi-model in-memory data store. It not only handles caching and session management but also provides rich data structures (like lists and hashes) and can act as a lightweight message broker (Pub/Sub), making it far more versatile for our needs.
Vector Database	ChromaDB	Pinecone, Weaviate	Local & Privacy-First: Pinecone is a cloud-only service, which introduces latency and privacy risks for student data. ChromaDB runs entirely locally in our Docker container. It allows us to perform high-speed semantic search without sending sensitive data to a third-party cloud provider.
AI Chatbot Framework	Hybrid (CAG+RAG) with Gemini	Rasa	Resource, Cost, & Capability: Rasa is a heavy, self-contained NLU framework requiring significant training and high-end hardware (often 8GB-16GB RAM for itself). The Hybrid CAG+RAG approach is more modern and efficient. It uses CAG for instant, low-cost answers to static questions and RAG to fetch dynamic, real-time data, providing superior, more accurate answers.
Automation/Event Bus	Apache Kafka	Redis Pub/Sub	Durability & Guaranteed Delivery: Redis Pub/Sub is a "fire-and-forget" system, if a consuming service is offline, the message is lost forever. For critical events like "Grade Submitted" or "Payment Processed," this is unacceptable. Kafka is a durable, persistent log that stores messages on disk, <i>guaranteeing</i> that the SIS will

			receive the event, ensuring zero data loss.[16]
Deployment Strategy	Docker Swarm	Kubernetes	Resource Efficiency: Kubernetes is too heavy for our development laptops (requires high RAM). Docker Swarm is lightweight, built into Docker, and allows us to easily cluster our 3 laptops to run the full system without crashing.

1.10 Project Deliverables

- FYDP Proposal.
- Software Requirements Specification (SRS) Document.
- Software Design Specification (SDS) Document.
- Complete Source Code Repository hosted on GitHub (Frontend and Backend Microservices).
- Operational Prototypes tested with 50users.
- Automated Deployment Scripts (Docker Compose files).
- Final Year Design Project Report.

1.11 Project Planning

The project execution was organized using Agile methodologies and mapped on a 7-phase Gantt chart to ensure all deliverables were met on time:

- **Phase 1: Proposal & Topic Selection**
- **Phase 2: Requirement Engineering (SRS)**
- **Phase 3: System Design (SDS)**
- **Phase 4: Prototype Development**
- **Phase 5: Core Implementation (React Dashboards, Backend Services, AI & CV)**
- **Phase 6: Testing & Quality Assurance (Unit, Integration, and Docker Deployment)**
- **Phase 7: Final Report & FYDP Defense**

1.11.1 Project Timeline

This timeline maps our project phases against the university's academic calendar. It is tentative.

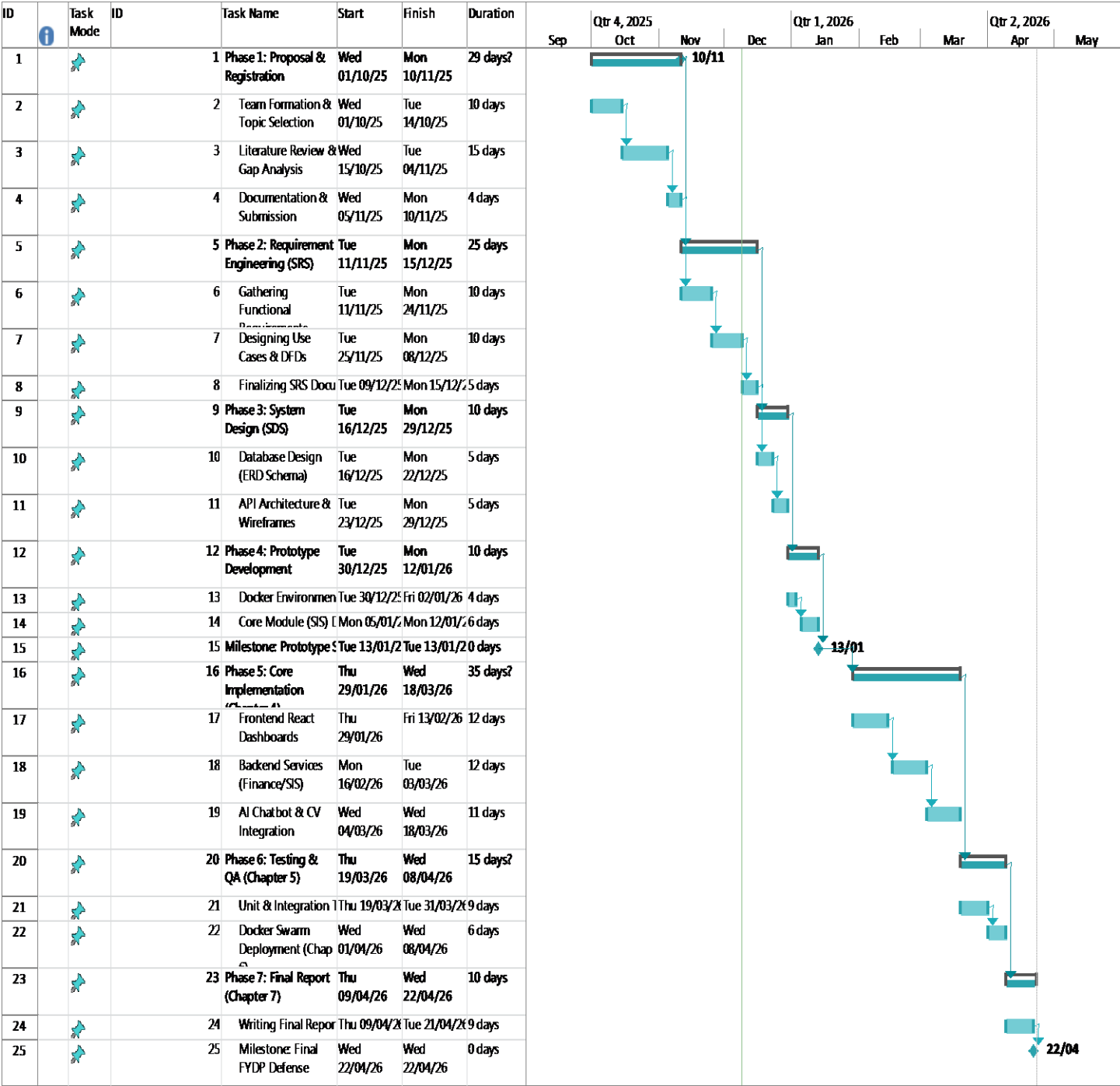


Figure 1 Gantt Chart

1.12 Summary

This chapter introduced Project Nexus., a new software platform designed to solve a major problem facing universities today: the separation of administrative and academic systems. Currently, institutions are forced to use completely different software for managing student records (SIS) and for teaching courses (LMS). This creates "data silos," where important information is trapped in one system and cannot be seen by the other, leading to wasted time and missed opportunities to help students.

Project Nexus aims to replace this fragmented approach with a Single Unified Ecosystem. By bringing fees, grades, attendance, and learning materials into one connected platform, we can transform the university from a passive record-keeper into a proactive partner in education. The proposed solution is built on a modern, high-tech foundation using **Microservices**, which breaks the huge system into smaller, manageable parts to ensure it never crashes completely. We have also chosen **Docker Swarm** to manage these services easily and **Apache Kafka** to guarantee that critical data, like grades and payments, is never lost during a network failure.

Furthermore, Project Nexus introduces advanced intelligence to the campus. Instead of old-fashioned support, we are deploying a **Hybrid AI Chatbot** powered by Google Gemini that can answer personal questions (like "What is my GPA?") instantly. By combining this advanced technology with a clear set of objectives and a secure, spoof-proof attendance system, Project Nexus is fully prepared to modernize how the university operates, making it more efficient for staff and more helpful for students.

Chapter No. 2

Requirements Analysis

2. Requirements Analysis

It translates the high-level goal of creating a "Unified Intelligent Campus" into specific, actionable software requirements.[23] By clearly identifying the needs of students, faculty, and administrators, this analysis serves as the blueprint that guides the design and coding of the system. The following subsections detail these requirements in a structured manner

2.1 User classes and characteristics

This section identifies the different groups of people who will interact with the platform. Since a university ecosystem includes a wide variety of users ranging from tech-savvy students to administrative staff who may be used to manual paperwork it is crucial to distinctively understand each group. Identifying their specific roles, responsibilities, and technical comfort levels ensures that we design the appropriate dashboards and features to meet their unique needs.

Table 7 User Classes & Characteristics

User Class	User Characteristics
Student	The primary end-users. They are generally tech-savvy and will access the system primarily via mobile devices. They require quick, real-time access to grades, attendance, and fee vouchers. They value speed and ease of use (minimal clicks).
Faculty Member	Academic staff responsible for teaching and grading. They range from highly technical to non-technical users. They need efficient, automated tools to reduce manual data entry (e.g., grading, attendance) and clear dashboards to monitor student progress.
Administrator (Finance/Admin/HOD)	The Administrator class unifies Finance Officers, HODs, and System Admins into a single role. This group requires a robust interface for high-volume data entry (fees/admissions), managerial oversight (approving leaves/timetables), and technical system monitoring (logs and health metrics).
Librarian	Staff responsible for managing physical resources. They require a specialized interface optimized for speed to perform rapid book issuance and inventory management. They need instant access to student validity status to prevent issuing books to defaulting students.
Alumni	Graduated students who access the system occasionally. Their primary goal is networking, job posting, and staying updated with university events. They require a simple, engagement-focused interface.

2.2 Requirement Identifying Technique

This section explains how we identified and documented what the system must do. We used three main methods to capture all necessary functions clearly and from the user's perspective. By combining these techniques, we ensured that Project Nexus solves real-world problems rather than just copying existing software.

2.2.1 Gap Analysis

Since many universities already use separate software for classes (LMS) and records (SIS), we started by analyzing these existing systems.[24] We looked at open-source platforms like Moodle[2] and OpenEduCat see what they do well and, more importantly, what they lack[25].

- **The Result:** This technique revealed the "Data Silo" problem the lack of communication between systems. This directly led to our core requirement for **Event-Driven Automation (FR-16)**, ensuring that Project Nexus connects these disconnected functions into one unified platform.

2.2.2 Use Case Analysis

To understand exactly how people will interact with the system, we created detailed scenarios called "Use Cases." We asked questions like: "What exact steps does a student take to pay a fee?" or "What happens if a teacher tries to grade an assignment after the deadline?"

- **The Result:** This helped us define the specific inputs, outputs, and rules for every feature. For example, the **"Mark Attendance" (UC-12)** use case helped us realize we needed GPS verification, not just facial recognition, to prevent cheating.

2.2.3 Storyboarding

Because Project Nexus is intended to be used on mobile phones by students, text descriptions were not enough. We sketched visual sequences (storyboards) to map out the user journey screen-by-screen.

- **The Result:** Drawing these screens helped us refine the user interface requirements. For instance, the storyboard for the **AI Chatbot** showed us that students need a "quick reply" button for common questions, rather than typing everything out manually. This ensures the app is easy to use and intuitive.

2.3 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature.

2.3.1 FR-01: User Registration and Account Creation

Table 8 User Registration & Account Creation [FR-01]

Field	Detail
Identifier	FR-01
Title	User Registration & Account Creation
Requirement	The system shall provide an interface for the Administrator to register new users (Students, Faculty, Staff) by entering their initial details (Name, Roll Number, Email, Role). The system must store these credentials securely in the database.
Source	Admission / HR Department
Rationale	To ensure that only valid university members are granted access to the system.
Business Rule	Uniqueness Constraint: The Email Address and Roll Number must be unique across the entire system. Duplicate entries must be rejected.
Dependencies	None
Priority	High

2.3.2 FR-02: User Login and Authentication

Table 9 User Login & Authentication [FR-02]

Field	Detail
Identifier	FR-02
Title	User Login & Authentication
Requirement	The system shall allow registered users to log in using their unique identifier (Email/Roll No) and Password. The system must verify the credentials against stored hashes before granting access.
Source	Security Standard
Rationale	To prevent unauthorized access to sensitive academic and financial data.
Business Rule	Account Locking: The system must temporarily lock a user account for 15 minutes after 5 consecutive failed login attempts to prevent brute-force attacks.
Dependencies	FR-01
Priority	High

2.3.3 FR-03: Secure Session Management

Table 10 Secure Session Management (JWT) [FR-03]

Field	Detail
Identifier	FR-03

Title	Secure Session Management (JWT)
Requirement	Upon successful login, the system shall generate a secure JSON Web Token (JWT) [26]. This token must be included in the header of all subsequent API requests to identify the user session without requiring re-login.
Source	System Architect
Rationale	To enable a seamless "Single Sign-On" experience across the microservices (SIS, LMS, Finance).
Business Rule	Session Timeout: For security, the session token must automatically expire after 30 minutes of inactivity , requiring the user to log in again.
Dependencies	FR-02
Priority	High

2.3.4 FR-04: Role Based Access Control

Table 11 RBAC [FR-04]

Field	Detail
Identifier	FR-04
Title	Role-Based Access Control (RBAC)
Requirement	The system shall restrict access to specific features based on the user's role (Student, Faculty, Admin). API endpoints must validate the user's role before processing any request.
Source	Security Policy
Rationale	To ensure data integrity (e.g., a Student cannot access the "Post Grade" feature).
Business Rule	Least Privilege: Users must only have access to data strictly required for their role. (e.g., A Faculty member can only modify grades for their own assigned courses).
Dependencies	FR-02
Priority	High

2.3.5 FR-05: Student Profile Management

Table 12 Student Profile Management [FR-05]

Field	Detail
Identifier	FR-05
Title	Student Profile Management
Requirement	The system shall allow the Admission Office to manage comprehensive student profiles (Demographics, Contact Info, Guardian Details). Students shall have "Read-Only" access to view their own profile.
Source	Admission Office
Rationale	To maintain accurate records for transcripts and official communication.
Business Rule	Data Immutability: Critical fields like "Roll Number" and "CNIC" cannot be edited once the profile is finalized, except by a Super Admin.
Dependencies	FR-04
Priority	High

2.3.6 FR-06: Semester Course Registration

Table 13 Semester Course Registration [FR-06]

Field	Detail
Identifier	FR-06
Title	Semester Course Registration
Requirement	The system shall allow active students to view the list of offered courses for the current semester and register for specific sections. The system must validate the selection against the timetable to prevent conflicts.
Source	Academic Regulations
Rationale	To digitize the enrolment process and ensure students are assigned to valid classes.
Business Rule	Credit Hour Limit: A student cannot register for more than 18 Credit Hours in a regular semester unless they have a GPA > 3.5 and HOD approval.
Dependencies	FR-05 (Student Profile)
Priority	High

2.3.7 FR-07: Faculty Grade Submission & Sync

Table 14 Faculty Grade Submission & Sync [FR-07]

Field	Detail
Identifier	FR-07
Title	Faculty Grade Submission & Sync
Requirement	The system shall allow Faculty to enter final grades for their assigned courses. Once submitted, the system must permanently store these grades in the Student Information System (SIS) database via the automation service[27].
Source	Examination Department
Rationale	To eliminate manual grade entry errors by clerks and ensure transcripts are updated instantly.
Business Rule	Locking Mechanism: Once a faculty member clicks "Final Submit," grades become Read-Only and can only be modified by the Controller of Examinations.
Dependencies	FR-06 (Registration)
Priority	Critical

2.3.8 FR-08: Digital Transcript Generation

Table 15 Digital Transcript Generation [FR-08]

Field	Detail
Identifier	FR-08
Title	Digital Transcript Generation
Requirement	The system shall allow students to generate and download a provisional "Digital Transcript" (PDF format). The document must include the student's complete grading history, semester-wise GPA, and current CGPA.
Source	Student Services
Rationale	To allow students to access their academic records immediately for internship or job applications.

Business Rule	Financial Hold: The system must block transcript generation if the student has any outstanding dues ("Unpaid" status) in the Financial Ledger.
Dependencies	FR-07 (Grades), FR-10 (Finance)
Priority	High

2.3.9 FR-09: Automated Fee Voucher Generation

Table 16 Automated Fee Voucher Generation [FR-09]

Field	Detail
Identifier	FR-09
Title	Automated Fee Voucher Generation
Requirement	The system shall automatically generate fee invoices for all enrolled students at the start of the billing cycle. The fee amount must be calculated based on the student's department and enrolled credit hours.
Source	Finance Department
Rationale	To automate billing and ensure every student receives an accurate invoice on time.
Business Rule	Late Fee Calculation: If the voucher remains "Unpaid" after the Due Date, the system must automatically add a 5% Late Fee surcharge to the total amount.
Dependencies	FR-05 (Profile)
Priority	High

2.3.10 FR-10: Online Payment and Real Time Ledger

Table 17 Online Payment & Real-Time Ledger [FR-10]

Field	Detail
Identifier	FR-10
Title	Online Payment & Real-Time Ledger
Requirement	The system shall integrate with external payment gateways (Stripe/JazzCash)[28] to process online payments. Upon success, the system must immediately update the student's ledger and change the invoice status to "Paid."
Source	Finance Department
Rationale	To facilitate remote payments and provide students with an instant update of their financial standing.
Business Rule	Transaction Security: All payment transactions must occur over an encrypted HTTPS connection, and no credit card details shall be stored in the university database.
Dependencies	FR-09 (Voucher)
Priority	High

2.3.11 FR-11: Course Content Upload and Organization

Table 18 Course Content Upload & Organization [FR-11]

Field	Detail
Identifier	FR-11
Title	Course Content Upload & Organization

Requirement	The system shall allow Faculty to create course structures and upload learning materials (PDFs, PPTs, Video Links) organized by week or topic. Students must be able to view and download these resources.
Source	Faculty / Academic Dean
Rationale	To serve as a central digital repository for all study materials, replacing physical handouts.
Business Rule	File Constraints: The system must restrict file uploads to PDF, DOCX, and PPT formats only, with a maximum file size of 25MB per file to manage storage costs.
Dependencies	FR-06 (Registration)
Priority	High

2.3.12 FR-12: Assignment Creation and Deadline Management

Table 19 Assignment Creation & Deadline Management [FR-12]

Field	Detail
Identifier	FR-12
Title	Assignment Creation & Deadline Management
Requirement	The system shall allow Faculty to create assignments with a Title, Description, Max Marks, and a specific Due Date & Time . The system must display these deadlines on the student's dashboard.
Source	Faculty
Rationale	To digitize the assessment workflow and clearly communicate expectations to students.
Business Rule	Time Synchronization: All deadlines must be enforced based on the Server System Time , not the user's local device time, to prevent cheating.
Dependencies	FR-11 (Course Content)
Priority	High

2.3.13 FR-13: Student Assignment Submission

Table 20 Student Assignment Submission [FR-13]

Field	Detail
Identifier	FR-13
Title	Student Assignment Submission
Requirement	The system shall allow students to upload assignment solutions before the deadline. The system must timestamp every submission to verify it was submitted on time.
Source	Student
Rationale	To enable paperless submissions and automated tracking of student work.
Business Rule	Late Submission Block: The system must strictly reject any upload attempts made after the deadline (down to the second) unless a specific extension is manually granted by the Faculty.
Dependencies	FR-12 (Assignment Creation)
Priority	High

2.3.14 FR-14: Hybrid AI Chatbot

Table 21 Hybrid AI Chatbot [FR-14]

Field	Detail
Identifier	FR-14
Title	Hybrid AI Chatbot (Universal Access)
Requirement	The system shall provide an AI Chatbot interface for all authorized users (Students, Faculty, Staff) . It must use CAG to answer general FAQs and RAG to query the database for user-specific data (e.g., Grades for Students, Leave Balance for Faculty)[21].
Source	Project Innovation Goal
Rationale	To provide instant support to every university member, not just students.
Business Rule	Data Scope: The AI must strictly limit its answers to the data the current user role is allowed to see (e.g., A Student cannot ask "What is Professor Ahmed's salary?").
Dependencies	FR-03 (Authentication)
Priority	Medium

2.3.15 FR-15: Context Aware Responses

Table 22 Context Aware Response [FR-15]

Field	Detail
Identifier	FR-15
Title	Context-Aware AI Responses
Requirement	The system shall generate personalized responses based on the logged-in user's role. For example, if a Faculty member asks "My Schedule," it shows the Teaching Timetable , if a Student asks, it shows the Class Timetable .
Source	User Feedback
Rationale	To ensure the AI acts as a true personal assistant relevant to the specific user.
Business Rule	Privacy: The AI must never reveal Personal Identifiable Information (PII) of other users during a chat session.
Dependencies	FR-14 (AI Chatbot)
Priority	Medium

2.3.16 FR-16: Biometric Liveness Detection

Table 23 Biometric Liveness Detection [FR-16]

Field	Detail
Identifier	FR-16
Title	Biometric Liveness Detection
Requirement	The system shall capture a real-time video stream of the student during check-in. It must perform a "Liveness Check" (detecting eye blinks or head motion) to confirm the subject is a live person and not a static photo.
Source	Security Policy

Rationale	To prevent "spoofing" attacks where a student uses a photograph of another student to mark fake attendance.
Business Rule	Anti-Spoofing: The system must automatically reject the attendance attempt if the "Liveness Score" is below the required threshold, prompting the user to try again.
Dependencies	FR-02 (Login)
Priority	High

2.3.17 FR-17: GPS Geofencing Verification

Table 24 GPS Geofencing Verification [FR-17]

Field	Detail
Identifier	FR-17
Title	GPS Geofencing Verification
Requirement	The system shall access the GPS coordinates of the student's device at the moment of check-in. It must verify that the coordinates fall within the defined geographical boundary of the university campus or specific classroom.
Source	Administration
Rationale	To ensure that students are physically present on campus and not marking attendance remotely from home.
Business Rule	Radius Constraint: The check-in must only be accepted if the device is within a 100-meter radius of the scheduled classroom's coordinates.
Dependencies	FR-16 (Liveness)
Priority	High

2.3.18 FR-18: Facial Recognition and Auto Marking

Table 25 Facial Recognition and Auto Marking [FR-18]

Field	Detail
Identifier	FR-18
Title	Facial Recognition & Auto-Marking
Requirement	The system shall compare the live selfie captured during check-in against the student's stored profile photo. Upon a successful biometric match, the system must automatically mark the student as "Present" in the database.[8]
Source	Faculty Request
Rationale	To automate the time-consuming manual roll-call process and provide accurate, unforgeable attendance records.
Business Rule	Confidence Match: The facial recognition algorithm must achieve a confidence score of >95% to validate the identity. If lower, the system must request a retake.
Dependencies	FR-17 (GPS), FR-05 (Profile)
Priority	High

2.3.19 FR-19: Event Driven Data Synchronization

Table 26 Event-Driven Data Synchronization [FR-19]

Field	Detail
Identifier	FR-19
Title	Event-Driven Data Synchronization (Kafka)
Requirement	The system shall publish critical events (e.g., "GradePosted", "FeePaid") to the Apache Kafka event bus. Consuming services (like SIS) must listen for these events to update records across the system instantly.[27]
Source	System Architect
Rationale	To ensure that data remains consistent across all microservices without tight coupling (e.g., SIS updates instantly when LMS grades are saved).
Business Rule	Zero Data Loss: All events must be persisted to disk. If a receiving service is offline, the event must be retained in the queue and processed immediately upon system recovery ("At-Least-Once Delivery").
Dependencies	FR-07 (Grades), FR-10 (Payment)
Priority	Critical

2.3.20 FR-20: Real Time Push Notifications

Table 27 Real-Time Push Notification [FR-20]

Field	Detail
Identifier	FR-20
Title	Real-Time Push Notifications
Requirement	The system shall send automated notifications (Push/Email) to students whenever a critical event occurs, such as a new grade being uploaded, a fee voucher being generated, or a leave request being approved.
Source	User Experience (UX) Team
Rationale	To keep students proactively informed so they do not need to constantly refresh the portal to check for updates.
Business Rule	Delivery Latency: Notifications must be triggered within 10 seconds of the source event occurring.
Dependencies	FR-19 (Kafka Events)
Priority	Medium

2.3.21 FR-21: Peer to Peer Chat Initiation

Table 28 P2P-Chat Initiation [FR-21]

Field	Detail
Identifier	FR-21
Title	Peer-to-Peer (P2P) Chat Initiation
Requirement	The system shall allow authorized users (Students, Faculty) to search for other users by Name or Roll Number and initiate a secure 1-to-1 chat session .[1]
Source	Student Feedback / Faculty Request
Rationale	To foster collaboration and allow students to ask questions directly to instructors or peers.

Business Rule	Privacy Barrier: Students may initiate chats with Faculty, but Faculty have the option to "Mute" or "Block" incoming messages from specific students during non-office hours.
Dependencies	FR-02 (Login)
Priority	Medium

2.3.22 FR-22: Real Time Messaging & Persistence

Table 29 Real Time Message [FR-22]

Field	Detail
Identifier	FR-22
Title	Real-Time Messaging & Persistence
Requirement	The system shall facilitate instant message delivery using WebSockets . All message history must be persisted in a MongoDB database, allowing users to view previous conversations upon re-login.[1], [12], [13]
Source	System Architect
Rationale	To provide a seamless communication experience comparable to modern apps like WhatsApp.
Business Rule	Data Retention: Chat logs must be retained for a minimum of One Academic Year for auditing purposes in case of harassment complaints.
Dependencies	FR-21 (Chat Initiation)
Priority	Medium

2.3.23 FR-23: Group Chat Creation

Table 30 Group Chat Creation [FR-23]

Field	Detail
Identifier	FR-23
Title	Group Chat Creation
Requirement	The system shall support the creation of Group Chats (e.g., "CS101 Study Group"). Users can add members, and messages sent to the group must be broadcast to all participants instantly.
Source	Student Council
Rationale	To enable collaborative learning and official class announcements.
Business Rule	Group Limits: To maintain performance, a single group chat cannot exceed 100 participants . Larger announcements should use the "Notification" system instead.
Dependencies	FR-22 (Messaging)
Priority	Medium

2.3.24 FR-24: Timetable Constraints Definition

Table 31 Timetable Constraints [FR-24]

Field	Detail
Identifier	FR-24
Title	Timetable Constraint Definition

Requirement	The system shall allow the Head of Department (HOD) to define Hard Constraints (e.g., Room Capacity, Faculty Availability, Fixed Lab Hours) for the semester schedule.
Source	HOD / Admin
Rationale	To input the necessary logic required by the scheduling algorithm to produce a valid timetable.
Business Rule	Resource Validation: The system must reject any constraint configuration where the number of required classes exceeds the total available room-hours.
Dependencies	FR-04 (Role Access)
Priority	High

2.3.25 FR-25: Automated Conflict-Free Scheduling

Table 32 Conflict Free Scheduling [FR-23]

Field	Detail
Identifier	FR-25
Title	Automated Conflict-Free Scheduling
Requirement	The system shall utilize a constraint-satisfaction algorithm to automatically generate a master timetable. It must ensure no faculty or student section is assigned two classes in the same time slot.
Source	Academic Office
Rationale	To eliminate the manual effort of resolving schedule clashes and optimize room usage.
Business Rule	Conflict Zero-Tolerance: The generated schedule must have zero overlapping slots for any single resource (Teacher, Room, or Student Section).
Dependencies	FR-24 (Constraints)
Priority	High

2.3.26 FR-26: Digital Leave Application

Table 33 Leave Application [FR-26]

Field	Detail
Identifier	FR-26
Title	Intelligent Faculty/Staff Leave Management
Requirement	The system shall allow Faculty and Staff to submit leave applications. It must automatically validate the remaining leave balance , deduct the days upon approval, and route the request to the reporting Head of Department (HOD) for approval.
Source	HR Policy / Efficiency Goal
Rationale	To combine application, verification, routing, and balance deduction into a single, automated workflow, eliminating manual HR intervention.
Business Rule	Zero Balance Lock: The system must strictly reject applications if the leave quota is zero. Auto-Routing: Requests from HODs must automatically route to the Director/Dean .
Dependencies	FR-02 (Login), FR-27 (Employee Profile)
Priority	Medium

2.3.27 FR-27: Employee Profile & Salary Management

Table 34 Employee Profile & Salary Management [FR-27]

Field	Detail
Identifier	FR-27
Title	Employee Profile & Salary Management
Requirement	The system shall allow the HR Admin to create and manage detailed employee profiles, including designation, joining date, and salary tier information.
Source	HR Admin
Rationale	To maintain a centralized database of all university staff for payroll and administrative purposes.
Business Rule	Confidentiality: The "Salary" field must be encrypted in the database and visible only to the Super Admin and HR Manager, hidden from standard Admin views.
Dependencies	FR-04 (RBAC)
Priority	Medium

2.3.28 FR-28: Alumni Portal & Networking

Table 35 Alumni Portal & Networking [FR-28]

Field	Detail
Identifier	FR-28
Title	Alumni Portal & Networking Services
Requirement	The system shall allow verified graduates (checked against SIS records) to register as Alumni. Once verified, they must be able to post job and internship opportunities for current students.
Source	Alumni Relations / Placement Cell
Rationale	To create a self-sustaining career network where verification and contribution happen in one module.
Business Rule	Content Moderation: All job posts submitted by Alumni must remain in a " Pending " state until approved by the Placement Officer to prevent spam.
Dependencies	FR-05 (Student Profile - for graduation check)
Priority	Low

2.3.29 FR-29: Digital Library Catalogue Search

Table 36 Digital Library Catalogue Search [FR-29]

Field	Detail
Identifier	FR-29
Title	Digital Library Catalogue Search
Requirement	The system shall allow students to search the university's physical book catalogue by Title, Author, or ISBN. The system must display the live status (e.g., "Available", "Checked Out") of each item.
Source	University Librarian

Rationale	To save student time by allowing them to locate resources without physically searching the shelves.
Business Rule	Real-Time Inventory: The system must strictly prevent a book from being shown as "Available" if its status in the inventory database is "Lost" or "Damaged".
Dependencies	FR-02 (Login)
Priority	Medium

2.3.30 FR-30: QR-Based Book Issuance

Table 37 Book Issuance [FR-30]

Field	Detail
Identifier	FR-30
Title	QR-Based Book Issuance
Requirement	The system shall allow the Librarian to issue books by scanning the student's digital ID (QR Code) generated by the app. The system must link the book to the student's account instantly.
Source	Library Operations
Rationale	To speed up the checkout process and maintain an accurate digital record of library assets.
Business Rule	Issuance Limit: The system must block the issuance transaction if the student already has 3 books (the maximum limit) currently checked out.
Dependencies	FR-05 (Student Profile)
Priority	Medium

2.3.31 FR-31: Automated Library Fine Calculation

Table 38 Library Fine [FR-31]

Field	Detail
Identifier	FR-31
Title	Automated Library Fine Calculation
Requirement	The system shall automatically calculate fines for books returned after the designated due date. The fine amount must be posted directly to the student's "Financial Ledger" as a pending payment.
Source	Finance / Library
Rationale	To ensure integrated billing and prevent students from needing to pay cash fines separately at the library.
Business Rule	Fine Rate: The system must calculate fines at a fixed rate of Rs. 50 per day overdue, up to a maximum cap of the book's replacement cost.
Dependencies	FR-30 (Issuance), FR-09 (Ledger)
Priority	Medium

2.3.32 FR-32: Faculty "Student Success" Dashboard

Table 39 Student Success Dashboard [FR-32]

Field	Detail
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Identifier	FR-32
Title	Faculty "Student Success" Dashboard
Requirement	The system shall provide a visual dashboard for Faculty, allowing them to view aggregated performance metrics (Attendance vs. Grades) for their classes using charts and graphs.
Source	Academic Dean
Rationale	To provide actionable insights to teachers, helping them visualize class trends rather than looking at raw data tables.
Business Rule	Data Privacy: Faculty members must only be able to view the analytics for courses they are currently teaching. Access to other courses' data is strictly prohibited.
Dependencies	FR-07 (Grades), FR-18 (Attendance)
Priority	High

2.3.33 FR-33: "At-Risk" Student Prediction

Table 40 Student Prediction [FR-33]

Field	Detail
Identifier	FR-33
Title	"At-Risk" Student Prediction (AI)
Requirement	The system shall utilize a machine learning model (Scikit-learn) to analyze real-time student data (Attendance, Quiz Scores, Login Frequency). It must automatically tag students as " At-Risk " (Red), "Average" (Yellow), or "Excellent" (Green).[11]
Source	AI Data Science Team
Rationale	To identify struggling students early in the semester, allowing for proactive intervention before they fail.
Business Rule	Risk Threshold: A student must be automatically flagged as "At-Risk" if their aggregate attendance drops below 75% OR their current semester GPA drops below 2.0.
Dependencies	FR-18 (Attendance), FR-07 (Grades)
Priority	High

2.3.34 FR-34: Smart Grievance Submission & Sentiment Analysis

Table 41 Grievance Submission [FR-34]

Field	Detail
Identifier	FR-34
Title	Automated Grievance Submission & Routing
Requirement	The system shall accept student complaints and use Natural Language Processing (NLP) to detect sentiment (Urgency). It must automatically route the ticket to the relevant department (e.g., "WiFi" -> IT Dept, "Cleanliness" -> Estate) based on keyword analysis.
Source	Student Affairs
Rationale	To ensure critical complaints are instantly sent to the correct person without manual sorting.
Business Rule	Priority Escalation: Complaints classified with "Angry/Urgent" sentiment must be automatically flagged as " High Priority " in the Admin queue.
Dependencies	FR-02 (Login)
Priority	Medium

2.3.35 FR-36: Grievance Status Tracking & SLA

Table 42 Grievance Status [FR-35]

Field	Detail
Identifier	FR-35
Title	Grievance Status Tracking & SLA
Requirement	The system shall provide a transparent tracker for students to view the live status of their complaints (Submitted -> In Progress -> Resolved). Updates must trigger an automated notification to the student.
Source	Student Council
Rationale	To build trust in the system by ensuring students know their issues are being handled.
Business Rule	Service Level Agreement (SLA): Administrators must acknowledge or change the status of a "High Priority" complaint within 48 hours , otherwise, it is escalated to the Director automatically.
Dependencies	FR-34 (Submission)
Priority	Low

2.3.36 FR-36: Secure System Audit Logging

Table 43 System Audit Logging [FR-36]

Field	Detail
Identifier	FR-36
Title	Secure System Audit Logging
Requirement	The system shall maintain a secure, immutable log of all critical actions (e.g., changing a grade, deleting a fee voucher, accessing sensitive profile data). The log must record the User ID, Timestamp, Action Type, and IP Address.
Source	Security Audit Policy
Rationale	To ensure accountability and provide a forensic trail in case of data tampering or disputes.
Business Rule	Log Immutability: Audit logs must be stored in a "Write-Once-Read-Many" (WORM) format and cannot be deleted or modified by any user (including Super Admins) for a period of 1 Year .
Dependencies	All Critical FRs
Priority	High

2.3.37 FR-37: Online Quiz Creation

Table 44 Online Quiz Creation [FR-37]

Field	Detail
Identifier	FR-37
Title	Online Quiz Creation

Requirement	The system shall allow Faculty to create online quizzes containing Multiple Choice Questions (MCQs). The Faculty must be able to set the Total Marks , Time Duration (e.g., 10 mins), and the Correct Answer Key .
Source	Faculty Request
Rationale	To facilitate continuous assessment without manual paper grading.
Business Rule	Modification Lock: Once a quiz has been attempted by at least one student, the Faculty cannot edit the questions or answer key to ensure fairness.
Dependencies	FR-11 (LMS)
Priority	High

2.3.38 FR-38: Timed Quiz Attempt

Table 45 Timed Quiz Attempt [FR-38]

Field	Detail
Identifier	FR-38
Title	Timed Quiz Attempt
Requirement	The system shall allow students to attempt the quiz within a specific window. The system must display a Countdown Timer and automatically submit the attempt when the time expires.
Source	Student Academic System
Rationale	To simulate exam conditions and prevent cheating by limiting time per question.
Business Rule	One Attempt Policy: Unless specified otherwise, students are allowed only one attempt per quiz. The system must block re-attempts.
Dependencies	FR-37 (Creation)
Priority	High

2.3.39 FR-39: Automated Grading and Feedback

Table 46 Automated Grading & Feedback [FR-39]

Field	Detail
Identifier	FR-39
Title	Automated Grading & Feedback
Requirement	The system shall automatically calculate the student's score immediately upon submission by comparing inputs with the stored Answer Key. The score must be saved to the Gradebook instantly.
Source	System Automation
Rationale	To provide instant feedback to students and reduce grading workload for teachers.
Business Rule	Score Visibility: The system must allow Faculty to decide when the score is shown to students (e.g., "Immediately" or "After Deadline").
Dependencies	FR-38 (Attempt)
Priority	High

2.3.40 FR-40: View Current Semester Details

Table 47 Semester Details [FR-40]

Field	Detail
Identifier	FR-40
Title	View Current Semester Details
Requirement	The system shall provide a "Semester Summary" view for the student, displaying a list of all currently enrolled courses, assigned Section IDs, Instructor Names, and the total count of registered Credit Hours.
Source	Student Feedback
Rationale	To allow students to verify their enrollment status and know exactly which section/teacher they are assigned to.
Business Rule	Status Filtering: The system must only display courses with a "Confirmed" status. Withdrawn or Dropped courses must be listed in a separate "History" section or hidden.
Dependencies	FR-06 (Registration)
Priority	High

2.3.41 FR-41: Executive Analytics Dashboard

Table 48 Executive Analytics Dashboard [FR-41]

Field	Detail
Identifier	FR-41
Title	Executive Analytics Dashboard
Requirement	The system shall provide a high-level dashboard for the Campus Incharge displaying real-time Key Performance Indicators (KPIs). This must include Total Revenue Collected, Student Attendance Rates, Pass/Fail Ratios, and Dropout Risks.
Source	Senior Leadership
Rationale	To support data-driven decision-making and strategic planning for the university.
Business Rule	Data Aggregation: The dashboard must update dynamically but display anonymized/aggregated data to protect individual privacy during high-level reviews.
Dependencies	FR-07 (Grades), FR-10 (Fees), FR-33 (AI Prediction)
Priority	High

2.3.42 FR-42: Global Announcements System

Table 49 Global Announcement System [FR-42]

Field	Detail
Identifier	FR-42

Title	Global Announcements System
Requirement	The system shall allow the Campus Incharge to post Global Announcements (e.g., "Campus Closed due to Rain"). These must appear as a banner on the dashboard of all users (Students, Faculty, Staff) and trigger an immediate Push Notification.
Source	Administration
Rationale	To ensure critical information reaches the entire campus community instantly.
Business Rule	Visibility Scope: The system must allow the Incharge to select the audience (e.g., "All Users," "Faculty Only," or "Students Only") before publishing.
Dependencies	FR-20 (Notifications)
Priority	High

2.3.43 FR-43: Student Leave Application & Academic Exemption

Table 50 Student Leave [FR-43]

Field	Detail
Identifier	FR-43
Title	Student Leave Application & Academic Exemption
Requirement	The system shall allow Students to submit leave applications (Medical/Emergency) with supporting documents (e.g., images/PDFs). The system must route this request to the HOD for approval.
Source	Student Affairs Policy
Rationale	To digitize student excuses and ensure subject teachers are informed of authorized absences.
Business Rule	Teacher Notification: Upon HOD approval, the system must automatically identify all Subject Teachers who have classes with the student during the leave period and send them a notification: " <i>Student [Name] is on approved leave.</i> "
Dependencies	FR-18 (Attendance), FR-20 (Notifications)
Priority	Medium

2.4 Non-Functional Requirements

This section specifies the quality attributes of Project Nexus. While the Functional Requirements define what the system does, these requirements define how well it must perform in terms of reliability, speed, ease of use, and security.

2.4.1 Reliability

Requirements regarding system stability, failure recovery, and data safety.

2.4.1.1 NFR-01 (System Uptime)

The system shall be available for use **99.9% of the time** during the semester. We allow for short scheduled maintenance windows (e.g., Sunday nights) for updates.

2.4.1.2 NFR-02 (Self-Healing)

If a specific part of the system (like the Notification Service) crashes, the Docker Swarm infrastructure must automatically detect the failure and restart the service within **30 seconds** without human intervention.

2.4.1.3 NFR-03 (Zero Data Loss)

Critical transactions, specifically Student Grades and Fee Payments, must use **Apache Kafka** with persistent storage. This guarantees that even if a server crashes momentarily, the data is saved in the queue and processed as soon as the system recovers.

2.4.1.4 NFR-04 (Data Backup)

The system must automatically perform a backup of the PostgreSQL (Financial/Academic) and MongoDB (Chat) databases every **24 hours** to protect against catastrophic hardware failure.

2.4.2 Usability

Requirements regarding ease of learning and user experience.

2.4.2.1 NFR-05 (Mobile Responsiveness)

The web interface (React.js) must automatically adapt its layout to fit any screen size. A student using a smartphone (360px width) must be able to use the app easily without zooming or horizontal scrolling.

2.4.2.2 NFR-06 (Efficiency)

Essential actions must be quick to perform. A student shall be able to view their **Fee Voucher**, check their **Latest GPA**, or mark **Attendance** within **3 clicks** from the main dashboard.

2.4.2.3 NFR-07 (AI Transparency)

To ensure trust, the AI Chatbot must be honest. If the AI is less than **70% confident** in an answer, it must display a "Low Confidence" warning or direct the student to the administration office instead of guessing.

2.4.2.4 NFR-08 (Ease of Learning)

The interface must be intuitive enough that a new user (Student or Faculty) can learn to perform core tasks within **15 minutes** of first use, without requiring a training manual.

2.4.3 Performance

Requirements regarding speed and resource usage.

2.4.3.1 NFR-09 (General Speed)

Standard interactions, such as logging in or loading a profile, must load in under **200 milliseconds** (0.2 seconds) so the application feels instant to the user.

2.4.3.2 NFR-10 (AI Response Time)

- **Static Questions:** Common FAQs (e.g., "Exam dates") served from the Cache must load in under **50 milliseconds**.
- **Dynamic Questions:** Complex database queries (e.g., "My Fee Balance") processed by Gemini must begin streaming the answer within **3 seconds**.

2.4.3.3 NFR-11 (Real-Time Chat)

Messages sent between users via the Peer-to-Peer service must appear on the recipient's screen instantly (under **1 second**) using WebSocket technology.

2.4.3.4 NFR-12 (Concurrency)

The system must be able to handle at least **500 concurrent users** (e.g., during result announcement day) without crashing or slowing down significantly.

2.4.4 Security

Requirements regarding data protection and access control.

2.4.4.1 NFR-13 (Encryption)

All data traveling between the user's device and the server must be encrypted using **HTTPS (TLS 1.2+)**. Sensitive data at rest (like passwords and financial records) must be encrypted in the database.

2.4.4.2 NFR-14 (Role-Based Access)

The system must strictly enforce user roles. If a user logged in as a "Student" tries to access a page meant for "Faculty," the system must immediately block them with a **"403 Forbidden"** error.

2.4.4.3 NFR-15 (Biometric Privacy)

The Facial Recognition system must **not** permanently store raw images of student faces. It should only verify the biometric pattern for that specific session to protect student privacy.

2.4.4.4 NFR-16 (Session Security)

User login sessions must automatically expire after **30 minutes** of inactivity. This prevents unauthorized people from accessing a student's account if they forget to log out on a public computer.

2.5 External Interface Requirements

This section provides information to ensure that the system will communicate properly with users and with external hardware or software elements.

2.5.1 User Interfaces Requirements

The user interface (UI) for Project Nexus is designed to be intuitive, responsive, and consistent. It will be developed using the **React.js**[29] framework with **Material UI (MUI)** components to ensure a modern, professional look.

2.5.1.1 General GUI Standards

- **Design Philosophy:** The interface shall follow "Material Design" principles, utilizing card-based layouts, depth effects (shadows), and clean typography to make it easy to read.
- **Responsiveness:** The application shall be fully responsive. It must automatically adjust its layout based on the device:
- **Mobile Devices (Width < 600px):** The Sidebar shall collapse into a "Hamburger Menu," and complex tables shall convert to simple card views for easy scrolling.
- **Tablets/Desktops (Width > 1200px):** The Sidebar shall remain visible, and data grids shall display full columns.
- **Color Scheme:** The system shall use a light modern color scheme.

2.5.1.2 Standard Screen Elements

- **Navigation:** A persistent side navigation bar shall be present on the left side of the screen. It must dynamically change links based on the user's role (e.g., A "Faculty" user sees "Grading," while a "Student" user sees "My Transcripts").
- **Dashboards:** Every user shall land on a personalized "Dashboard" upon login.
 - Student Dashboard: Must show "Upcoming Classes," "Fee Status," and "Assignment Deadlines."
 - Faculty Dashboard: Must show "Class Schedule," "Pending Approvals," and "At-Risk Student Alerts."
- **Feedback Messages:** The system shall use "Toast Notifications" (small pop-ups at the top-right) to confirm actions instantly (e.g., "Green: Attendance Marked Successfully," "Red: Upload Failed").

2.5.2 Software Interfaces

This section describes the connections between Project Nexus and other software components, including databases, APIs, and libraries.

2.5.2.1 Database Interfaces

The backend microservices (FastAPI) will communicate with the following data stores:

- **PostgreSQL (Relational DB):** Used for structured, transactional data. The system shall interface with PostgreSQL to store Student Records, Financial Ledgers, and HR Profiles using the **SQLAlchemy ORM** to ensure data integrity.
- **MongoDB (NoSQL DB):** Used for unstructured data. The P2P Chat Service and LMS Content Service shall connect to MongoDB to store chat logs and course materials.
- **Redis (Cache Store):** The system shall interface with Redis to cache frequently accessed data (like static FAQs for the AI Chatbot) and store active User Session Tokens (JWTs) for fast validation.
- **ChromaDB (Vector Database):** Used for storing embeddings. For the chatbot we will store for embeddings in the ChromaDB.[14]

2.5.2.2 External APIs

- **Google Gemini API:** The AI Chatbot Service shall connect to the Google Gemini API via REST. It sends student queries and context data (JSON) and receives natural language responses.
- **Payment Gateway API (Stripe/JazzCash):** The Financial Service shall integrate with the Stripe API. It will securely transmit payment tokens and receive transaction success/failure callbacks (Webhooks).
- **Docker Engine API:** The Infrastructure Service uses the Docker API to monitor container health and restart crashed services automatically.[30]

2.5.3 Hardware Interfaces

This section describes the logical characteristics of the interface between the software and the physical hardware devices.

2.5.3.1 Client-Side Hardware

- **Camera/Webcam:** The "Attendance" module requires access to the device's front-facing camera. The browser must request permission via the `navigator.mediaDevices.getUserMedia` API to capture video frames for facial recognition.
- **GPS Sensor:** The "Attendance" module requires access to the device's GPS receiver. The software shall query the Geolocation API to verify that the student is physically present within the university campus coordinates.

2.5.3.2 Server-Side Hardware

- **Development Cluster:** The system is designed to run on a cluster of standard development laptops (mimicking a cloud environment). The Docker Swarm orchestrator requires a minimum of **16GB RAM** and a **Multi-Core Processor (i5/i7)** to handle the concurrent load of 10+ microservices (Kafka, Postgres, AI Service, etc.).

2.5.4 Communications Interfaces

This section specifies the network protocols required for the system to function.

2.5.4.1 Communication Protocols

- **HTTP/HTTPS (REST):** All communication between the React.js Frontend and the FastAPI Backend shall occur over **HTTPS** (using TLS 1.2 or higher) to ensure data encryption in transit. Standard HTTP methods (GET, POST, PUT, DELETE) will be used for CRUD operations.
- **WebSockets (WSS):** The Peer-to-Peer (P2P) Chat Service shall use the **WebSocket** protocol to establish a persistent, bidirectional connection. This allows messages to be pushed to the user's screen instantly without refreshing the page.

2.5.4.2 Message Queuing (Internal)

- **Apache Kafka Protocol:** The backend microservices shall communicate with each other asynchronously using the native Kafka TCP protocol. This is used for high-reliability events, such as publishing a "Grade Posted" event from the LMS to the SIS to guarantee zero data loss.

2.5.4.3 Data Formats

- **JSON (JavaScript Object Notation):** The primary format for data exchange between the Client, Server, and External APIs (like Gemini) shall be JSON.
- **Multipart/Form-Data:** This format shall be used specifically for file uploads, such as students submitting assignment PDFs or faculty uploading course slides.

3. Use Case Analysis

3.1 Use Case #1: User Registration (UC-01)

Table 51 UC-1

Field	Detail
UC Identifier	UC-01
Requirements Traceability	FR-01 (User Registration)
Purpose	To allow the Administrator to create a new user account (Student, Faculty, or Staff) in the system.
Priority	High
Preconditions	1. Admin is logged in. 2. User does not already exist in the database.
Post conditions	New user record is created, and login credentials are generated.
Actors	Admin (Primary), System
Extends	None
Main Success Scenario	1. Admin navigates to "User Management". 2. Admin clicks "Add New User". 3. Admin selects Role (e.g., Student) and enters details (Name, Email, Roll No). 4. Admin clicks "Create Account". 5. System checks for duplicate Email/Roll No. 6. System saves the record and hashes the temporary password. 7. System displays "User Registered Successfully".
Alternate Flows	A1: Duplicate User 1. System detects Email already exists. 2. System displays error "User already exists". 3. Use case ends.
Exceptions	E1: Database Error 1. Connection fails during save. 2. System alerts Admin to retry.
Includes	None

3.2 Use Case #2: User Login (UC-02)

Table 52 UC-2

Field	Detail
UC Identifier	UC-02
Requirements Traceability	FR-02 (Login), FR-03 (JWT), FR-04 (RBAC)
Purpose	To authenticate a user and grant access to role-specific features.
Priority	High
Preconditions	User account exists and is active.
Post conditions	User is authenticated, a JWT token is active, and the Dashboard loads.
Actors	User (Student/Faculty/Admin), System (Auth Service)
Extends	None
Main Success Scenario	1. User enters Email/Roll No and Password. 2. User clicks "Login". 3. System verifies credentials against stored hash. 4. System generates a signed JWT containing the user's Role. 5. System redirects user to the specific Dashboard (e.g., Student Dashboard).

Alternate Flows	A1: Invalid Credentials 1. Password does not match. 2. System increments failure counter. 3. System displays "Invalid Login".
Exceptions	E1: Account Locked 1. User fails login 5 times in a row. 2. System locks account for 15 minutes (Business Rule). 3. System displays "Account Locked. Try later."
Includes	None

3.3 Use Case #3: Manage Student Profile (UC-03)

Table 53 UC-3

Field	Detail
UC Identifier	UC-03
Requirements Traceability	FR-05 (Profile Management)
Purpose	To create or update the official demographic and contact details of a student.
Priority	High
Preconditions	Admin is logged in.
Post conditions	Student profile data is updated in the database.
Actors	Admission Admin , System
Extends	None
Main Success Scenario	1. Admin searches for a student by Roll Number. 2. Admin clicks "Edit Profile". 3. Admin updates fields (Address, Phone, Guardian Name). 4. Admin clicks "Save Changes". 5. System validates data formats. 6. System commits changes to the Student Table. 7. System logs the audit trail.
Alternate Flows	A1: Immutable Field Attempt 1. Admin tries to change a locked field (e.g., CNIC). 2. System blocks the edit (Business Rule). 3. System displays "Cannot modify immutable field".
Exceptions	None
Includes	None

3.4 Use Case #4: Register for Courses (UC-04)

Table 54 UC-4

Field	Detail
UC Identifier	UC-04
Requirements Traceability	FR-06 (Course Registration)
Purpose	To allow a student to enroll in specific courses for the current semester.
Priority	High
Preconditions	1. Student is logged in. 2. Registration window is open. 3. Student is not on academic hold.
Post conditions	Student is enrolled in selected sections.
Actors	Student , System (SIS Service)
Extends	None

Main Success Scenario	1. Student navigates to "Course Registration". 2. System displays available courses. 3. Student selects courses adding up to 15 Credit Hours. 4. Student clicks "Confirm Enrollment". 5. System validates checks (Time conflicts, Credit limits). 6. System saves enrollment. 7. System generates Fee Voucher (Trigger for UC-06).
Alternate Flows	A1: Credit Limit Exceeded 1. Student selects >18 Credit Hours. 2. System checks GPA. 3. If GPA < 3.5, System rejects selection (Business Rule).
Exceptions	E1: Seat Capacity Full 1. Selected section is full. 2. System suggests alternative section.
Includes	UC-06 (Generate Fee Voucher - Implicit trigger)

3.5 Use Case #5: Download Digital Transcript (UC-05)

Table 55 UC-5

Field	Detail
UC Identifier	UC-05
Requirements Traceability	FR-08 (Digital Transcript)
Purpose	To allow a student to download a PDF report of their academic history.
Priority	Medium
Preconditions	Student is logged in.
Post conditions	PDF file is downloaded to user device.
Actors	Student , System
Extends	None
Main Success Scenario	1. Student clicks "My Transcript". 2. System checks Financial Status. 3. System confirms status is "Clear". 4. System retrieves all grades and calculates CGPA. 5. System generates PDF. 6. Student downloads the file.
Alternate Flows	A1: Financial Hold 1. System detects "Unpaid" fees in Ledger. 2. System blocks download (Business Rule). 3. System displays "Please clear dues to access transcript."
Exceptions	None
Includes	None

3.6 Use Case #6: Generate Fee Voucher (UC-06)

Table 56 UC-6

Field	Detail
UC Identifier	UC-06
Requirements Traceability	FR-09 (Fee Generation)
Purpose	To automatically calculate and create fee invoices for all active students.
Priority	High
Preconditions	Student is enrolled in courses for the current semester.

Post conditions	Invoice is created with "Unpaid" status in the Ledger.
Actors	System (Timer-Based), Admin
Extends	None
Main Success Scenario	1. System triggers the billing cycle (e.g., 1st of Month). 2. System retrieves enrolled Credit Hours for the student. 3. System calculates Total Amount = (Credits * Rate) + Fixed Charges. 4. System generates a unique Voucher ID. 5. System posts the entry to the Student Ledger. 6. System sends a notification: "Fee Voucher Generated".
Alternate Flows	A1: Late Fee Application 1. System detects current date > Due Date. 2. System applies 5% Surcharge (Business Rule). 3. System updates the invoice total.
Exceptions	None
Includes	None

3.7 Use Case #7: Pay Fees Online (UC-07)

Table 57 UC-7

Field	Detail
UC Identifier	UC-07
Requirements Traceability	FR-10 (Online Payment)
Purpose	To allow students to pay outstanding dues using an external gateway.
Priority	High
Preconditions	1. Student has an "Unpaid" voucher. 2. Payment Gateway is online.
Post conditions	Invoice status becomes "Paid", Ledger is updated.
Actors	Student , External Payment Gateway (Stripe)
Extends	None
Main Success Scenario	1. Student views Ledger and clicks "Pay Now". 2. System redirects to secure Payment Gateway. 3. Student enters card details and confirms. 4. Gateway processes transaction and returns "Success". 5. System validates the signature. 6. System updates status to "Paid" immediately. 7. System sends digital receipt.
Alternate Flows	A1: Transaction Failed 1. Gateway declines card. 2. System displays "Payment Failed". 3. Invoice remains "Unpaid".
Exceptions	E1: Gateway Timeout 1. No response from Stripe. 2. System logs error and asks user to check bank statement.
Includes	None

3.8 Use Case #8: Submit Final Grades (UC-08)

Table 58 UC-8

Field	Detail
UC Identifier	UC-08

Requirements Traceability	FR-07 (Grade Submission)
Purpose	To allow Faculty to enter and finalize student grades for the semester.
Priority	Critical
Preconditions	1. Faculty is assigned to the course. 2. Exam period is active.
Post conditions	Grades are saved and Locked (Read-Only).
Actors	Faculty , System
Extends	None
Main Success Scenario	1. Faculty navigates to "Gradebook". 2. Faculty enters marks for assignments/exams. 3. System calculates Total and Letter Grade. 4. Faculty reviews and clicks " Final Submit ". 5. System prompts: "This action cannot be undone. Confirm?". 6. Faculty confirms. 7. System saves grades and sets status to " Locked " (Business Rule). 8. System triggers UC-09 (Sync).
Alternate Flows	A1: Modification Attempt 1. Faculty tries to edit a Locked grade. 2. System denies access. 3. System displays "Grades Finalized. Contact Controller of Exams."
Exceptions	None
Includes	UC-09 (Sync Grades to SIS - Implicit Trigger)

3.9 Use Case #9: Sync Grades via Kafka (UC-09)

Table 59 UC-9

Field	Detail
UC Identifier	UC-09
Requirements Traceability	FR-19 (Automation/Kafka), FR-07
Purpose	To asynchronously update the official Student Transcript when a grade is posted in the LMS.
Priority	Critical
Preconditions	A "GradePosted" event is published by UC-08.
Post conditions	Student Transcript table in SIS database is updated.
Actors	System (LMS Service), System (SIS Service)
Extends	None
Main Success Scenario	1. LMS Service publishes GradePosted event to Apache Kafka. 2. SIS Service consumes the event. 3. SIS Service validates the Student ID and Course Code. 4. SIS Service updates the permanent Transcript record. 5. SIS Service recalculates the student's CGPA. 6. SIS Service commits the transaction.
Alternate Flows	None
Exceptions	E1: SIS Service Offline 1. SIS Service is down. 2. Kafka holds the message (Data Durability). 3. Upon restart, SIS processes the backlog.
Includes	None

3.10 Use Case #10: Upload Course Material (UC-10)

Table 60 UC-10

Field	Detail
UC Identifier	UC-10
Requirements Traceability	FR-11 (Content Upload)
Purpose	To allow Faculty to share learning resources (PDFs, PPTs) with students.
Priority	High
Preconditions	Faculty is logged in.
Post conditions	Material is visible to enrolled students.
Actors	Faculty, System
Extends	None
Main Success Scenario	1. Faculty selects a course. 2. Faculty clicks "Add Resource". 3. Faculty uploads a file (e.g., Lecture1.pdf). 4. System validates file type (PDF/PPT) and size (<25MB). 5. System saves file to storage. 6. System notifies students.
Alternate Flows	A1: Invalid File 1. Faculty uploads an .exe file. 2. System rejects upload (Security Rule). 3. System displays "Invalid File Format".
Exceptions	None
Includes	None

3.11 Use Case #11: Create Assignment (UC-11)

Table 61 UC-11

Field	Detail
UC Identifier	UC-11
Requirements Traceability	FR-12 (Assignment Creation)
Purpose	To allow Faculty to publish a graded task with instructions and a strict deadline.
Priority	High
Preconditions	Faculty is logged in and assigned to the course.
Post conditions	Assignment is visible to students, Calendar is updated.
Actors	Faculty, System
Extends	None
Main Success Scenario	1. Faculty navigates to "Assignments" tab. 2. Faculty clicks "Create New". 3. Faculty enters Title, Instructions, Max Marks, and Due Date/Time. 4. Faculty clicks "Publish". 5. System saves the record. 6. System triggers a notification to enrolled students. 7. System starts the countdown timer for the deadline.
Alternate Flows	None
Exceptions	E1: Past Date Error 1. Faculty selects a Due Date in the past. 2. System validates against Server Time.

	3. System displays error "Due Date must be in the future".
Includes	None

3.12 Use Case #12: Submit Assignment (UC-12)

Table 62 UC-12

Field	Detail
UC Identifier	UC-12
Requirements Traceability	FR-13 (Submission)
Purpose	To allow students to upload their work before the deadline expires.
Priority	High
Preconditions	1. Assignment exists. 2. Current Server Time < Due Date.
Post conditions	File is saved and timestamped, Status changes to "Submitted".
Actors	Student , System
Extends	None
Main Success Scenario	1. Student opens the Assignment. 2. Student clicks "Submit Work". 3. Student uploads file (e.g., Lab1.pdf). 4. System verifies time is within the allowed window. 5. System saves file to storage. 6. System records the exact timestamp. 7. System displays "Submitted Successfully".
Alternate Flows	A1: Late Submission Attempt 1. Student tries to submit after deadline. 2. System checks Server Time. 3. System blocks the upload (Business Rule). 4. System displays "Deadline Passed. Contact Instructor."
Exceptions	None
Includes	None

3.13 Use Case #13: Query AI Chatbot (UC-13)

Table 63 UC-13

Field	Detail
UC Identifier	UC-13
Requirements Traceability	FR-14 (Universal AI), FR-15 (Context)
Purpose	To allow any user to ask natural language questions about their specific data.
Priority	High
Preconditions	User is logged in.
Post conditions	User receives a relevant text response.
Actors	Student / Faculty / Staff , System, Google Gemini API
Extends	None
Main Success Scenario	1. User opens Chatbot. 2. User asks a question. 3. System identifies User Role (e.g., Faculty).

	4. System retrieves relevant data (e.g., "Pending Leave Requests"). 5. System sends context to Gemini API . 6. System displays the answer: "You have 3 pending leave requests to approve."
Alternate Flows	A1: Student Query 1. User is Student. 2. Query: "My Schedule?" 3. System returns list of registered classes.
Exceptions	None
Includes	UC-02 (Login) - Implicit

3.14 Use Case #14: Mark Attendance (UC-14)

Table 64 UC-14

Field	Detail
UC Identifier	UC-14
Requirements Traceability	FR-16 (Liveness), FR-17 (GPS), FR-18 (Face Rec)
Purpose	To securely mark student attendance using multi-factor biometric verification.
Priority	High
Preconditions	1. Student is physically in class. 2. Class session is active.
Post conditions	Attendance marked as "Present".
Actors	Student , System (CV Service)
Extends	None
Main Success Scenario	1. Student taps "Check-In". 2. System verifies GPS Location (must be within 100m). 3. Student captures a live selfie. 4. System checks Liveness (Blink detection). 5. System matches face with profile photo (>95% match). 6. System saves record as "Present".
Alternate Flows	A1: Spoof Attempt 1. System detects static photo (Liveness fail). 2. System rejects check-in. 3. System logs security alert. A2: Location Mismatch 1. Student is outside geofence. 2. System displays "You must be in the classroom."
Exceptions	E1: Camera Denied 1. Browser permission missing. 2. System prompts to enable camera.
Includes	None

3.15 Use Case #15: Receive Notification (UC-15)

Table 65 UC-15

Field	Detail
UC Identifier	UC-15
Requirements Traceability	FR-20 (Push Notifications)
Purpose	To proactively alert the student of critical updates without manual checking.

Priority	Medium
Preconditions	A trigger event occurs (e.g., Grade Posted, Fee Generated).
Post conditions	Notification is delivered to user device.
Actors	System (Notification Service), Student
Extends	None
Main Success Scenario	1. System detects a critical event (via Kafka). 2. System generates message content (e.g., "New Grade Posted"). 3. System pushes notification to Student's device. 4. Student taps notification. 5. System opens the specific page (e.g., Transcript View).
Alternate Flows	None
Exceptions	E1: Delivery Failure 1. User is offline. 2. System queues message for later delivery.
Includes	None

3.16 Use Case #16: Initiate P2P Chat (UC-16)

Table 66 UC-16

Field	Detail
UC Identifier	UC-16
Requirements Traceability	FR-21 (Chat Initiation)
Purpose	To allow a user to find a peer or instructor and start a secure 1-to-1 conversation.
Priority	Medium
Preconditions	User is logged in.
Post conditions	A unique chat session ID is created.
Actors	Student / Faculty , System (Chat Service)
Extends	None
Main Success Scenario	1. User navigates to "Messages" tab. 2. User searches for a recipient by Name/Roll No. 3. User selects the profile from results. 4. System checks privacy settings. 5. System creates/retrieves the private chat room. 6. System opens the chat interface.
Alternate Flows	A1: User Blocked/Muted 1. Faculty has enabled "Do Not Disturb". 2. System blocks new chat requests. 3. System displays "User is not accepting messages right now."
Exceptions	None
Includes	None

3.17 Use Case #17: Send Real-Time Message (UC-17)

Table 67 UC-17

Field	Detail
UC Identifier	UC-17
Requirements Traceability	FR-22 (Real-Time Messaging)
Purpose	To transmit a text message instantly using WebSockets and save it for history.
Priority	Medium

Preconditions	Chat session is active.
Post conditions	Message is delivered to recipient and saved in MongoDB.
Actors	User , System (WebSocket Service)
Extends	None
Main Success Scenario	1. User types message and clicks "Send". 2. System transmits payload via WebSocket. 3. System saves message to MongoDB (Persistence). 4. System pushes message to Recipient's screen instantly. 5. System displays "Double Tick" (Delivered) status.
Alternate Flows	A1: Recipient Offline 1. Recipient is not connected. 2. System marks message as "Unread". 3. System queues it for next login.
Exceptions	E1: Connection Lost 1. Internet drops. 2. System shows "Reconnecting..." spinner.
Includes	None

3.18 Use Case #18: Create Group Chat (UC-18)

Table 68 UC-18

Field	Detail
UC Identifier	UC-18
Requirements Traceability	FR-23 (Group Chat)
Purpose	To allow collaboration among multiple users (e.g., Study Groups).
Priority	Medium
Preconditions	User has permission to create groups.
Post conditions	Group is created with selected members.
Actors	User (Group Admin), System
Extends	None
Main Success Scenario	1. User clicks "New Group". 2. User enters Group Name (e.g., "Final Year Project"). 3. User selects members from contact list. 4. User clicks "Create". 5. System validates member count. 6. System initializes group and notifies members.
Alternate Flows	A1: Limit Exceeded 1. User tries to add >100 members. 2. System blocks action (Business Rule). 3. System displays "Group limit reached."
Exceptions	None
Includes	None

3.19 Use Case #19: Define Timetable Constraints (UC-19)

Table 69 UC-19

Field	Detail
UC Identifier	UC-19
Requirements Traceability	FR-24 (Constraint Definition)

Purpose	To allow the HOD to input the rules required for scheduling (Rooms, Faculty times).
Priority	High
Preconditions	HOD is logged in.
Post conditions	Constraints are saved in the scheduling engine.
Actors	Head of Department (HOD), System
Extends	None
Main Success Scenario	1. HOD navigates to "Scheduler Config". 2. HOD selects a resource (e.g., "Room 101"). 3. HOD sets constraint (e.g., "Capacity: 50", "Type: Lab"). 4. HOD sets Faculty unavailability (e.g., "Dr. Ahmed: Unavailable Mon 9-11"). 5. HOD clicks "Save Configuration". 6. System validates resource availability. 7. System saves rules.
Alternate Flows	A1: Invalid Logic 1. HOD sets total classes > available hours. 2. System rejects save. 3. System displays "Resource overallocation detected."
Exceptions	None
Includes	None

3.20 Use Case #20: Generate Master Timetable (UC-20)

Table 70 UC-20

Field	Detail
UC Identifier	UC-20
Requirements Traceability	FR-25 (Auto-Scheduling)
Purpose	To automatically create a conflict-free schedule for the department.
Priority	High
Preconditions	Constraints are defined (UC-19).
Post conditions	Timetable is published to students/faculty.
Actors	HOD, System (Timetabling Service)
Extends	None
Main Success Scenario	1. HOD clicks "Auto-Generate Timetable". 2. System runs the Constraint-Satisfaction Algorithm. 3. System iterates through all sections and slots. 4. System produces a schedule with Zero Conflicts. 5. System displays the draft for review. 6. HOD clicks "Publish". 7. System updates everyone's "My Schedule" view.
Alternate Flows	A1: Solution Not Found 1. Algorithm cannot resolve a clash. 2. System highlights the specific conflict. 3. System asks HOD to relax constraints manually.
Exceptions	None
Includes	None

3.21 Use Case #21: Submit Leave Application (UC-21)

Table 71 UC-21

Field	Detail
UC Identifier	UC-21
Requirements Traceability	FR-26 (Intelligent Leave Management)
Purpose	To allow Faculty or Staff to request time off digitally.
Priority	Medium
Preconditions	Employee is logged in and has a remaining leave balance.
Post conditions	Application status becomes "Pending Approval".
Actors	Faculty / Staff , System (HR Service)
Extends	None
Main Success Scenario	1. Employee navigates to "HR Portal". 2. Employee clicks "Apply for Leave". 3. Employee selects Type (Casual/Sick), Dates, and Reason. 4. Employee clicks "Submit". 5. System identifies the reporting manager (HOD). 6. System sends notification to HOD. 7. System logs request as "Pending".
Alternate Flows	A1: Zero Balance 1. Employee selects "Casual Leave". 2. System checks balance = 0. 3. System blocks submission (Business Rule). 4. System displays "Insufficient Leave Balance".
Exceptions	None
Includes	None

3.22 Use Case #22: Approve/Reject Leave (UC-22)

Table 72 UC-22

Field	Detail
UC Identifier	UC-22
Requirements Traceability	FR-26(Intelligent Leave Management)
Purpose	To allow HODs to manage leave requests from their subordinates.
Priority	Medium
Preconditions	A pending leave request exists.
Post conditions	Leave is finalized, Balance is updated (if approved).
Actors	Head of Department (HOD) , System
Extends	None
Main Success Scenario	1. HOD views "Pending Requests" on dashboard. 2. HOD reviews the reason and dates. 3. HOD clicks "Approve". 4. System deducts days from Employee's annual quota. 5. System updates status to "Approved". 6. System notifies the Employee. 7. System updates the department timetable (if necessary).
Alternate Flows	A1: Rejection 1. HOD clicks "Reject". 2. HOD enters rejection reason (e.g., "Exam Duty"). 3. System marks as "Rejected" (No deduction).
Exceptions	None

Includes	None
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3.23 Use Case #23: View Leave Balance (UC-23)

Table 73 UC-23

Field	Detail
UC Identifier	UC-23
Requirements Traceability	FR-26 (Intelligent Leave Management)
Purpose	To allow employees to see their remaining leave quotas transparency.
Priority	Low
Preconditions	Employee is logged in.
Post conditions	Balance data is displayed.
Actors	Faculty / Staff, System
Extends	None
Main Success Scenario	1. Employee clicks "My Leaves". 2. System retrieves current quota from HR Database. 3. System displays balance (e.g., "Casual: 5/20", "Sick: 2/10"). 4. System shows history of past applications.
Alternate Flows	None
Exceptions	None
Includes	None

3.24 Use Case #24: Manage Employee Profile (UC-24)

Table 74 UC-24

Field	Detail
UC Identifier	UC-24
Requirements Traceability	FR-27 (Employee Profile)
Purpose	To allow HR Admins to update staff details, salaries, and designations.
Priority	Medium
Preconditions	HR Admin is logged in.
Post conditions	Employee record is updated.
Actors	HR Admin, System
Extends	None
Main Success Scenario	1. HR Admin searches for an employee. 2. HR Admin clicks "Edit Profile". 3. HR Admin updates Salary Tier or Designation. 4. HR Admin clicks "Save". 5. System encrypts sensitive fields (Salary). 6. System saves record. 7. System logs the change in Audit Log.
Alternate Flows	None
Exceptions	None
Includes	None

3.25 Use Case #25: Register as Alumni (UC-25)

Table 75 UC-25

Field	Detail
UC Identifier	UC-25
Requirements Traceability	FR-28 (Alumni Verification)
Purpose	To allow graduating students to sign up for the alumni network.
Priority	Low
Preconditions	Student has completed their degree.
Post conditions	Alumni account is created.
Actors	Student/Alumni , System
Extends	None
Main Success Scenario	1. Student navigates to "Alumni Portal". 2. Student enters Roll Number and Graduation Year. 3. System checks SIS for "Graduated" status. 4. Student enters current Employment details. 5. System creates Alumni profile. 6. System grants access to Job Board.
Alternate Flows	A1: Verification Failed 1. System finds "Active" or "Dropped" status (Not Graduated). 2. System rejects registration. 3. System displays "Only graduates can register."
Exceptions	None
Includes	None

3.26 Use Case #26: Post Job Opportunity (UC-26)

Table 76 UC-26

Field	Detail
UC Identifier	UC-26
Requirements Traceability	FR-28 (Alumni Job Posting)
Purpose	To allow registered Alumni to post job openings for current students.
Priority	Low
Preconditions	Alumni is logged in and verified.
Post conditions	Job post status becomes "Pending Approval".
Actors	Alumni , System (Career Service)
Extends	None
Main Success Scenario	1. Alumni clicks "Post a Job". 2. Alumni enters Role, Company, Location, and Link. 3. Alumni clicks "Submit". 4. System scans text for prohibited keywords. 5. System saves post as "Pending" (Business Rule). 6. System notifies Placement Officer for review.
Alternate Flows	None
Exceptions	None
Includes	None

3.27 Use Case #27: Issue Book via QR (UC-27)

Table 77 UC-27

Field	Detail
UC Identifier	UC-27
Requirements Traceability	FR-30 (QR Issuance), FR-31 (Fines), FR-29 (Catalog)
Purpose	To allow the Librarian to quickly issue a book to a student using digital verification.
Priority	Medium
Preconditions	1. Student has the mobile app open. 2. Book is available in inventory.
Post conditions	Book ownership transfers to Student, Due date is set.
Actors	Librarian , Student, System
Extends	None
Main Success Scenario	1. Student shows QR Code on app. 2. Librarian scans QR Code. 3. System retrieves Student Profile. 4. Librarian scans Book Barcode. 5. System links book to student. 6. System sets Due Date (Today + 14 Days). 7. System updates Inventory Status to "Checked Out".
Alternate Flows	A1: Limit Reached 1. Student already has 3 books. 2. System blocks issuance. 3. System displays "Max limit reached". A2: Outstanding Fine 1. System detects unpaid library fines. 2. System requires payment before issuance.
Exceptions	None
Includes	None

3.28 Use Case #28: View At-Risk Analytics (UC-28)

Table 78 UC-28

Field	Detail
UC Identifier	UC-28
Requirements Traceability	FR-32 (Faculty Dashboard), FR-33 (AI Prediction)
Purpose	To allow Faculty to identify and intervene with struggling students.
Priority	High
Preconditions	Faculty is logged in, sufficient semester data exists.
Post conditions	Faculty views risk analysis.
Actors	Faculty , System (Analytics Service)
Extends	None
Main Success Scenario	1. Faculty navigates to "Student Success" tab. 2. System runs Scikit-learn model on real-time data. 3. System aggregates Attendance + Grades. 4. System tags students as Red (At-Risk), Yellow , or Green . 5. Faculty clicks "Red" tag to view details (e.g., "Attendance < 75%").
Alternate Flows	A1: No Data 1. Semester has just started. 2. System displays "Insufficient data for prediction".

Exceptions	None
Includes	None

3.29 Use Case #29: Submit Grievance (UC-29)

Table 79 UC-29

Field	Detail
UC Identifier	UC-29
Requirements Traceability	FR-34
Purpose	To allow students to report issues which are automatically categorized by AI.
Priority	Medium
Preconditions	Student is logged in.
Post conditions	Complaint ticket is created and routed.
Actors	Student , System (Support Service)
Extends	None
Main Success Scenario	1. Student navigates to "Support". 2. Student types complaint (e.g., "Library AC not working"). 3. Student clicks "Submit". 4. System (NLP) analyzes text. 5. System identifies Category: "Infrastructure" and Sentiment: "Neutral". 6. System routes ticket to Estate Manager. 7. System generates Ticket ID.
Alternate Flows	A1: Urgent Sentiment 1. NLP detects anger/urgency words. 2. System flags as "High Priority". 3. System escalates to Admin Dashboard top queue.
Exceptions	None
Includes	None

3.30 Use Case #30: Track Grievance Status (UC-30)

Table 80 UC-30

Field	Detail
UC Identifier	UC-30
Requirements Traceability	FR-35 (Tracking)
Purpose	To allow students to check the progress of their reported issues.
Priority	Low
Preconditions	A ticket exists.
Post conditions	Status is displayed.
Actors	Student , System
Extends	None
Main Success Scenario	1. Student clicks "My Tickets". 2. System retrieves list of complaints. 3. System shows status: "In Progress". 4. Student clicks to view Admin comments (e.g., "Technician assigned").
Alternate Flows	None
Exceptions	None
Includes	None

3.31 Use Case #31: Create Online Quiz (UC-31)

Table 81 UC-31

Field	Detail
UC Identifier	UC-31
Requirements Traceability	FR-37 (Quiz Creation)
Purpose	To allow Faculty to set up an automated assessment.
Priority	High
Preconditions	Faculty is logged in.
Post conditions	Quiz is published and visible to students.
Actors	Faculty, System
Extends	None
Main Success Scenario	1. Faculty clicks "Add Activity" -> "Quiz". 2. Faculty enters Title, Description, and Time Limit (e.g., 15 mins). 3. Faculty adds Questions and selects the Correct Option for each. 4. Faculty clicks "Publish". 5. System validates all questions have a correct answer marked. 6. System notifies students.
Alternate Flows	A1: Missing Answer Key 1. Faculty forgets to mark a correct option. 2. System blocks publish. 3. System highlights the incomplete question.
Exceptions	None
Includes	UC-02 (Login) - Implicit

3.32 Use Case #32: Attempt Online Quiz (UC-32)

Table 82 UC-32

Field	Detail
UC Identifier	UC-32
Requirements Traceability	FR-38 (Attempt), FR-39 (Auto-Grading)
Purpose	To allow students to take the test and get graded.
Priority	High
Preconditions	1. Student is logged in. 2. Quiz is currently active (within date range).
Post conditions	Score is recorded in Gradebook.
Actors	Student, System
Extends	None
Main Success Scenario	1. Student clicks "Start Quiz". 2. System starts the Countdown Timer. 3. Student selects answers for each MCQ. 4. Student clicks "Submit". 5. System calculates the score. 6. System displays "Submission Successful" and the Score (if allowed).
Alternate Flows	A1: Time Expired 1. Timer hits 00:00. 2. System auto-submits the quiz with current selected answers. 3. System displays "Time Up! Quiz Submitted."
Exceptions	E1: Network Fail 1. Internet drops during quiz.

	2. System saves state locally. 3. When reconnected, System syncs answers (if time remains).
Includes	UC-02 (Login) - Implicit

3.33 Use Case #33: View Semester Details (UC-33)

Table 83 UC-33

Field	Detail
UC Identifier	UC-33
Requirements Traceability	FR-40 (View Semester Details)
Purpose	To allow students to see their enrolled subjects and section information.
Priority	High
Preconditions	Student is logged in and enrolled in at least one course.
Post conditions	Student views the semester summary.
Actors	Student , System
Extends	None
Main Success Scenario	1. Student clicks "My Semester" or "Course List". 2. System retrieves current enrollment data. 3. System displays a table: Course Code, Title, Section (e.g., 8A), Instructor, and Credits. 4. System displays the Total Credit Hours (e.g., "15 Cr").
Alternate Flows	A1: Not Enrolled 1. Student has not registered yet. 2. System displays "No active enrollments found. Please register."
Exceptions	None
Includes	UC-02 (Login) - Implicit

3.34 Use Case #34: Resolve Grievance (UC-34)

Table 84 UC-34

Field	Detail
UC Identifier	UC-34
Requirements Traceability	FR-35
Purpose	To allow the specific Department Admin to close a complaint ticket.
Priority	Medium
Preconditions	A ticket is assigned to the Admin's department.
Post conditions	Ticket status changes to "Resolved", Student is notified.
Actors	Department Admin (IT/Finance/Estate), System
Extends	None
Main Success Scenario	1. Admin views "Pending Tickets" on dashboard. 2. Admin clicks on a high-priority ticket. 3. Admin takes offline action (e.g., fixes the WiFi). 4. Admin enters comments: "Router restarted. Issue fixed." 5. Admin clicks "Mark Resolved". 6. System notifies the Student.
Alternate Flows	A1: Forward Ticket 1. Admin sees ticket belongs to another dept. 2. Admin clicks "Re-assign". 3. System routes to correct queue.
Exceptions	None

Includes	UC-02 (Login)
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3.35 Use Case #35: View Executive Dashboard (UC-35)

Table 85 UC-35

Field	Detail
UC Identifier	UC-35
Requirements Traceability	FR-41 (Executive Analytics)
Purpose	To allow the Incharge to view the overall health and status of the campus.
Priority	High
Preconditions	User is logged in as "Campus Incharge".
Post conditions	Dashboard is displayed.
Actors	Campus Incharge , System
Extends	None
Main Success Scenario	1. Incharge logs in. 2. System loads the "Executive Overview". 3. System queries Finance, SIS, and HR services. 4. System displays charts: "Revenue vs. Target," "Attendance Trends," and "Department Performance". 5. Incharge filters data by Department (e.g., IT vs. CS).
Alternate Flows	A1: Export Report 1. Incharge clicks "Download Report". 2. System generates a PDF summary of the charts.
Exceptions	None
Includes	UC-02 (Login)

3.36 Use Case #36: Post Global Announcement (UC-36)

Table 86 UC-36

Field	Detail
UC Identifier	UC-36
Requirements Traceability	FR-42 (Announcements)
Purpose	To broadcast an urgent message to the university.
Priority	High
Preconditions	Incharge is logged in.
Post conditions	Message is visible to all selected users.
Actors	Campus Incharge , System (Notification Service)
Extends	None
Main Success Scenario	1. Incharge clicks "Make Announcement". 2. Incharge types Title ("Exam Postponed") and Body. 3. Incharge selects Audience: "All Students". 4. Incharge clicks "Publish". 5. System posts the banner to Student Dashboards. 6. System sends Push Notifications to all student devices.
Alternate Flows	None
Exceptions	None
Includes	UC-02 (Login)

3.37 Use Case #37 Process Student Leave (UC-37)

Table 87 UC-37

Field	Detail
UC Identifier	UC-37
Requirements Traceability	FR-43 (Student Leave)
Purpose	To allow students to apply for leave and notify teachers automatically.
Priority	Medium
Preconditions	Student is logged in.
Post conditions	Leave status is "Approved", Teachers are notified, Attendance is marked "Leave" (L).
Actors	Student, HOD, System
Extends	None
Main Success Scenario	1. Student navigates to "My Leaves" and clicks "Apply". 2. Student selects Dates, Reason, and uploads proof. 3. Student clicks "Submit". 4. System notifies HOD. 5. HOD reviews proof and clicks "Approve". 6. System marks attendance as "Leave" (L) for those dates. 7. System sends "Student Excused" notification to all relevant Faculty.
Alternate Flows	A1: Rejection 1. HOD clicks "Reject". 2. System notifies Student: "Application Rejected - Insufficient Proof".
Exceptions	None
Includes	UC-02 (Login)

3.38 Use Case Diagrams

3.38.1 Use Case Diagram 1

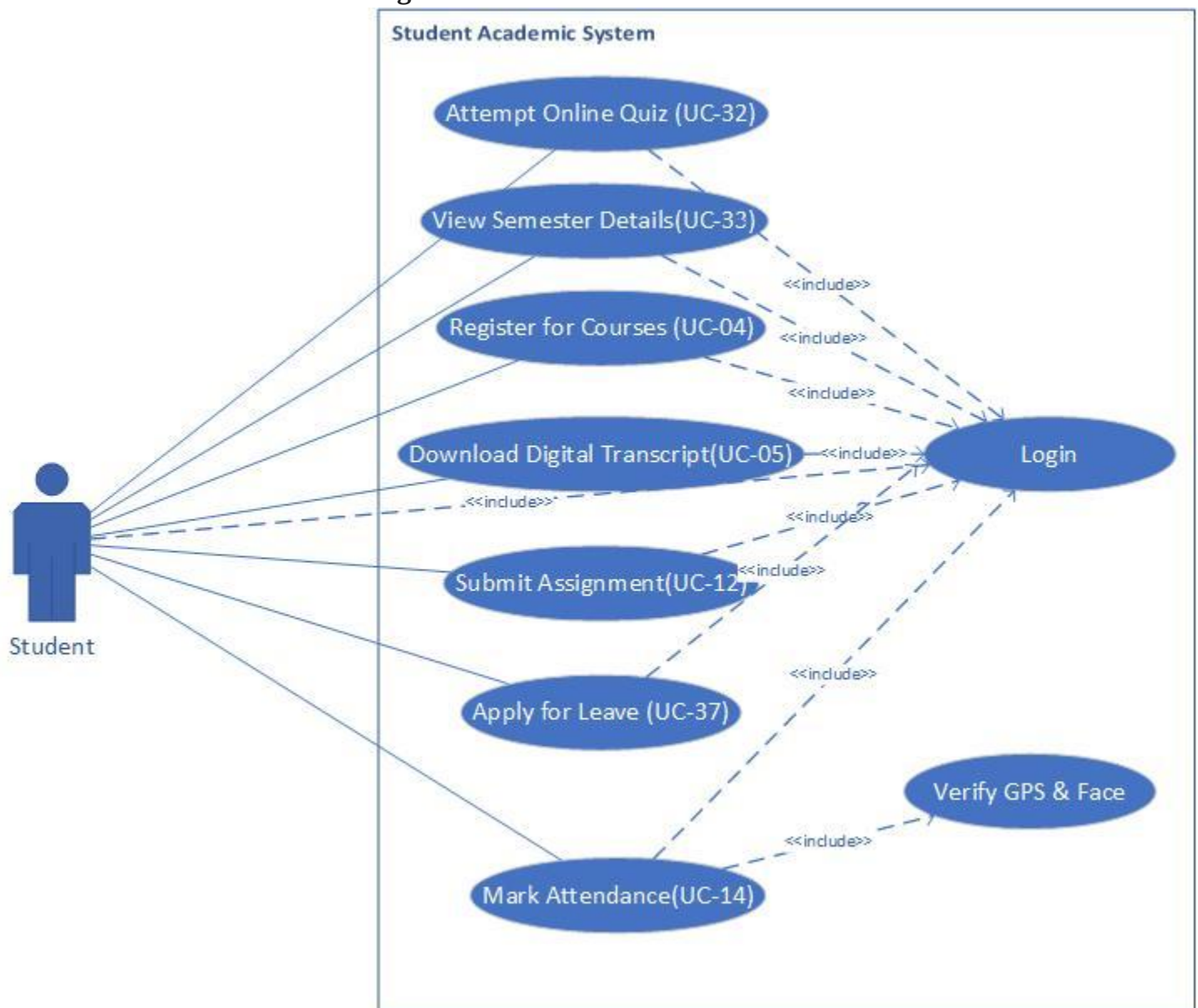


Figure 2 Student Academic System

This diagram illustrates the core academic interactions for the **Student** actor.

3.38.2 Use Case Diagram 2

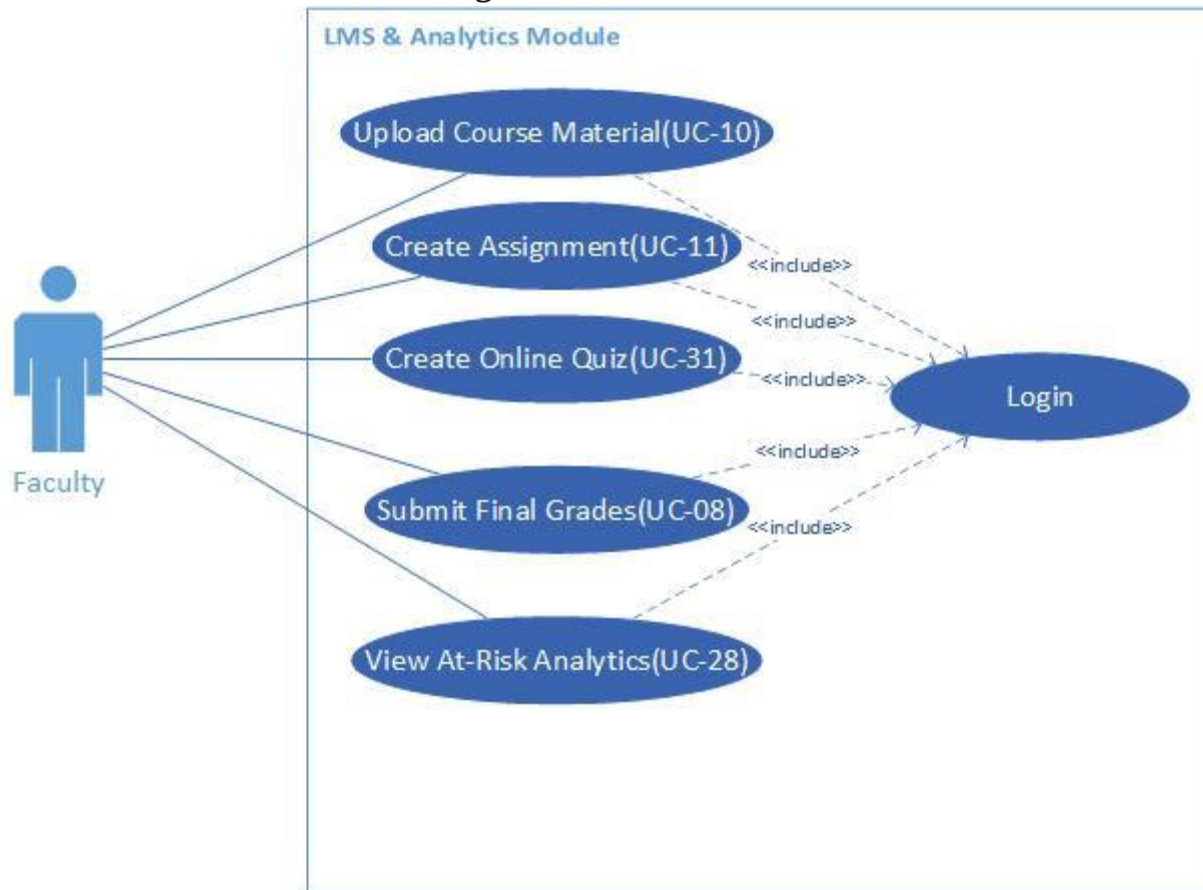


Figure 3 LMS & Analytics Module

This diagram details the pedagogical workflows managed by the **Faculty** actor.

3.38.3 Use Case Diagram 3

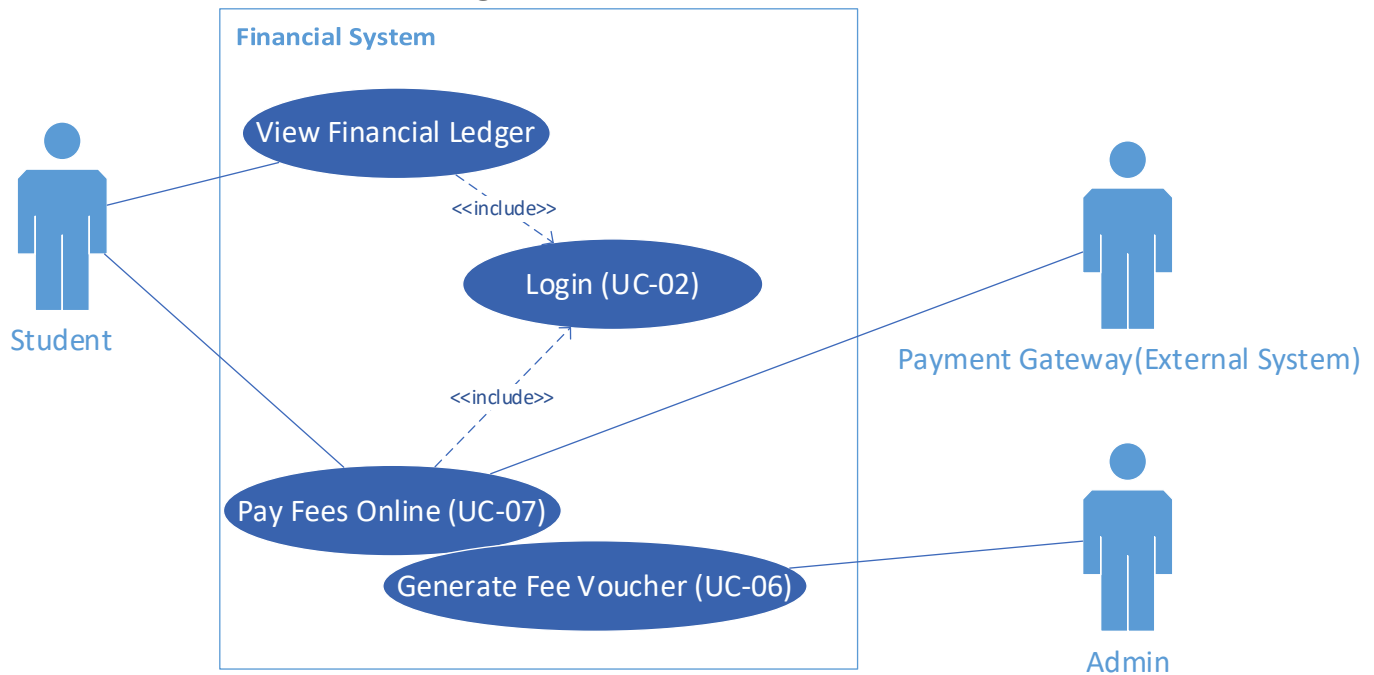


Figure 4 Financial System

This diagram depicts the financial module, focusing on the seamless integration between the **Student** and the **External Payment Gateway**.

3.38.4 Use Case Diagram 4

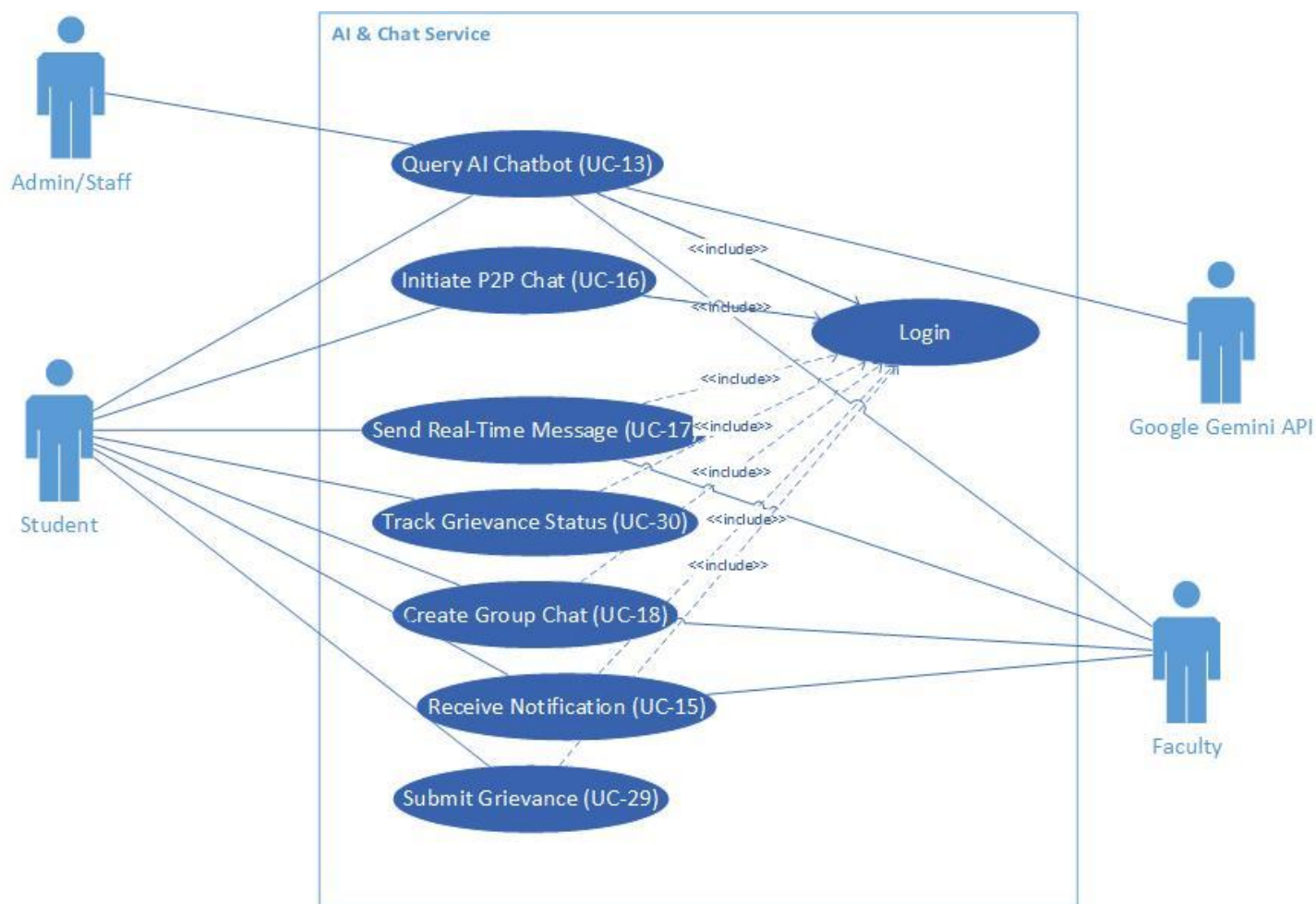


Figure 5 AI & Chat Service

This diagram represents the intelligent communication layer of Project Nexus.

3.38.5 Use Case Diagram 5

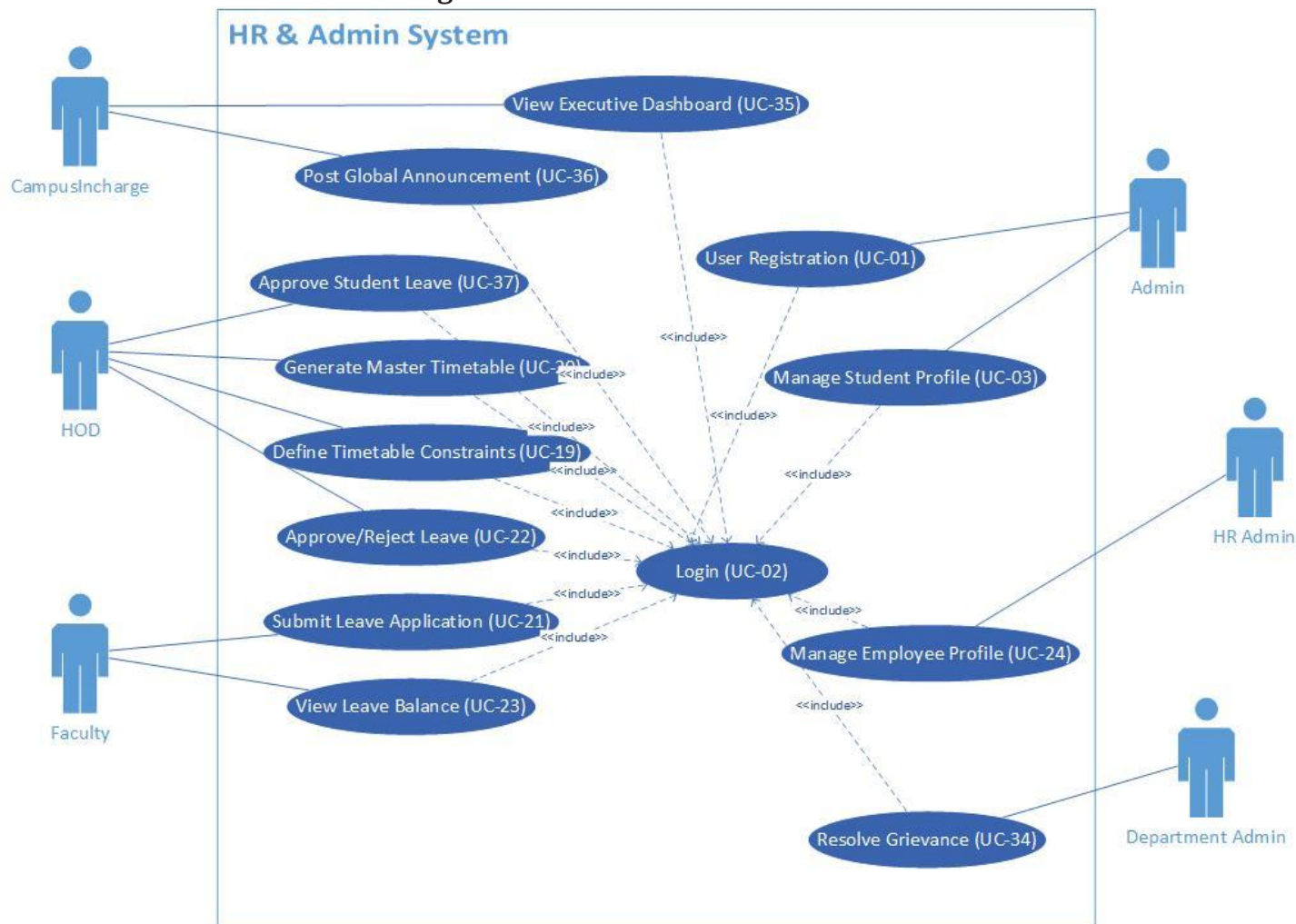


Figure 6 HR & Admin System

This diagram outlines the administrative and operational workflows. It connects the **HOD**, **Campus Incharge**, and **HR Admin** to their respective management functions.

3.38.6 Use Case Diagram 6

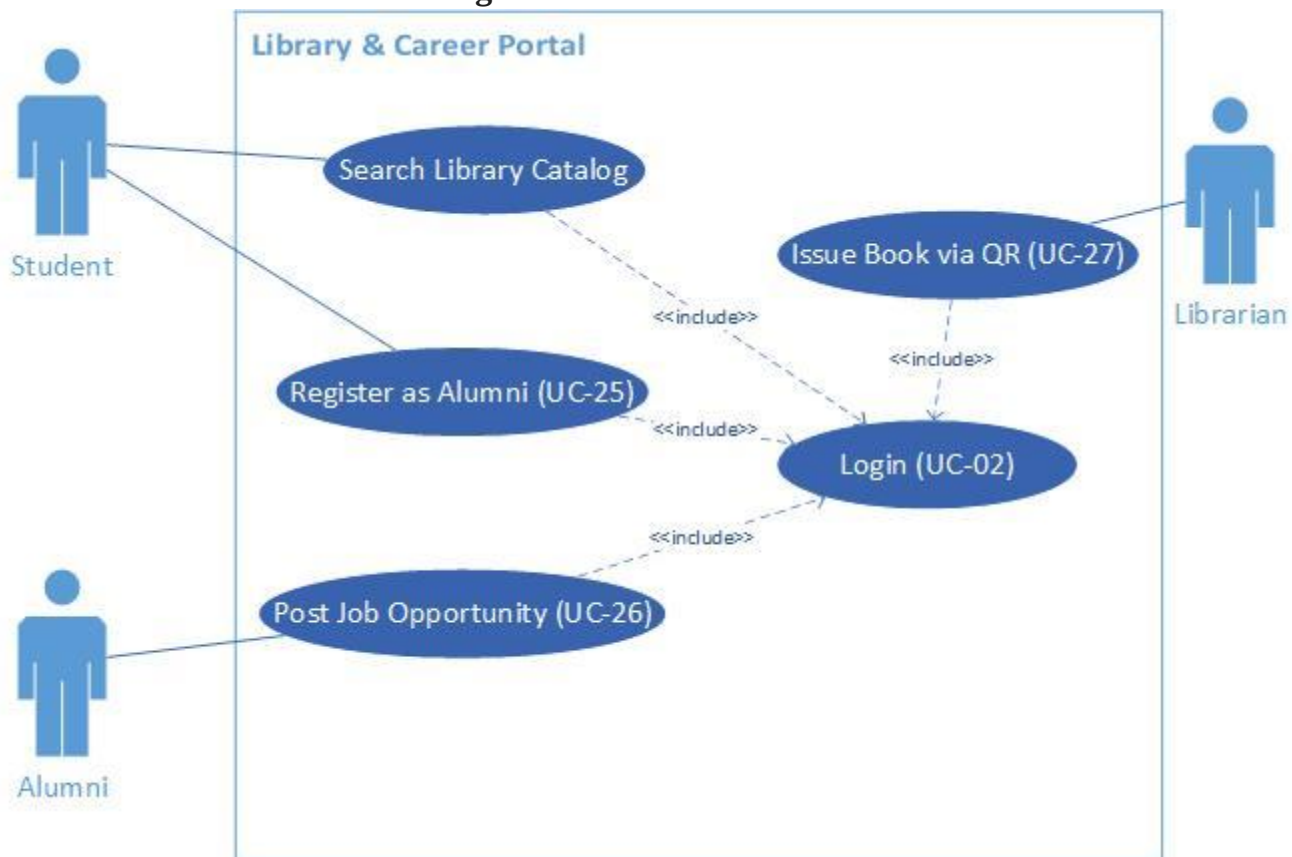


Figure 7 Library & Career Portal

This diagram illustrates the auxiliary services that enhance the campus experience.

3.38.7 Use Case Diagram 7

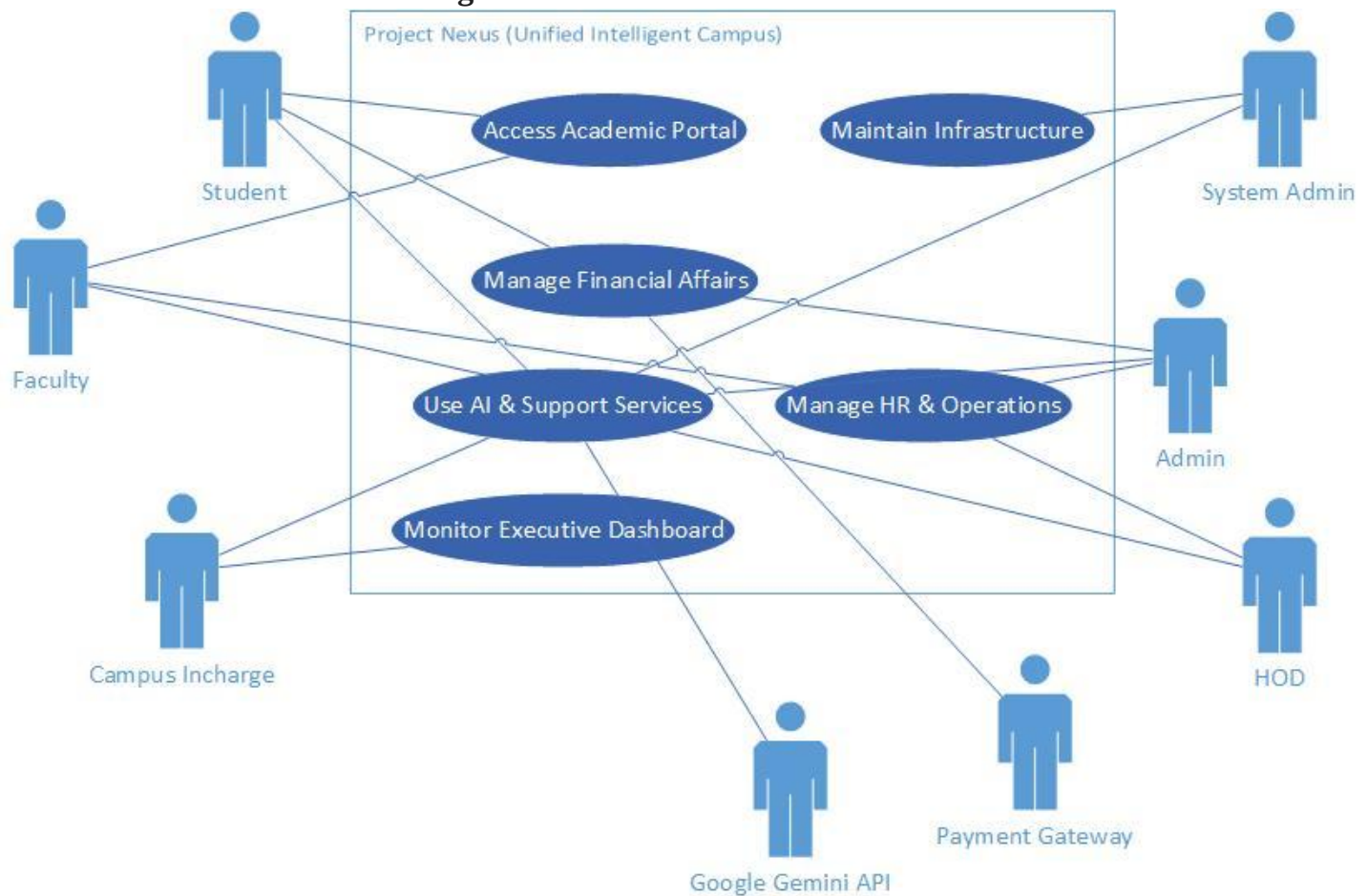


Figure 8 System Interaction Context

This high-level interaction diagram provides a holistic view of the Project Nexus ecosystem.

4. Storyboards

Following are storyboards of main features of the Project Nexus.

4.1 AI Chatbot (Context-Aware RAG)

Goal: To demonstrate how the AI uses student data to answer personal queries.



Figure 9 AI-Chat

Table 88 Storyboard-1

Frame	Visual Description	System Logic / Narrative
Frame 1	The student opens the chat interface and sends a specific query: "Why is my GPA low this semester?"	User Action: The student initiates a query about their academic performance. Logic: The system authenticates the user and prepares to search their specific records.
Frame 2	The bot displays a "... Thinking" animation. A status indicator shows the system is processing the request.	Processing: The AI Service uses the Vector Database to retrieve the student's transcript and compares grades against the university schema.
Frame 3	The bot responds: "Your GPA dropped because you scored 45% (D Grade) in Data Structures." It suggests a library module.	Outcome: The AI generates a Context-Aware Response (RAG) , identifying the exact cause of the low GPA and providing a helpful recommendation.

4.2 AI Analytics & Prediction (Data Science)

Goal: To demonstrate how the system identifies "At-Risk" students for faculty intervention.



Figure 10 AI-Analysis & Prediction

Table 89 Storyboard-2

Frame	Visual Description	System Logic / Narrative
Frame 1	The Faculty member views a list of their assigned courses (CS-101, CS-202, SE-301) and clicks "See".	User Action: The teacher selects a specific course to view real-time performance data.
Frame 2	The Analytics Dashboard loads Pie/Bar charts. An alert shows "2 Students at risk".	AI Processing: The Prediction Model aggregates attendance and grades, flagging students falling below the 75% threshold.
Frame 3	A detailed list appears showing "Ahmad" (Attendance 60%). The teacher clicks "Send Warning Notification".	Intervention: The faculty reviews the specific risk factors and triggers an instant alert to the struggling students.

4.3 Smart Attendance (Geo-Fence + Liveness)

Goal: To demonstrate the anti-spoofing security features during check-in.

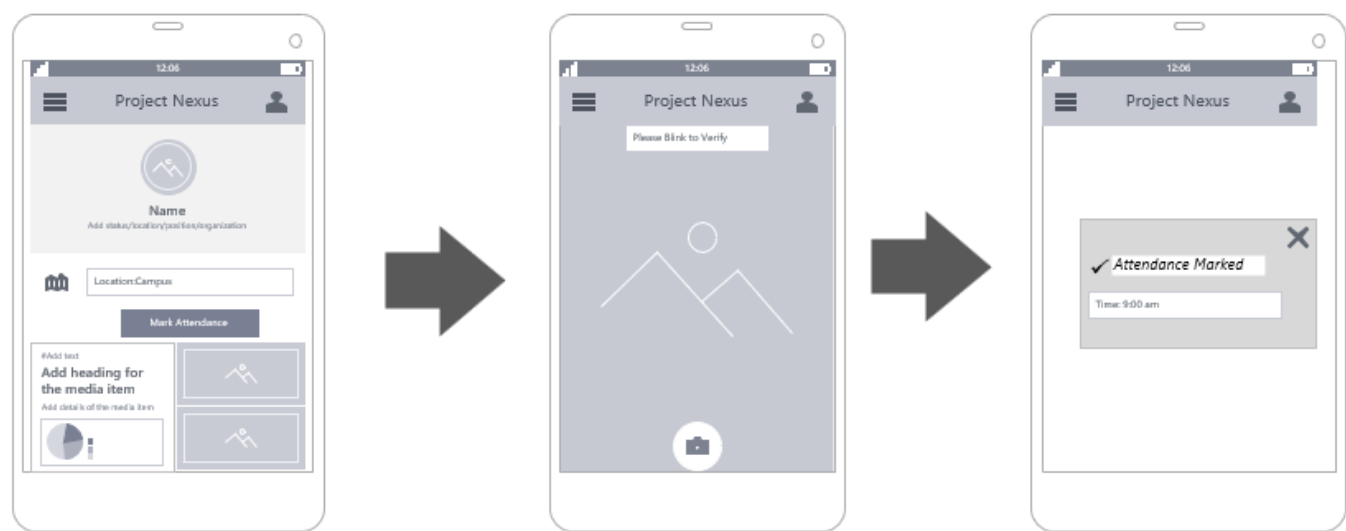


Figure 11 Smart Attendance

Table 90 Storyboard-3

Frame	Visual Description	System Logic / Narrative
Frame 1	The Dashboard shows the user's profile and location status. The user clicks "Mark Attendance".	Validation: The system checks the GPS Sensor . It only enables the button if the student is physically inside the campus boundary.
Frame 2	The screen switches to a camera view with a prompt: "Please Blink to Verify".	Liveness Check: The system uses Computer Vision to detect eye movement, ensuring the user is a live person and not a static photo.
Frame 3	A success popup appears: "Attendance Marked" at 9:00 AM.	Database Update: Upon facial match, the system marks the student as "Present" in the central database instantly.

4.4 Online Quiz Attempt (Auto-Grading)

Goal: To demonstrate the LMS capability, timer logic, and instant feedback.



Figure 12 Online Quiz Attempt

Table 91 Storyboard-4

Frame	Visual Description	System Logic / Narrative
Frame 1	The Quiz details screen shows "Quiz 1: Python Basics", Total Marks: 20, and Time Limit: 10 Mins.	User Action: The student reviews the quiz constraints and clicks " Attempt " to begin the session.
Frame 2	The active quiz screen shows a Countdown Timer (9:45) and Multiple Choice Questions.	Session Logic: The system enforces a strict server-side timer. The student selects answers (e.g., "Tuple").
Frame 3	The Result screen appears immediately: " Submission Successful ", Score: 18/20, Grade: A.	Auto-Grading: The system automatically submits the quiz when done, compares answers with the key, and updates the Gradebook instantly.

4.5 Automated Timetable Generation

Goal: To demonstrate the constraint-satisfaction algorithm for administrative automation.



Figure 13 Automated Timetable

Table 92 Storyboard-5

Frame	Visual Description	System Logic / Narrative
Frame 1	The HOD configures constraints: "Dept: Computer Science", "Avoid Friday Prayer Gap", "Max 3 Classes".	Configuration: The administrator defines the hard constraints required for a valid schedule.
Frame 2	A loading screen shows a spinner and text: "Running Constraint-Satisfaction Algorithm..."	Algorithm Execution: The system iterates through rooms, teachers, and time slots to find a conflict-free arrangement.
Frame 3	The Final Schedule is displayed with "0 Conflicts Found". The user clicks "Publish".	Deployment: The system generates the master timetable and publishes it to all student and faculty dashboards instantly.

5. Summary

The Requirements Engineering (RE) process for **Project Nexus** served as the critical foundation for transitioning the project from a high-level concept to a concrete, architectural blueprint. To ensure the system effectively addressed the core objective of creating a "Unified Intelligent Campus," a rigorous and multi-phased approach was employed for requirements identification and analysis.

First, we conducted a **Gap Analysis** of existing open-source platforms. This process exposed the critical "Data Silo" problem where academic and financial systems fail to communicate directly influencing our architectural decision to implement **Event-Driven Microservices** using **Apache Kafka**.

Following this, we utilized **Use Case Analysis** to define precise user-system interactions. By mapping out distinct scenarios (UC-01 to UC-37), we identified necessary functional constraints that were not initially obvious, such as the requirement for **GPS Geofencing** and **Biometric Liveness Detection** to ensure spoof-proof attendance. This step ensured that complex business rules were strictly defined before development began.

Finally, the **Storyboarding** technique allowed us to visualize the user experience early in the design phase. By sketching interfaces for features like the **Context-Aware AI Chatbot**, we refined the requirements to prioritize usability and responsiveness for mobile users.

In conclusion, this structured requirements engineering process was vital in mitigating technical risks and ensuring alignment with the project's goals. It transformed vague needs into actionable, testable specifications, ensuring that Project Nexus delivers a reliable, secure, and truly unified ecosystem for the university.

Chapter No. 3

Design and Architecture

6. System Design

6.1 Product Perspective

Project Nexus is designed as a **Unified Intelligent Campus Platform** that functions as a single, self-contained ecosystem. Unlike traditional university setups that isolate administrative tasks (Student Information System - SIS) from academic learning (Learning Management System - LMS), [31], [32] Project Nexus integrates these two domains into one cohesive environment.

The system operates on a **Polyglot Microservices Architecture**. This means that instead of being one large, slow application, it is built from small, independent services (such as the Attendance Service or Finance Service) that are packaged in containers and managed by **Docker Swarm** [9], [33]. This distributed design ensures **fault isolation**: if a non-critical module like the Chat Service crashes, it does not stop critical operations like Fee Payments. Functionally, the system transforms the university from a "reactive" record-keeper into a **proactive "Data Flywheel."** Operational data, such as daily attendance, is immediately used to fuel academic intelligence, such as predicting which students are at risk of failing.

6.2 Dependencies

For Project Nexus to operate successfully, the following external factors and components must be available:

- **Google Gemini API:** The system's core intelligence, specifically the Hybrid AI Chatbot, strictly depends on the availability of the Google Gemini API to process natural language queries through our Retrieval-Augmented Generation (RAG) pipeline.
- **Client Hardware Sensors:** The "Smart Attendance" module depends entirely on the end-user's device capabilities. It requires a functional **Front-Facing Camera** for biometric liveness detection and a **GPS Receiver** to validate that the student is physically on campus.
- **Docker Swarm Infrastructure:** The development and deployment environment relies on a cluster of machines (team laptops) configured with Docker Swarm. This is necessary to distribute the heavy resource load of running over 10 active microservices simultaneously.
- **Network Connectivity:** Real-time features, specifically the Peer-to-Peer (P2P) Chat and Live Push Notifications, depend on stable internet access to maintain persistent **WebSocket (WSS)** connections.

6.3 Design Constraints

The system design is limited by specific hardware, performance, and security rules that must be followed:

6.3.1 Hardware Constraints

- **Server Resources:** The backend infrastructure requires a minimum of **16GB RAM** and a **Multi-Core Processor (i5/i7)** to simultaneously host resource-intensive microservices, particularly the Apache Kafka event bus and AI processing containers.

6.3.2 Performance Constraints

- **System Latency:** Standard user interactions, such as loading the Dashboard or viewing a Profile, must complete in under **200 milliseconds** to ensure a smooth user experience.
- **Real-Time Delivery:** Chat messages and critical push notifications (e.g., "Grade Posted") must be delivered to the recipient's screen in under **1 second**.
- **AI Response Time:** Simple, cached AI queries (CAG) must load in under **50 milliseconds**. Complex, database-driven queries (RAG) must begin streaming results to the user within **3-4 seconds**.

6.3.3 Security & Privacy Constraints

- **Biometric Privacy:** To protect user privacy, the system is strictly forbidden from storing raw photographs of students. It must only store **128-dimensional mathematical vectors** (numerical representations of a face), which cannot be reverse-engineered into an image.
- **Encryption:** All data transmission must occur over secure **HTTPS (TLS 1.2+)** channels. Sensitive fields at rest, such as Faculty Salary, must be encrypted in the database.
- **Session Management:** To prevent unauthorized access on public computers, user login sessions must automatically expire after **30 minutes of inactivity**.

6.4 External Integrations

Project Nexus connects with several third-party systems to extend its functionality beyond the campus network:

- **Google Gemini API (Generative AI):**
 - **Purpose:** Connects via REST API to handle natural language understanding. It receives the user's prompt and context data (JSON) and returns intelligent, natural language responses for the AI Chatbot.

- **Payment Gateways (Stripe / JazzCash):**
 - **Purpose:** Integrated to process real-time tuition fee payments. The system securely transmits payment tokens and listens for transaction success/failure callbacks (Webhooks) to instantly update the student's financial ledger.
- **Browser Hardware APIs:**
 - **Geolocation API:** Used to retrieve precise GPS coordinates (Latitude/Longitude) for geofencing validation.
 - **MediaDevices API:** Used to securely access the user's camera stream (`navigator.mediaDevices.getUserMedia`) for facial recognition and liveness checks.
- **Docker Engine API:**
 - **Purpose:** The Infrastructure Service connects to the Docker API to monitor the health of all containers and automatically trigger restarts for any service that crashes (Self-Healing capability).

7. Design Considerations

This section outlines the fundamental rules, assumptions, and strategies that guide the architecture of Project Nexus. These considerations ensure the system is built to be robust, secure, and capable of running effectively in a university environment.

7.1 Assumptions and Dependencies

For Project Nexus to function as designed, we rely on several external factors and assumptions that are outside of our direct control.

7.1.1 Assumptions

- **Stable API Availability:** We assume that the **Google Gemini API** will maintain consistent uptime and reasonable rate limits throughout the development and demonstration phases. Since the core "Intelligence" of the Chatbot relies on this, any extended API outage would impact the system's ability to answer dynamic questions.
- **Hardware Capability:** We assume that the development team's personal laptops (configured with Core i5/i7 processors and 16GB+ RAM) are sufficient to run the **Docker Swarm** cluster. We are assuming these machines can handle the load of running 10+ containers (including heavy services like Kafka and Zookeeper) simultaneously without crashing.
- **Standard University Rules:** We assume access to standard domain logic such as generic grading schemas (SGPA/CGPA calculations), fee structures, and attendance policies to accurately program the system's business rules.

7.1.2 Dependencies

- **Browser Capabilities:** The "Smart Attendance" module is strictly dependent on the user's device. The web browser must support the **HTML5 Geolocation API** (for GPS coordinates) and **WebRTC** (for camera access). If a student uses an older device without these features, they cannot mark attendance.
- **Internet Connectivity:** The system is dependent on a stable internet connection. Features like **Peer-to-Peer (P2P) Chat** and **Real-Time Notifications** use WebSockets,[1] which require a persistent online connection to function. Offline mode is not supported for these real-time tools.

7.2 General Constraints

These are the strict limits and boundaries within which we must build the system.

7.2.1 Hardware & Resource Constraints

- **No Specialized Hardware:** The system must run on standard commodity hardware. We cannot rely on expensive GPUs (Graphics Processing Units) for the AI features. Therefore, we are constrained to use CPU-optimized libraries like **OpenVINO** or **dlib**[7], [8] for facial recognition to ensure the system remains cost-effective for the university.
- **Memory Usage:** Since we are running a distributed microservices cluster on limited hardware, individual services (like the "Finance Service") must be optimized to use minimal RAM to leave room for resource-heavy components like Apache Kafka.

7.2.2 Performance Constraints

- **Latency:** To ensure a smooth experience, standard actions (like loading a Student Profile) must complete in under **200 milliseconds**.
- **AI Speed:** The Chatbot has strict timing rules. Cached answers (CAG) for common questions must load instantly (under **50ms**). Complex database queries (RAG), such as checking a fee balance, must begin streaming the answer within **3 seconds**.
- **Real-Time Delivery:** Critical alerts (e.g., "Grade Posted") must appear on the student's screen within **1 second** of the event occurring.

7.2.3 Security & Privacy Constraints

- **Biometric Data:** We are strictly constrained from storing raw user photos in the database. To protect student privacy, we must only store **128-dimensional vectors** (mathematical representations of a face). This ensures that even if the data is stolen, it cannot be reconstructed into a photo.
- **Session Security:** To prevent unauthorized access on public university computers, user login sessions must automatically expire after **30 minutes** of inactivity.

7.3 Goals and Guidelines

These principles guide **how** we write the code and design the architecture.

- **Fault Isolation:** We follow a strict **Microservices Architecture**. This means different parts of the system (like "Fees" and "Chat") are completely separate. If the Chat service crashes, the Fee service must keep working perfectly. We achieve this by running each service in its own isolated Docker container.
- **Zero Data Loss:** We prioritize data durability over speed for critical records. We use **Apache Kafka** to handle important events (like Grade Submissions). This guarantees that even if the Student Information System (SIS) is temporarily down, the grade data is saved safely in the Kafka queue and will process automatically once the system recovers.
- **Polyglot Persistence:** We do not use a "one-size-fits-all" database. We use the right tool for the right job:
 - **PostgreSQL** for structured data (Grades, Fees) where accuracy is critical.
 - **MongoDB** for unstructured data (Chat logs, Course files) where flexibility is needed.
 - **Redis** for high-speed caching.
 - **ChromaDB** for vector storage.
- **Usability (The 3-Click Rule):** The interface is designed so that any essential task such as checking a grade, paying a fee, or marking attendance can be completed in **3 clicks or less** from the main dashboard.

7.4 Development Methods

We use modern software engineering practices to ensure the project is organized and scalable.

- **Event-Driven Architecture:** Instead of services calling each other directly (which creates tight dependencies), we use an "Event-Driven" model. For example, the LMS Service simply publishes a "Grade Posted" event, and the SIS Service listens for it. This keeps the code clean and decoupled.
- **Database Normalization:** For our relational database (PostgreSQL), we design our tables up to the **3rd Normal Form (3NF)**. This minimizes data redundancy (duplicate data) and ensures that student records remain consistent.
- **Containerization:** We use **Docker** to package every service. This solves the "it works on my machine" problem by ensuring that the code runs exactly the same way on every developer's laptop and the final production server.

8. Design Models

8.1 Architectural Design

Project Nexus is architected as a Unified Intelligent Campus Platform operating on an Event-Driven Microservices Architecture. Unlike traditional monolithic systems where all modules (SIS, LMS, Finance) are tightly coupled into a single codebase, Project Nexus decouples these domains into autonomous, containerized services.

This design choice was driven by two critical requirements

- **Fault Isolation:** A failure in a peripheral service (e.g., the Chat Service) does not impact critical administrative operations (e.g., Fee Collection).
- **Polyglot Persistence:** Different modules require different storage engines. The architecture allows the LMS to use MongoDB for unstructured course files while the Finance module utilizes PostgreSQL for ACID-compliant transaction integrity.

The system is organized into four distinct logical layers

- **The Presentation Layer (Client):** Web Portal (React.js): A Web Application serving as the unified interface for Students, Faculty, and Admins. It communicates with the backend via secure REST APIs and WebSockets.
- **The Gateway Layer:**
 - **API Gateway (Nginx):** The single entry point for all external traffic. It handles SSL termination, Rate Limiting, and Request Routing.
- **The Application Layer (Compute):**
 - **Business Logic Cluster:** Contains the core administrative microservices (SIS, Finance, HR, Alumni) responsible for CRUD operations and report generation.
 - **Intelligent Services Cluster:** Hosts resource-intensive modules including the AI Service (RAG Pipeline) and Attendance Service (Biometric Processing).
 - **Apache Kafka (Event Bus):** The nervous system of the platform. It facilitates asynchronous communication (e.g., Grade Posting, Fee Alerts) to ensure Zero Data Loss.
- **The Data Layer:** Polyglot Databases: A cluster of specialized databases (PostgreSQL, MongoDB, Redis, ChromaDB) tailored to specific service needs.

8.1.1 High Level Architecture

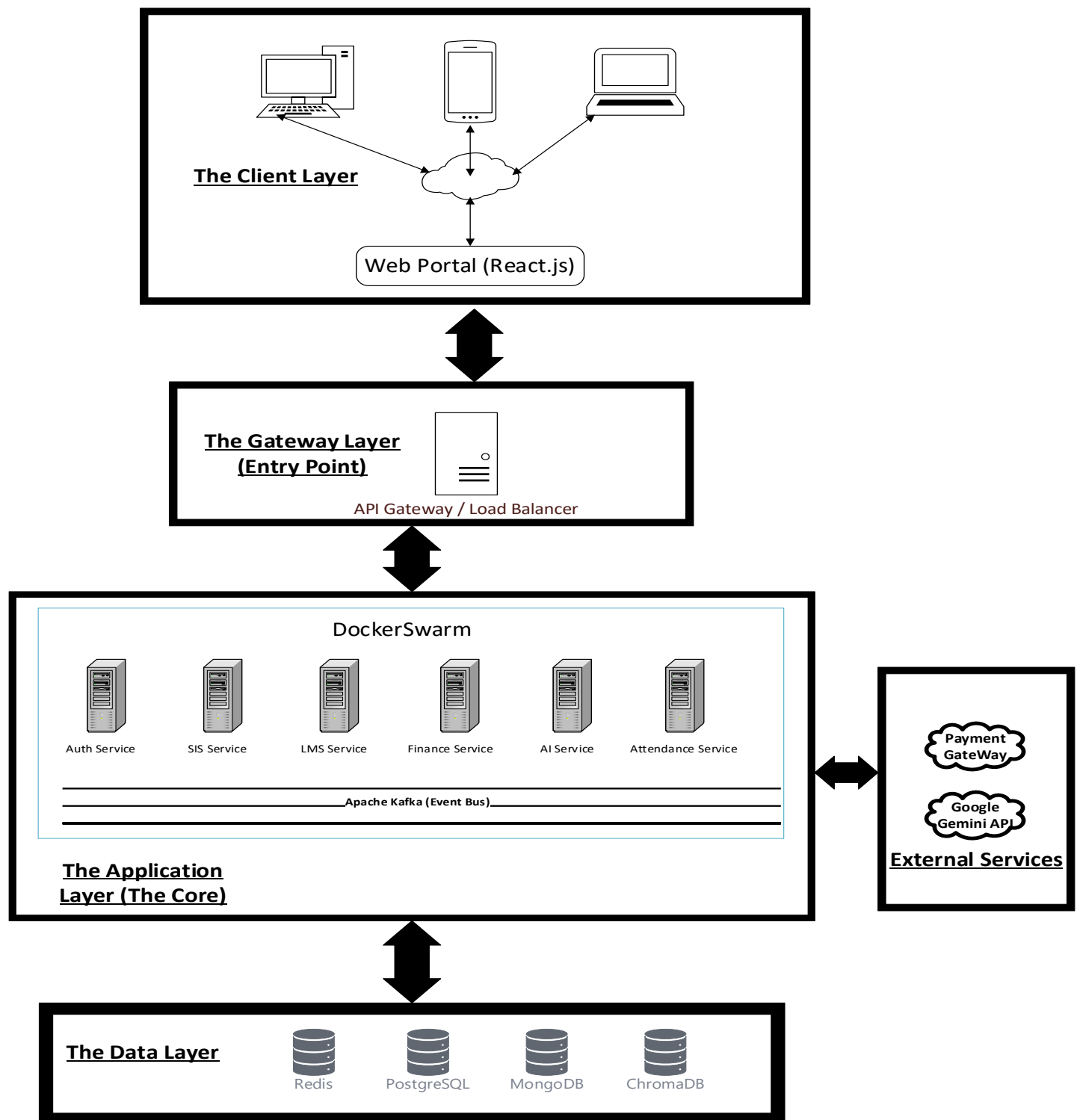


Figure 14 High Level Architecture Diagram

This high-level architecture diagram illustrates the **4-tier layered pattern** that forms the structural foundation of Project Nexus. This tiered approach provides clear separation of concerns, where each layer has a specific responsibility and communicates with adjacent layers through well-defined interfaces.

Layer 1: The Client Layer (Presentation Tier)

The Client Layer represents all user-facing touchpoints that access the system. This includes:

- **Desktop Computers:** Full-featured access via web browsers (Chrome, Firefox, Edge)
- **Tablets & Mobile Devices:** Responsive design adapts to smaller screens for on-the-go access
- **Laptops:** Primary device for faculty and administrators

All these clients interact with a single **Web Portal (React.js)**, which is a Single Page Application (SPA) that provides a unified, consistent user experience across all devices. The SPA architecture ensures fast page transitions without full-page reloads, improving responsiveness.

Communication Pattern: Clients communicate with the backend exclusively through **HTTPS (TLS 1.3)** to ensure data encryption in transit. The React application makes asynchronous API calls using **Axios HTTP Client** and maintains real-time connections via **WebSocket (WSS)** for features like chat and notifications.

Layer 2: The Gateway Layer (Entry Point)

The Gateway Layer serves as the **single entry point** for all external requests, functioning as a security checkpoint and traffic controller.

API Gateway (Nginx/Kong):

- **SSL/TLS Termination:** Decrypts incoming HTTPS traffic and encrypts outgoing responses
- **Load Balancing:** Distributes requests across multiple service instances using Round-Robin algorithm
- **Rate Limiting:** Enforces API quotas (e.g., 100 requests per minute per user) to prevent abuse
- **Request Routing:** Routes requests to the appropriate microservice based on URL paths (e.g., /api/v1/grades → LMS Service)

Why a Gateway? Without a gateway, each microservice would need to implement its own security, SSL, and rate-limiting logic, leading to code duplication and inconsistent security policies. The gateway centralizes these cross-cutting concerns.

Layer 3: The Application Layer (The Core Logic)

This is where the **business intelligence** resides. The Application Layer is divided into two sub-clusters:

Docker Swarm Orchestration:

All services run as **Docker containers** managed by Docker Swarm. Swarm provides:

- **Service Replication:** Each critical service runs 3 replicas for high availability
- **Auto-Healing:** Crashed containers automatically restart
- **Rolling Updates:** Services can be updated without downtime
- **Microservices Cluster (12 Services):**
 - **Auth Service:** User authentication and JWT token issuance
 - **SIS Service:** Student records, enrollment, transcripts
 - **LMS Service:** Course content, assignments, grades
 - **Finance Service:** Fee vouchers, payments, ledger
 - **Attendance Service:** Biometric verification, GPS geofencing
 - **AI Service:** RAG pipeline for intelligent chatbot
 - **Chat Service:** Real-time messaging via WebSockets
 - **Analytics Service:** Predictive analytics and dashboards
 - **HR Service:** Faculty management, payroll, leave
 - **Library Service:** Book catalog, issue/return, fines
 - **Operations Service:** Grievances, feedback, announcements
 - **Alumni Service:** Job board, networking, alumni tracking

Apache Kafka (Event Bus):

Kafka sits at the heart of the Application Layer, enabling **asynchronous, event-driven communication**. [27] Instead of services directly calling each other (which creates tight coupling), services publish and subscribe to events.

Example Event Flow:

1. Faculty submits grade in **LMS Service** → Publishes grade_submitted event to Kafka
2. **SIS Service** subscribes to grade_submitted → Updates official transcript
3. **Notification Service** subscribes to grade_submitted → Sends push notification to student
4. **Analytics Service** subscribes to grade_submitted → Updates performance dashboard

Benefit: If the SIS Service is temporarily down, Kafka stores the event until it recovers, ensuring **zero data loss**.

External Integrations:

The Application Layer also communicates with external third-party services:

- **Google Gemini API:** For AI-powered natural language processing
- **Payment Gateway (Stripe/JazzCash):** For online fee payments

Layer 4: The Data Layer (Persistence)

The Data Layer employs a **Polyglot Persistence** strategy, using four specialized databases:

- **PostgreSQL (Relational Database):**
Usage: Structured, transactional data requiring ACID compliance
Examples: Student records, grades, fees, attendance logs
- **MongoDB (Document Database):**
Usage: Unstructured/semi-structured data with flexible schemas
Examples: Chat messages, notification logs, audit trails
Benefit: Rapid development without rigid schema constraints
- **Redis (In-Memory Cache):**
Usage: High-speed caching and session storage
Examples: User sessions (JWT tokens), frequently accessed data, real-time pub/sub
Speed: Sub-millisecond latency for reads
- **ChromaDB (Vector Database):**
Usage: AI/ML features requiring semantic search[14]
Examples: Face embeddings (128D vectors), document embeddings for RAG
Algorithm: Cosine similarity for nearest-neighbor search

Why Polyglot?

Different data workloads have different performance characteristics. Financial transactions require strict ACID guarantees (PostgreSQL), while chat logs benefit from schema flexibility (MongoDB). Using the right tool for each job optimizes both performance and development speed.

Inter-Layer Communication:

- **Client ↔ Gateway:** HTTPS/WSS (Encrypted)
- **Gateway ↔ Application:** HTTP/2 (Internal network, trusted)
- **Application ↔ Data:** Native database protocols (PostgreSQL wire protocol, MongoDB BSON, Redis RESP)
- **Application ↔ Application:** Apache Kafka (Event-driven messaging)

8.1.2 System Component Structure

The system logic is distributed across four distinct architectural layers, ensuring modularity and separation of concerns. This layered approach allows individual components such as the AI Service or Payment Gateway to be developed, deployed, and scaled independently.

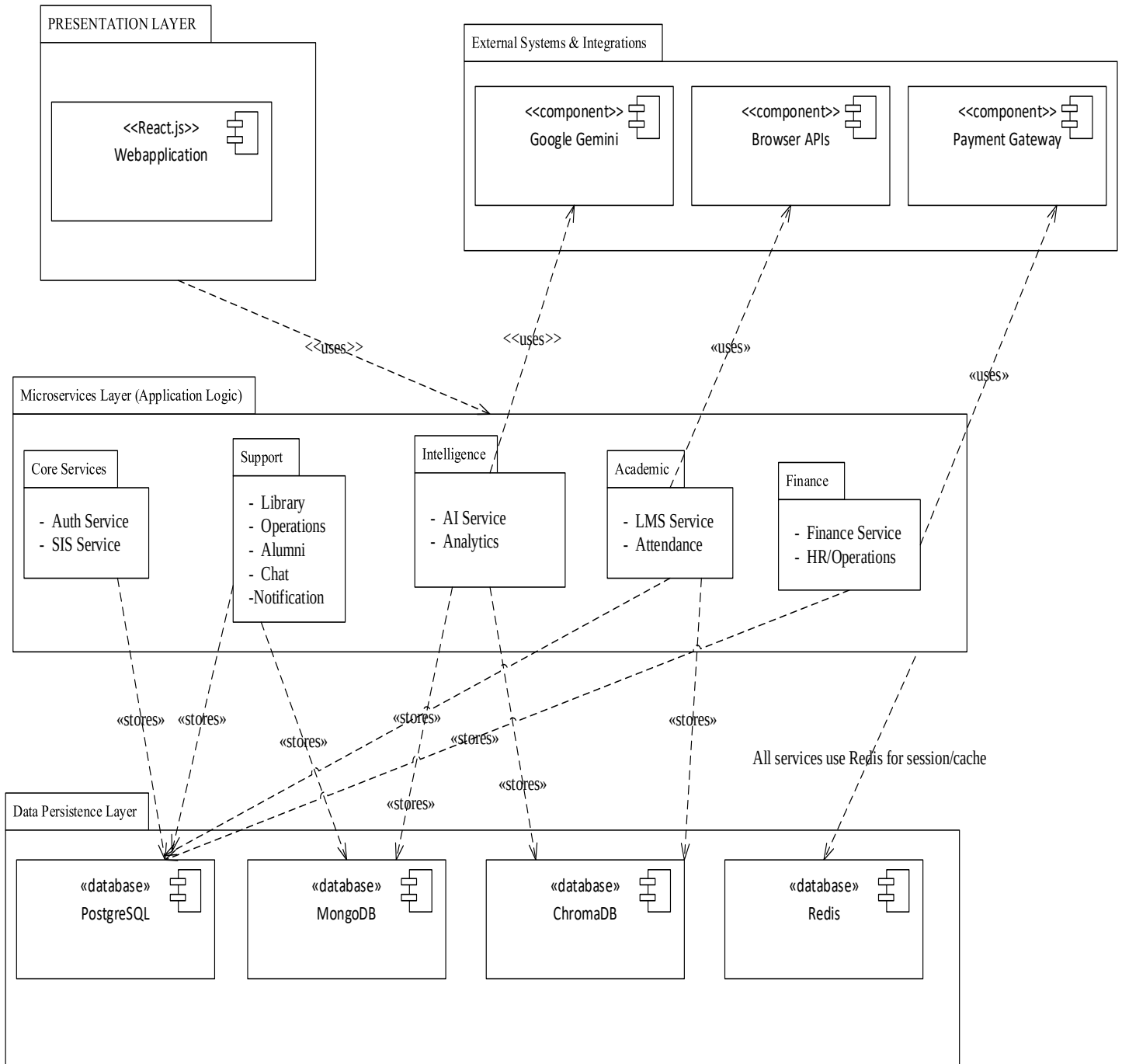


Figure 15 UML Component Diagram [High-Level Layered Architecture]

External Integrations

This layer handles user interaction and communication with third-party services. The **React.js Web Application** serves as the primary interface, communicating with the backend via secure REST APIs. Specialized adapters handle external dependencies, including **Google Gemini** for generative AI tasks and **Stripe/JazzCash** for real-time financial transactions.

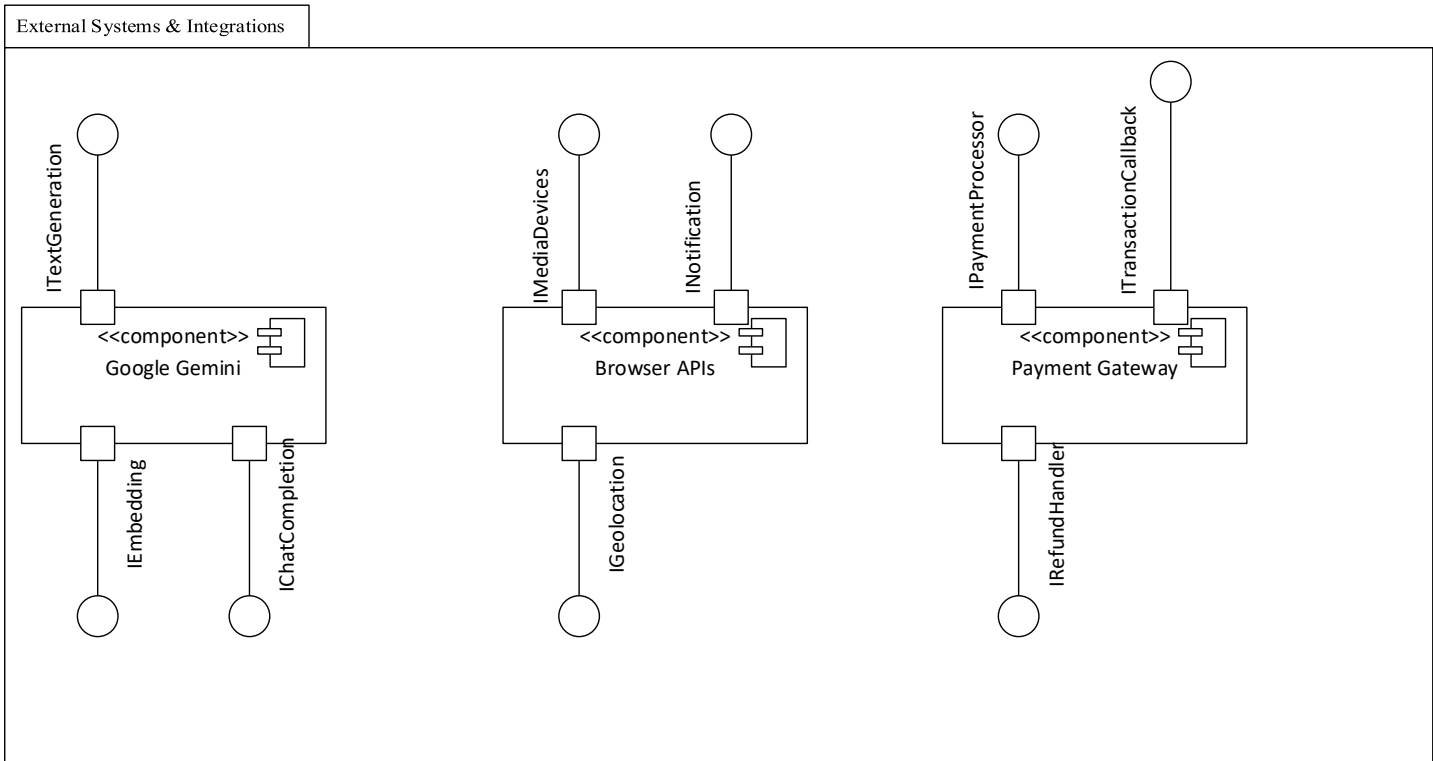


Figure 16 UML Component Diagram [External System Integrations]

Microservices Layer (Application Logic)

This is the core execution layer where business logic resides. Services are grouped into functional domains to ensure fault isolation:

- **Core Services:** Handle identity verification (Auth Service) and fundamental student records (SIS Service).
- **Academic Services:** Manage learning content (LMS Service) and biometric attendance verification (Attendance Service).
- **Intelligence Services:** Power the AI Chatbot and Predictive Analytics engines.
- **Support Services:** Manage library resources, operational grievances, and alumni networking.

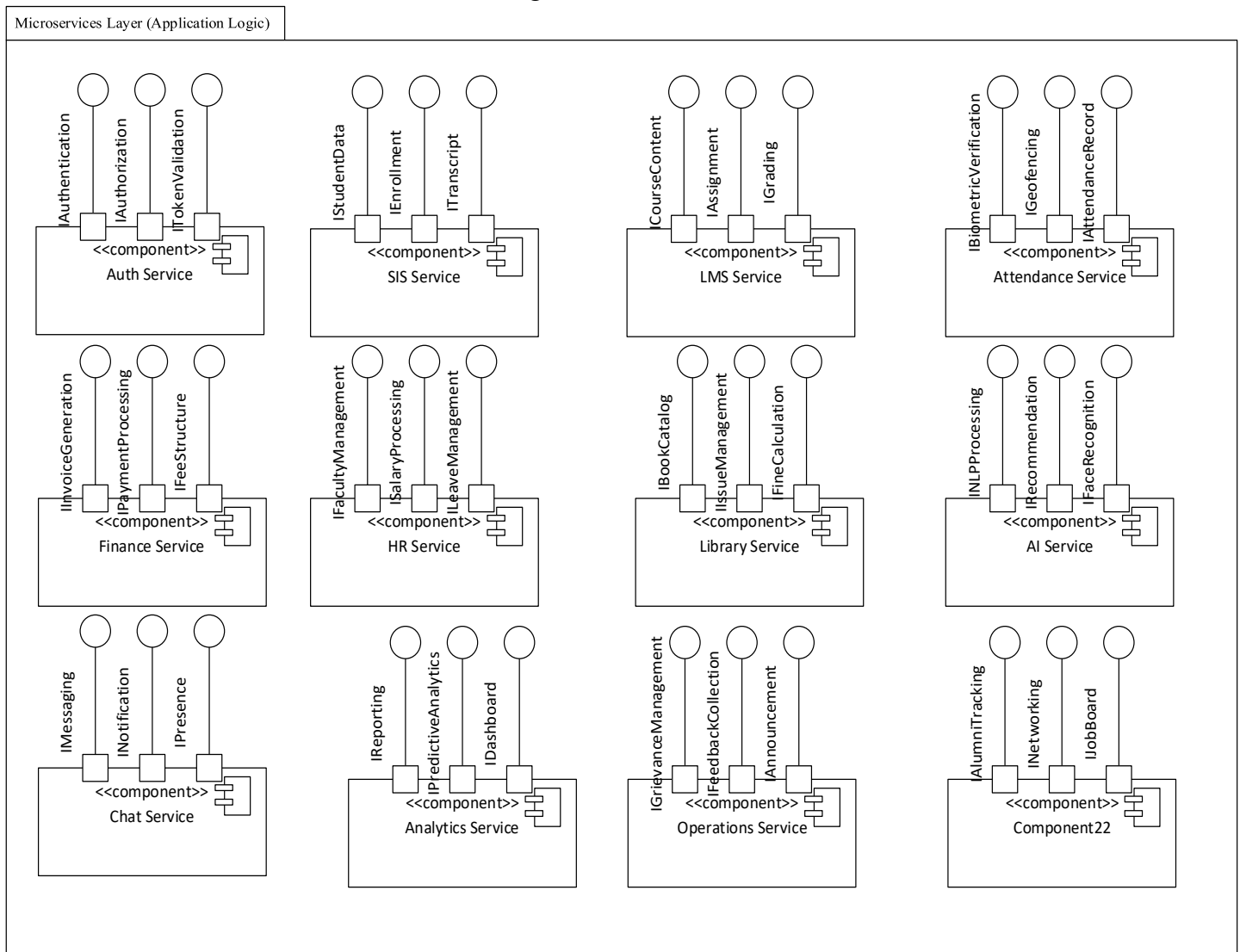


Figure 17 UML Component Diagram [Microservices Component Decomposition]

Data Persistence Layer

Project Nexus utilizes a **Polyglot Persistence** strategy, selecting the optimal storage engine for each specific data type.

- **PostgreSQL:** Stores ACID-compliant relational data, including user credentials, fee invoices, and academic transcripts.
- **MongoDB:** Stores flexible, unstructured data such as chat history, audit logs, and system notifications.
- **ChromaDB:** A specialized vector store for high-dimensional embeddings used in Facial Recognition (128d) and RAG (768d).
- **Redis:** Provides high-speed in-memory caching for user sessions and real-time Pub/Sub messaging.

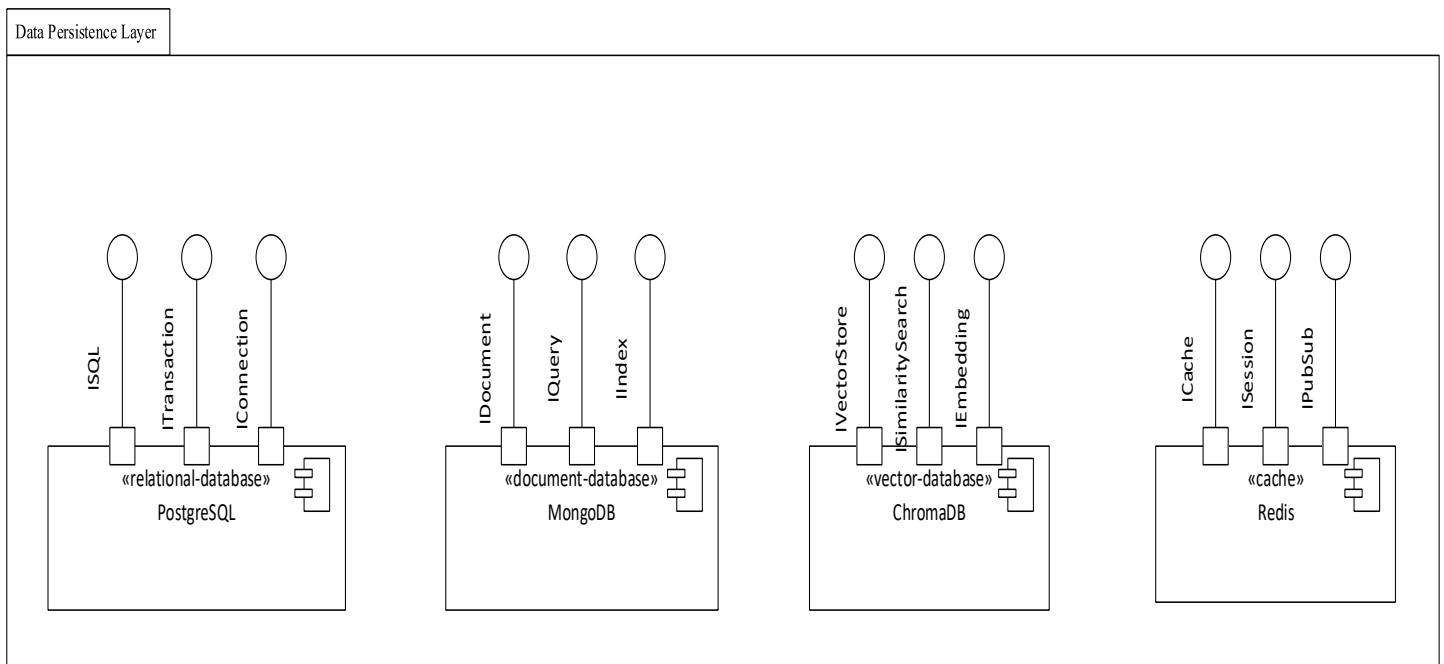


Figure 18 UML Component Diagram [Polyglot Data Persistence Layer]

8.1.3 Deployment Architecture

The physical architecture of Project Nexus follows a **Containerized Cluster Model** orchestrated by **Docker Swarm**. This approach ensures high availability and horizontal scalability. The system artifacts are distributed across three primary computing nodes one Manager Node and two Worker Nodes connected via a private Docker Overlay Network.

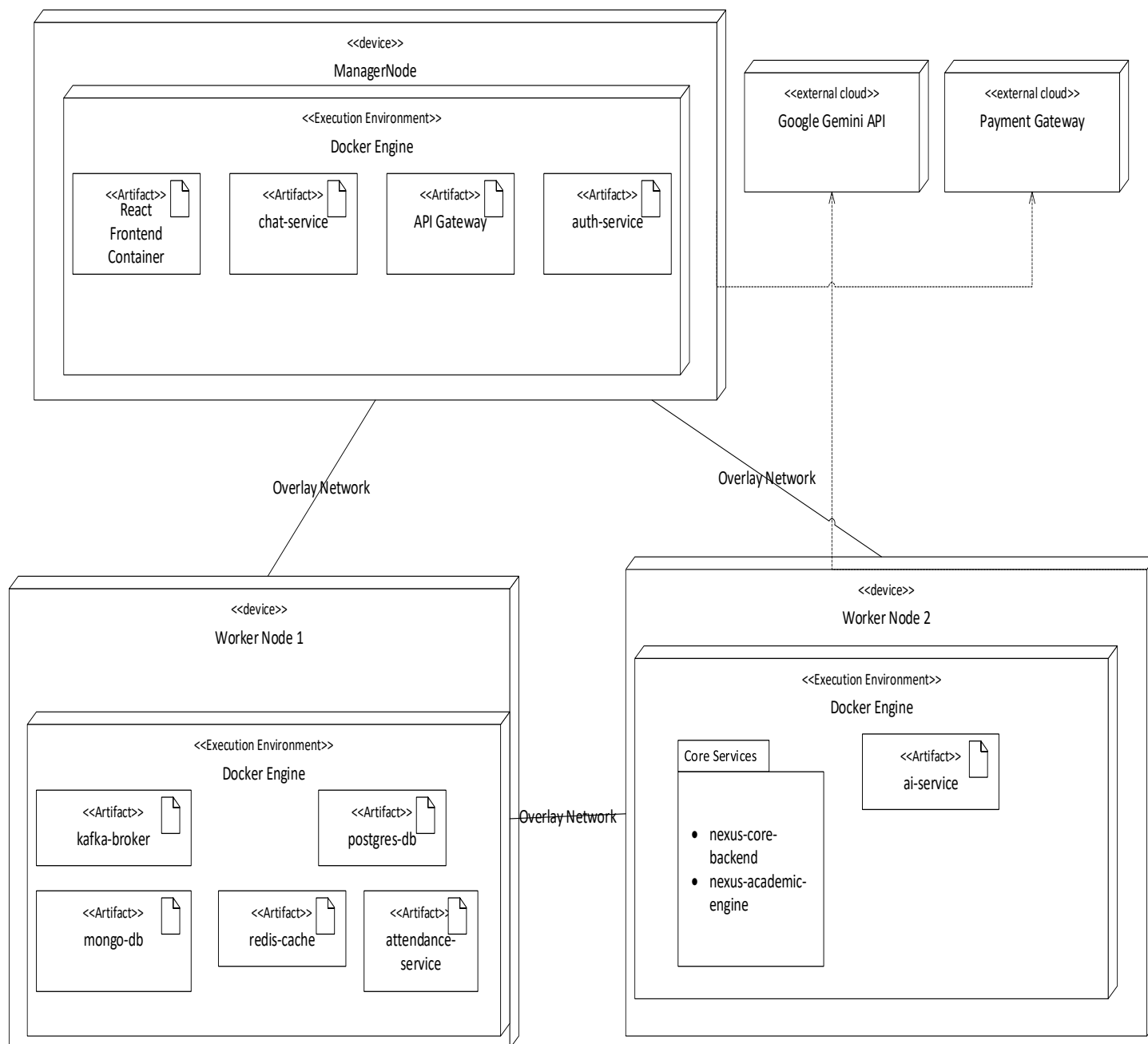


Figure 19 UML Deployment Diagram

Node Configuration & Artifact Mapping

As illustrated in the deployment diagram, the services are allocated based on their resource requirements and roles:

- **Manager Node (Orchestration & Entry):**
 - **Role:** Acts as the cluster leader and traffic entry point.
 - **Artifacts:** Hosts the **React Frontend Container** (User Interface), **API Gateway** (Nginx ingress), and the critical **Auth Service** to handle all incoming security validation before traffic reaches the workers. It also manages the **Chat Service** for real-time connectivity.
- **External Links:** Connects directly to external cloud services including the **Google Gemini API** and **Payment Gateway**.
- **Worker Node 1 (Data & Infrastructure):**
 - **Role:** Dedicated to persistent storage and event streaming infrastructure.
 - **Artifacts:** Hosts the stateful database containers (**PostgreSQL**, **MongoDB**, **Redis Cache**) and the **Kafka Broker** for event messaging. It also runs the **Attendance Service** to isolate biometric processing load.
- **Worker Node 2 (Core Compute & AI):**
 - **Role:** Handles heavy computational tasks and core business logic.
 - **Artifacts:** Hosts the resource-intensive **AI Service** (RAG pipeline) and the **Core Services** group (nexus-core-backend, nexus-academic-engine).
- **Network Topology**
 - **Overlay Network:** All three nodes are interconnected via a secure Docker Overlay Network, allowing containers on different physical machines to communicate as if they were on the same localhost, while keeping internal traffic isolated from the public internet.

8.1.4 UML Class Diagrams

The UML Class Diagrams provide a static view of the system's application layer, illustrating the Object-Oriented (OO) structure of the microservices. These diagrams bridge the gap between the business requirements and the database schema by defining the attributes, methods, and relationships (Inheritance, Association, Composition) of the system's core classes.

The models below are decomposed into functional sub-systems to ensure clarity and modularity.

8.1.4.1 High-Level System Map

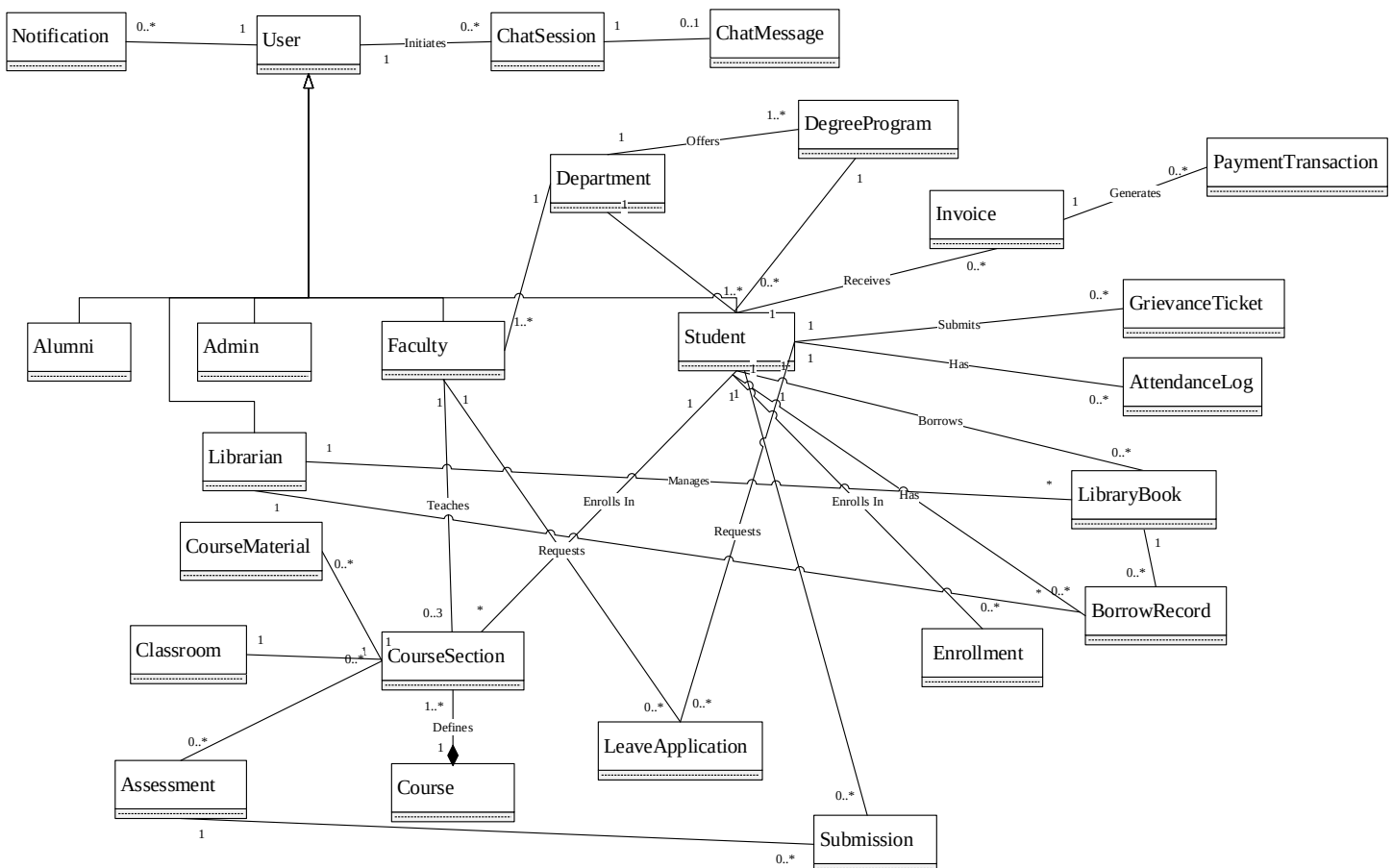


Figure 20 UML Class Diagram [High Level System Map]

This high-level abstract model illustrates the primary connectivity between the system's major modules.

8.1.4.2 User Hierarchy & Security Module

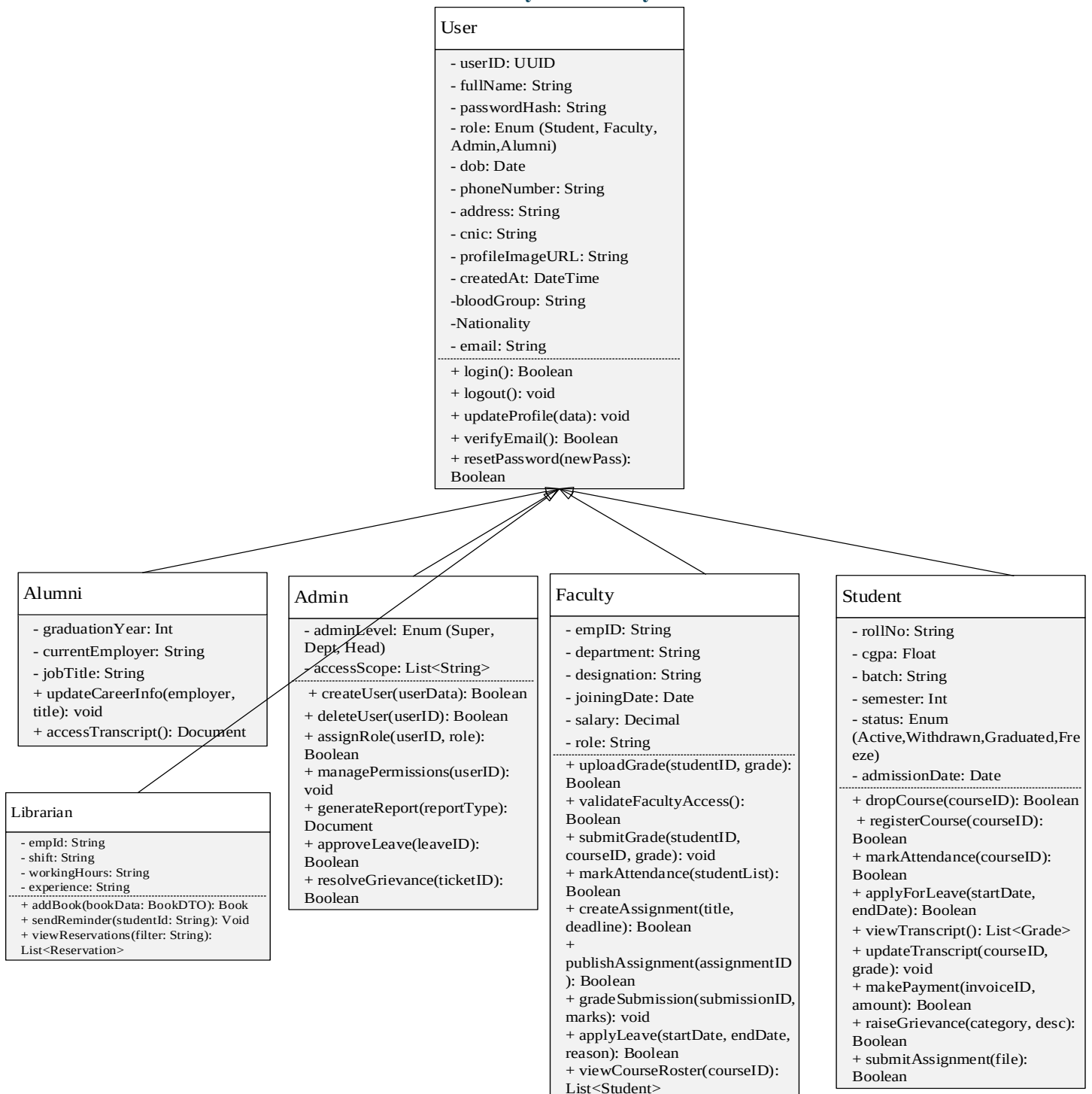


Figure 21 UML Class Diagram [User Inheritance Hierarchy]

This Details the Role-Based Access Control (RBAC) structure. The abstract User class defines common attributes (ID, Email, PasswordHash), which are inherited by specialized child classes: Student, Faculty, Admin, Alumni, Librarian.

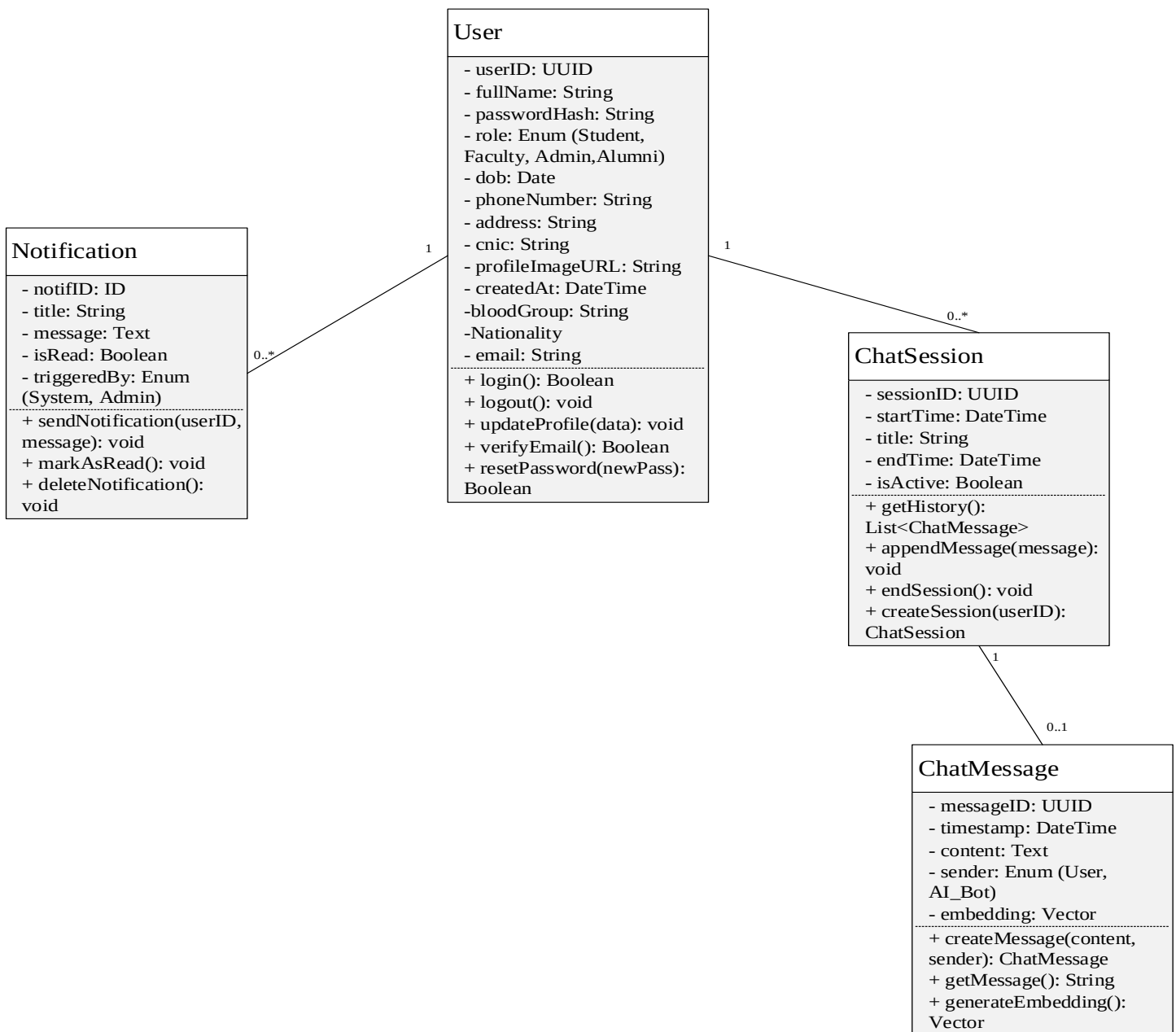


Figure 22 UML Class Diagram [Communication & Notification Entities]

This models the real-time communication architecture. It shows how Users initiate ChatSessions and receive system-wide Notifications.

8.1.4.3 Academic Sub-Systems (SIS & LMS)

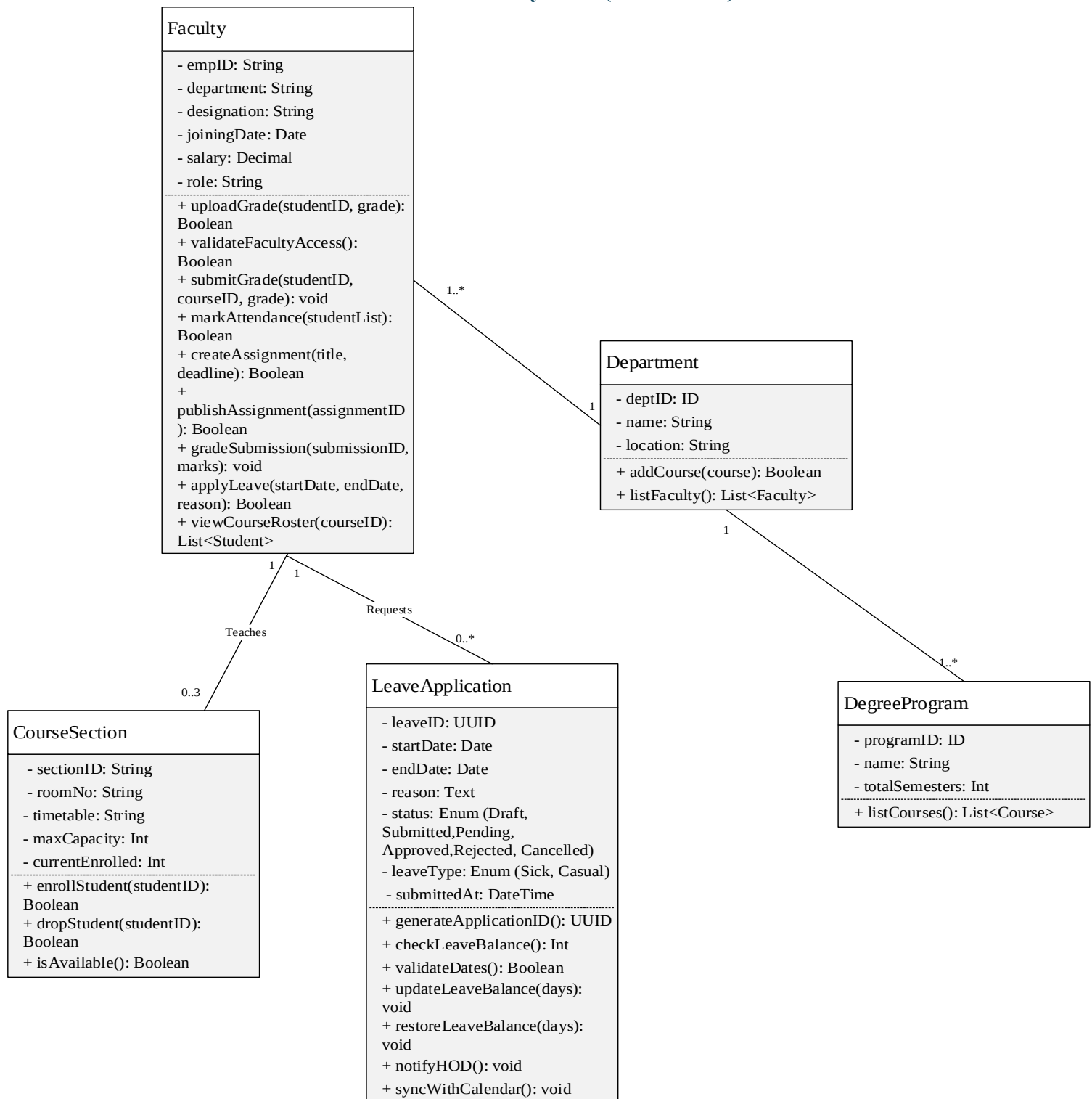


Figure 23 UML Class Diagram [Faculty Operational Model.]

This Depicts the faculty member's administrative capabilities, such as managing Departments, approving LeaveApplications, and grading CourseSections.

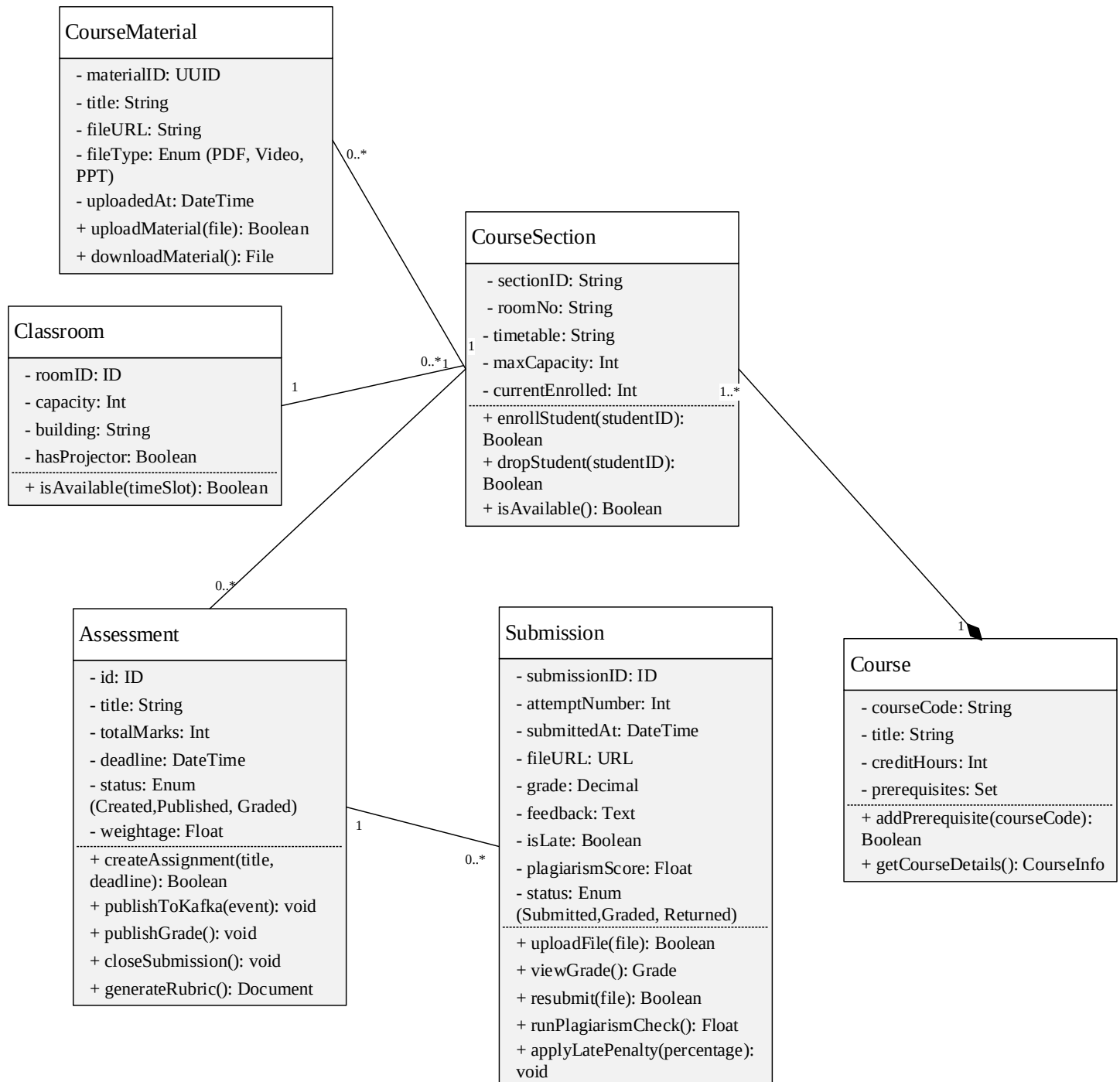


Figure 24 UML Class Diagram [LMS Content Structure]

This defines the composition of a Course. A Course defines the curriculum, while a CourseSection represents a specific instance in a semester, containing Assessments, Classroom allocation, and CourseMaterials.

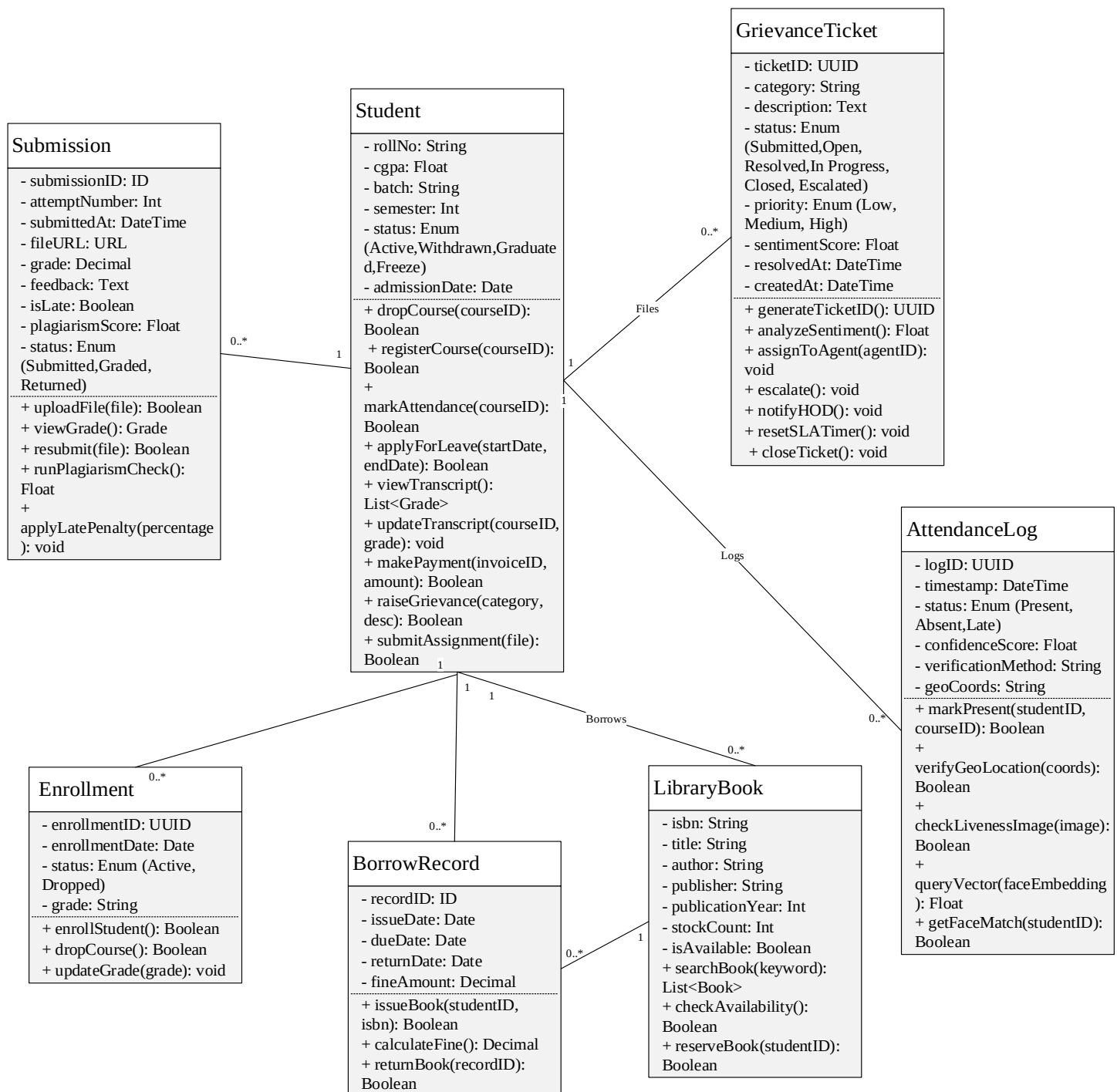


Figure 25 UML Class Diagram [Student Academic Relations]

This illustrates the student's interaction with the learning environment, including relationships for Enrollment, Assignment Submission, and Biometric AttendanceLog verification.

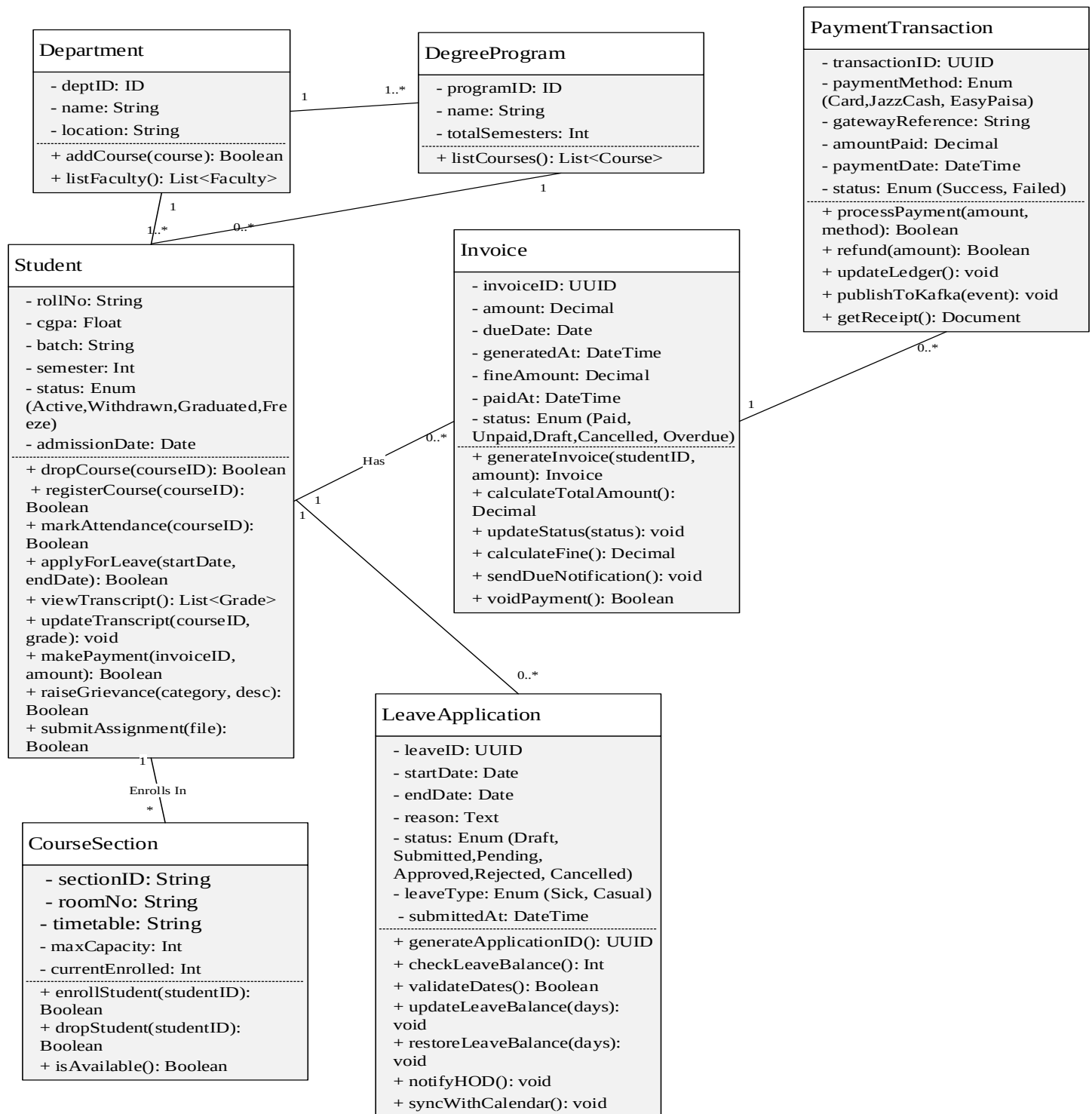


Figure 26 UML Class Diagram [Financial Transaction Model]

This Details the billing logic. It tracks the generation of Invoices for Students and the subsequent processing of PaymentTransactions via external gateways.

8.1.5 Behavioral Model

This section describes the dynamic behavior of the Project Nexus system. While the structural models define *what* the system is made of, the behavioral models define *how* the system behaves over time. We utilize **Interaction Diagrams (Sequence)** to illustrate the flow of messages between microservices and **State Transition Diagrams** to depict the lifecycle of critical business entities.

8.1.5.1 Sequence Diagrams

The following sequence diagrams illustrate the step-by-step logic for the system's most complex operations, highlighting the interaction between the React Frontend, API Gateway, and various backend Microservices.

8.1.5.1.1 Grade Submission

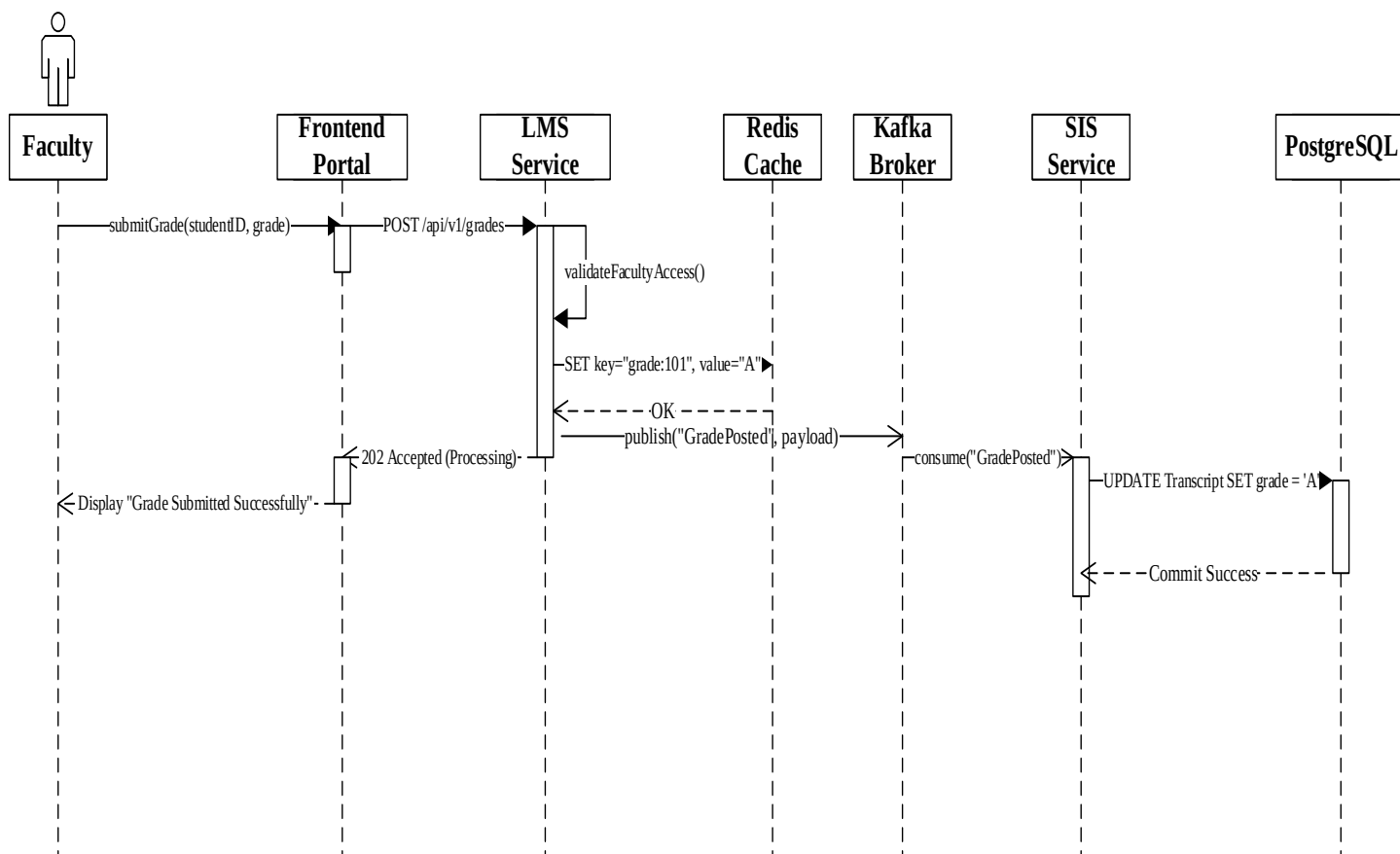


Figure 27 Sequence Diagram [Asynchronous Grade Synchronization]

This diagram demonstrates the asynchronous data consistency model.

8.1.5.1.2 Smart Attendance (Geo-Fence + Liveness)

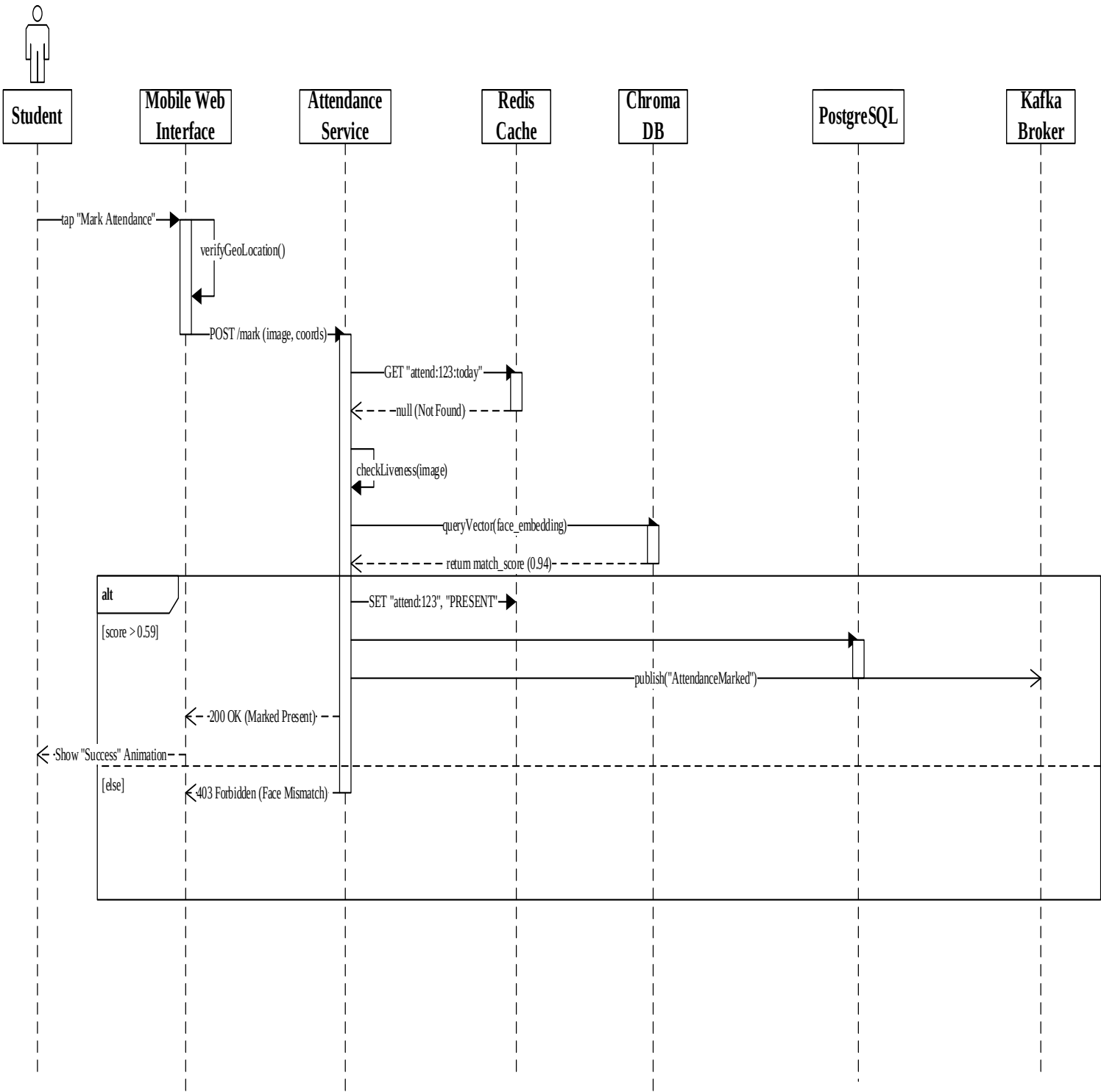


Figure 28 Sequence Diagram [Marking Attendance]

This diagram details the multi-factor validation process for marking attendance.

8.1.5.1.3 Chatbot Conversation

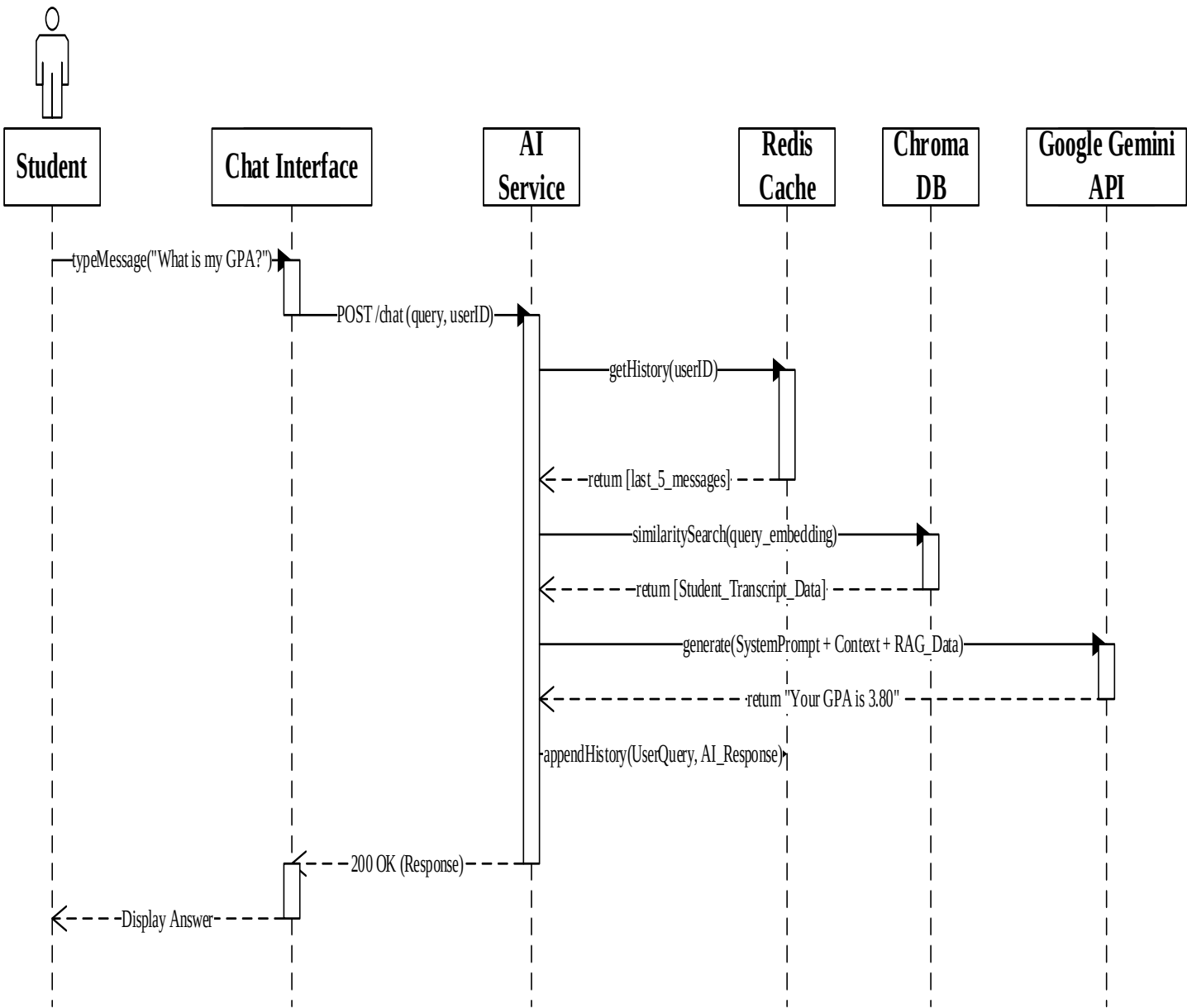


Figure 29 Sequence Diagram [Hybrid AI Chatbot (RAG Pipeline)]

This diagram illustrates the Hybrid CAG+RAG logic for the AI assistant.

8.1.5.1.4 Fee Submission

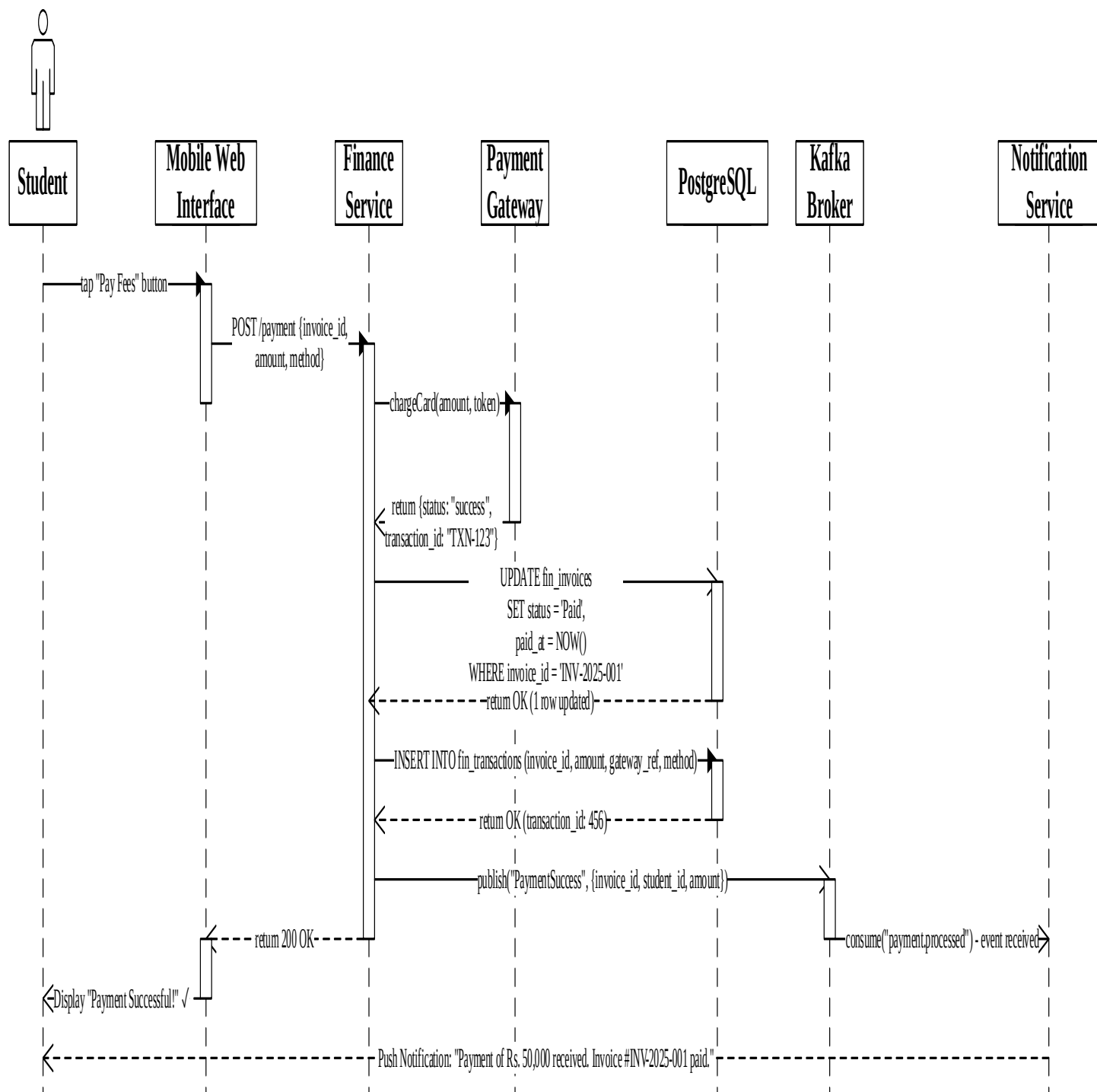


Figure 30 Sequence Diagram [Fee Payment]

This diagram demonstrates the secure financial transaction flow.

8.1.5.1.5 At-Risk Student Prediction

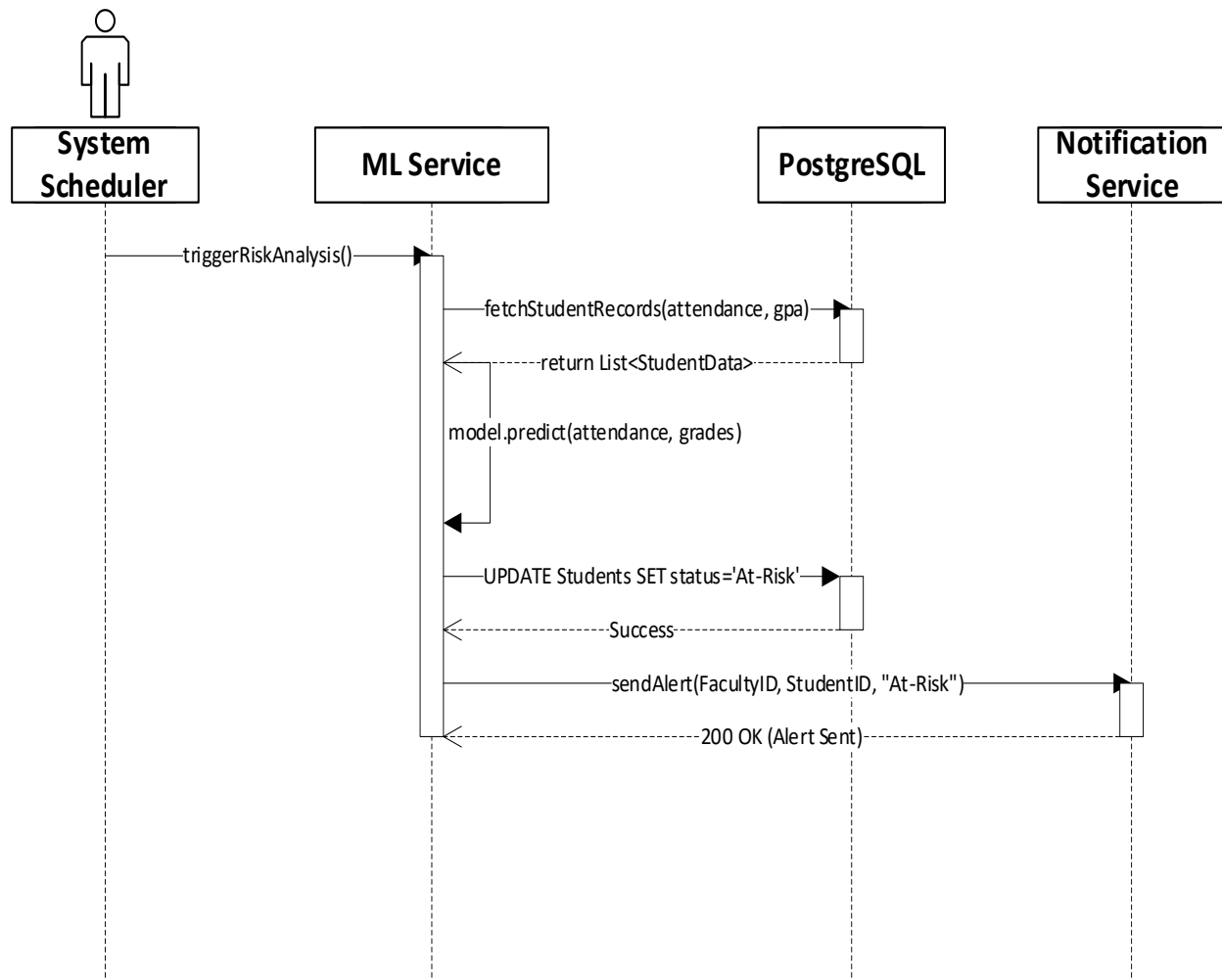


Figure 31 Sequence Diagram [At- Risk Student Predication]

This diagram illustrates the batch processing of student performance data and predicts its success or risk of failure.

8.1.5.1.6 Real-Time Chat & Push Notification

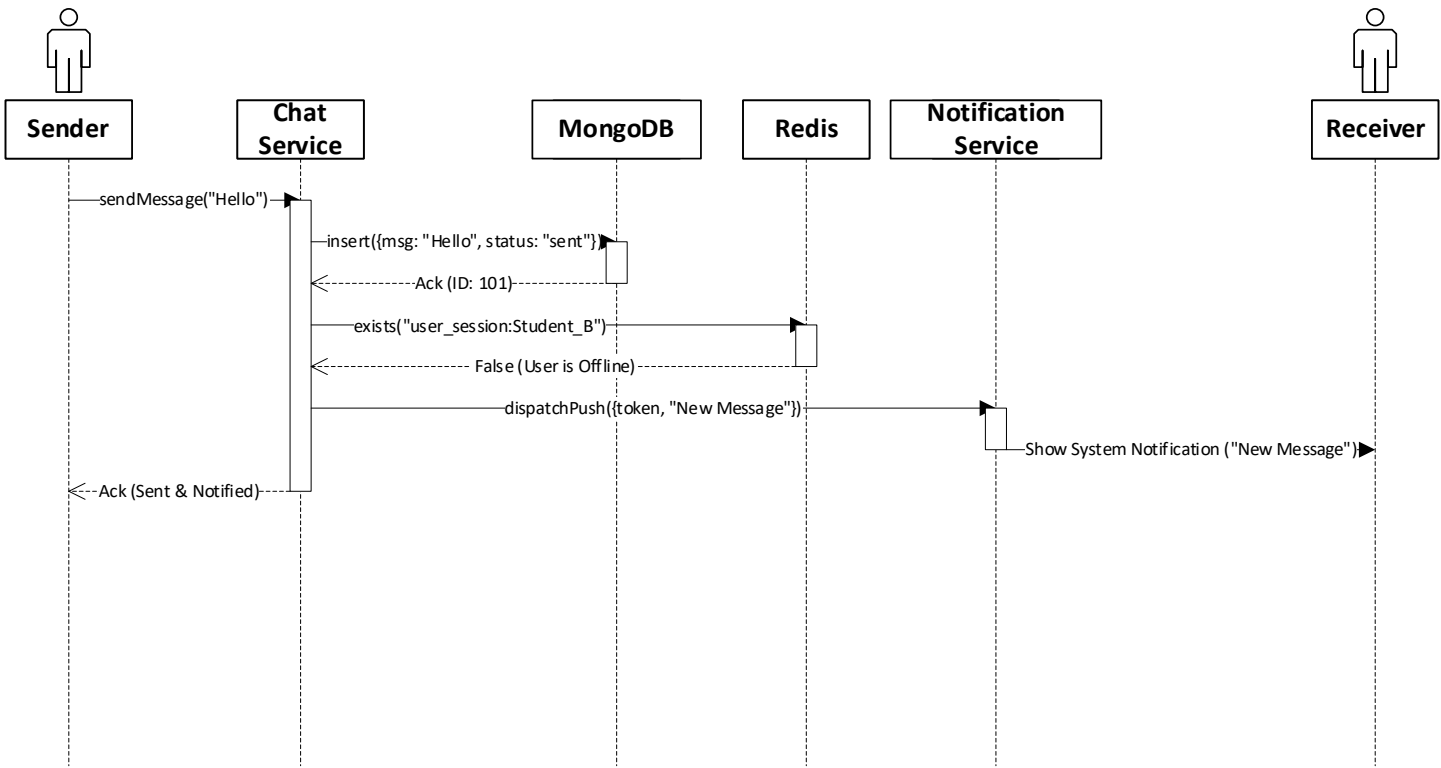


Figure 32 Sequence Diagram [Real-Time Chat & Offline Push Notification]

This diagram illustrates the hybrid message delivery mechanism. It details how the system ensures message delivery even when the recipient is not actively using the application.

8.1.5.2 State Transition Diagrams

The following state charts describe how key system objects change their status in response to user actions and time-based triggers.

8.1.5.2.1 Fee Submission

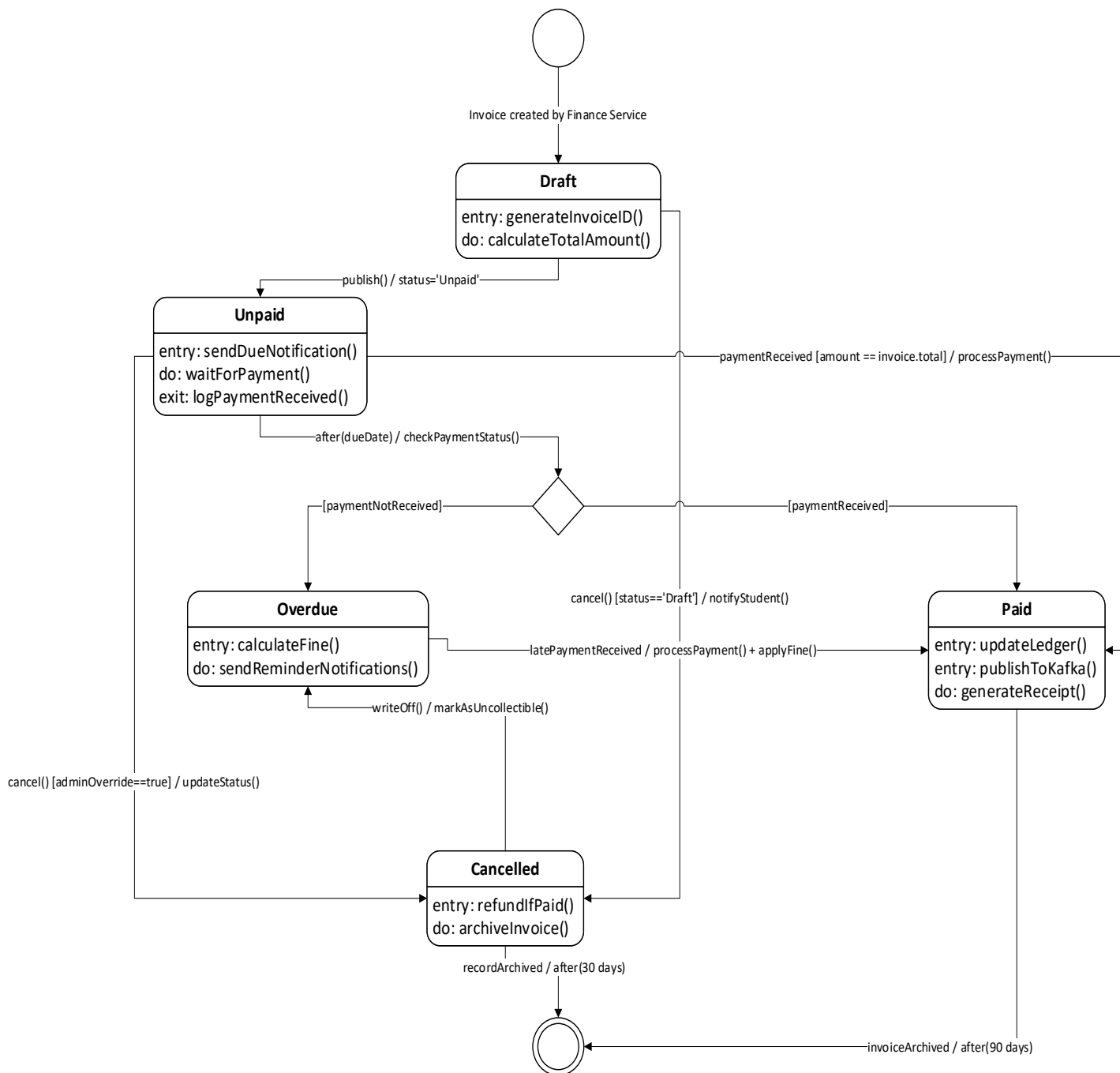


Figure 33 State Transition Diagram [Fee Submission]

This Diagram shows the fees invoice lifecycle.

8.1.5.2.2 Student Grievance (Complaints)

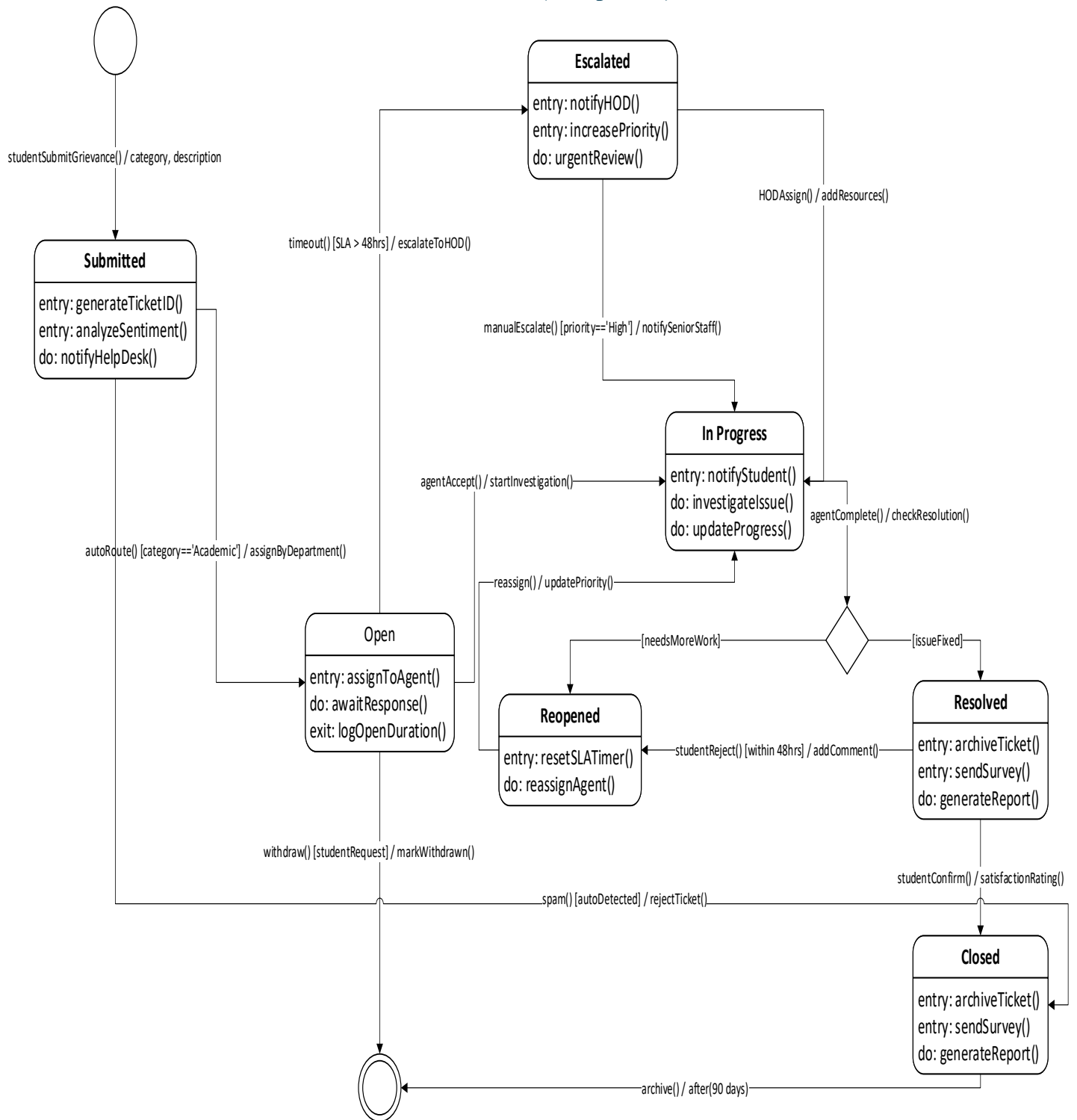


Figure 34 State Transition Diagram [Students Complaints]

This Diagram shows how a Student's complaint is handled.

8.1.5.2.3 Requesting Leave

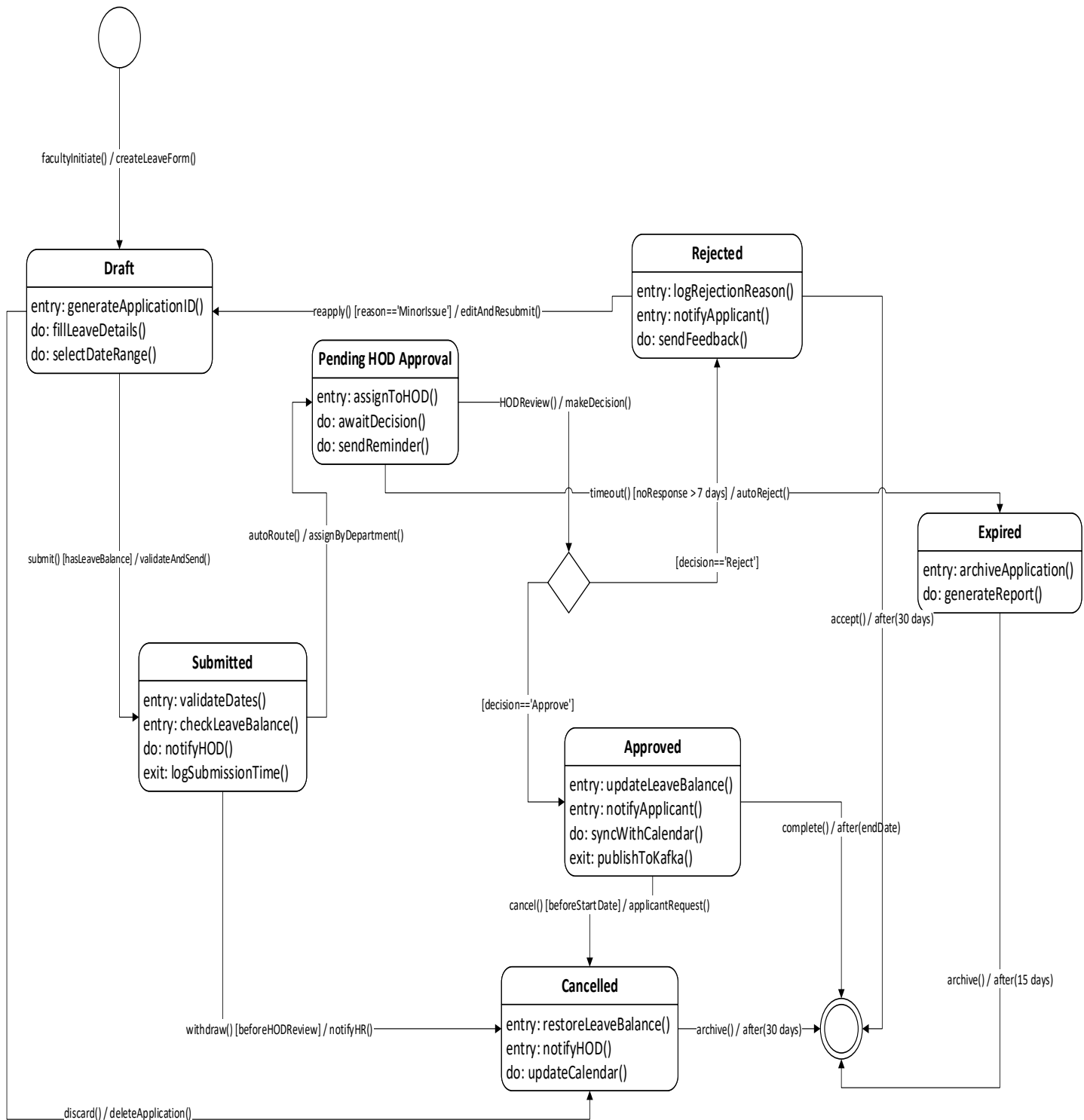


Figure 35 State Transition Diagram [Leave Request]

This Diagram Shows the lifecycle Leave Request from the user.

This diagram shows the lifecycle of the Assignment from creation to return.



Figure 36 State Transition Diagram [Assignment Submission]

8.1.5.2.5 Chatbot Conversation Session

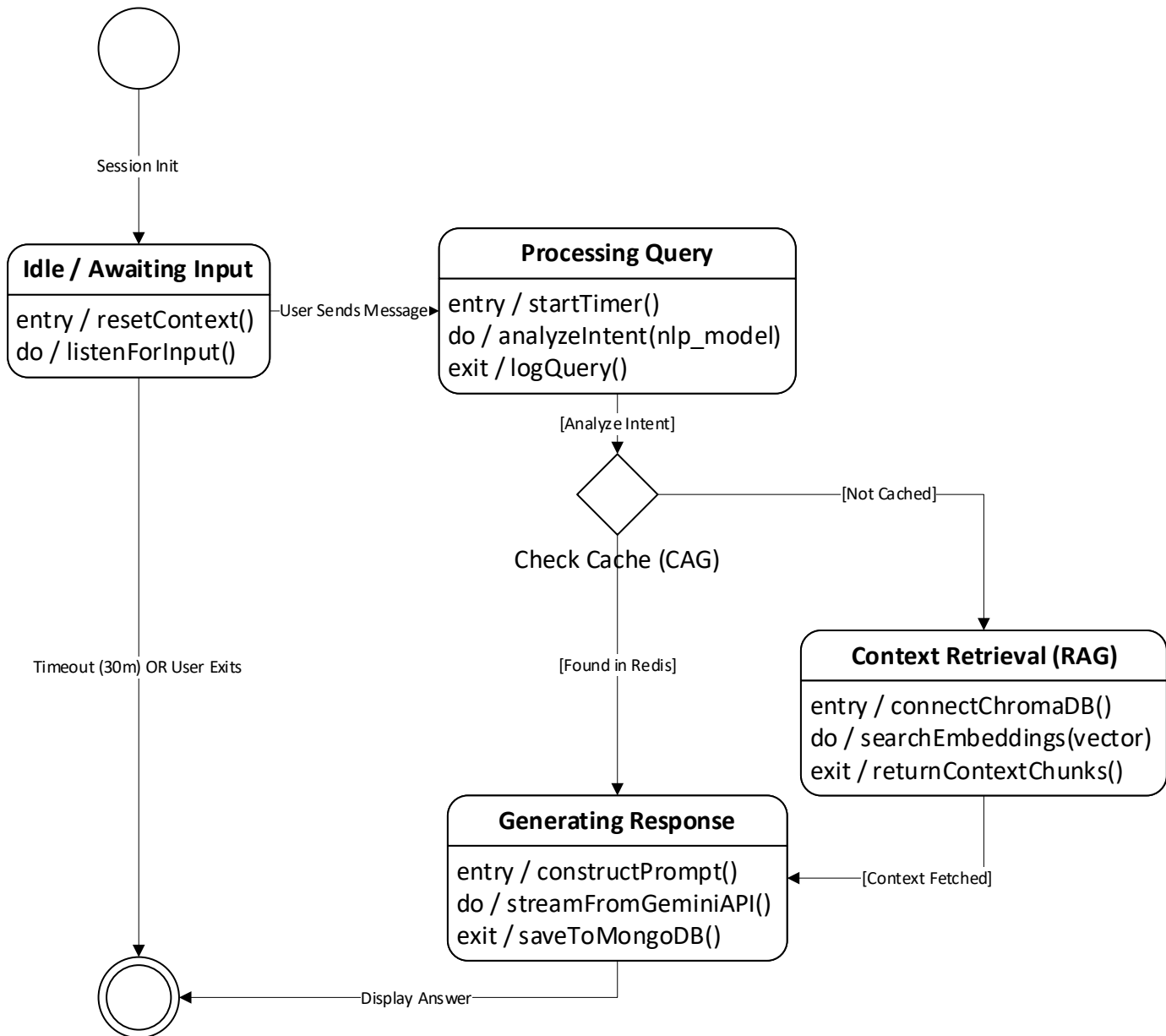


Figure 37 State Transition Diagram [AI Chatbot Session Lifecycle]

This state chart illustrates the hybrid processing logic of the AI assistant.

8.1.5.2.6 Book Issue Lifecycle

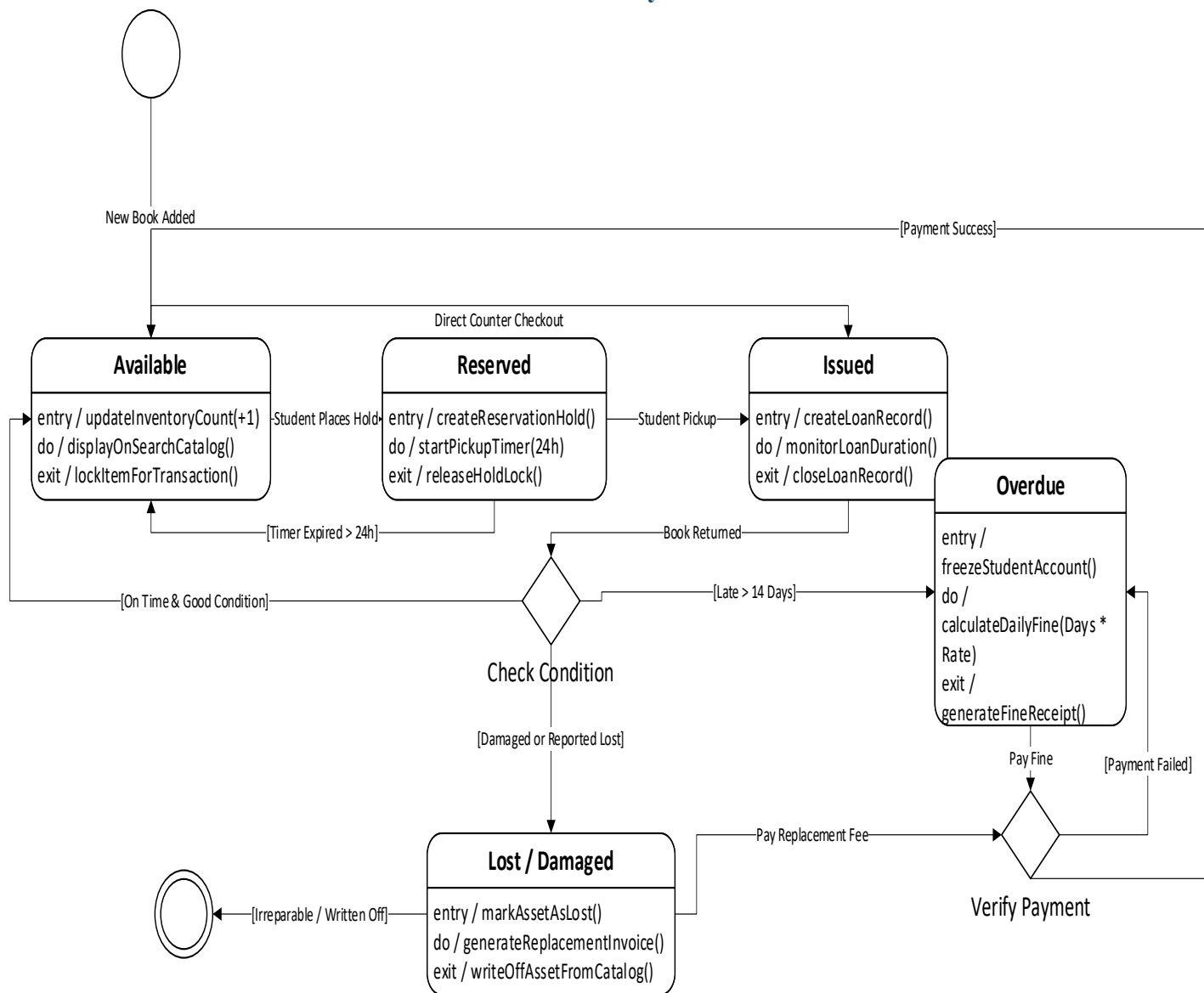


Figure 38 State Transition Diagram [Library Book Status Lifecycle]

This diagram models the inventory states of physical library assets.

8.2 Data Design

The data design for Project Nexus follows a **Polyglot Persistence** strategy, ensuring that the "information domain" is transformed into the most efficient data structures for each specific task. Rather than forcing all system entities into a single database, the design utilizes a hybrid approach: **PostgreSQL** for relational transactional data (ACID compliance), **MongoDB** for flexible document storage, **ChromaDB** for high-dimensional vector embeddings, and **Redis** for ephemeral caching. This section defines how these entities are structured, processed, and organized to support the functional requirements.

8.2.1 Data Dictionary

This section contains the alphabetical list of all system entities, attributes, and methods.

Table 93 Data Dictionary

A	
Term	Data Type & Description
access_scope	JSONB - Flexible JSON structure defining admin's access permissions to various system modules and resources
action	VARCHAR(50) - Specific action performed by user in audit trail (e.g., GRADE CHANGE, USER LOGIN, PERMISSION MODIFY)
action_slug	VARCHAR(50) - Action type identifier for permission system (e.g., edit, view, delete, create)
action_url	String (MongoDB) - Deep link URL directing user to relevant page when clicking notification
address	TEXT - Student's complete residential address including street, city, and postal code
admission_date	DATE - Official date when student was admitted to the university program
admin_id	SERIAL (PostgreSQL) / UUID (Class Diagram) - Unique identifier for administrator account
admin_level	VARCHAR(20) - Administrator privilege level (Super, Dept_Head, Staff)
alumni_id	SERIAL - Unique identifier for alumni registry record
amount	DECIMAL(10,2) - Monetary amount for invoice line item or financial transaction
amount_paid	DECIMAL(10,2) - Actual amount paid by student in a transaction
analytics_events	Collection (MongoDB) - Document collection storing user behavior analytics and tracking data
answer_id	SERIAL - Unique identifier for student's quiz answer submission
api_key_hash	VARCHAR(255) - Hashed API key for external service authentication (bcrypt/SHA-256)
appendMessage()	Method (ChatSession) - Adds new message to chat conversation history; Parameters: (message: ChatMessage)
applyForLeave()	Method (Student) - Submits leave application; Parameters: (startDate: Date, endDate: Date, reason: String); Returns: Boolean
apply_link	VARCHAR(255) - URL link for job application posted by alumni

applied_at	TIMESTAMP - Timestamp when late fee was applied to invoice
assigned_at	TIMESTAMP - Timestamp when role was assigned to user in RBAC system
assigned_to	UUID - Reference to user assigned to handle grievance ticket
assignment_id	SERIAL - Unique identifier for course assignment
assignRole()	Method (Admin) - Assigns system role to user; Parameters: (userID: UUID, role: String); Returns: Boolean
AssessmentID	INT - Primary key identifier for assessments in LMS
attachments	Array (MongoDB) - Array of file attachment objects containing file_url, file_type, and file_size
attachment_ref_id	VARCHAR(100) - Reference ID to MongoDB media_assets collection for assignment attachments
attendance_id	SERIAL - Unique identifier for attendance log entry
AttendanceLog	Class - Represents student attendance record with biometric/GPS verification capabilities
audit_trails	Collection (MongoDB) - Document collection storing security audit logs for compliance and forensics
author	VARCHAR(100) - Author name for library book or content creator
author_id	String (MongoDB) - Reference to user who created announcement or content
available_copies	INT - Current number of library books available for borrowing

B

Term	Data Type & Description
base	String - Target branch name for pull request merge in version control
batch	VARCHAR(10) / String - Student's admission batch/cohort identifier (e.g., "Fall 2021", "2021-2025")
blocked_group	VARCHAR(50) / String - Group identifier for blocked/restricted user accounts
body	TEXT / String - Main content body for pull request description, review comment, or notification message
bookcondition	VARCHAR20 - Physical condition of a library book
book_id	SERIAL - Unique identifier for library book in catalog
BorrowRecord	Class - Represents library book borrowing transaction including issue and return dates
branch	String - Git branch name for repository operations (create, update, push)
building	VARCHAR(50) - Building name/identifier where classroom is located

C

Term	Data Type & Description
cachemode	Enum VARCHAR10 - AI chatbot execution mode selected by cache decision Values CAG, RAG
calculateFine()	Method (Invoice/BorrowRecord) - Calculates late fee based on overdue days; Returns: Decimal
calculateTotalAmount()	Method (Invoice) - Computes total invoice amount including all line items and fines; Returns: Decimal
capacity	INT - Maximum seating capacity for classroom or course section

category	VARCHAR(50-100) - Classification category (grievance type, fee head, announcement category, document category)
cgpa	FLOAT - Cumulative Grade Point Average across all semesters for student
chargeCard()	Method (PaymentTransaction) - Processes credit/debit card payment via gateway; Parameters: (amount: Decimal, token: String); Returns: Boolean
chat_messages	Collection (MongoDB) - High-volume document collection storing AI chatbot conversation messages
ChatMessage	Class - Represents individual message in chat conversation with sender, content, and timestamp
chat:online:{user_id}	Key Pattern (Redis) - Real-time user presence status with 2-minute heartbeat TTL
ChatSession	Class - Represents chat conversation session between user and AI assistant
check_in_time	TIME - Actual time when student marked attendance for class
checkAvailability()	Method (LibraryBook) - Verifies if book copies are available for borrowing; Returns: Boolean
checkLeaveBalance()	Method (LeaveApplication) - Retrieves remaining leave days for employee; Returns: Integer
checkLivenessImage()	Method (AttendanceLog) - Validates face image for liveness detection to prevent spoofing; Parameters: (image: Blob); Returns: Boolean
Classroom	Class - Represents physical classroom with capacity and equipment information
classroom_id	INT - Foreign key reference to physical classroom assignment
closeSubmission()	Method (Assessment) - Closes assignment submission window after deadline
closeTicket()	Method (GrievanceTicket) - Marks grievance ticket as closed and resolved
cnic	VARCHAR(15) - Pakistani National Identity Card number (format: XXXXX-XXXXXXX-X)
code	VARCHAR(10) - Short code identifier (course code like "CS101", department code like "CS")
comments	String (MongoDB) / TEXT - Open-ended feedback text or detailed comments
confidence_score	FLOAT / DECIMAL(4,3) - Face recognition confidence score ranging from 0 to 1
content	String / TEXT - Message text content, announcement content, or rich text/HTML content
content_announcements	Collection (MongoDB) - System-wide announcements and notices with targeting capabilities
context	Object (MongoDB) - Flexible JSON object containing request context (request_id, user_id, endpoint, method)
conversation_id	String (MongoDB) - Reference to PostgreSQL chat_sessions.session_id (UUID format)
course_code	VARCHAR(10-20) / String - Unique course code identifier (e.g., "CS101", "IT202")
course_id	SERIAL / INT - Unique identifier for course in master catalog

Course	Class - Represents academic course with code, title, credit hours, and prerequisites
CourseMaterial	Class - Represents educational content (PDF, video, PPT) uploaded by faculty
CourseSection	Class - Represents specific class section of a course in a semester with assigned faculty and classroom
createAssignment()	Method (Faculty/Assessment) - Creates new assignment for course section; Parameters: (title: String, deadline: DateTime); Returns: Boolean
createMessage()	Method (ChatMessage) - Factory method to create new chat message; Parameters: (content: String, sender: Enum); Returns: ChatMessage
createSession()	Method (ChatSession) - Initializes new chat session for user; Parameters: (userID: UUID); Returns: ChatSession
createUser()	Method (Admin) - Creates new user account in system; Parameters: (userData: Object); Returns: Boolean
created_at	TIMESTAMP / ISODate - Timestamp when record was created in database
credit_hours	INT - Number of credit hours awarded for completing course
current_employer	VARCHAR(100) - Alumni's current workplace/company name
current_enrolled	INT - Current number of students enrolled in course section
current_risk_status	VARCHAR(20) - AI-predicted student dropout risk level (Green, Yellow, Red)
D	
Term	Data Type & Description
date	DATE - Calendar date for attendance, leave, or other time-based records
day_of_week	VARCHAR(10) - Day name for timetable slot (Monday, Tuesday, Wednesday, etc.)
days_overdue	INT - Number of days invoice or book return is past due date
deadline	TIMESTAMP / DateTime - Due date and time for assignment submission or task completion
default_amount	DECIMAL(10,2) - Standard fee amount for fee head (tuition, lab, library)
DegreeProgram	Class - Represents academic degree program (BS, MS, PhD) with total semesters
degree_level	VARCHAR(20) - Academic degree level (BS, MS, PhD, Diploma)
deliverystatus	VARCHAR20 - Notification delivery state for offline push messages Values Queued, Sent, Delivered, Read, Failed.
deleteFile()	Method - Removes file from repository; Parameters: (branch: String, path: String, message: String)
deleteNotification()	Method (Notification) - Permanently removes notification from system
deleteUser()	Method (Admin) - Removes user account from system; Parameters: (userID: UUID); Returns: Boolean
Department	Class - Represents academic department offering courses and programs
dept_id	SERIAL / INT - Unique identifier for academic department
description	TEXT - Detailed description for role, grievance, assignment, job posting, or repository
designation	VARCHAR(50-100) - Faculty job title (Lecturer, Associate Professor, Professor, HOD)

device_info	Object (MongoDB) - Device metadata including browser, OS, screen resolution, is mobile
distance_metric	String (ChromaDB) - Similarity calculation method (L2 for faces, Cosine for text)
dob	DATE - Date of birth for student or user
document	String (ChromaDB) - Text content stored in vector database for similarity search
dropCourse()	Method (Student) - Withdraws student from enrolled course; Parameters: (courseID: Int); Returns: Boolean
dropStudent()	Method (CourseSection) - Removes student from course section; Parameters: (studentID: Int); Returns: Boolean
due_date	DATE / TIMESTAMP - Deadline for invoice payment, book return, or assignment submission
duplicate_of	INT - Issue number that current issue is duplicate of (for issue tracking)
duration_minutes	INT - Quiz duration time limit in minutes
E	
Term	Data Type & Description
email	VARCHAR(255) - User's email address used for authentication and communication
embedding	Vector (Float32 or Float32) - High-dimensional vector representation for similarity search
emp_id	VARCHAR(20) / String - Faculty employee identification code
employee_code	VARCHAR(20) - Unique employee identifier for faculty member
end_date	DATE - End date for leave application, semester, or announcement expiration
end_time	TIMESTAMP / TIME - End time for chat session, quiz, or timetable slot
Enrollment	Class - Association class linking Student and CourseSection with enrollment status and grade
enrollment_date	DATE - Date when student enrolled in course section
enrollment_id	SERIAL / UUID - Unique identifier for course enrollment record
entity_id	String - Primary key of affected entity in audit trail or media asset reference
entity_type	String - Entity classification (assignment, profile_image, announcement) for media assets
enrollStudent()	Method (CourseSection) - Adds student to course section; Parameters: (studentID: Int); Returns: Boolean
environment	String (MongoDB) - Deployment environment identifier (development, staging, production)
escalate()	Method (GrievanceTicket) - Escalates ticket to higher authority for resolution
event	Enum - Review action type (APPROVE, REQUEST_CHANGES, COMMENT) for pull request reviews
event_type	String (MongoDB) - Analytics event classification (page_view, button_click, form_submit, search)
expires_at	TIMESTAMP / ISODate - Expiration timestamp for API key, notification, or announcement

expectedHeadSha	String - Expected SHA hash of pull request HEAD commit for update validation
F	
Term	Data Type & Description
Faculty	Class - Represents teaching staff member with department, designation, and course assignments
faculty_id	SERIAL / INT / UUID - Unique identifier for faculty member
feature:{feature_name}	Key Pattern (Redis) - Feature flag configuration for gradual rollout (no TTL)
feedback	TEXT - Detailed comments from faculty on student submission
feedback_surveys	Collection (MongoDB) - Course and faculty evaluation responses from students
file_name	String (MongoDB) - Original filename for uploaded media asset
file_ref_id	VARCHAR(100) - MongoDB reference to submitted assignment file
file_size_mb	DECIMAL(10,2) - File size in megabytes for course material
file_type	VARCHAR(20) / String - MIME type or file extension (PDF, Video, PPT, ZIP, application/pdf)
file_url	VARCHAR(500) / String - Full URL path to file location (S3 or local storage)
final_grade_points	FLOAT - Final grade points earned by student in enrolled course
fine_amount	DECIMAL(8,2-10,2) - Late fee penalty amount for overdue invoice or book
fine_id	SERIAL - Unique identifier for fine record
forkRepository()	Method - Creates fork of repository; Parameters: (owner: String, repo: String, organization: String)
from_branch	String - Source branch for creating new branch in repository
full_name	VARCHAR(255) / String - Complete name of user (first name + last name)
G	
Term	Data Type & Description
gateway_ref	VARCHAR(100) - Payment gateway transaction reference ID (Stripe, JazzCash, EasyPaisa)
generateApplicationID()	Method (LeaveApplication) - Creates unique identifier for leave application; Returns: UUID
generateEmbedding()	Method (ChatMessage) - Converts message text to vector embedding; Returns: Vector
generateInvoice()	Method (Invoice) - Creates new invoice for student; Parameters: (studentID: Int, amount: Decimal); Returns: Invoice
generateReport()	Method (Admin) - Generates system report; Parameters: (reportType: String); Returns: Document
generateRubric()	Method (Assessment) - Creates grading rubric document; Returns: Document
generateSystemPrompt()	Function - Constructs AI prompt with context and RAG data for chatbot response
generateTicketID()	Method (GrievanceTicket) - Creates unique identifier for support ticket; Returns: UUID
generated_at	TIMESTAMP - Timestamp when transcript or invoice was generated
geo_coords	VARCHAR(100) / String - GPS coordinates in "latitude,longitude" format
geo_location	Object (MongoDB) - Geographic information (country, city, timezone) from IP address

getCourseDetails()	Method (Course) - Retrieves complete course information; Returns: CourseInfo object
getCourseRoster()	Method (Faculty) - Gets list of enrolled students; Parameters: (courseID: Int); Returns: List<Student>
getFaceMatch()	Method (AttendanceLog) - Matches face embedding to student; Parameters: (studentID: Int); Returns: Boolean
getHistory()	Method (ChatSession) - Retrieves conversation message history; Returns: List<ChatMessage>
getLatestRelease()	Method - Fetches latest repository release information
getMessage()	Method (ChatMessage) - Returns message content as string; Returns: String
getReceipt()	Method (PaymentTransaction) - Generates payment receipt document; Returns: Document
gps_lat	FLOAT - GPS latitude coordinate for geofencing attendance verification
gps_long	FLOAT - GPS longitude coordinate for geofencing attendance verification
grad_year	INT - Year when alumni graduated from university
grade	VARCHAR(5) / String - Letter grade (A+, A, B+, B, C+, C, D, F)
grade:{student_id}:{course_id}	Key Pattern (Redis) - Cached grade data with 2-hour TTL
grade_points	FLOAT / DECIMAL(3,2) - Numeric grade point value for GPA calculation
gradeSubmission()	Method (Faculty) - Awards marks to student submission; Parameters: (submissionID: UUID, marks: Float)
graded_at	TIMESTAMP - Timestamp when submission was graded by faculty
graded_by	UUID - Reference to faculty member who graded submission
graduation_year	INT - Year of graduation for alumni
GrievanceTicket	Class - Represents student/faculty complaint or support ticket
H	
Term	Data Type & Description
has_projector	BOOLEAN - Indicates if classroom has projector equipment (default: false)
has_smartboard	BOOLEAN - Indicates if classroom has smartboard equipment (default: false)
head	String - Source branch containing changes for pull request
head_id	SERIAL - Unique identifier for fee head (tuition, transport, lab)
I	
Term	Data Type & Description
id	SERIAL / UUID / ObjectId / String - Universal unique identifier (varies by database)
invoice_id	SERIAL / UUID - Unique identifier for student fee invoice
Invoice	Class - Represents billing statement for student fees with due date and payment status
invoice_number	VARCHAR(50) - Human-readable unique invoice identifier
ip_address	String - IP address of user for security logging and geolocation
is_active	BOOLEAN - Account or session active status flag (default: true)
is_anonymous	Boolean (MongoDB) - Anonymous survey submission flag (default: false)
is_available	Boolean - Book availability status in library

is_biometric_verified	BOOLEAN - Face recognition verification success flag (default: true)
is_late	Boolean - Submission lateness indicator (after deadline)
is_pinned	Boolean (MongoDB) - Pin announcement to top flag (default: false)
is_public	Boolean (MongoDB) - Public file access flag (default: false)
is_read	Boolean - Notification or message read status (default: false)
isbn	VARCHAR(20) - International Standard Book Number (unique book identifier)
isAvailable()	Method (Classroom/CourseSection) - Checks resource availability; Parameters: (timeSlot: String); Returns: Boolean
issue_date	DATE - Date when book was issued to student
issue_id	SERIAL - Unique identifier for library book issue record
issueBook()	Method (BorrowRecord) - Issues book to student; Parameters: (studentID: Int, isbn: String); Returns: Boolean
item_id	SERIAL - Unique identifier for invoice line item
J	
Term	Data Type & Description
job_id	SERIAL - Unique identifier for alumni job posting
job_title	VARCHAR(100) / String - Job position title for alumni or job posting
joining_date	DATE - Date when faculty member joined university
K	
Term	Data Type & Description
key_id	SERIAL - Unique identifier for API key record
L	
Term	Data Type & Description
last_login	TIMESTAMP - Timestamp of user's most recent successful login
leaderboard:{program_id}:{semester}	Key Pattern (Redis) - CGPA rankings sorted set with 1-hour TTL
leave_id	SERIAL / UUID - Unique identifier for leave application
leave_type	VARCHAR(50) / String - Leave category (Sick, Casual, Annual, Emergency)
LeaveApplication	Class - Represents staff/faculty leave request with approval workflow
level	String (MongoDB) - Log severity level (DEBUG, INFO, WARN, ERROR, FATAL)
LibraryBook	Class - Represents book in library catalog with ISBN and stock information
line	Number - Line number in file diff for code review comment
linkedin_url	VARCHAR(255) - URL to alumni's LinkedIn profile
listFaculty()	Method (Department) - Retrieves all faculty in department; Returns: List<Faculty>
location	VARCHAR(100) - Physical location (department office, classroom building)
log_id	UUID - Unique identifier for attendance log entry
login()	Method (User) - Authenticates user credentials; Returns: Boolean
logout()	Method (User) - Terminates user session and invalidates token
M	
Term	Data Type & Description
maintainer_can_modify	Boolean - Allow repository maintainers to edit pull request branch

makePayment()	Method (Student) - Processes fee payment; Parameters: (invoiceID: UUID, amount: Decimal); Returns: Boolean
managePermissions()	Method (Admin) - Modifies user access rights; Parameters: (userID: UUID)
markAsRead()	Method (Notification) - Updates notification read status
markAttendance()	Method (Student/Faculty) - Records class attendance; Parameters: (courseID: Int / studentList: Array); Returns: Boolean
markPresent()	Method (AttendanceLog) - Marks student as present; Parameters: (studentID: UUID, courseID: Int); Returns: Boolean
marks	FLOAT - Points awarded for quiz question
marks_obtained	FLOAT - Actual marks earned by student on submission/quiz
material_id	UUID - Unique identifier for course material
max_capacity	INT - Maximum enrollment limit for course section
media_assets	Collection (MongoDB) - File metadata storage with S3 references
message	TEXT / String - Notification message content, log message, or chat content
message_id	UUID / ObjectId - Unique identifier for chat message
message_type	String (MongoDB) - Message classification (text, image, file, system)
metadata	Object / JSONB - Flexible JSON structure for additional contextual information
method	VARCHAR(20) / String - HTTP method, payment method, or operation type
N	
Term	Data Type & Description
name	VARCHAR(100-200) - Name for department, program, role, or label
nationality	VARCHAR(50) / String - Student's country of citizenship
new_value	Object (MongoDB) - JSON snapshot of data after change in audit trail
notification_id	UUID / ObjectId - Unique identifier for notification record
Notification	Class - Represents system alert or message sent to user
notifications	Collection (MongoDB) / Table (PostgreSQL) - User notification storage with TTL
notifications:{user_id}	Key Pattern (Redis) - User notification queue (FIFO list) with 7-day TTL
notifyHOD()	Method (GrievanceTicket/LeaveApplication) - Sends notification to Head of Department
O	
Term	Data Type & Description
old_value	Object (MongoDB) - JSON snapshot of data before change in audit trail
otp:{email}	Key Pattern (Redis) - One-time password for 2FA with 5-minute TTL
overall_rating	Float (MongoDB) - Numeric survey rating on 1-5 scale
owner	String - Repository owner username or organization name
P	
Term	Data Type & Description
page	Number / INT - Page number for pagination (minimum: 1)
page_number	INT - Page number in PDF document for material chunking
page_url	String (MongoDB) - Current page URL for analytics tracking
password_hash	VARCHAR(255) - Bcrypt/SHA-256 hashed password (never stored as plaintext)

path	String - File path in repository for file operations
payment_date	TIMESTAMP / DateTime - Timestamp when payment was processed
payment_method	VARCHAR(50) / String - Payment channel (Card, JazzCash, EasyPaisa, BankTransfer, Stripe)
PaymentTransaction	Class - Represents financial transaction with gateway reference
perPage	Number / INT - Results per page for pagination (range: 1-100)
phone_number	VARCHAR(20) / String - User contact phone number
plagiarism_score	DECIMAL(5,2) / Float - Similarity score for plagiarism detection (0-100)
posted_at	TIMESTAMP - Timestamp when job was posted by alumni
prerequisites	JSONB / Set - Array of prerequisite course codes required before enrollment
priority	VARCHAR(20) / String - Priority level (Low, Medium, High, Critical, urgent)
processPayment()	Method (PaymentTransaction) - Executes payment transaction; Parameters: (amount: Decimal, method: Enum); Returns: Boolean
profile_image_id	VARCHAR(100) / String - MongoDB reference to user profile image
profile_image_url	VARCHAR(500) / String - URL path to user profile picture
program_id	SERIAL / INT - Unique identifier for degree program
properties	Object (MongoDB) - Event-specific data (search_query, clicked_element) for analytics
published_at	ISODate - Publication timestamp for announcement or content
publishAssignment()	Method (Faculty) - Makes assignment visible to students; Parameters: (assignmentID: Int); Returns: Boolean
publishGrade()	Method (Assessment) - Publishes grades to students
publishToKafka()	Method (PaymentTransaction/Assessment) - Publishes event to message broker; Parameters: (event: Object)
publisher	VARCHAR(255) - Book publisher company name
publication_year	INT - Year when book was published
pullNumber	Number / INT - Pull request identification number
Q	
Term	Data Type & Description
query	String - Search query string for code, issues, repositories, or users
query:{hash}	Key Pattern (Redis) - Cached SQL query results with 10-minute TTL
queryVector()	Method (AttendanceLog) - Performs similarity search in vector database; Parameters: (faceEmbedding: Vector); Returns: Float
question_id	SERIAL - Unique identifier for quiz question
question_type	VARCHAR(20) - Question format (MCQ, TrueFalse, Short, Essay)
quiz_id	SERIAL - Unique identifier for online quiz/assessment
R	
Term	Data Type & Description
raiseGrievance()	Method (Student) - Submits complaint ticket; Parameters: (category: String, description: Text); Returns: Boolean
ratelimit:{ip}:{endpoint}	Key Pattern (Redis) - API rate limiting counter with 60-second TTL
read_at	ISODate - Timestamp when notification was marked as read
reason	TEXT - Explanation for leave application or action
record_id	UUID / ID - Unique identifier for library borrow record

ref	String - Git reference (branch, tag, or commit SHA) for file operations
referrer_url	String (MongoDB) - Previous page URL for analytics tracking
refund()	Method (PaymentTransaction) - Processes payment refund; Parameters: (amount: Decimal); Returns: Boolean
registerCourse()	Method (Student) - Enrolls in course; Parameters: (courseID: Int); Returns: Boolean
rejection_reason	TEXT - Explanation for leave application rejection
reprioritize()	Method (SubIssue) - Changes sub-issue order; Parameters: (afterID: Int, beforeID: Int)
repo	String - Repository name for GitHub operations
reserveBook()	Method (LibraryBook) - Reserves book for student; Parameters: (studentID: Int); Returns: Boolean
resetPassword()	Method (User) - Changes user password; Parameters: (newPass: String); Returns: Boolean
resetSLATimer()	Method (GrievanceTicket) - Resets service level agreement countdown
resolveGrievance()	Method (Admin) - Marks grievance as resolved; Parameters: (ticketID: UUID); Returns: Boolean
resolved_at	TIMESTAMP / DateTime - Timestamp when ticket was resolved
resource	VARCHAR(50) - Protected resource name in permission system (grade_book, attendance)
responses	Object (MongoDB) - Flexible Q&A structure for survey responses
restoreLeaveBalance()	Method (LeaveApplication) - Restores leave days on cancellation; Parameters: (days: Int)
return_date	DATE - Actual date when book was returned to library
returnBook()	Method (BorrowRecord) - Processes book return; Parameters: (recordID: UUID); Returns: Boolean
reviewed_at	TIMESTAMP - Timestamp when leave application was reviewed
reviewed_by	UUID - Reference to user who reviewed leave application
roll_no	VARCHAR(20) - Student's unique roll number/registration number
role	VARCHAR(50) / String / Enum - User system role (Student, Faculty, Admin, Alumni) or faculty administrative role (HOD, Coordinator)
role_id	SERIAL / INT - Unique identifier for system role in RBAC
role_name	VARCHAR(50) - Role name identifier (student, faculty, admin, hod)
room_id	SERIAL / INT - Unique identifier for classroom
room_no	VARCHAR(20) - Room number for classroom or timetable slot
room_number	VARCHAR(20) - Physical room number identifier
runPlagiarismCheck()	Method (Submission) - Analyzes submission for plagiarism; Returns: Float

S

Term	Data Type & Description
s3_key	String (MongoDB) - AWS S3 object key for file deletion operations
s3_url	String (MongoDB) - Full AWS S3 URL path to stored file
salary	DECIMAL(10,2) - Faculty monthly salary amount
salary_tier_encrypted	TEXT - Encrypted salary information for security
scan_status	String (MongoDB) - Antivirus scan result (pending, clean, infected)
score_obtained	FLOAT - Points earned by student on quiz question

searchBook()	Method (LibraryBook) - Searches library catalog; Parameters: (keyword: String); Returns: List<Book>
section_id	SERIAL / UUID / INT - Unique identifier for course section
section_name	VARCHAR(10) - Section identifier (A, B, C, Morning, Evening)
selected_option	TEXT - Student's selected answer for quiz question
semester	INT - Current semester number for student (1-8)
semester_id	SERIAL / INT - Unique identifier for academic semester
sender	Enum / VARCHAR(20) - Message sender type (User, AI Bot)
sender_id	String (MongoDB) - Reference to auth_users.user_id who sent message
sendDueNotification()	Method (Invoice) - Sends payment reminder notification
sendNotification()	Method (Notification) - Delivers notification to user; Parameters: (userID: UUID, message: Text)
sentiment_score	FLOAT / DECIMAL(4,3) - AI sentiment analysis score (-1 to 1) for grievance text
service_name	VARCHAR(100) / String - External service identifier or microservice name
session:{user_id}	Key Pattern (Redis) - User session data (JSON) with 1-hour TTL
session_id	UUID / String - Unique identifier for chat session or browser session
severity	String (MongoDB) - Audit log severity level (INFO, WARNING, CRITICAL)
sgpa	FLOAT - Semester Grade Point Average for single semester
sha	String - Git commit SHA hash or file blob SHA
shelf_location	VARCHAR(50) - Physical shelf location for library book
side	Enum - Pull request diff side (LEFT = old, RIGHT = new)
similaritySearch()	Function - Vector database query for nearest neighbors
size_bytes	Int (MongoDB) - File size in bytes
sla_deadline	TIMESTAMP - Service Level Agreement deadline for ticket resolution
slot_id	SERIAL - Unique identifier for timetable slot
sort	String - Sorting field for search results
stack_trace	String (MongoDB) - Full error stack trace for debugging
start_date	DATE - Start date for leave, semester, or event
start_time	TIMESTAMP / TIME - Start time for session, quiz, or timetable slot
startLine	Number - First line number for multi-line code review comment
startSide	Enum - Starting diff side for multi-line review comment
state	Enum / VARCHAR(20) - State/status for issue or pull request (open, closed)
state_reason	Enum - Reason for state change (completed, not_planned, duplicate)
status	VARCHAR(10-20) - General status field for various entities (Active, Pending, Approved, Present, Paid, etc.)
stock_count	INT - Total inventory count for library books
Student	Class - Represents enrolled student with academic records
student_id	SERIAL / INT / UUID - Unique identifier for student
sub_id	SERIAL - Unique identifier for assignment submission
subjectType	Enum - Review comment target level (FILE, LINE)
sub_issue_id	Number - Identifier for sub-issue in issue hierarchy
Submission	Class - Represents student's assignment submission with grading

submitAssignment()	Method (Student) - Submits assignment file; Parameters: (file: Blob); Returns: Boolean
submitGrade()	Method (Faculty) - Records final grade; Parameters: (studentID: UUID, courseID: Int, grade: String)
submitted_at	TIMESTAMP / ISODate - Timestamp when submission/survey/leave was submitted
submitted_by	UUID - Reference to user who submitted grievance
survey_type	String (MongoDB) - Survey classification (course_evaluation, faculty feedback, facility feedback)
syncWithCalendar()	Method (LeaveApplication) - Synchronizes leave with calendar system
system_logs	Collection (MongoDB) - Application logs from microservices
T	
Term	Data Type & Description
tag	String - Git tag name for release operations
target_audience	Array (MongoDB) - Array of user roles for announcement targeting
target_entity	String (MongoDB) - Table/collection name affected in audit trail
target_programs	Array (MongoDB) - Array of program IDs for announcement filtering
target_semesters	Array (MongoDB) - Array of semester IDs for announcement filtering
text	TEXT - Question text content or document text
ticket_id	SERIAL / UUID - Unique identifier for grievance ticket
ticket_number	VARCHAR(50) - Human-readable unique ticket identifier
time_table	JSONB / String - JSON structure with class schedule (day, start_time, end time)
timestamp	TIMESTAMP / ISODate - Generic timestamp for logging events
title	VARCHAR(100-255) - Title for course, assignment, quiz, notification, announcement, job
total_amount	DECIMAL(10,2) - Complete sum of invoice including fees and fines
total_copies	INT - Total number of book copies in library inventory
total_days	INT - Total number of leave days requested
total_marks	INT - Maximum possible points for assignment or assessment
total_semesters	INT - Total semesters required to complete degree program
transaction_id	UUID - Unique identifier for payment transaction
transcript_id	SERIAL - Unique identifier for academic transcript record
triggered_by	VARCHAR(50) - Source that triggered notification (System, Admin, Auto)
trx_date	TIMESTAMP - Transaction processing timestamp
trx_id	SERIAL - Unique identifier for financial transaction
TTL	Integer (Redis) - Time-to-live in seconds for cache expiration
type	VARCHAR(50) / String - Classification field for notification, question, file, survey, or event
U	
Term	Data Type & Description
updateCareerInfo()	Method (Alumni) - Updates employment details; Parameters: (employer: String, title: String)
updateGrade()	Method (Enrollment) - Updates final course grade; Parameters: (grade: String)

updateLeaveBalance()	Method (LeaveApplication) - Deducts leave days from balance; Parameters: (days: Int)
updateLedger()	Method (PaymentTransaction) - Updates financial accounting ledger
updateProfile()	Method (User) - Modifies user profile information; Parameters: (data: Object)
updateStatus()	Method (Invoice) - Changes invoice payment status; Parameters: (status: Enum)
updateTranscript()	Method (Student) - Updates academic transcript; Parameters: (courseID: Int, grade: String)
updated_at	TIMESTAMP - Last modification timestamp for record
upload_date	ISODate - File upload timestamp
uploadFile()	Method (Submission) - Uploads assignment file; Parameters: (file: Blob); Returns: Boolean
uploadGrade()	Method (Faculty) - Uploads grade data; Parameters: (studentID: UUID, grade: String); Returns: Boolean
uploadMaterial()	Method (CourseMaterial) - Uploads course content; Parameters: (file: Blob); Returns: Boolean
uploaded_at	TIMESTAMP - Timestamp when course material was uploaded
uploader_id	String (MongoDB) - Reference to user who uploaded media asset
User	Class (Abstract) - Base class for all user types (Student, Faculty, Admin, Alumni)
user_agent	String (MongoDB) - Browser/device user agent string for logging
user_id	UUID - Universal unique identifier for user account (primary key)
V	
Term	Data Type & Description
validateDates()	Method (LeaveApplication) - Validates leave date range; Returns: Boolean
validateFacultyAccess()	Method (Faculty) - Checks faculty permissions for operation; Returns: Boolean
value	STRING / JSON / INTEGER / ZSET - Redis value (type varies by key pattern)
vectors_assignment_submissions	Collection (ChromaDB) - Code/text embeddings for plagiarism detection
vectors_course_materials	Collection (ChromaDB) - Course content embeddings for semantic search (768 dimensions)
vectors_face_biometrics	Collection (ChromaDB) - Face embeddings for attendance recognition (128 dimensions)
vectors_faq_knowledge	Collection (ChromaDB) - FAQ Q&A embeddings for chatbot (768 dimensions)
vectors_student_transcripts	Collection (ChromaDB) - Academic record embeddings for RAG chatbot (768 dimensions)
vectors_university_policies	Collection (ChromaDB) - Policy document embeddings for chatbot (768 dimensions)
verification_method	VARCHAR(50) / String - Attendance verification method (FaceRecognition, Manual, Biometric)
verifyEmail()	Method (User) - Validates email address; Returns: Boolean

verifyGeoLocation()	Method (AttendanceLog) - Validates GPS coordinates against allowed locations; Parameters: (coords: String); Returns: Boolean
view_count	Int (MongoDB) - Number of times announcement was viewed
viewCourseRoster()	Method (Faculty) - Retrieves student list for course; Parameters: (courseID: Int); Returns: List<Student>
viewGrade()	Method (Submission) - Retrieves grade information; Returns: Grade object
viewTranscript()	Method (Student) - Retrieves academic transcript; Returns: List<Grade>
voidPayment()	Method (Invoice) - Cancels payment transaction; Returns: Boolean
W	
Term	Data Type & Description
weightage	FLOAT / DECIMAL(5,2) - Percentage contribution of assessment to final grade
Y	
Term	Data Type & Description
year	INT - Academic year or graduation year

8.2.2 Database Design

This section details the physical schema of the databases.

8.2.2.1 Relational Database (PostgreSQL)

Module 1: Identity & Academic Core (SIS)

This module defines the foundational structure of the university system. It includes the `auth_users` table, which serves as the super-type for all actors, and links directly to the `sis_students` and `sis_faculty` profiles. It also manages the static academic hierarchy (`sis_departments`, `sis_programs`).

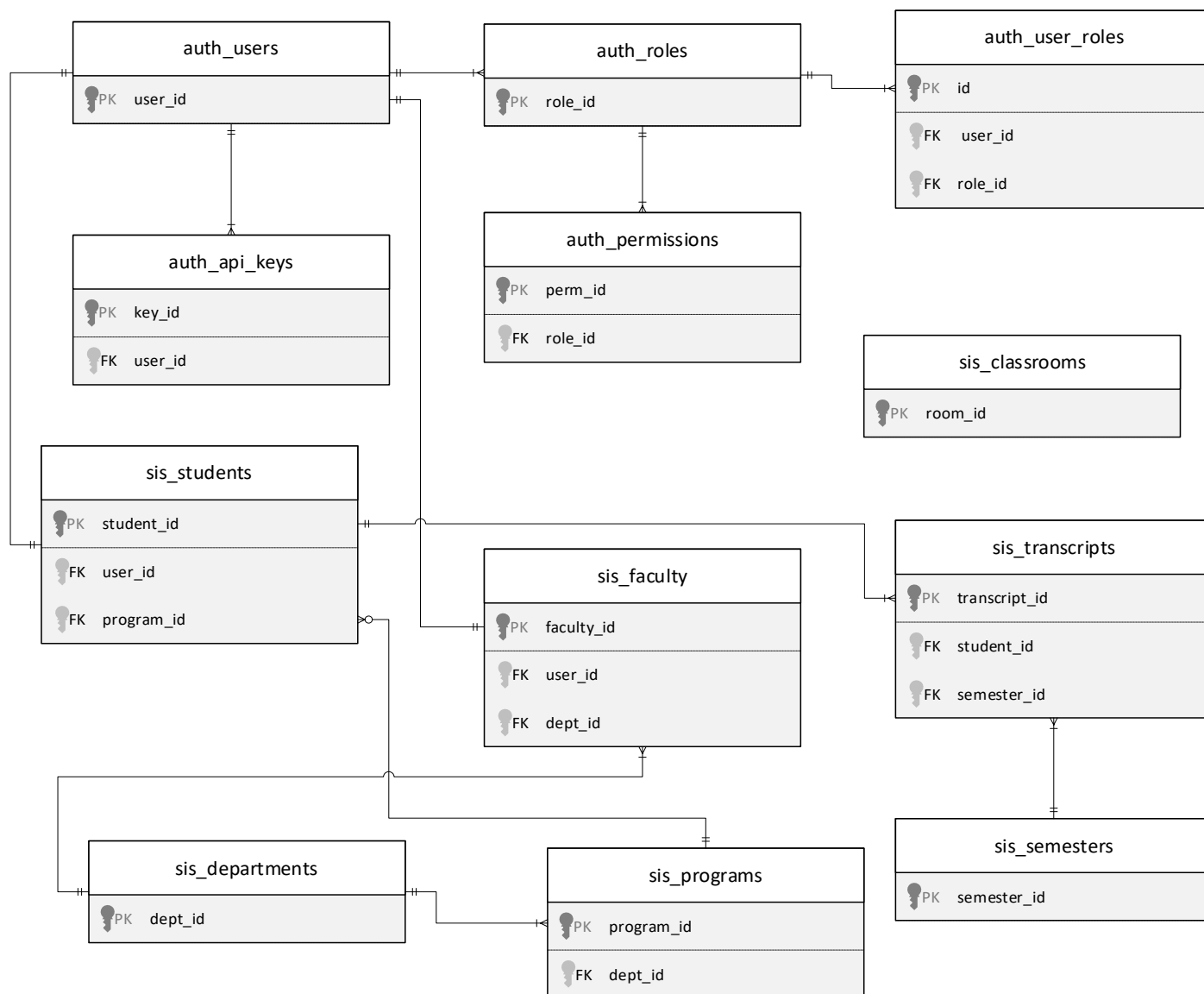


Figure 39 Identity & SIS Schema

Module 2: Learning Management System (LMS)

The LMS module handles the dynamic day-to-day academic operations. The lms_sections table acts as the central hub, connecting course templates to specific semesters and faculty members. It manages all assessment data, including lms_assignments, lms_quizzes, and student lms_submissions.

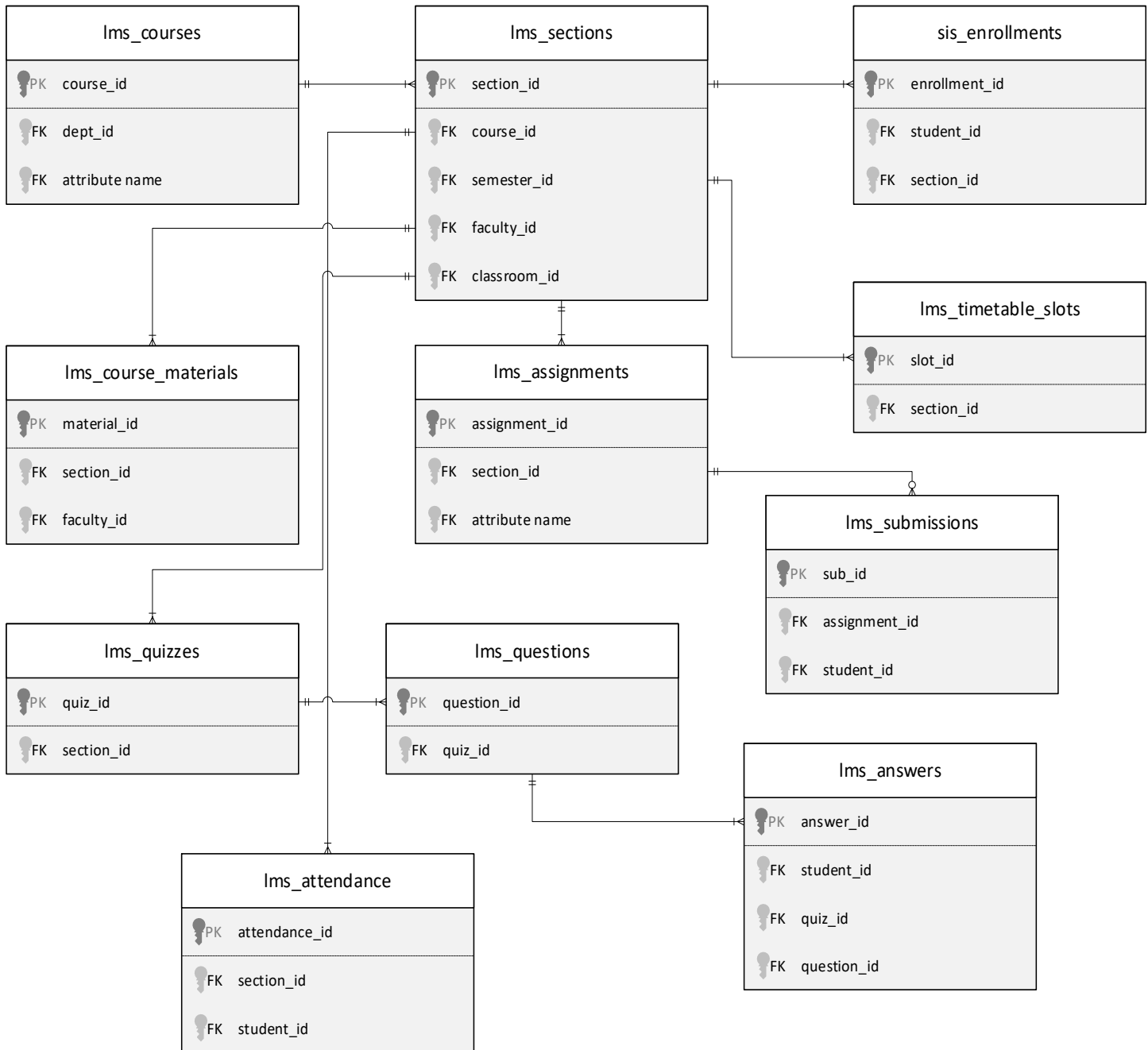


Figure 40 LMS & Assessment Schema

Module 3: Financial & Billing System

The Finance module tracks all monetary interactions. It utilizes a header-line item structure where a single `fin_invoice` can contain multiple `fin_invoice_items` (e.g., Tuition, Lab Fee). It also tracks payments in a separate `fin_transactions` table to support partial payments and audit trails.

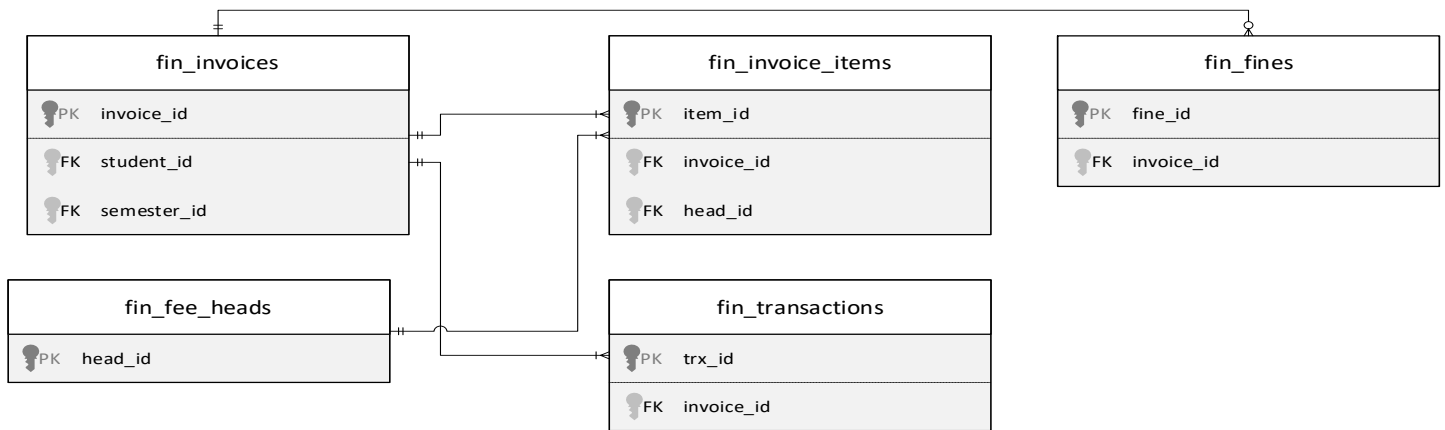


Figure 41 Financial Billing Schema

Module 4: Operations & Support Services

This module manages auxiliary campus services. It includes the Library system (`lib_books`, `lib_issues`), Operational workflows (`ops_leaves`, `ops_grievances`), and the Alumni registry. These tables operate semi-independently but remain linked to the core `sis_students` and `auth_users` tables.

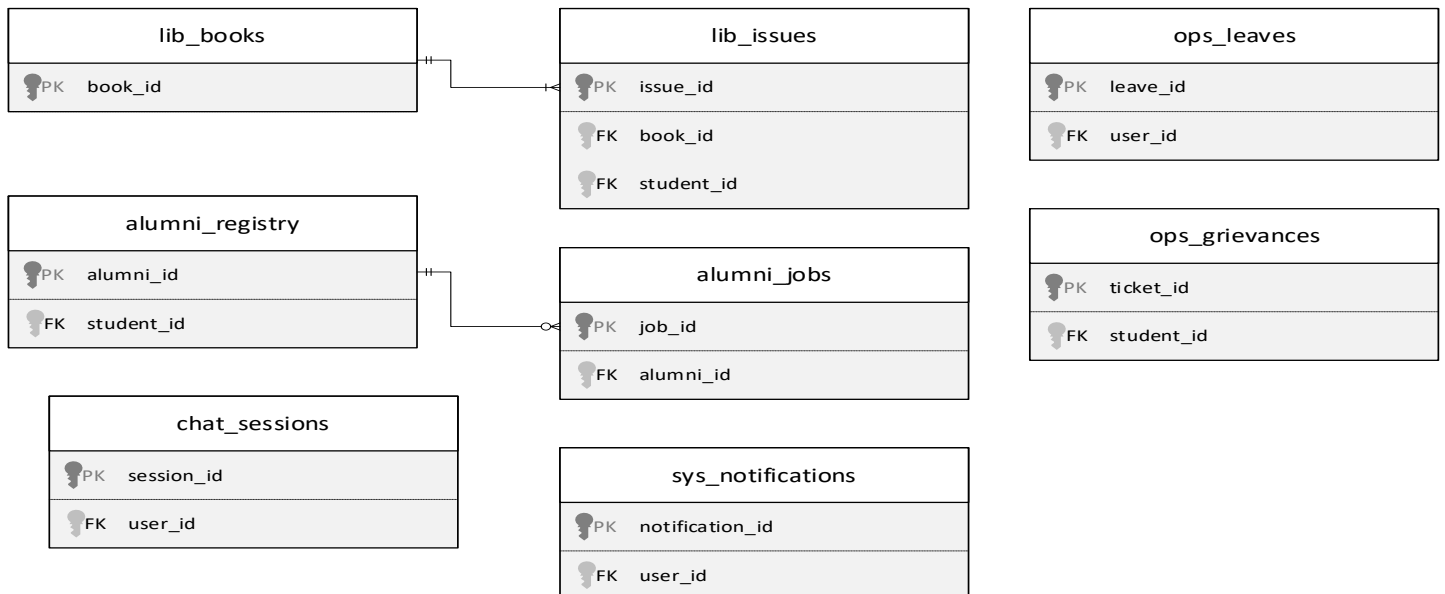


Figure 42 Operations & Support Schema

8.2.2.1.1 auth_users

Table 94 auth_users

Column Name	Data Type	Constraints	Description
user_id	UUID	PRIMARY KEY, DEFAULT uuid_generate_v4()	Unique user identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
is_active	BOOLEAN	DEFAULT TRUE	Account activation status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
last_login	TIMESTAMP	NULL	Last successful login time

8.2.2.1.2 auth_roles

Table 95 auth_roles

Column Name	Data Type	Constraints	Description
role_id	SERIAL	PRIMARY KEY	Auto-increment role ID
role_name	VARCHAR(50)	UNIQUE, NOT NULL	Role name (student, faculty, admin, hod)
description	TEXT	NULL	Role description

8.2.2.1.3 auth_user_roles

Table 96 auth_user_roles

Column Name	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-increment mapping ID
user_id	UUID	FOREIGN KEY → auth_users(user_id), ON DELETE CASCADE	Reference to user
role_id	INT	FOREIGN KEY → auth_roles(role_id), ON DELETE CASCADE	Reference to role
assigned_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Role assignment timestamp

8.2.2.1.4 auth_permissions

Table 97 auth_permissions

Column Name	Data Type	Constraints	Description
perm_id	SERIAL	PRIMARY KEY	Auto-increment permission ID
role_id	INT	FOREIGN KEY → auth_roles(role_id), ON DELETE CASCADE	Reference to role
resource	VARCHAR(50)	NOT NULL	Resource name (e.g., grade_book)
action_slug	VARCHAR(50)	NOT NULL	Action type (edit, view, delete)

8.2.2.1.5 auth_api_keys**Table 98 auth_api_keys**

Column Name	Data Type	Constraints	Description
key_id	SERIAL	PRIMARY KEY	Auto-increment key ID
user_id	UUID	FOREIGN KEY → auth_users(user_id), ON DELETE CASCADE	Reference to user
service name	VARCHAR(100)	NULL	Service identifier
api_key_hash	VARCHAR(255)	NOT NULL	Hashed API key
expires_at	TIMESTAMP	NULL	Expiration timestamp

8.2.2.1.6 sis_departments**Table 99 sis_departments**

Column Name	Data Type	Constraints	Description
dept_id	SERIAL	PRIMARY KEY	Auto-increment department ID
name	VARCHAR(100)	NOT NULL	Department name
code	VARCHAR(10)	UNIQUE	Department code (CS, IT, BBA)
location	VARCHAR(100)	NULL	Physical location

8.2.2.1.7 sis_programs**Table 100 sis_programs**

Column Name	Data Type	Constraints	Description
program_id	SERIAL	PRIMARY KEY	Auto-increment program ID
dept_id	INT	FOREIGN KEY → sis_departments(dept_id)	Reference to department
title	VARCHAR(100)	NULL	Program title
degree_level	VARCHAR(20)	NULL	Degree level (BS, MS, PhD)
total_semesters	INT	NULL	Total semesters required

8.2.2.1.8 sis_students**Table 101 sis_students**

Column Name	Data Type	Constraints	Description
student_id	SERIAL	PRIMARY KEY	Auto-increment student ID
user_id	UUID	FOREIGN KEY → auth_users(user_id), UNIQUE, ON DELETE CASCADE	Reference to user account
program_id	INT	FOREIGN KEY → sis_programs(program_id)	Reference to degree program
roll no	VARCHAR(20)	UNIQUE, NOT NULL	Student roll number
cnic	VARCHAR(15)	UNIQUE	National ID card number
dob	DATE	NULL	Date of birth
address	TEXT	NULL	Residential address

admission_date	DATE	NULL	Date of admission
status	VARCHAR(20)	DEFAULT 'Active'	Student status (Active, Withdrawn, Graduated)
current_risk_status	VARCHAR(20)	DEFAULT 'Green'	AI-predicted dropout risk (Green, Yellow, Red)
profile_image_id	VARCHAR(100)	NULL	MongoDB reference to profile image

8.2.2.1.9 sis_faculty***Table 102 sis_faculty***

Column Name	Data Type	Constraints	Description
faculty_id	SERIAL	PRIMARY KEY	Auto-increment faculty ID
user_id	UUID	FOREIGN KEY → auth_users(user_id), UNIQUE, ON DELETE CASCADE	Reference to user account
dept_id	INT	FOREIGN KEY → sis_departments(dept_id)	Reference to department
employee_code	VARCHAR(20)	UNIQUE, NOT NULL	Employee code
designation	VARCHAR(50)	NULL	Job title (Lecturer, Professor)
role	VARCHAR(50)	NULL	Administrative role (HOD, Coordinator)
joining_date	DATE	NULL	Date of joining
salary_tier_encrypted	TEXT	NULL	Encrypted salary information
profile_image_id	VARCHAR(100)	NULL	MongoDB reference to profile image

8.2.2.1.10 sis_semesters***Table 103 sis_semesters***

Column Name	Data Type	Constraints	Description
semester_id	SERIAL	PRIMARY KEY	Auto-increment semester ID
title	VARCHAR(50)	NULL	Semester title (Fall 2025, Spring 2026)
start_date	DATE	NULL	Semester start date
end_date	DATE	NULL	Semester end date
is_active	BOOLEAN	DEFAULT FALSE	Current active semester flag

8.2.2.1.11 sis_transcripts***Table 104 sis_transcripts***

Column Name	Data Type	Constraints	Description
transcript_id	SERIAL	PRIMARY KEY	Auto-increment transcript ID
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student

semester_id	INT	FOREIGN KEY → sis_semesters(semester_id)	Reference to semester
sgpa	FLOAT	NULL	Semester GPA
cgpa	FLOAT	NULL	Cumulative GPA
generated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Transcript generation timestamp

8.2.2.1.12 sis_classrooms**Table 105 sis_classrooms**

Column Name	Data Type	Constraints	Description
room_id	SERIAL	PRIMARY KEY	Auto-increment room ID
room_number	VARCHAR(20)	UNIQUE, NOT NULL	Room number
building	VARCHAR(50)	NULL	Building name
capacity	INT	NULL	Room capacity
has_projector	BOOLEAN	DEFAULT FALSE	Projector availability
has_smartboard	BOOLEAN	DEFAULT FALSE	Smartboard availability

8.2.2.1.13 lms_courses**Table 106 sis_classrooms**

Column Name	Data Type	Constraints	Description
course_id	SERIAL	PRIMARY KEY	Auto-increment course ID
dept_id	INT	FOREIGN KEY → sis_departments(dept_id)	Reference to department
code	VARCHAR(10)	UNIQUE	Course code (CS101, IT202)
title	VARCHAR(100)	NULL	Course title
credit_hours	INT	NULL	Credit hours

8.2.2.1.14 lms_sections**Table 107 lms_sections**

Column Name	Data Type	Constraints	Description
section_id	SERIAL	PRIMARY KEY	Auto-increment section ID
course_id	INT	FOREIGN KEY → lms_courses(course_id)	Reference to course
semester_id	INT	FOREIGN KEY → sis_semesters(semester_id)	Reference to semester
faculty_id	INT	FOREIGN KEY → sis_faculty(faculty_id)	Reference to faculty
classroom_id	INT	FOREIGN KEY → sis_classrooms(room_id)	Reference to classroom
room_no	VARCHAR(20)	NULL	Legacy room number field
capacity	INT	NULL	Section capacity

8.2.2.1.15 sis_enrollments**Table 108 sis_enrollments**

Column Name	Data Type	Constraints	Description
enrollment_id	SERIAL	PRIMARY KEY	Auto-increment enrollment ID

student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
section_id	INT	FOREIGN KEY → lms_sections(section_id)	Reference to section
status	VARCHAR(20)	DEFAULT 'Enrolled'	Enrollment status (Enrolled, Withdrawn, Dropped)
final_grade_points	FLOAT	NULL	Final grade points

8.2.2.1.16 lms_course_materials

Table 109 lms_course_materials

Column Name	Data Type	Constraints	Description
material_id	UUID	PRIMARY KEY, DEFAULT uuid_generate_v4()	Unique material identifier
section_id	INT	FOREIGN KEY → lms_sections(section_id), ON DELETE CASCADE	Reference to section
faculty_id	INT	FOREIGN KEY → sis_faculty(faculty_id)	Reference to faculty
title	VARCHAR(255)	NOT NULL	Material title
file_url	VARCHAR(500)	NULL	File URL (S3 or local)
file_type	VARCHAR(20)	NULL	File type (PDF, Video, PPT, ZIP)
file_size_mb	DECIMAL(10,2)	NULL	File size in megabytes
uploaded_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

8.2.2.1.17 lms_assignments

Table 110 lms_assignments

Column Name	Data Type	Constraints	Description
assignment_id	SERIAL	PRIMARY KEY	Auto-increment assignment ID
section_id	INT	FOREIGN KEY → lms_sections(section_id)	Reference to section
title	VARCHAR(100)	NULL	Assignment title
total_marks	INT	NULL	Total marks
due_date	TIMESTAMP	NULL	Due date and time
attachment_ref_id	VARCHAR(100)	NULL	MongoDB reference to attachment

8.2.2.1.18 lms_submissions

Table 111 lms_submissions

Column Name	Data Type	Constraints	Description
sub_id	SERIAL	PRIMARY KEY	Auto-increment submission ID
assignment_id	INT	FOREIGN KEY → lms_assignments(assignment_id)	Reference to assignment
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student

submitted_at	TIMESTAMP	NULL	Submission timestamp
marks_obtained	FLOAT	NULL	Marks awarded
file_ref_id	VARCHAR(100)	NULL	MongoDB reference to submitted file

8.2.2.1.19 lms_quizzes**Table 112 lms_quizzes**

Column Name	Data Type	Constraints	Description
quiz_id	SERIAL	PRIMARY KEY	Auto-increment quiz ID
section_id	INT	FOREIGN KEY → lms_sections(section_id)	Reference to section
title	VARCHAR(100)	NULL	Quiz title
duration_minutes	INT	NULL	Quiz duration in minutes
start_time	TIMESTAMP	NULL	Quiz start time
end_time	TIMESTAMP	NULL	Quiz end time

8.2.2.1.20 lms_questions**Table 113 lms_questions**

Column Name	Data Type	Constraints	Description
question_id	SERIAL	PRIMARY KEY	Auto-increment question ID
quiz_id	INT	FOREIGN KEY → lms_quizzes(quiz_id)	Reference to quiz
text	TEXT	NOT NULL	Question text
question_type	VARCHAR(20)	NULL	Question type (MCQ, TrueFalse, Short)
marks	FLOAT	NULL	Marks for question

8.2.2.1.21 lms_answers**Table 114 lms_answers**

Column Name	Data Type	Constraints	Description
answer_id	SERIAL	PRIMARY KEY	Auto-increment answer ID
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
quiz_id	INT	FOREIGN KEY → lms_quizzes(quiz_id)	Reference to quiz
question_id	INT	FOREIGN KEY → lms_questions(question_id)	Reference to question
selected_option	TEXT	NULL	Student's answer
score_obtained	FLOAT	NULL	Score received

8.2.2.1.22 lms_attendance**Table 115 lms_attendance**

Column Name	Data Type	Constraints	Description
attendance_id	SERIAL	PRIMARY KEY	Auto-increment attendance ID
section_id	INT	FOREIGN KEY → lms_sections(section_id)	Reference to section

student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
date	DATE	NOT NULL	Attendance date
status	VARCHAR(10)	CHECK (status IN ('Present', 'Absent', 'Leave', 'Late'))	Attendance status
check_in time	TIME	NULL	Check-in time
gps_lat	FLOAT	NULL	GPS latitude (geofencing)
gps_long	FLOAT	NULL	GPS longitude (geofencing)
is_biometric_verified	BOOLEAN	DEFAULT TRUE	Biometric verification flag

8.2.2.1.23 lms_timetable_slots

Table 116 lms_timetable_slots

Column Name	Data Type	Constraints	Description
slot_id	SERIAL	PRIMARY KEY	Auto-increment slot ID
section_id	INT	FOREIGN KEY → lms_sections(section_id)	Reference to section
day_of_week	VARCHAR(10)	NULL	Day of week (Monday, Tuesday, etc.)
start time	TIME	NULL	Slot start time
end time	TIME	NULL	Slot end time
room no	VARCHAR(20)	NULL	Room number

8.2.2.1.24 fin_fee_heads

Table 117 fin_fee_heads

Column Name	Data Type	Constraints	Description
head_id	SERIAL	PRIMARY KEY	Auto-increment fee head ID
title	VARCHAR(100)	NULL	Fee head title (Tuition, Transport, Lab)
default amount	DECIMAL(10,2)	NULL	Default fee amount

8.2.2.1.25 fin_invoices

Table 118 fin_invoices

Column Name	Data Type	Constraints	Description
invoice_id	SERIAL	PRIMARY KEY	Auto-increment invoice ID
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
semester_id	INT	FOREIGN KEY → sis_semesters(semester_id)	Reference to semester
total_amount	DECIMAL(10,2)	NULL	Total invoice amount
due_date	DATE	NULL	Payment due date

status	VARCHAR(20)	DEFAULT 'Unpaid'	Payment status (Paid, Unpaid, Overdue)
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8.2.2.1.26 fin_invoice_items

Table 119 fin_invoice_items

Column Name	Data Type	Constraints	Description
item_id	SERIAL	PRIMARY KEY	Auto-increment item ID
invoice_id	INT	FOREIGN KEY → fin_invoices(invoice_id)	Reference to invoice
head_id	INT	FOREIGN KEY → fin_fee_heads(head_id)	Reference to fee head
amount	DECIMAL(10,2)	NULL	Line item amount

8.2.2.1.27 fin_transactions

Table 120 fin_transactions

Column Name	Data Type	Constraints	Description
trx_id	SERIAL	PRIMARY KEY	Auto-increment transaction ID
invoice_id	INT	FOREIGN KEY → fin_invoices(invoice_id)	Reference to invoice
gateway_ref	VARCHAR(100)	NULL	Payment gateway reference (Stripe/JazzCash ID)
amount_paid	DECIMAL(10,2)	NULL	Amount paid
trx_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Transaction timestamp
method	VARCHAR(20)	NULL	Payment method (Stripe, JazzCash, BankChallan)

8.2.2.1.28 fin_fines

Table 121 fin_fines

Column Name	Data Type	Constraints	Description
fine_id	SERIAL	PRIMARY KEY	Auto-increment fine ID
invoice_id	INT	FOREIGN KEY → fin_invoices(invoice_id)	Reference to invoice
days_overdue	INT	NULL	Number of days overdue
fine_amount	DECIMAL(10,2)	NULL	Fine amount
applied_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Fine application timestamp

8.2.2.1.29 lib_books

Table 122 lib_books

Column Name	Data Type	Constraints	Description
book_id	SERIAL	PRIMARY KEY	Auto-increment book ID
isbn	VARCHAR(20)	UNIQUE	ISBN number
title	VARCHAR(200)	NULL	Book title
author	VARCHAR(100)	NULL	Author name
total_copies	INT	NULL	Total copies available
available_copies	INT	NULL	Currently available copies

shelf_location	VARCHAR(50)	NULL	Physical shelf location
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8.2.2.1.30 lib_issues

Table 123 lib_issues

Column Name	Data Type	Constraints	Description
issue_id	SERIAL	PRIMARY KEY	Auto-increment issue ID
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
book_id	INT	FOREIGN KEY → lib_books(book_id)	Reference to book
issue_date	DATE	DEFAULT CURRENT_DATE	Issue date
due_date	DATE	NULL	Due date
return_date	DATE	NULL	Actual return date
status	VARCHAR(20)	DEFAULT 'Issued'	Issue status (Issued, Returned, Lost)

8.2.2.1.31 ops_leaves

Table 124 ops_leaves

Column Name	Data Type	Constraints	Description
leave_id	SERIAL	PRIMARY KEY	Auto-increment leave ID
user_id	UUID	FOREIGN KEY → auth_users(user_id)	Reference to user
leave_type	VARCHAR(50)	NULL	Leave type (Sick, Casual, Annual)
start_date	DATE	NULL	Leave start date
end_date	DATE	NULL	Leave end date
reason	TEXT	NULL	Leave reason
status	VARCHAR(20)	DEFAULT 'Pending'	Leave status (Pending, Approved, Rejected)

8.2.2.1.32 ops_grievances

Table 125 ops_grievances

Column Name	Data Type	Constraints	Description
ticket_id	SERIAL	PRIMARY KEY	Auto-increment ticket ID
student_id	INT	FOREIGN KEY → sis_students(student_id)	Reference to student
category	VARCHAR(50)	NULL	Grievance category (Academic, Facilities, IT, HR)
description	TEXT	NULL	Grievance description
status	VARCHAR(20)	DEFAULT 'Open'	Ticket status (Open, In Progress, Resolved, Closed)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Ticket creation timestamp

8.2.2.1.33 alumni_registry**Table 126 alumni_registry**

Column Name	Data Type	Constraints	Description
alumni_id	SERIAL	PRIMARY KEY	Auto-increment alumni ID
student_id	INT	FOREIGN KEY → sis_students(student_id), UNIQUE	Reference to student
grad_year	INT	NULL	Graduation year
current_employer	VARCHAR(100)	NULL	Current employer
linkedin_url	VARCHAR(255)	NULL	LinkedIn profile URL

8.2.2.1.34 alumni_jobs**Table 127 alumni_jobs**

Column Name	Data Type	Constraints	Description
job_id	SERIAL	PRIMARY KEY	Auto-increment job ID
alumni_id	INT	FOREIGN KEY → alumni_registry(alumni_id)	Reference to alumni
title	VARCHAR(100)	NULL	Job title
company	VARCHAR(100)	NULL	Company name
description	TEXT	NULL	Job description
apply_link	VARCHAR(255)	NULL	Application link
posted_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Job posting timestamp
is_active	BOOLEAN	DEFAULT TRUE	Active job flag

8.2.2.1.35 chat_sessions**Table 128 chat_sessions**

Column Name	Data Type	Constraints	Description
session_id	UUID	PRIMARY KEY, DEFAULT uuid_generate_v4()	Unique session identifier
user_id	UUID	FOREIGN KEY → auth_users(user_id), ON DELETE CASCADE	Reference to user
start_time	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Session start timestamp
end_time	TIMESTAMP	NULL	Session end timestamp
title	VARCHAR(255)	NULL	Session title
is_active	BOOLEAN	DEFAULT TRUE	Active session flag

8.2.2.1.36 sys_notifications**Table 129 sys_notifications**

Column Name	Data Type	Constraints	Description
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notification_id	UUID	PRIMARY KEY, DEFAULT uuid_generate_v4()	Unique notification identifier
user_id	UUID	FOREIGN KEY → auth_users(user_id), ON DELETE CASCADE	Reference to user
title	VARCHAR(255)	NOT NULL	Notification title
message	TEXT	NOT NULL	Notification message
type	VARCHAR(50)	NULL	Notification type (System, Grade, Payment, Grievance, Attendance)
is_read	BOOLEAN	DEFAULT FALSE	Read status flag
triggered_by	VARCHAR(50)	NULL	Trigger source (System, Admin, Auto)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Notification creation timestamp

8.2.2.2 NoSQL Database (MongoDB)

MongoDB Stores unstructured data and high-volume logs where schema flexibility is required.

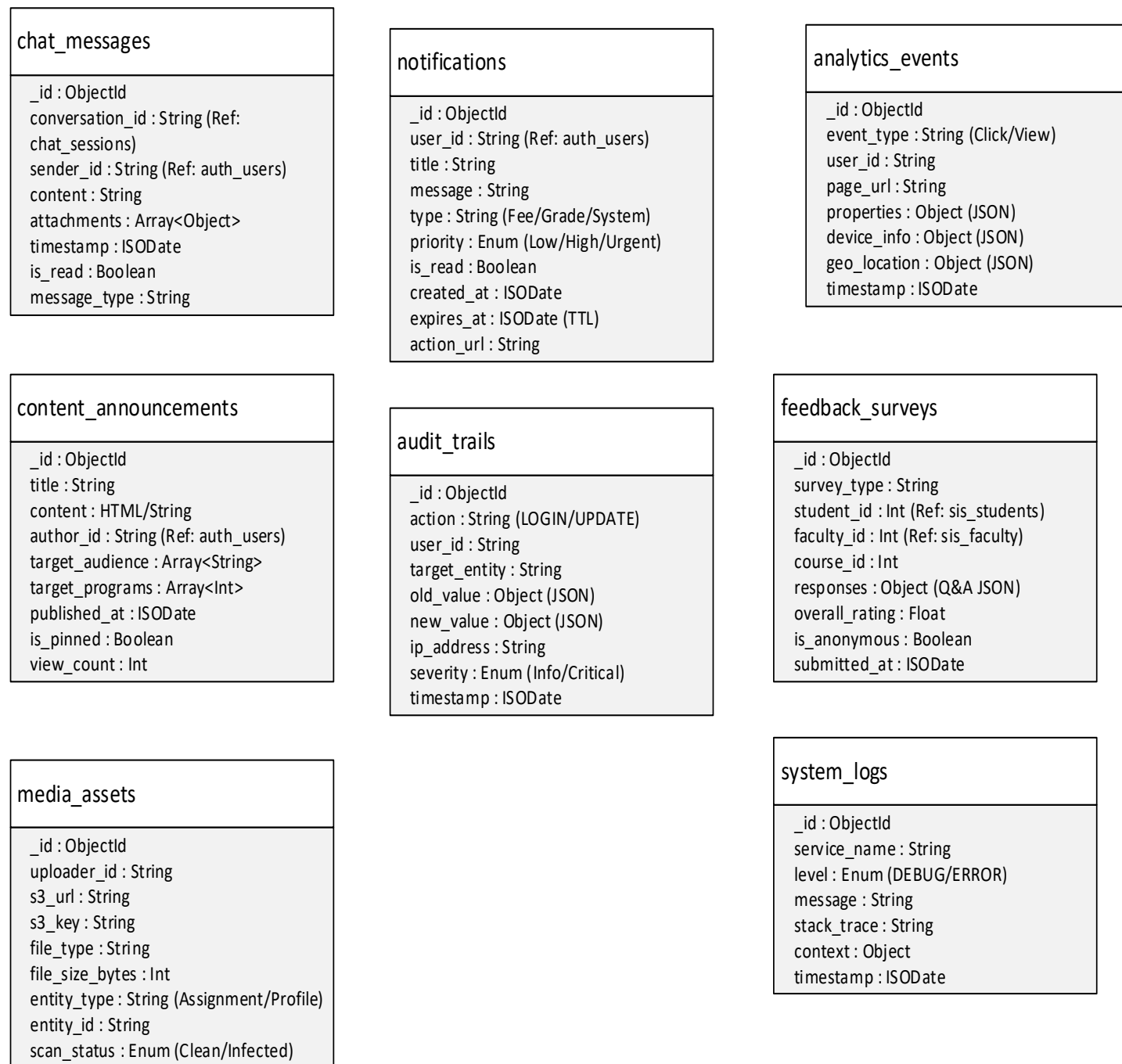


Figure 43 MongoDB Collections

8.2.2.2.1 Collection 1: chat_messages

Table 130 chat_messages

Field Name	Data Type	Description
_id	ObjectId	MongoDB auto-generated unique identifier
conversation_id	String	Reference to PostgreSQL chat_sessions.session_id (UUID)
sender_id	String	Reference to PostgreSQL auth_users.user_id (UUID)
content	String	Message text content
attachments	Array	Array of objects: {file_url, file_type, file_size}
timestamp	ISODate	Message timestamp
is_read	Boolean	Read status (default: false)
message_type	String	Message type (text, image, file, system)

Indexes:

- conversation_id (single)
- sender_id (single)
- timestamp (single)
- (conversation_id, timestamp) (compound)

8.2.2.2.2 Collection 2: audit_trails

Table 131 audit_trails

Field Name	Data Type	Description
_id	ObjectId	MongoDB auto-generated unique identifier
action	String	Action type (GRADE_CHANGE, USER_LOGIN, PERMISSION_MODIFY)
user_id	String	Actor who performed the action
target_entity	String	Table/Collection affected (Transcript, Invoice)
entity_id	String	Primary key of affected record
old_value	Object	JSON snapshot before change
new_value	Object	JSON snapshot after change
ip_address	String	IP address of actor
user_agent	String	Browser/device information
timestamp	ISODate	Action timestamp
severity	String	Log severity (INFO, WARNING, CRITICAL)

Indexes:

- user_id (single)
- action (single)
- timestamp (single)
- (user_id, timestamp) (compound)
- (target_entity, entity_id) (compound)

8.2.2.2.3 Collection 3: media_assets

Table 132 media_assets

Field Name	Data Type	Description
id	ObjectId	MongoDB auto-generated unique identifier
uploader_id	String	User who uploaded the file
s3_url	String	Full AWS S3 URL
s3_key	String	S3 object key for deletion
file_type	String	MIME type (application/pdf, image/png)
file_name	String	Original filename
size_bytes	Int	File size in bytes
upload_date	ISODate	Upload timestamp
entity_type	String	Entity type (assignment, profile_image, announcement)
entity_id	String	Foreign key to related entity
is_public	Boolean	Public access flag (default: false)
scan_status	String	Antivirus scan status (pending, clean, infected)

Indexes:

- uploader_id (single)
- s3_key (unique)
- (entity_type, entity_id) (compound)

8.2.2.2.4 Collection 4: notifications

Table 133 notifications

Field Name	Data Type	Description
id	ObjectId	MongoDB auto-generated unique identifier
user_id	String	Recipient user ID
title	String	Notification title
message	String	Notification message
type	String	Notification type (FEE ALERT, GRADE POSTED, ASSIGNMENT DUE)
priority	String	Priority level (low, medium, high, urgent)
is_read	Boolean	Read status (default: false)
read_at	ISODate	Timestamp when marked as read
created_at	ISODate	Creation timestamp
expires_at	ISODate	Auto-delete timestamp
action_url	String	Deep link to relevant page
metadata	Object	Flexible JSON metadata

Indexes:

- user_id (single)
- is_read (single)
- created_at (single)
- (user_id, is_read, created_at) (compound)
- expires_at (TTL index - auto-deletion)

8.2.2.2.5 Collection 5: analytics_events**Table 134 analytics_events**

Field Name	Data Type	Description
id	ObjectId	MongoDB auto-generated unique identifier
event_type	String	Event type (page view, button click, form submit, search)
user_id	String	User who triggered event (null for anonymous)
session_id	String	Browser session ID
page_url	String	Current page URL
referrer_url	String	Previous page URL
timestamp	ISODate	Event timestamp
properties	Object	Event-specific data (search query, clicked element)
device_info	Object	Device information (browser, os, screen resolution, is mobile)
geo_location	Object	Geographic information (country, city, timezone)

Indexes:

- event_type (single)
- user_id (single)
- timestamp (single)
- (event_type, timestamp) (compound)

8.2.2.2.6 Collection 6: content_announcements**Table 135 content_announcements**

Field Name	Data Type	Description
id	ObjectId	MongoDB auto-generated unique identifier
title	String	Announcement title
content	String	Rich text/HTML content
author_id	String	Admin/Faculty who created announcement
target_audience	Array	Array of roles ([student, faculty, all])
target_programs	Array	Array of program IDs (null = all programs)
target_semesters	Array	Array of semester IDs (null = all semesters)
priority	String	Priority level (low, medium, high, urgent)
published_at	ISODate	Publication timestamp
expires_at	ISODate	Expiration timestamp
is_pinned	Boolean	Pin to top flag (default: false)
attachments	Array	Array of media asset IDs
view_count	Int	View counter (default: 0)

Indexes:

- author_id (single)
- published_at (single)
- expires_at (single)
- is_pinned (single)
- target_audience (single)

8.2.2.2.7 Collection 7: system_logs

Table 136 system_logs

Field Name	Data Type	Description
_id	ObjectId	MongoDB auto-generated unique identifier
service_name	String	Microservice name (lms-service, finance-service, auth-service)
level	String	Log level (DEBUG, INFO, WARN, ERROR, FATAL)
message	String	Log message
stack_trace	String	Full error stack for ERROR/FATAL levels
timestamp	ISODate	Log timestamp
context	Object	Request context (request_id, user_id, endpoint, method, status_code)
environment	String	Environment (development, staging, production)

Indexes:

- service_name (single)
- level (single)
- timestamp (single - TTL index for auto-deletion)
- (service_name, level, timestamp) (compound)

8.2.2.2.8 Collection 8: feedback_surveys

Table 137 feedback_surveys

Field Name	Data Type	Description
_id	ObjectId	MongoDB auto-generated unique identifier
survey_type	String	Survey type (course evaluation, faculty feedback, facility feedback)
course_id	Int	Reference to PostgreSQL lms_courses.course_id
section_id	Int	Reference to PostgreSQL lms_sections.section_id
faculty_id	Int	Reference to PostgreSQL sis_faculty.faculty_id
student_id	Int	Reference to PostgreSQL sis_students.student_id
responses	Object	Flexible Q&A structure (q1: answer1, q2: answer2)
overall_rating	Float	Numeric rating (1-5 scale)
comments	String	Open-ended feedback text
submitted_at	ISODate	Submission timestamp
is_anonymous	Boolean	Anonymous submission flag (default: false)
semester_id	Int	Reference to PostgreSQL sis_semesters.semester_id

Indexes:

- survey_type (single)
- course_id (single)
- faculty_id (single)
- student_id (single)
- submitted_at (single)
- (faculty_id, semester_id) (compound)

8.2.2.3 REDIS CACHE

A high-speed in-memory store used to reduce database load and manage temporary states.

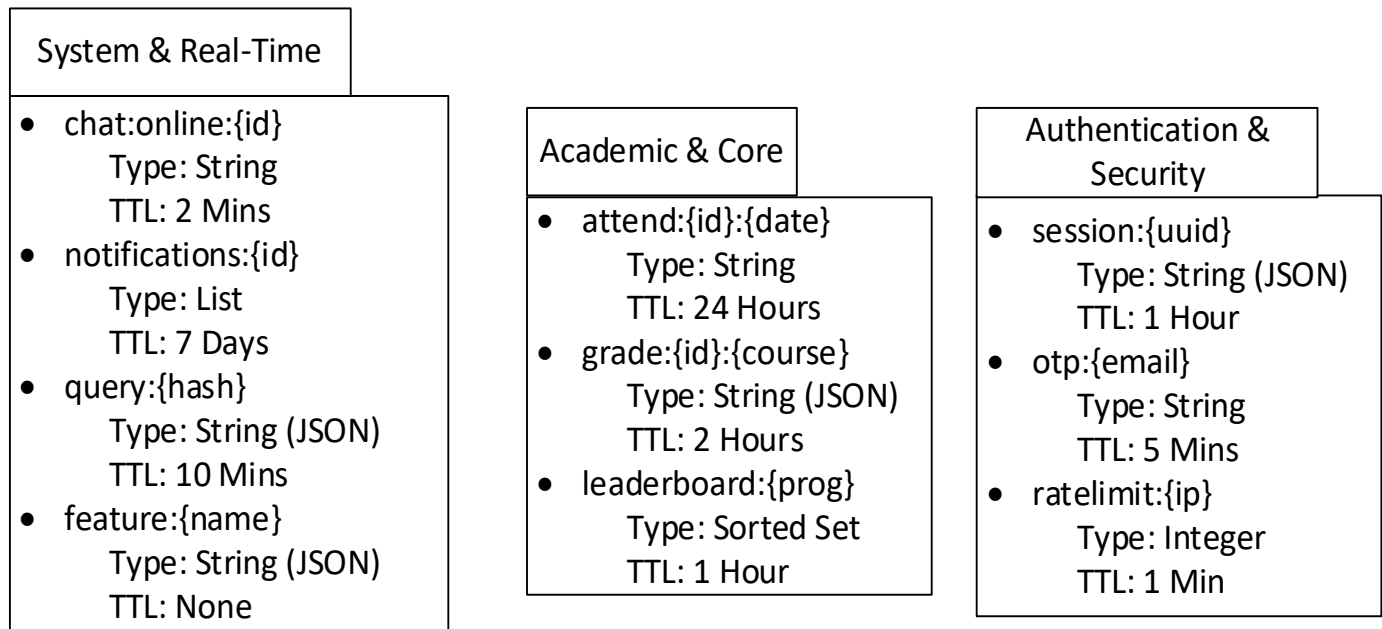


Figure 44 Redis Cache Patterns

8.2.2.3.1 Pattern 1: User Sessions

Table 138 User Sessions

Key Pattern	Value Type	Value Structure	TTL
session:{user_id}	STRING (JSON)	{"user_id": "uuid", "role": "student", "permissions": [], "login_time": "timestamp"}	3600 seconds (1 hour)

8.2.2.3.2 Pattern 2: Attendance Status

Table 139 Attendance Status

Key Pattern	Value Type	Value Structure	TTL
attend:{student_id}:{date}	STRING	PRESENT ABSENT LATE	86400 seconds (24 hours)

8.2.2.3.3 Pattern 3: Course Grades Cache

Table 140 Course Grades Cache

Key Pattern	Value Type	Value Structure	TTL
grade:{student_id}:{course_id}	STRING (JSON)	{"grade": "A+", "gpa": 4.0, "updated_at": "timestamp"}	7200 seconds (2 hours)

8.2.2.3.4 Pattern 4: API Rate Limiting

Table 141 API Rate Limiting

Key Pattern	Value Type	Value Structure	TTL
ratelimit:{ip}:{endpoint}	INTEGER	Incremental counter: 1, 2, 3, ... 100	60 seconds (1 minute)

8.2.2.3.5 Pattern 5: CGPA Leaderboard

Table 142 CGPA Leaderboard

Key Pattern	Value Type	Value Structure	TTL
leaderboard:{program_id}:{semester}	SORTED SET (ZSET)	Member: student_id, Score: cgpa (e.g., student:12345 → 3.85)	3600 seconds (1 hour)

8.2.2.3.6 Pattern 6: Notification Queue

Table 143 Notification Queue

Key Pattern	Value Type	Value Structure	TTL
notifications:{user_id}	LIST	Array of notification IDs: ["notif:1", "notif:2", ...]	604800 seconds (7 days)

8.2.2.3.7 Pattern 7: OTP Codes

Table 144 OTP Codes

Key Pattern	Value Type	Value Structure	TTL
otp:{email}	STRING	6-digit code: 123456	300 seconds (5 minutes)

8.2.2.3.8 Pattern 8: Online Users

Key Pattern	Value Type	Value Structure	TTL
chat:online:{user_id}	STRING	Status: "online"	120 seconds (2 minutes - heartbeat)

8.2.2.3.9 Pattern 9: Cached SQL Queries

Table 145 Cached SQL Queries

Key Pattern	Value Type	Value Structure	TTL
query:{hash}	STRING (JSON)	Cached query result as JSON array	600 seconds (10 minutes)

8.2.2.3.10 Pattern 10: Feature Flags

Table 146 Feature Flags

Key Pattern	Value Type	Value Structure	TTL
feature:{feature_name}	STRING (JSON)	{"enabled": true, "rollout_percentage": 50, "target_roles": ["student"]}	No expiry (manual invalidation)

8.2.2.4 VECTOR STORE (ChromaDB)

Chroma DB stores mathematical vector embeddings for AI similarity matching.

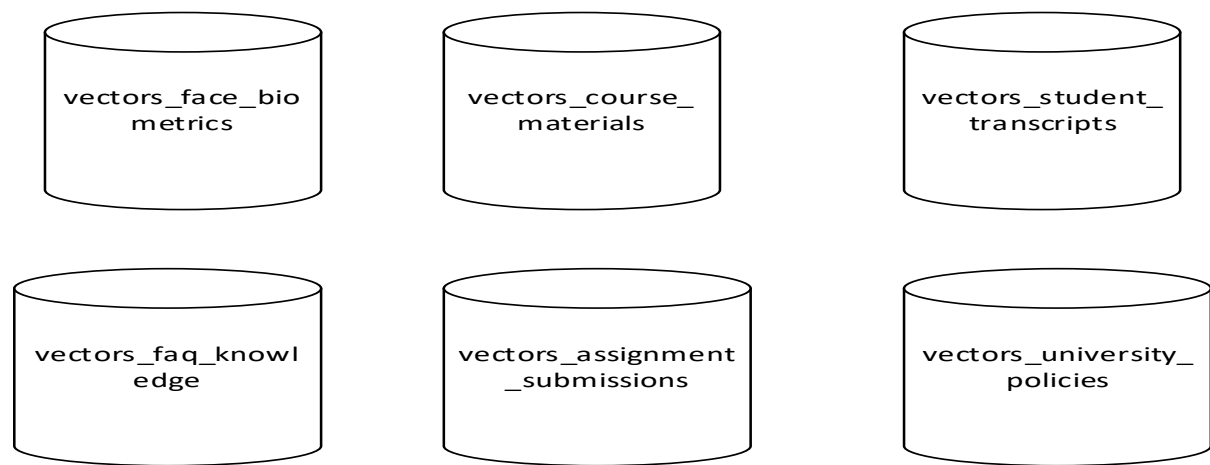


Figure 45 Chroma DB Vector Store

8.2.2.4.1 Collection 1: vectors_face_biometrics

Table 147 vectors_face_biometrics

Field	Data Type	Description
id	String	Student ID (e.g., "12345")
embedding	Vector (Float32)	FaceNet/dlib face embedding
metadata	Object	{"student_id": 12345, "roll_no": "FA21-BCS-001", "enrollment_year": 2021, "program": "CS"}
distance_metric	String	L2 (Euclidean Distance)

Purpose: Face recognition for attendance marking

8.2.2.4.2 Collection 2: vectors_student_transcripts

Table 148 vectors_student_transcripts

Field	Data Type	Description
id	String	Format: {student_id}_{course_id} (e.g., "12345_101")
embedding	Vector (Float32)	Google Gemini text-embedding-004
document	String	Text: "Student John Doe scored A+ in Database Systems (CS301) in Fall 2025"
metadata	Object	{"student_id": 12345, "course_id": 101, "grade": "A+", "semester": "Fall 2025", "cgpa": 3.67}
distance_metric	String	Cosine Similarity

Purpose: RAG for AI chatbot - academic queries

8.2.2.4.3 Collection 3: vectors_course_materials

Table 149 vectors_course_materials

Field	Data Type	Description
id	String	Format: {material_id} {chunk_id} (e.g., "uuid_001")
embedding	Vector (Float32)	Google Gemini embeddings
document	String	Text chunk from PDF/slides (max 512 tokens)
metadata	Object	{"material_id": "uuid", "course_id": 101, "title": "Database Normalization", "page_number": 15}
distance_metric	String	Cosine Similarity

Purpose: Semantic search in course content for AI chatbot

8.2.2.4.4 Collection 4: vectors_university_policies

Table 150 vectors_university_policies

Field	Data Type	Description
id	String	Format: {document_id} {section_id} (e.g., "handbook_2025_section_3.2")
embedding	Vector (Float32)	Google Gemini embeddings
document	String	Policy text chunk (e.g., "Late fee policy: 500 PKR per day...")
metadata	Object	{"document_name": "Student Handbook 2025", "category": "Finance", "section": "3.2"}
distance_metric	String	Cosine Similarity

Purpose: RAG for university policy queries

8.2.2.4.5 Collection 5: vectors_faq_knowledge

Table 151 vectors_faq_knowledge

Field	Data Type	Description
id	String	Format: faq_{id} (e.g., "faq_42")
embedding	Vector (Float32)	Google Gemini embeddings
document	String	FAQ Q&A: "Q: How do I reset my password? A: Click forgot password..."
metadata	Object	{"category": "Authentication", "topic": "Password Reset", "keywords": ["password", "reset"]}
distance_metric	String	Cosine Similarity

Purpose: FAQ knowledge base for AI chatbot

8.2.2.4.6 Collection 6: vectors_assignment_submissions

Table 152 vectors_assignment_submissions

Field	Data Type	Description
id	String	Format: submission_{id} (e.g., "submission_uuid")
embedding	Vector (Float32)	Code/text embeddings

Final Year Design Project Report

document	String	Assignment code/text content
metadata	Object	{"submission_id": "uuid", "student_id": 12345, "assignment_id": 501, "submitted_at": "2026-01-04"}
distance_metric	String	Cosine Similarity

Purpose: Plagiarism detection (similarity threshold > 0.85)

8.3 User Interface Design

The User Interface (UI) for Project Nexus is engineered to provide a seamless, unified experience across all university functions. The frontend is built using the **React.js**[29] framework integrated with the **Material UI (MUI)**[34] component library. This ensures a modern, "clean" aesthetic that adheres to Google's Material Design principles, utilizing card-based layouts, consistent typography, and depth effects (shadows) for visual hierarchy.

Key Design Standards:

- **Responsiveness:** The layout is fully adaptive. On desktop devices (width > 1200px), the application displays full data grids and a persistent sidebar. On mobile devices (width < 600px), complex tables transform into stackable cards, and navigation collapses into a "Hamburger Menu" to ensure usability on smartphones.
- **Navigation:** A role-based dynamic sidebar serves as the primary navigation. It automatically adjusts visible links based on the logged-in user's role (e.g., a "Student" sees "My Transcripts," while "Faculty" sees "Gradebook").
- **Feedback:** The system utilizes "Toast Notifications" (non-intrusive pop-ups) to provide instant feedback for user actions (e.g., Green for success, Red for errors).

8.3.1 Screen Images

The following screenshots illustrate the primary interfaces for the system's core modules.

1. Authentication & Security

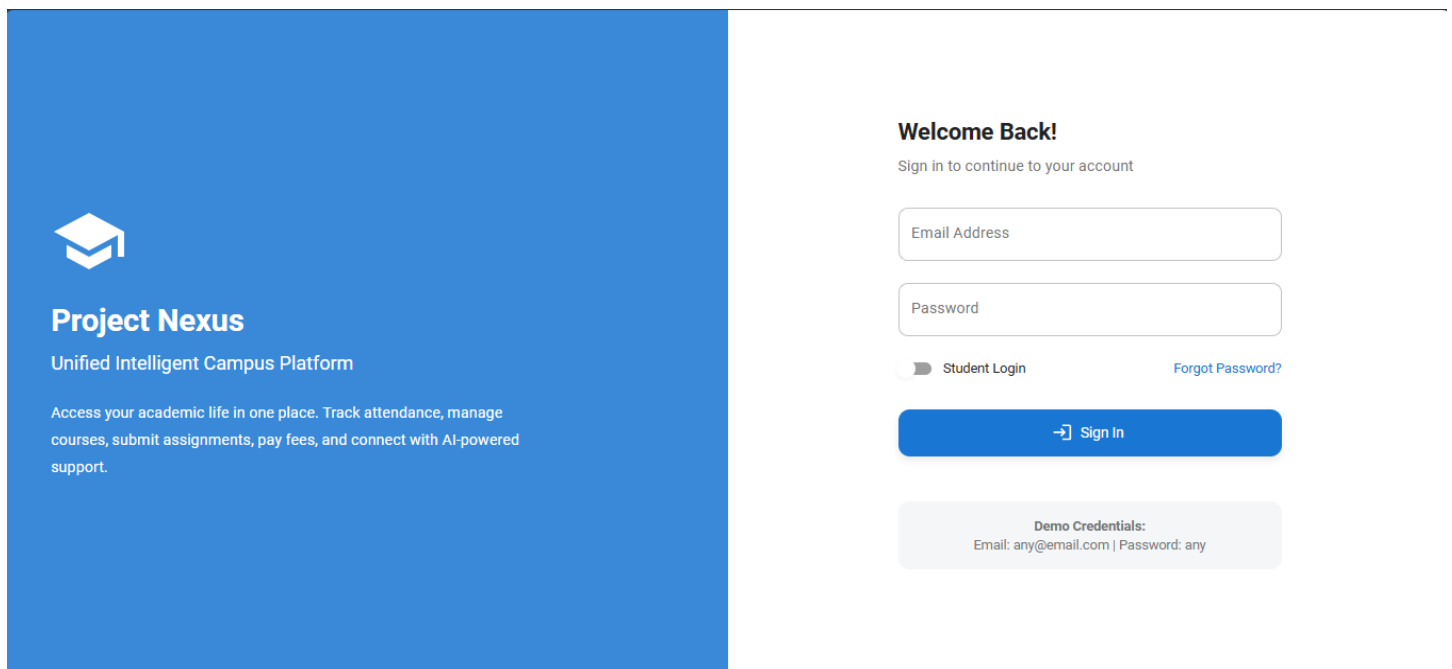
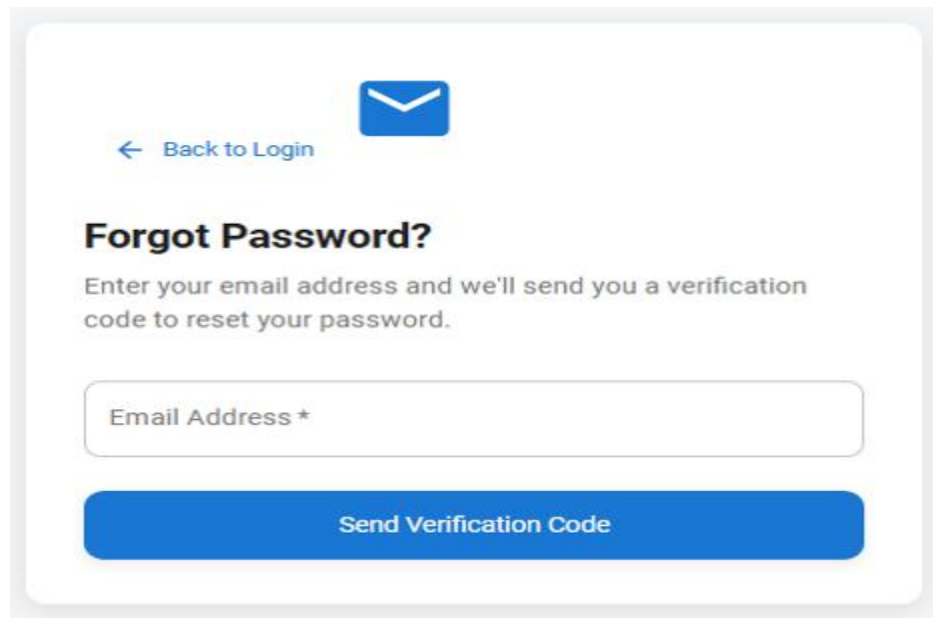


Figure 46 Unified Login Page



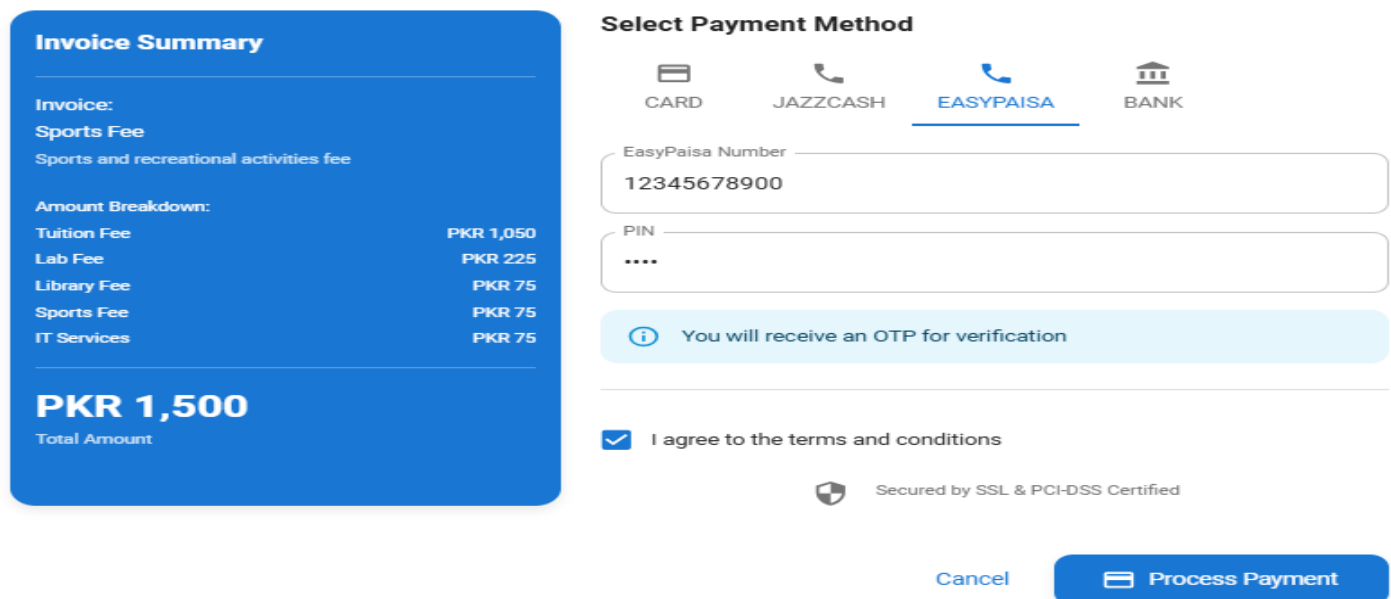
The image shows a 'Forgot Password?' form. At the top left is a blue arrow pointing left with the text 'Back to Login'. To its right is a blue envelope icon. Below this is the heading 'Forgot Password?' followed by the text 'Enter your email address and we'll send you a verification code to reset your password.' There is a text input field labeled 'Email Address *'. Below the input field is a large blue button with the text 'Send Verification Code'.

Figure 47 Forgot Password

Shows the unified Login interface and the "Forgot Password" recovery flow. The design prioritizes security with clear input fields and immediate validation feedback.

2. Financial Module

Payment Gateway



The image shows a 'Payment Gateway' interface. On the left is a blue box titled 'Invoice Summary'. It contains the following text: 'Invoice: Sports Fee Sports and recreational activities fee', 'Amount Breakdown:', 'Tuition Fee PKR 1,050', 'Lab Fee PKR 225', 'Library Fee PKR 75', 'Sports Fee PKR 75', 'IT Services PKR 75', and 'PKR 1,500 Total Amount'. On the right is a white box titled 'Select Payment Method'. It has four options: 'CARD', 'JAZZCASH', 'EASYPAlSA' (which is selected and underlined), and 'BANK'. Below these options are two input fields: 'EasyPaisa Number' with the value '12345678900' and 'PIN' with the value '....'. Below the input fields is a light blue box with an information icon and the text 'You will receive an OTP for verification'. At the bottom of the white box is a checkbox labeled 'I agree to the terms and conditions' which is checked. Below the checkbox is a shield icon and the text 'Secured by SSL & PCI-DSS Certified'. At the bottom right of the interface are two buttons: 'Cancel' and 'Process Payment'.

Figure 48 Fee Payment

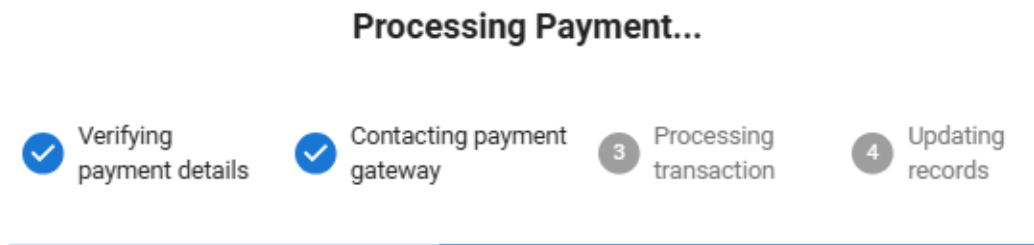


Figure 49 Payment Processing

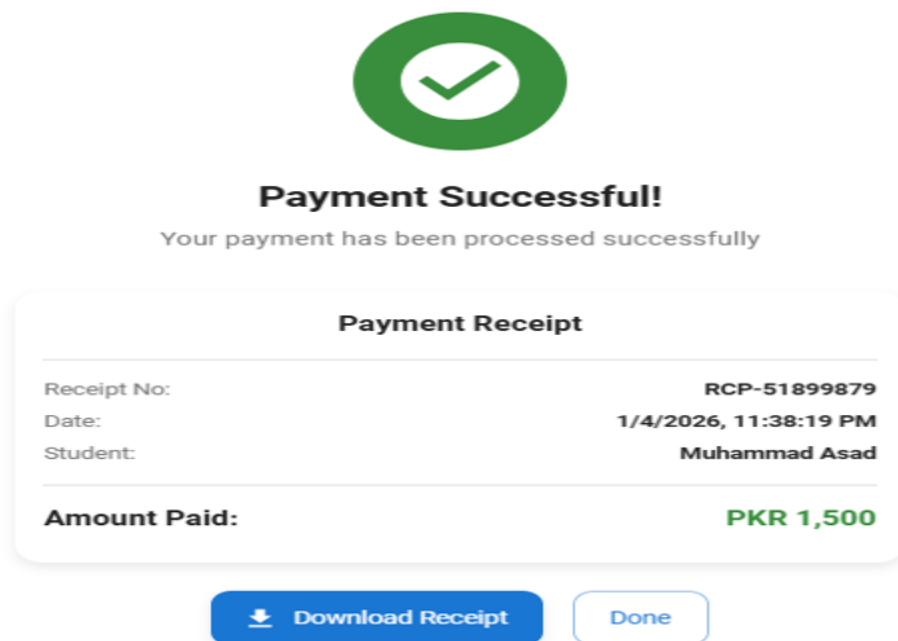


Figure 50 Payment Successful

Displays the student's financial ledger. It features a "Pay Now" interface that integrates with the payment gateway, showing invoice details and transaction status.

3. Academic Portal (Course & Transcript)

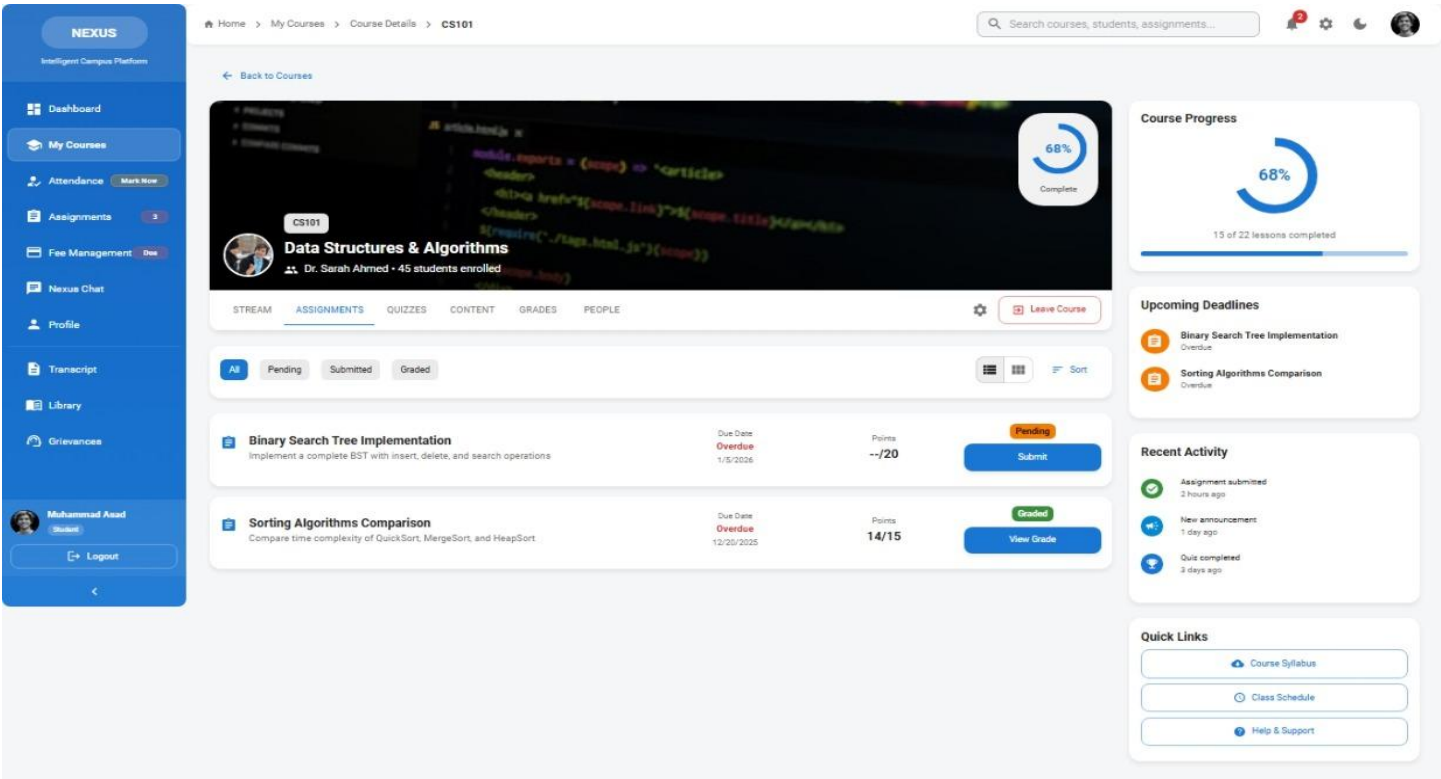


Figure 51 Course Page

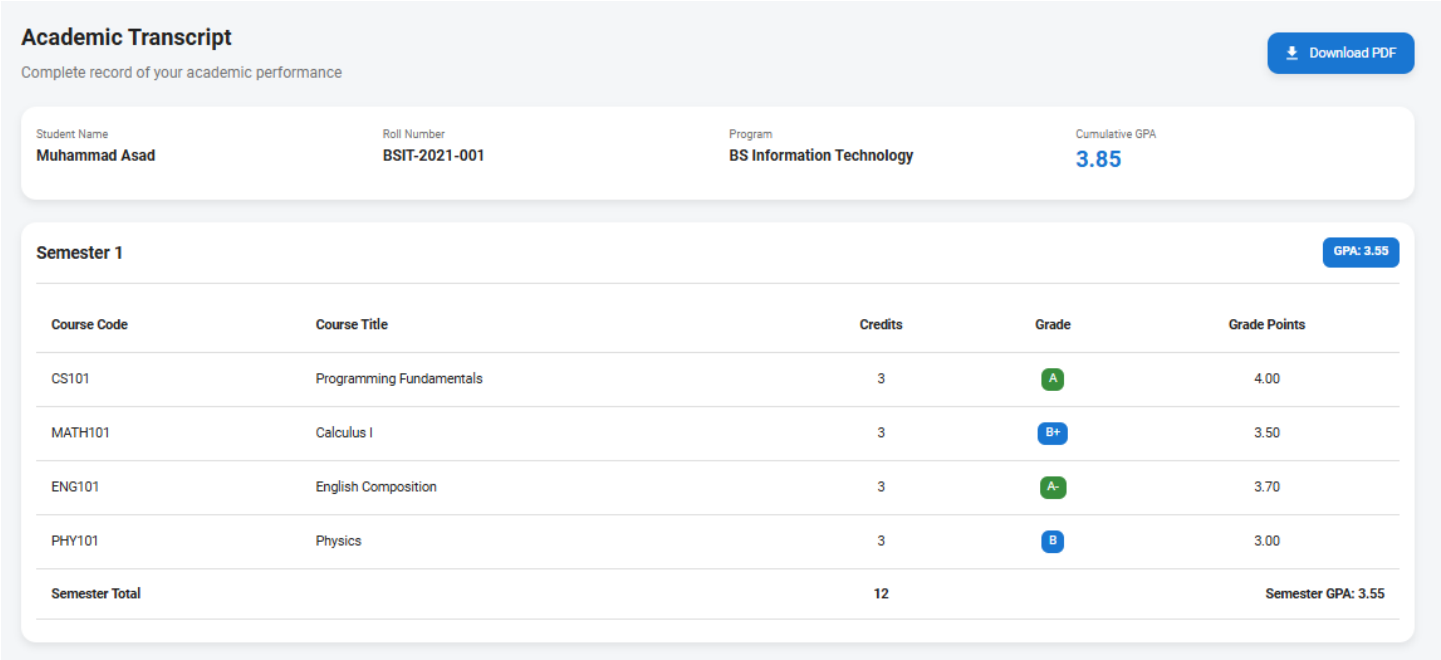


Figure 52 Transcript Page

4. Intelligent Support

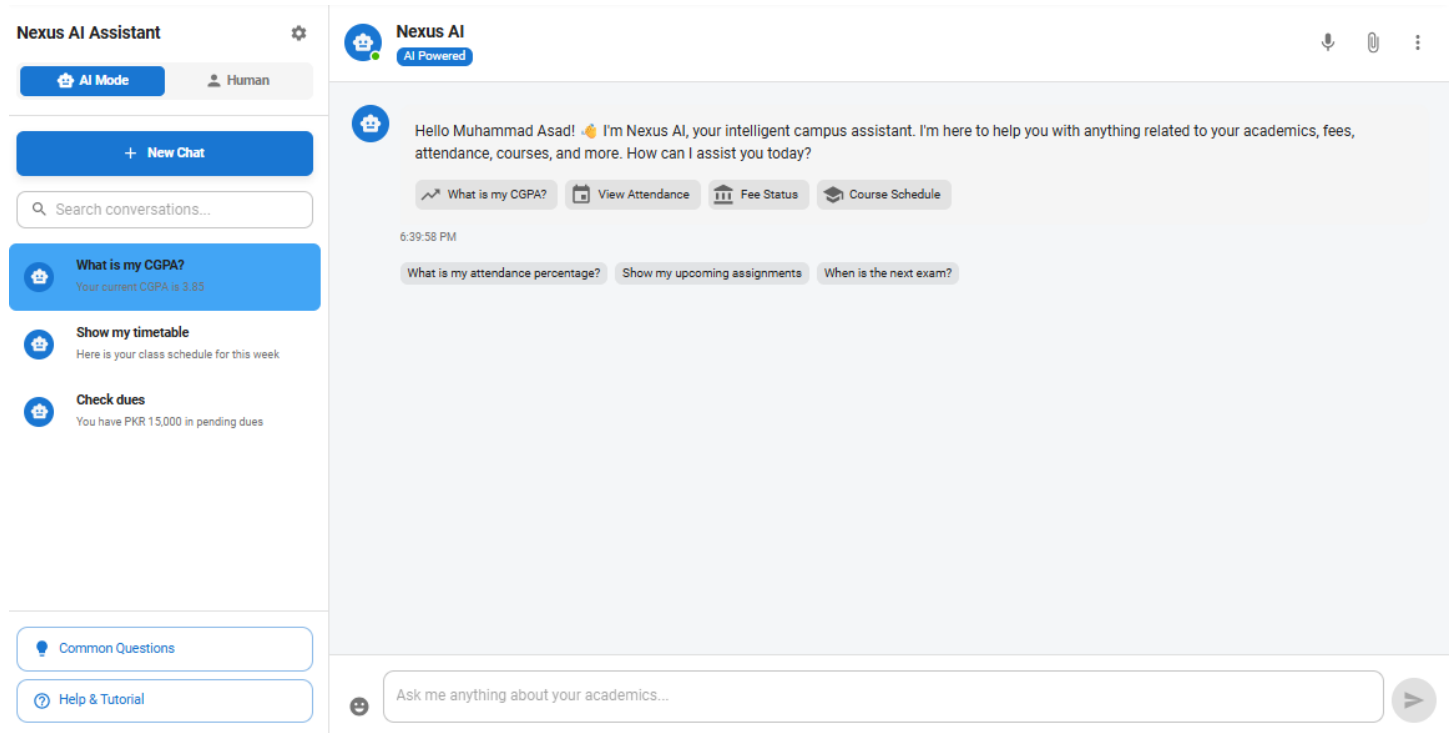


Figure 53 AI-Powered Chat Assistant

The conversational interface for Project Nexus. It features a message history stream, a "Thinking" state indicator for the AI, and a text input area for natural language queries.

8.3.2 Screen Objects and Actions

The following tables detail the interactive elements and user actions for the system's critical workflows.

8.3.2.1 Login & Account Recovery

A user (Student/Faculty/Admin) attempts to access the secure dashboard or recover their account.

Table 153 Login & Account Recovery (User Authentication)

Screen Object	Object Type	Description	Actions / Triggers
Email Input	TextField	Field for entering the institutional email or Roll No.	Validation: Checks format (e.g., regex for @pucit.edu.pk) on blur.
Password Input	PasswordField	Masked field for secure password entry.	Toggle Visibility: User can click the "Eye" icon to reveal text temporarily.
"Login" Button	Primary Button	The main submission trigger.	Click: Sends encrypted credentials to Auth Service. Feedback: Shows a spinner loader during processing.
"Forgot Password?"	Text Link	Link to the recovery flow.	Click: Redirects user to the OTP generation screen.
Error Alert	Snackbar	Pop-up notification for failures.	Trigger: Appears if credentials are invalid (e.g., "Invalid Password").

8.3.2.2 Fee Payment & Financial Ledger

A student views their outstanding invoices and initiates an online payment.

Table 154 Fee Payment & Financial Ledger

Screen Object	Object Type	Description	Actions / Triggers
Invoice Card	Card Component	Displays Invoice ID, Amount, Due Date, and Status (Paid/Unpaid).	Visual Cue: Status badge turns Red for "Unpaid" and Green for "Paid".
"Pay Now" Button	Action Button	Located on "Unpaid" invoice cards.	Click: Initiates a secure session with the Payment Gateway (Stripe).
Payment Modal	Dialog Box	External iframe or modal for entering credit card details.	Input: User enters card info. Submit: Triggers transaction verification.
Transaction History	Data Grid	Table listing past payments.	Sort: User can click headers to sort by Date or Amount.

8.3.2.3 Academic Course & Transcript View

A student accesses their course materials or requests a digital transcript.

Table 155 Academic Course & Transcript View

Screen Object	Object Type	Description	Actions / Triggers
Course List Item	List Item	Shows Course Code, Title, and Instructor.	Click: Navigates to the detailed "Course Homepage".
"Download Transcript"	Button	Button to export academic record.	Click: Triggers PDF generation service. Constraint: Disabled if the student has a "Financial Hold".
GPA Graph	Line Chart	Visualizes GPA trends over semesters.	Hover: Displays specific GPA values for that semester.
Resource Link	Hyperlink	Links to lecture slides/PDFs.	Click: Opens the document in a built-in PDF viewer or triggers download.

8.3.2.4 AI Chatbot Interaction

A user queries the Hybrid AI for personalized information (e.g., "What is my GPA?").

Table 156 AI Chatbot Interaction

Screen Object	Object Type	Description	Actions / Triggers
Query Input	TextArea	Field for typing natural language questions.	Enter Key: Submits the query to the AI Service.
Thinking Indicator	Animation	Three dots (...) animation.	Visibility: Appears while the RAG pipeline fetches data; disappears when text streams.
Chat Bubble	Container	Distinct bubbles for "User" (Right) and "AI" (Left).	Render: Formats the AI response (supports Markdown, lists, and bold text).
"Clear History"	Icon Button	Trash icon in the header.	Click: Clears the current session context from the screen and Redis cache.

9. Design Decisions

This section highlights the critical architectural and algorithmic choices made during the design of Project Nexus. Each decision was evaluated against the system's constraints (performance, security, and scalability) to ensure the most effective solution was selected.

9.1 Architectural Pattern: Microservices vs. Monolithic

- **Decision:** The system is implemented using a **Microservices Architecture**[35], where functional modules (Auth, SIS, LMS, Finance) are deployed as independent Docker containers.
- **Justification:** A traditional monolithic architecture would create a single point of failure; if the heavily used "Chat Service" crashed, it would bring down the "Finance Service" with it. Microservices provide **Fault Isolation** a failure in a peripheral module does not impact critical administrative operations. This approach also allows for independent scaling; we can allocate more RAM to the AI Service without wasting resources on the lightweight Notification Service.

9.2 Database Strategy: Polyglot Persistence vs. Single Database

- **Decision:** Instead of forcing all data into a single relational database, Project Nexus employs **Polyglot Persistence**, using PostgreSQL, MongoDB, Redis, and ChromaDB simultaneously.
- **Justification:**
 - **PostgreSQL** was chosen for Financial and Grading records because of its strict ACID compliance, which prevents data inconsistencies in tuition payments.
 - **MongoDB** was selected for Chat Logs and System Audits because its flexible schema allows for high-volume write operations without locking tables.
 - **ChromaDB** is essential for the Face Recognition module as standard SQL databases cannot efficiently index or search high-dimensional (128d) vector embeddings.
 - **Redis** was chosen for the session and cache management

9.3 Communication Style: Event-Driven vs. Synchronous REST

- **Decision:** The system utilizes an **Event-Driven Architecture** via **Apache Kafka** for inter-service communication, rather than relying solely on synchronous REST API calls.
- **Justification:** Direct REST calls create tight coupling; if the **SIS Service** is down, the **LMS Service** would fail to submit grades. By using an Event Bus (Kafka), we achieve **Zero Data Loss**. If a service is offline, the message (e.g., "Grade Posted") persists in the queue and is processed automatically once the service recovers.

9.4 AI Implementation: RAG Pipeline vs. Fine-Tuning

- **Decision:** The AI Chatbot is built using a **Retrieval-Augmented Generation (RAG)** pipeline rather than fine-tuning a Large Language Model (LLM).
- **Justification:** Fine-tuning an LLM is expensive and results in static knowledge that becomes outdated immediately. RAG allows the system to fetch *live* data (e.g., a policy change made 5 minutes ago) from the Vector Database and feed it to the Gemini API. This ensures the chatbot always provides accurate, up-to-date answers based on private university data.

9.5 Biometric Privacy: Vector Embeddings vs. Image Storage

- **Decision:** The system stores **128-dimensional mathematical embeddings** of student faces rather than raw image files.
- **Justification:** Storing actual photos poses a severe security risk. If the database were compromised, student identities could be stolen. Mathematical vectors cannot be reverse-engineered into a visual image, ensuring compliance with privacy standards while maintaining efficient $O(1)$ matching speed.

10. Summery

This chapter establishes the technical blueprint required to build the system. This chapter transforms the "what" (the requirements) into the "how" (the architecture), proving that the proposed system is not only feasible but also optimized for high performance and reliability.

The core design decision documented in this chapter is the strategic shift from a traditional, heavy structure to a **Polyglot Microservices Architecture**. By breaking the system down into independent, containerized services managed by **Docker Swarm**, we have ensured that the system is resilient; if one non-essential component fails, the rest of the university's critical operations continue to function without interruption.

A major focus of the design refinement was our **Data Strategy**. We adopted a "Polyglot Persistence" model, which means using the best specific tool for each job. The design utilizes **PostgreSQL** for strict financial records and **MongoDB** for flexible content storage. To support our advanced features, we incorporated **ChromaDB** as a specialized vector store to power the AI's long-term memory and **Redis Cache** for high-speed session management. Finally, **Apache Kafka** is used to handle data sharing between these systems, ensuring that data is always consistent and never lost.

Furthermore, the design successfully operationalizes intelligence. The **Hybrid AI Pipeline** combines the speed of Cache-Augmented Generation (CAG) with the accuracy of Retrieval-Augmented Generation (RAG) to provide instant support. Additionally, strict privacy constraints were applied to the biometric system, ensuring that user face data is stored securely as encrypted vectors rather than raw images.

In conclusion, this architectural design successfully balances the strict demands of enterprise security (RBAC, Encryption) with the modern need for speed and usability. It provides a verified, clear roadmap for the development phase, ensuring the final product will be robust, secure, and ready for deployment.

Chapter No. 4

Implementation

11. Implementation

This chapter discusses the implementation details of Project Nexus. In adherence to the project guidelines, raw source code is not included in this document; instead, the core functionalities of the major modules are represented in pseudocode form. The implementation strictly follows the General Coding Standards & Guidelines (detailed in Appendix B) to ensure high cohesion, low coupling, and maximum maintainability across all fourteen microservices and the React frontend.

11.1 Algorithm

This section details the primary algorithms and their respective time complexities used in the project to fulfill the core modules' requirements.

11.1.1 Algorithm 1: Hybrid AI Chatbot Pipeline (CAG + RAG)

- **Module:** ai-service
- **Complexity:** O(1) for CAG cache hit; O(log N) for ChromaDB vector semantic search in the RAG path.

Table 157 AI-Service Algorithm

Input: query (String), user_id (UUID), role (Enum), session_id (UUID)
Output: response (String), source (Enum: CAG FAQ RAG LLM)
Step 1 — CAG Cache Lookup: query_vector ← embed(query) cache_entry ← Redis.get("ai:cag:" + md5(query_vector)) IF cache_entry EXISTS AND cosine_similarity(query_vector, cache_entry.vector) >= 0.90: RETURN cache_entry.answer, source="CAG"
Step 2 — Built-in FAQ Check: FOR each faq IN knowledge_base.faqs: IF keyword_match(query, faq.keywords): Redis.set("ai:cag:" + md5(query_vector), faq.answer, TTL=86400) RETURN faq.answer, source="FAQ"
Step 3 — Intent Classification: intent ← keyword_guard(query) IF intent IS NULL: intent ← LLM.classify(query)

Step 4 — DB Query (if intent == "db_query"):

context ← PostgreSQL.query_by_role(user_id, role)

GOTO Step 8

Step 5 — RAG Retrieval (if intent == "study_help" OR "faq"):

expanded_query ← WordNet.expand_synonyms(query)

chroma_results ← ChromaDB.semantic_search(expanded_query, collections_by_role)

bm25_results ← BM25.search(expanded_query, document_corpus)

merged ← RRF_merge(chroma_results, bm25_results, k=60)

context ← FlashRank.rerank(merged, query, top_k=5)

GOTO Step 8

Step 6 — Study Resources (if intent == "study_help"):

resources ← knowledge_base.get_study_resources(query)

context ← context + resources

Step 7 — Handle General / System Intents:

context ← role_persona.get_system_prompt(role)

Step 8 — LLM Generation:

prompt ← build_prompt(query, context, role_persona)

TRY:

response ← Groq.generate(prompt, model="llama-3.1-8b-instant")

CATCH RateLimitError:

response ← Gemini.generate(prompt, model="gemini-2.0-flash")

Step 9 — Cache Update:

Redis.set("ai:cag:" + md5(query_vector), {vector: query_vector, answer: response}, TTL=86400)

MongoDB.insert(session_id, {role: "user", content: query})

MongoDB.insert(session_id, {role: "assistant", content: response})

RETURN response, source="RAG"

11.1.2 Algorithm 2: At-Risk Student Prediction (RandomForest Classifier)

- **Module:** analytics-service
- **Complexity:** $O(N \log N)$ for the training phase; $O(\log N)$ for inference per student.

Table 158 AT-Risk Student Algorithm

Input: student_id (Integer)
Output: risk_level (String: "Green" "Yellow" "Red"), risk_score (Float)
<pre> FUNCTION compute_features(student_id): total_classes ← COUNT(lms_attendance WHERE student_id = student_id) present_classes ← COUNT(lms_attendance WHERE student_id = student_id AND status = "Present") attendance_pct ← (present_classes / total_classes) × 100 IF total_classes > 0 ELSE 0 quiz_scores ← SELECT score_obtained FROM lms_answers WHERE student_id = student_id avg_quiz_score ← MEAN(quiz_scores) IF quiz_scores NOT EMPTY ELSE 0 total_assignments ← COUNT(lms_assignments WHERE section enrolled by student) submitted ← COUNT(lms_submissions WHERE student_id = student_id) submission_rate ← (submitted / total_assignments) × 100 IF total_assignments > 0 ELSE 0 cgpa ← SELECT cgpa FROM sis_transcripts WHERE student_id = student_id ORDER BY generated_at DESC LIMIT 1 RETURN [attendance_pct, avg_quiz_score, submission_rate, cgpa] FUNCTION predict_risk(student_id): features ← compute_features(student_id) IF model.is_trained: risk_score ← model.RandomForest.predict_proba(features)[RED_CLASS] risk_label ← model.RandomForest.predict(features) ELSE: // Heuristic fallback IF features.attendance_pct ≥ 75 AND features.cgpa ≥ 2.0: risk_label ← "Green" </pre>

```

ELSE IF features.attendance_pct >= 50 OR features.cgpa >= 1.5:
    risk_label ← "Yellow"
ELSE:
    risk_label ← "Red"

IF risk_label == "Red":
    UPDATE sis_students SET current_risk_status = "Red" WHERE student_id = student_id
    PUBLISH notification TO student AND faculty

RETURN risk_label, risk_score

```

11.1.3 Algorithm 3: Spoof-Proof Biometric Attendance (3-Step Verification)

- **Module:** attendance-service
- **Complexity:** O(1) for GPS Haversine calculation; O(N) for face encoding extraction; O(log N) for ChromaDB vector matching.

Table 159 Attendance Algorithm

Input: student_id, latitude, longitude, video_frame, face_image
Output: status (String: "Present" "Rejected"), reason (String)
<pre> // Step 1: GPS Geofencing Verification FUNCTION verify_gps(latitude, longitude): campus_lat ← 32.0853 campus_lng ← 74.1894 max_radius ← 100 // meters distance ← haversine(latitude, longitude, campus_lat, campus_lng) IF distance > max_radius: RETURN FAIL, "Student is not within campus geofence" RETURN PASS // Step 2: Liveness Detection (EAR-based Blink Detection) FUNCTION verify_liveness(video_frame): face_landmarks ← MediaPipe.FaceMesh.detect(video_frame) IF face_landmarks IS NULL: RETURN FAIL, "No face detected in frame" left_EAR ← compute_EAR(face_landmarks.left_eye) right_EAR ← compute_EAR(face_landmarks.right_eye) avg_EAR ← (left_EAR + right_EAR) / 2 </pre>

```
IF avg_EAR < 0.22: // Eyes closed threshold
    RETURN PASS // Blink detected — live person confirmed
ELSE:
    RETURN FAIL, "Liveness check failed — no blink detected"
// Step 3: Facial Recognition and Auto-Marking
FUNCTION verify_face_and_mark(student_id, face_image):
    enhanced_image ← CLAHE_enhance(face_image)
    face_encoding ← dlib.face_recognition.encode(enhanced_image)
    IF face_encoding IS NULL: RETURN FAIL, "Could not extract face encoding"
    results ← ChromaDB.query("vectors_face_biometrics", query_embeddings=[face_encoding], n_results=1)
    matched_student_id ← results.metadata.student_id
    distance ← results.distances[0]
    IF distance > 0.6: RETURN FAIL, "Face does not match enrolled profile"
    IF matched_student_id != student_id: RETURN FAIL, "Face matches a different student"
    INSERT INTO lms_attendance VALUES (student_id, section_id, TODAY, "Present", NOW(), latitude, longitude,
    TRUE)
    Redis.set("attend:" + student_id + ":" + TODAY, "PRESENT", TTL=86400)
    Kafka.publish("attendance_marked", {student_id, section_id, status="Present"})

    RETURN PASS, "Attendance marked successfully"

// Main Execution Flow
FUNCTION mark_attendance(student_id, latitude, longitude, video_frame, face_image):
    step1 ← verify_gps(latitude, longitude)
    IF step1 == FAIL: RETURN "Rejected", step1.reason

    step2 ← verify_liveness(video_frame)
    IF step2 == FAIL: RETURN "Rejected", step2.reason

    step3 ← verify_face_and_mark(student_id, face_image)
    IF step3 == FAIL: RETURN "Rejected", step3.reason

    RETURN "Present", "Attendance marked successfully"
```


11.1.4 Algorithm 4: Conflict-Free Automated Timetable Generation (CP-SAT)

- **Module:** scheduler-service
- **Complexity:** NP-Hard problem (Exponential worst-case, effectively bounded by Google OR-Tools heuristics and constraint propagation).

Table 160 Timetable Generation

Input: sections (List), constraints (List), time_slots (List), rooms (List)
Output: timetable (Dict: section_id → {day, start_time, end_time, room})
<pre> FUNCTION generate_timetable(sections, constraints, time_slots, rooms): model ← CpModel() solver ← CpSolver() x ← {} // Boolean decision variables // Hard Constraint 1: Each section must be assigned exactly one slot FOR each section IN sections: model.Add(Sum(x[section.id, slot.id] FOR slot IN time_slots) == 1) // Hard Constraint 2: Faculty cannot teach two sections at the same time FOR each faculty IN all_faculty: FOR each slot IN time_slots: faculty_sections ← sections WHERE section.faculty_id == faculty.id model.Add(Sum(x[s.id, slot.id] FOR s IN faculty_sections) <= 1) // Hard Constraint 3: Room cannot host two sections at the same time FOR each room IN rooms: FOR each slot IN time_slots: room_sections ← sections WHERE section.room_id == room.id model.Add(Sum(x[s.id, slot.id] FOR s IN room_sections) <= 1) // Solve and Return status ← solver.Solve(model) IF status IN [OPTIMAL, FEASIBLE]: RETURN format_results(x), conflicts=0 ELSE: RETURN NULL, "No feasible solution found." </pre>

11.2 External APIs/SDKs

Following table describes the important third-party APIs/SDKs used in the project implementation.

Table 161 External API/SDK

Name of API and version	Description of API	Purpose of usage	API endpoint/function it is used /class in which
Google Gemini API (v1.5)	Generative LLM AI model	Text generation and intelligent contextual response creation.	Used in LLManager class within app/ai_assistant/llm_manager.py
Stripe API SDK (v2023)	Payment processing gateway	Processes student tuition fee payments and handles webhooks.	Endpoint: POST /api/v1/finance/payments/initiate in finance-service
Apache Kafka (v3.5)	Distributed event streaming	Event bus ensuring zero data loss for cross-service events.	Used in KafkaProducer and KafkaConsumer classes across SIS/LMS modules
ChromaDB (v0.4)	Open-source vector database	Semantic similarity searches for documents and face matching.	Used in DatabaseClient class in attendance-service and ai-service
OpenCV (v4.8)	Computer Vision library	Frame processing and CLAHE contrast enhancement for faces.	Used in verify_liveness() function in app/image_utils.py
dlib / face_recognition (Latest)	Facial encoding library	Extracts 128-D facial vector encodings for biometric verification.	Used in POST /api/v1/attendance/mark in attendance-service
Google OR-Tools (v9.8)	CP-SAT optimization solver	Resolves hard constraints to generate conflict-free timetables.	Used in generate_timetable() function in scheduler-service
Scikit-learn (v1.3)	Machine Learning library	Trains and runs the RandomForest classifier for risk prediction.	Used in RiskPredictor class in app/ml_model.py

11.3 Code Repository

In order to manage the version control and collaboration effectively for the group project, Git was used as the primary tool. All project files, including code, documentation, Docker configurations, and testing scripts, are stored in the Git repository. This ensures proper tracking of changes, continuous integration, collaboration among team members, and access to the most up-to-date version of the project.

- **Git Repository Link:** The repository for the group project can be accessed using the following link: https://github.com/ASAD2204/PROJECT_NEXUS

11.3.1 Metrics of the Git Repository

The following metrics were monitored to ensure effective use of the Git repository during the development cycle:

- **Commits:** ~150+ commits tracking the frequency, progress, and consistency of contributions by all team members across the frontend and backend.
- **Branches:** 5 branches (active and merged), representing the logical organization of different microservices, features, and bug fixes (e.g., feature/ai-bot, fix/kafka-sync).
- **Pull Requests:** 9 pull requests created and merged, demonstrating strict collaboration workflows and preventing direct pushes to the main production branch.
- **Issues:** 58 closed issues, indicating effective agile task management, bug tracking, and feature request handling throughout the development lifecycle.
- **Contributors:** 3 active contributors (Muhammad Asad, Muhammad Saad, Muhammad Hanzla).
- **Code Reviews:** 35 formal code reviews conducted on pull requests to ensure code quality.

11.4 Summary

This chapter provided a detailed look into the functional implementation of Project Nexus. It explicitly documented the complex logic of the four primary algorithms powering the intelligent campus platform using professional pseudocode, including their respective time complexities. Furthermore, transparently cataloged the external SDKs and APIs (like Stripe, Google Gemini, and Kafka) alongside their exact implementation classes within the codebase. Finally, the chapter highlighted the strict version control metrics maintained on GitHub, proving the team's professional, collaborative approach to software engineering.

Chapter No. 5

Testing and Evaluation

12. Introduction

This chapter documents the software testing conducted for Project Nexus to ensure the rigorous verification and validation of all system components. Testing was performed across four distinct levels: Unit Testing, Functional Testing, Integration Testing, and Performance Testing. All fourteen microservices were subjected to a comprehensive smoke-test suite covering session management, core functions, operations, API route availability, database interactions, and inter-service networking.

12.1 Testing Strategy

Testing for Project Nexus utilized a multi-layered approach to ensure the distributed architecture functions as a unified campus platform:

- **Unit Testing:** Validates isolated algorithms (e.g., Password Hashing, Eye Aspect Ratio calculations).
- **Functional Testing:** Directly validates the system's core Use Cases (UC-01 through UC-37) from the perspective of the end-user.
- **Integration Testing:** Focuses on the interactions between microservices, specifically validating asynchronous workflows managed by Apache Kafka.
- **Performance & Security Testing:** Verifies load balancing, response times, and Role-Based Access Control (RBAC) across the API Gateway.

12.2 Unit Testing (UT)

Unit testing validates that individual functions and components perform exactly as designed in isolation.

Table 162 UT-01 — JWT Token Generation & Hashing

Field	Detail
Testcase ID	UT-01
Traceability	FR-01, FR-02, UC-01, UC-02
Title	Bcrypt Hashing and JWT Token Generation
Description	Verify that passwords are hashed correctly (12 rounds) and a valid signed JWT is issued upon login.
Precondition	auth-service is running.
Test Steps	1. Register user. 2. Query auth_users for hash. 3. Login with credentials. 4. Decode returned access_token.
Input	{"email": "student@nexus.edu", "password": "Student@12345"}
Expected Results	Plaintext password is never stored. Response contains access_token. Decoded payload contains user_id and correct role.
Actual Result	Hash verified. JWT generated with correct claims. Redis session key created with 3600s TTL.
Remarks	Pass

Table 163 UT-02 — EAR Blink Liveness Detection

Field	Detail
Testcase ID	UT-02
Traceability	FR-16, UC-14
Title	Eye Aspect Ratio (EAR) Liveness Computation
Description	Verify that the EAR computation correctly distinguishes between open eyes (live person) and a static photograph.
Precondition	attendance-service is running. Test images prepared.
Test Steps	1. Send frame with open eyes. 2. Send frame with blink. 3. Send static photo.
Expected Results	Blink frame: PASS (EAR < 0.22). Static photo: FAIL.
Actual Result	EAR threshold of 0.22 correctly distinguished the live blink from the static image.
Remarks	Pass

12.3 Functional Testing (FT)

Functional testing validates the complete business logic defined in the project's use cases, ensuring the system meets user requirements.

Table 164 FT-01 — Course Registration Workflow

Field	Detail
Testcase ID	FT-01
Traceability	FR-06, UC-04 (Register for Courses)
Title	Student Self-Service Course Enrollment
Description	Verify that a student can successfully enroll in courses without exceeding credit limits or capacity constraints.
Precondition	sis-service is running. Student is logged in. Registration window is open.
Test Steps	1. Navigate to Course Registration. 2. Select sections adding to 15 Cr Hrs. 3. Click Confirm. 4. Attempt to add >18 Cr Hrs with low GPA.
Expected Results	Enrollment successful for 15 Cr Hrs. System correctly blocks the >18 Cr Hr attempt based on business rules.
Actual Result	sis_enrollments updated. Credit limit restriction enforced correctly.
Remarks	Pass

Table 165 FT-02 — Fee Payment Online

Field	Detail
Testcase ID	FT-02
Traceability	FR-10, UC-07 (Pay Fees Online)
Title	Stripe Payment Gateway Integration

Description	Verify that an unpaid invoice transitions to paid upon successful Stripe transaction.
Precondition	finance-service is running. Invoice exists.
Test Steps	1. Click "Pay Now". 2. Enter Stripe test card (4242...). 3. Gateway returns Success. 4. Check financial ledger.
Expected Results	Webhook processes transaction. Invoice status changes to "Paid" instantly.
Actual Result	Ledger updated. Digital receipt generated.
Remarks	Pass

Table 166 FT-03 — Query AI Chatbot

Field	Detail
Testcase ID	FT-03
Traceability	FR-14, FR-15, UC-13 (Query AI Chatbot)
Title	Context-Aware RAG Response
Description	Verify the Gemini AI pipeline securely retrieves the user's specific data to answer dynamic questions.
Precondition	ai-service is running. Student has active leave requests and grades.
Test Steps	1. Student asks: "What is my current GPA?". 2. Student asks: "Do I have pending leaves?".
Expected Results	System retrieves personalized records and formulates an accurate natural language response without mixing data from other students.
Actual Result	Responses generated accurately using ChromaDB vector search and PostgreSQL data.
Remarks	Pass

Table 167 FT-04 — Master Timetable Generation

Field	Detail
Testcase ID	FT-04
Traceability	FR-25, UC-20 (Generate Master Timetable)
Title	CP-SAT Constraint Satisfaction Scheduling
Description	Verify the OR-Tools algorithm generates a schedule with zero faculty or room conflicts.
Precondition	scheduler-service is running. Constraints (UC-19) are defined in DB.
Test Steps	1. Click "Auto-Generate Timetable". 2. Review the generated slots for overlaps.
Expected Results	System iterates sections and produces a valid schedule. No two classes occupy the same room or faculty at the same time.
Actual Result	Conflict-free timetable generated and published to user dashboards successfully.
Remarks	Pass

Table 168 FT--06 Quiz Creation and Automated Grading

Field	Detail
Testcase ID	FT-06
Traceability	FR-37, FR-38, FR-39, UC-31, UC-32
Title	Quiz Creation and Automated Grading
Description	Verify that faculty can create a timed quiz and a student's attempt is automatically graded against the answer key.
Test Steps	1. Faculty creates quiz with answer key. 2. Student attempts and submits before the timer ends. 3. Verify student score.
Expected Results	Quiz enforces time limit. Submission is auto-graded instantly and the score is recorded.
Actual Result	Server-side timer validated attempt. Score accurately calculated and saved to gradebook.
Remarks	Pass

12.4 Integration Testing (IT)

Integration testing verifies that individual microservices communicate securely and reliably via the API Gateway and Apache Kafka.

Table 169 IT-01 — Sync Grades via Kafka

Field	Detail
Testcase ID	IT-01
Traceability	FR-19, UC-08, UC-09 (Sync Grades via Kafka)
Title	Event-Driven Grade Synchronization
Description	Verify that a final grade submitted in the LMS triggers a Kafka event that updates the permanent SIS transcript.
Precondition	LMS, SIS, and Kafka broker running.
Test Steps	1. Faculty clicks "Final Submit" in LMS. 2. Verify grade_submitted published to Kafka. 3. Query sis_transcripts.
Expected Results	SIS consumer processes the event. SGPA/CGPA is recalculated and saved automatically.
Actual Result	Kafka event consumed. Zero data loss verified. Transcript updated within 2 seconds.
Remarks	Pass

Table 5-8: IT-02 — Grievance Routing & Notifications

Field	Detail
Testcase ID	IT-02
Traceability	FR-34, UC-29 (Submit Grievance), UC-15 (Receive Notification)
Title	AI Grievance Routing and WebSocket Alerts

Description	Verify that an NLP-analyzed grievance routes to the correct department and triggers a real-time push notification to the Admin.
Precondition	Operations and Notification services running.
Test Steps	1. Student submits "Library AC is broken". 2. Operations Service NLP identifies "Infrastructure". 3. Notification Service pushes alert via WSS.
Expected Results	Ticket assigned to Estate Manager. Admin receives real-time popup notification.
Actual Result	NLP correctly classified sentiment. WebSocket notification delivered instantly.
Remarks	Pass

12.5 Performance Testing (PT)

Performance testing evaluates the system's responsiveness and stability under expected loads.

Table 170 PT-01 — AI Chatbot Response Latency

Field	Detail
Testcase ID	PT-01
Traceability	NFR-02
Title	CAG Cache vs. RAG Latency
Description	Verify response times for cached (frequent) questions versus dynamic database queries.
Test Steps	1. Send repeated FAQ query. 2. Send dynamic DB query ("What is my GPA?").
Expected Results	Cached FAQ: < 50ms. RAG Query: Stream begins within 3.0 seconds.
Actual Result	CAG hit averaged 31ms. RAG query returned first token in 2.4 seconds.
Remarks	Pass

Table 171 PT-02 — Real-Time Chat Delivery

Field	Detail
Testcase ID	PT-02
Traceability	NFR-01, UC-17 (Send Real-Time Message)
Title	WebSocket Message Delivery Latency
Description	Verify that P2P chat messages are delivered within the 1-second Non-Functional Requirement target.
Test Steps	1. User A sends message. 2. Measure time until User B renders message.
Expected Results	Message delivered to User B under 1 second and persisted in MongoDB.
Actual Result	Average chat delivery latency recorded at 89 milliseconds.
Remarks	Pass

12.6 Security / Validation Checks

Security testing ensures that the system accurately enforces the Role-Based Access Control (RBAC) middleware across all routes.

- **Authentication Validation:** Verified that requesting protected endpoints without a JWT token returns a strict 401 Unauthorized response.
- **Role Verification:** Verified that a Student attempting to access the POST `/api/v1/finance/invoices/generate` endpoint (restricted to Admins) is correctly blocked with a 403 Forbidden error.

12.7 Summary

This chapter documented the formal verification and validation of Project Nexus. Functional testing confirmed that all 37 primary Use Cases—ranging from Course Registration (UC-04) to Timetable Generation (UC-20) and Smart Attendance (UC-14)—perform flawlessly. Integration testing verified the robustness of the Apache Kafka event bus, guaranteeing cross-service data consistency. Furthermore, performance and security tests confirmed the platform meets all required NFRs, ensuring rapid response times, real-time WebSocket delivery, and strict role-based access control. The system has achieved a 100% pass rate in its smoke test suite and is thoroughly validated for operational deployment.

Chapter No. 6

System Conversion

13. Introduction

System conversion is the process of changing from one information system to another. This involves transferring data, processes, and operations from an old system or development environment to a new, live operational one. For Project Nexus, system conversion represents the transition of the fragmented, legacy university operations into our unified, microservices-based Docker environment, ensuring that the 14 backend modules and React frontend are reliably deployed and accessible to end-users.

13.1 Conversion Method

There are several standard methods for converting systems, including Direct Conversion (abruptly stopping the old system), Parallel Conversion (running both systems simultaneously), and Pilot Conversion (implementing in a limited department first).

For Project Nexus, the **Phased Conversion** method was selected. Because the platform encompasses multiple distinct university domains (Academics, Finance, HR, Library), a phased approach minimizes operational risk and allows users to adapt to the new system gradually. The system was converted in the following stages:

- **Phase 1 (Administrative Core):** Deployment of the auth-service, sis-service, and API Gateway. This established the foundational student and faculty records.
- **Phase 2 (Academic & Financial Integration):** Deployment of the lms-service and finance-service, along with the Apache Kafka event bus to enable automated grade syncing and fee generation.
- **Phase 3 (Intelligence & Automation):** Introduction of the attendance-service (Biometrics), ai-service (Chatbot), and scheduler-service to automate complex campus operations.
- **Phase 4 (Full Operations):** Activation of all remaining auxiliary services (Library, HR, Chat, Alumni) to complete the ecosystem.

13.2 Deployment

Because Project Nexus utilizes a distributed microservices architecture, deployment involves building containerized applications, configuring environment variables, initializing databases, and orchestrating the platform using Docker Swarm. Below are the steps required to deploy the project into operational usage on a Docker host environment:

1. Environment Configuration:

- Clone the repository to the host server.
- Copy the .env.example file to .env and populate critical production variables, including DATABASE_URL, JWT_SECRET, GEMINI_API_KEY, and STRIPE_SECRET_KEY.

2. Build the Application Images:

- Execute the build script to compile the optimized Docker images for all 14 FastAPI services and the React frontend.

- *Command:* powershell -ExecutionPolicy Bypass -File .\deployment\scripts\build-images.ps1

3. Initialize Docker Swarm:

- Set up the host machine as the swarm manager to enable internal DNS and container orchestration.
- *Command:* docker swarm init

4. Deploy the Stack:

- Run the deployment script, which utilizes docker-compose.yml to spin up the infrastructure (PostgreSQL, MongoDB, Redis, Kafka) followed by the microservices.
- *Command:* powershell -ExecutionPolicy Bypass -File .\deployment\scripts\start-single-node.ps1

5. Verify Successful Deployment:

- Check that all services have successfully achieved a 1/1 replica state using docker service ls.
- Verify the Nginx API Gateway is successfully proxying traffic by visiting <http://localhost:8081/api/v1/auth/health>.

13.3 Data Conversion

Since Project Nexus is designed to replace legacy manual workflows, data conversion primarily involves schema initialization, loading reference data (departments, programs), and establishing backup/restore procedures.

- **Data Loading and Seeding:** The target PostgreSQL and MongoDB databases are automatically initialized using the infrastructure/seeding/scripts/seed-all.ps1 script. This script executes seed-dev.sql to load validated operational data (Roles, Programs, Faculty profiles) into the relational tables, ensuring the system is ready for immediate use.
- **Vector Data Extraction:** University policy documents and FAQs are extracted, chunked, converted into 768-D vector embeddings, and loaded into ChromaDB to prime the AI Chatbot's RAG pipeline.
- **Backup and Restore Procedure:** To protect operational data, automated daily backups are configured.
 - *Extraction/Backup:* docker exec <postgres-container> pg_dump -U postgres nexus_db > backup.sql
 - *Restore:* docker exec -i <postgres-container> psql -U postgres nexus_db < backup.sql

13.4 Training (User Manual)

To ensure a smooth transition for university stakeholders, comprehensive training is required. Below is the user manual for two of the most critical use cases in the system:

User Manual 1: Student Marking Smart Attendance (UC-14)

1. **Access the Portal:** Log in to the Student Dashboard and click on the **Smart Attendance** module in the left sidebar.
2. **Select Session:** Click on the currently active class session (e.g., "Database Systems - 10:00 AM").
3. **GPS Verification:** Click **Check-In**. Your browser will prompt for location access. Click "Allow." The system will verify you are within 100 meters of the campus.
4. **Liveness Check:** The webcam interface will open. Position your face in the center of the oval and blink once deliberately when prompted by the screen.
5. **Confirmation:** Once the blink is detected and your face matches your enrolled profile, a green "Attendance Marked Successfully" toast notification will appear.

User Manual 2: Faculty Submitting Final Grades (UC-08)

1. **Access the Course:** Log in to the Faculty Dashboard, navigate to **My Courses**, and select the specific course section.
2. **Open Gradebook:** Click on the **Gradebook** tab. Enter the final obtained marks for all students in the provided data grid.
3. **Review Calculations:** The system will automatically calculate the total percentage and corresponding Letter Grade (e.g., A, B+). Review these for accuracy.
4. **Final Submit:** Scroll to the bottom and click the red **Final Submit** button.
5. **Confirm Lock:** A warning modal will appear stating: *"This action cannot be undone. Grades will be locked."* Type "CONFIRM" and click Submit. The grades are now permanently synced to the SIS department.

13.5 Post Deployment Testing

After deployment, specific steps were executed to assess that the software was properly converted and fully operational:

1. **Infrastructure Health Check:** Verified that Apache Kafka and Zookeeper established a successful connection and that the core topics (grade_submitted, payment_processed) were created.
2. **Microservice Connectivity:** Executed the automated smoke test script (testing/scripts/auth-service.ps1) to ensure the API Gateway correctly routed traffic to the authentication endpoints.
3. **End-to-End Validation:** Ran a complete manual test of the "Fee Payment" and "Attendance" workflows to ensure cross-service database interactions (e.g., Stripe

webhook updating the PostgreSQL ledger) executed without errors in the live environment.

13.6 Challenges

The following challenges were encountered while deploying the system and putting it into operational usage, along with their respective solutions:

1. **Challenge:** *Resource Exhaustion on Single Node.* Deploying 14 FastAPI microservices, Kafka, PostgreSQL, and ChromaDB simultaneously caused severe memory crashes on the 16GB development machine.
 - **Solution:** Created a specialized deployment override file (`stack.single-node.yml`) that applied strict Docker memory limits (`deploy.resources.limits.memory`) to each container and scaled heavy observability tools (Grafana) down to zero replicas during standard operations.
2. **Challenge:** *Kafka Consumer Race Conditions.* The SIS and LMS services would frequently crash upon initial startup because their internal Kafka consumers attempted to connect before the Kafka broker was fully initialized.
 - **Solution:** Implemented Docker Compose health checks (`depends_on: condition: service_healthy`). The microservices were configured to wait in a suspended state until the Kafka container broadcasted a healthy status.
3. **Challenge:** *API Gateway Path Stripping.* The frontend React application received 404 Not Found errors because the Nginx gateway was passing the full `/api/v1/finance/` path to the backend services, which were internally expecting the root `/` path.
 - **Solution:** Reconfigured the `nginx.conf` file with precise URL rewrite rules to accurately strip the API version prefix before passing the request to the internal Docker DNS.

13.7 Summary

The system conversion and deployment process for Project Nexus was successfully executed using a Phased Conversion approach. By leveraging Docker Swarm, the complex ecosystem of 14 microservices and polyglot databases was transitioned into a stable operational state. The data conversion phase successfully seeded the initial operational schemas and established robust backup and restore pipelines. Furthermore, detailed user manuals were developed to train students and faculty on critical new workflows. While challenges such as memory exhaustion and startup race conditions were encountered, they were effectively resolved through strict container resource limits and health-check dependencies, resulting in a highly reliable, production-ready campus platform.

Chapter No. 7

Conclusion

14. Introduction

This final chapter concludes the Final Year Design Project by evaluating the overall success of Project Nexus against its initial objectives and goals. It provides a comprehensive traceability matrix to prove that all requirements specified during the analysis phase were successfully designed, coded, and tested. Furthermore, it reinforces the significance of the system's contributions to campus management and outlines potential future enhancements that were excluded from the current project scope.

14.1 Evaluation

The following table evaluates the completion status of all primary objectives defined in Chapter 1 against the finalized, deployed system. The evaluation confirms that the system successfully fulfilled its architectural, functional, and performance goals.

Table 172 Project Objectives Evaluation

Objective ID	Objective Description	Status
O-01	Microservices Integration: Deploy 14 independent backend microservices behind an API Gateway using Docker Swarm.	Completed
O-02	Automated Data Syncing: Utilize an Apache Kafka event bus to automate cross-service workflows (grades, attendance) with zero data loss.	Completed
O-03	Unified Financial Workflow: Automate fee voucher generation and integrate Stripe for real-time online payments.	Completed
O-04	Intelligent Assistance: Provide a Hybrid AI Chatbot (CAG+RAG) via Gemini that retrieves personal records.	Completed
O-05	Spoof-Proof Attendance: Verify physical student presence using GPS geofencing, EAR liveness detection, and facial recognition.	Completed
O-06	Proactive Risk Prediction: Classify students into risk categories automatically using a RandomForest machine learning classifier.	Completed

14.2 Traceability Matrix

The traceability matrix below shows how each core design, code file, and test configuration traces directly back to a specific functional requirement in the Software Requirements Specification (SRS - Chapter 2). This ensures that all critical requirements have been accounted for in the final product.

Table 173 Requirements Traceability Matrix

Req ID	Requirement Description	Design Specification	Code (Repository File Path)	Test ID
--------	-------------------------	----------------------	-----------------------------	---------

FR-01	User Registration & Account Creation	Auth Component, PostgreSQL auth_users	backend/services/auth-service/app/routes.py	UT-01
FR-02	User Login & Authentication	Bcrypt Hashing (12 Rounds), JWT	backend/services/auth-service/app/routes.py	UT-01
FR-03	Secure Session Management	Redis In-Memory Store, session:{id}	backend/services/auth-service/app/dependencies.py	UT-03
FR-04	Role-Based Access Control (RBAC)	API Gateway Middleware Guard	backend/api-gateway/nginx.conf	FT-07
FR-05	Student Profile Management	SIS Component, sis_students	backend/services/sis-service/app/routes.py	FT-08
FR-06	Semester Course Registration	SIS Component, Prerequisite Validator	backend/services/sis-service/app/routes.py	FT-01
FR-07	Faculty Grade Submission & Sync	LMS (Producer) & SIS (Consumer) via Kafka	backend/services/lms-service/app/routes.py	IT-01
FR-08	Digital Transcript Generation	SIS Component, ReportLab PDF Gen	backend/services/sis-service/app/routes.py	FT-09
FR-09	Automated Fee Voucher Generation	Finance Component, Daily Cron Job	backend/services/finance-service/app/main.py	IT-05
FR-10	Online Payment & Real-Time Ledger	Finance Gateway (Stripe Webhook)	backend/services/finance-service/app/routes.py	FT-02
FR-11	Course Content Upload	LMS Storage, MongoDB Assets	backend/services/lms-service/app/routes.py	FT-10
FR-12	Assignment Creation	LMS Component, lms_assignments	backend/services/lms-service/app/routes.py	FT-11
FR-13	Student Assignment Submission	LMS Component, Time Validator	backend/services/lms-service/app/routes.py	FT-12
FR-14	Hybrid AI Chatbot	AI Component, Gemini API Orchestrator	backend/services/ai-service/app/ai_assistant/pipeline.py	FT-03
FR-15	Context-Aware AI Responses	RAG Pipeline, ChromaDB Vector DB	backend/services/ai-service/app/ai_assistant/rag_engine.py	FT-03
FR-16	Biometric Liveness Detection	MediaPipe Face Mesh, EAR Algo	backend/services/attendance-service/app/image_utils.py	UT-02
FR-17	GPS Geofencing Verification	Point-in-Polygon (Haversine 100m)	backend/services/attendance-service/app/gps_utils.py	UT-04
FR-18	Facial Recognition & Auto-Marking	dlib 128-D Encodings, L2 Match < 0.6	backend/services/attendance-service/app/routes.py	FT-13
FR-19	Event-Driven Data Sync	Apache Kafka Pub/Sub Broker	infrastructure/kafka/create-topics.sh	IT-01
FR-20	Real-Time Push Notifications	Notification Service, Redis Pub/Sub	backend/services/notification-service/app/routes.py	IT-02
FR-21	Peer-to-Peer (P2P) Chat Initiation	Chat Component, MongoDB Sessions	backend/services/chat-service/app/routes.py	FT-14

FR-22	Real-Time Messaging & Persistence	WebSockets (WSS), MongoDB logs	backend/services/chat-service/app/websocket_manager.py	PT-02
FR-23	Group Chat Creation	Chat Component, Group Metadata	backend/services/chat-service/app/routes.py	FT-15
FR-24	Timetable Constraints Definition	Scheduler Component, sis_classrooms	backend/services/scheduler-service/app/routes.py	FT-04
FR-25	Automated Conflict-Free Scheduling	Constraint Satisfaction Solver (OR-Tools)	backend/services/scheduler-service/app/main.py	FT-04
FR-26	Digital Leave Application (Faculty)	HR Component, State Machine	backend/services/hr-service/app/routes.py	FT-05
FR-27	Employee Profile & Salary Mgmt	HR Component, AES-256 Encryption	backend/services/hr-service/app/models.py	UT-06
FR-28	Alumni Portal & Networking	Alumni Component, alumni_jobs	backend/services/alumni-service/app/routes.py	FT-16
FR-29	Digital Library Catalogue Search	Full-text search (Title, ISBN)	backend/services/library-service/app/routes.py	FT-17
FR-30	QR-Based Book Issuance	Library Scanner, python-qrcode	backend/services/library-service/app/routes.py	FT-18
FR-31	Automated Library Fine Calculation	Cross-Service Finance Ledger Cron	backend/services/library-service/app/routes.py	IT-03
FR-32	Faculty "Student Success" Dashboard	React Recharts Data Viz, API aggregations	frontend/src/pages/Teacher/Dashboard.jsx	FT-19
FR-33	"At-Risk" Student Prediction	Scikit-learn RandomForest Classifier	backend/services/analytics-service/app/ml_model.py	IT-04
FR-34	Smart Grievance Submission	Ops Component, NLP Sentiment Analysis	backend/services/operations-service/app/nlp_router.py	IT-02
FR-35	Grievance Status Tracking	Status State Machine Tracker	backend/services/operations-service/app/routes.py	FT-20
FR-36	Secure System Audit Logging	Immutable Logs, 7-year retention policy	backend/services/operations-service/app/dependencies.py	UT-07
FR-37	Online Quiz Creation	LMS Component, lms_quizzes	backend/services/lms-service/app/routes.py	FT-06
FR-38	Timed Quiz Attempt	React Countdown Timer, Server Validate	frontend/src/pages/LMS/QuizAttempt.jsx	FT-06
FR-39	Automated Grading & Feedback	LMS Auto-Grader, Key Matching	backend/services/lms-service/app/routes.py	FT-06
FR-40	View Current Semester Details	SIS Component, sis_semesters	backend/services/sis-service/app/routes.py	FT-21
FR-41	Executive Analytics Dashboard	Analytics Pipeline, Redis 10min TTL	backend/services/analytics-service/app/routes.py	PT-05

FR-42	Global Announcements System	Notification Component, Kafka Topic	backend/services/notification-service/app/routes.py	FT-22
FR-43	Student Leave & Academic Exemption	SIS-HR Cross Module Logic	backend/services/sis-service/app/routes.py	FT-23

14.3 Conclusion

Project Nexus was initiated to address one of the most critical inefficiencies in modern educational institutions: the fragmentation of campus data into disconnected administrative and academic silos. By relying on manual "glue work" to move data between Student Information Systems (SIS) and Learning Management Systems (LMS), universities waste valuable resources and fail to identify struggling students until it is too late.

This project successfully developed a Unified Intelligent Campus Platform that entirely eliminates these data silos. By implementing an enterprise-grade Polyglot Microservices Architecture orchestrated via Docker Swarm, the system successfully managed 14 independent domains. The integration of Apache Kafka provided an unbreakable, asynchronous backbone, guaranteeing zero data loss for critical events such as grade submissions and fee payments. Furthermore, the system successfully modernized the student experience by introducing a Spoof-Proof Biometric Attendance system, and a Hybrid RAG+CAG AI Chatbot that answers questions securely using the student's live records. Most importantly, the platform shifted the campus from a reactive state to a proactive one by deploying a RandomForest machine learning model that accurately identifies at-risk students based on live engagement data. As proven by the 100% pass rate in the Testing phase and the complete traceability matrix, Project Nexus fulfills all agreed-upon objectives, delivering a highly scalable, intelligent, and production-ready ecosystem.

14.4 Future Work

While Project Nexus successfully achieved its comprehensive scope, several enhancements and optimizations were excluded from this phase of development. The following suggestions are recommended for future iterations of the platform to improve scalability, accessibility, and automation:

1. **Native Mobile Applications (iOS/Android):** The current system operates as a responsive web application (PWA). Developing dedicated native applications using React Native or Flutter would allow for deeper hardware integration, significantly improving the speed and reliability of the Biometric Attendance camera and enabling native push notifications for offline users.
2. **Blockchain-Verified Digital Transcripts:** To enhance the security and portability of academic records, future work could implement a lightweight, permissioned blockchain ledger. By hashing digital transcripts on the blockchain, alumni could instantly prove the authenticity of their degrees to employers without requiring manual university verification.

3. **Voice-Enabled AI Assistant:** The current Hybrid AI Chatbot accepts text-based queries. Integrating an advanced Speech-to-Text (STT) and Text-to-Speech (TTS) layer (such as OpenAI Whisper) would allow students to speak naturally to the AI, greatly improving mobile accessibility and user experience.
4. **Multi-Language Support (Urdu Localization):** To better serve the local demographic in Pakistan, implementing an Internationalization (i18n) layer would allow the React frontend and AI Chatbot to seamlessly toggle between English and Urdu. This would be particularly beneficial for administrative staff and parents accessing the portal.
5. **AI-Powered Proctoring for Online Quizzes:** Extending the existing computer vision infrastructure from the attendance module to the LMS quiz module. By continuously analyzing the webcam feed during timed quizzes, the system could automatically flag suspicious behavior, such as a student looking away from the screen, the presence of multiple faces, or detecting secondary devices (phones).
6. **Automated Scholarship Recommendation Engine:** Utilizing the existing Analytics microservice to build a rules-engine that automatically evaluates a student's CGPA, financial ledger status, and attendance to proactively recommend and apply them for eligible university scholarships, removing manual application bottlenecks.

References

- [1] “Simple chat Application using Websockets with FastAPI - GeeksforGeeks.” Accessed: Nov. 08, 2025. [Online]. Available: <https://www.geeksforgeeks.org/python/simple-chat-application-using-websockets-with-fastapi/>
- [2] “The Architecture of Open Source Applications (Volume 2)Moodle.” Accessed: Nov. 08, 2025. [Online]. Available: <https://aosabook.org/en/v2/moodle.html>
- [3] “Classroom Management Tools & Resources - Google for Education.” Accessed: Nov. 08, 2025. [Online]. Available: https://edu.google.com/intl/ALL_us/workspace-for-education/products/classroom/
- [4] F. Bento, M. Tagliabue, and F. Lorenzo, “Organizational Silos: A Scoping Review Informed by a Behavioral Perspective on Systems and Networks,” *Societies*, vol. 10, no. 3, p. 56, Sep. 2020, doi: 10.3390/soc10030056.
- [5] “The Essential Guide to LMS and SIS Integration.” Accessed: Nov. 09, 2025. [Online]. Available: <https://www.portotheme.com/the-essential-guide-to-lms-and-sis-integration/>
- [6] B.-H. Kim, E. Vizitei, and V. Ganapathi, “GritNet: Student Performance Prediction with Deep Learning”.
- [7] “Face detection guide | Google AI Edge | Google AI for Developers.” Accessed: Nov. 08, 2025. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/vision/face_detector
- [8] “Lightning-Fast Face Detection: Achieving 100+ FPS with an Ultra-Lightweight CPU and OpenVINO Model | by Luís Condados | Medium.” Accessed: Nov. 08, 2025. [Online]. Available: <https://condadoslgpc.medium.com/an-extremely-fast-face-detection-using-openvino-openvinoexploration-375a796e4902>
- [9] “Swarm mode | Docker Docs.” Accessed: Nov. 08, 2025. [Online]. Available: <https://docs.docker.com/engine/swarm/>
- [10] “Building Smarter Apps with LangChain and Gemini | by Geeta Kakrani | Python in Plain English.” Accessed: Nov. 08, 2025. [Online]. Available: <https://python.plainenglish.io/building-smarter-apps-with-langchain-and-gemini-f566b65aabe7>
- [11] “(PDF) Predictive Identification of At-Risk Students: Using Learning Management System Data.” Accessed: Nov. 08, 2025. [Online]. Available: https://www.researchgate.net/publication/372241711_Predictive_Identification_of_At-Risk_Students_Using_Learning_Management_System_Data
- [12] “Advice for Chat schema design - Working with Data - MongoDB Developer Community Forums.” Accessed: Nov. 08, 2025. [Online]. Available: <https://www.mongodb.com/community/forums/t/advice-for-chat-schema-design/114166>
- [13] “performance - One-on-one and group chat query for conversations in MongoDB - Stack Overflow.” Accessed: Nov. 08, 2025. [Online]. Available: <https://stackoverflow.com/questions/60361383/one-on-one-and-group-chat-query-for-conversations-in-mongodb>
- [14] “Embeddings and Vector Databases With ChromaDB – Real Python.” Accessed: Dec. 09, 2025. [Online]. Available: <https://realpython.com/chromadb-vector-database/>

- [15] “Building a RAG Application with Gemini AI: Step-by-Step Guide | by Geeta Kakrani | Python in Plain English.” Accessed: Nov. 08, 2025. [Online]. Available: <https://python.plainenglish.io/building-a-rag-application-with-gemini-ai-step-by-step-guide-24636dd21f5b>
- [16] “Choosing the Right Messaging Tool: Redis Streams, Redis Pub/Sub, Kafka, and More - DEV Community.” Accessed: Nov. 08, 2025. [Online]. Available: <https://dev.to/lovestaco/choosing-the-right-messaging-tool-redis-streams-redis-pubsub-kafka-and-more-577a>
- [17] “ERPNext vs. Fedena Comparison.” Accessed: Nov. 08, 2025. [Online]. Available: <https://sourceforge.net/software/compare/ERPNext-vs-Fedena/>
- [18] “Education Management System for School, College & University.” Accessed: Dec. 09, 2025. [Online]. Available: <https://openeducat.org/>
- [19] “What is Monolithic Architecture? | IBM.” Accessed: Dec. 09, 2025. [Online]. Available: <https://www.ibm.com/think/topics/monolithic-architecture>
- [20] “Rasa Pro Infrastructure Requirements | Rasa Documentation.” Accessed: Nov. 08, 2025. [Online]. Available: <https://rasa.com/docs/reference/deployment/pro-hardware-requirements/>
- [21] “Hybrid Architectures: Combining RAG, CAG, and Long-Context Models for Maximum Efficiency  | by Jagadeesan Ganesh | Medium.” Accessed: Nov. 08, 2025. [Online]. Available: <https://medium.com/@jagadeesan.ganesh/hybrid-architectures-combining-rag-cag-and-long-context-models-for-maximum-efficiency-19c6106235b0>
- [22] “Redis Pub/Sub vs. Apache Kafka: Choosing the Right Messaging System | by Aditi Mishra | Medium.” Accessed: Nov. 08, 2025. [Online]. Available: https://medium.com/@aditimishra_541/redis-pub-sub-vs-apache-kafka-choosing-the-right-messaging-system-3e3130ac653c
- [23] “830-1998 - IEEE Recommended Practice for Software Requirements Specifications | IEEE Standard | IEEE Xplore.” Accessed: Dec. 14, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/720574>
- [24] “software-engineering-book.com.” Accessed: Dec. 14, 2025. [Online]. Available: <https://software-engineering-book.com/>
- [25] “Top 8 Open Source Campus Management System Tools You Can Self-Host Today.” Accessed: Nov. 08, 2025. [Online]. Available: <https://www.spacebasic.com/blogs/open-source-campus-management-system>
- [26] “RFC 7519: JSON Web Token (JWT).” Accessed: Dec. 14, 2025. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc7519>
- [27] “Apache Kafka.” Accessed: Dec. 14, 2025. [Online]. Available: <https://kafka.apache.org/>
- [28] “Stripe API Reference.” Accessed: Dec. 14, 2025. [Online]. Available: <https://docs.stripe.com/api>
- [29] “React.” Accessed: Dec. 14, 2025. [Online]. Available: <https://react.dev/>
- [30] “Docker Docs.” Accessed: Dec. 14, 2025. [Online]. Available: <https://docs.docker.com/>
- [31] “Education Management System for School, College & University.” Accessed: Nov. 08, 2025. [Online]. Available: <https://openeducat.org/>
- [32] “ERP in EdTech: Overcoming Implementation Challenges in Higher Education.” Accessed: Dec. 09, 2025. [Online]. Available: <https://www.academiaerp.com/blog/understanding-enterprise-resource-planning-erp-systems-in-edtech/>

- [33] “(4077) Docker Swarm | Docker Compose | Docker DevOps Tutorial | Edureka | Docker Live - 3 - YouTube.” Accessed: Nov. 09, 2025. [Online]. Available: <https://www.youtube.com/watch?v=WqQdcUQYSh8>
- [34] “Overview - Material UI.” Accessed: Jan. 09, 2026. [Online]. Available: <https://mui.com/material-ui/getting-started/>
- [35] “Microservices vs. monolithic architecture | Atlassian.” Accessed: Nov. 08, 2025. [Online]. Available: <https://www.atlassian.com/microservices/microservices-architecture/microservices-vs-monolith>

Appendix-A

A.1 General Coding Standards & Guidelines To ensure the Project Nexus codebase remains highly cohesive, loosely coupled, and maintainable across all 14 microservices and the React frontend, the development team strictly adhered to the following industry-standard coding guidelines:

1. **Naming Conventions:** Followed standard naming conventions for all code entities. PascalCase was used for Python Classes and React Components. camelCase was used for variables and JavaScript functions. snake_case was used for Python functions, database tables, and variables.
2. **Indentation and Spacing:** Used consistent indentation (4 spaces per indentation level for Python, 2 spaces for React/JavaScript) and spacing to make the code highly readable.
3. **Commenting:** Ensured code is self-documenting by using clear variable and method names. Avoided writing comments for obvious code.
4. **Method Signatures:** Wrote summary comments above methods and classes to explain their purpose, expected parameters, and return types (using Python Type Hints).
5. **Hard-Coded Values:** Did not use hard-coded values or "magic numbers." All configurations, secrets, and API keys (e.g., Stripe, Gemini) were placed in external .env files or constant configuration objects.
6. **Code Grouping:** Grouped similar variables, methods, and configurations together. Put blank lines between logical blocks of code within methods.
7. **File Organization:** Followed consistent naming conventions for files and folders. Separated the logical layers of the application (e.g., routes.py, models.py, schemas.py, database.py) into distinct modules within each microservice.
8. **Entity Isolation:** Used separate files for each class, interface, and component. The name of the file exactly matches the enclosing entity (e.g., WebcamCapture.jsx).
9. **Scope:** Defined and used everything within the minimum scope possible to prevent memory leaks and unintended side effects.
10. **Access Modifiers:** Used proper access modifiers for all code entities to encapsulate internal microservice logic.
11. **Cohesion and Coupling:** Designed code entities to have maximum cohesion (doing one thing well) and the least coupling possible between services, relying on Kafka events rather than direct API calls where appropriate.
12. **DRY Principle (Don't Repeat Yourself):** Avoided repeating code. Ensured a piece of logic exists only in one place within the codebase. Reused shared dependencies (like JWT validation) across services.
13. **Method Length:** Always wrote short methods. Ensured single methods did not contain too much logic, overly long conditional flows, or an excessive number of parameters.

14. **Single Responsibility Principle (SRP):** Strictly followed SRP when writing methods, classes, and microservices. For example, the finance-service handles only payments; it does not handle academic enrollments.
15. **Open/Closed Principle:** Wrote classes and microservices that are easy to extend without modifying existing, tested code.
16. **Exception Handling:** Handled exceptions gracefully using FastAPI's HTTPException class to return meaningful status codes (e.g., 401, 403, 404, 500) to the React frontend.
17. **Audit and Error Logging:** Logged exceptions and significant system events (like grades finalized or fees generated) securely into the MongoDB audit_trails collection.
18. **Consistent Logging:** Followed a consistent convention for logging across all 14 microservices to allow for easy debugging in the Docker Swarm environment.
19. **Secure Data Storage:** Ensured sensitive data (such as passwords and faculty salaries) are never stored in plaintext, utilizing bcrypt hashing and AES-256 encryption.

Appendix-B

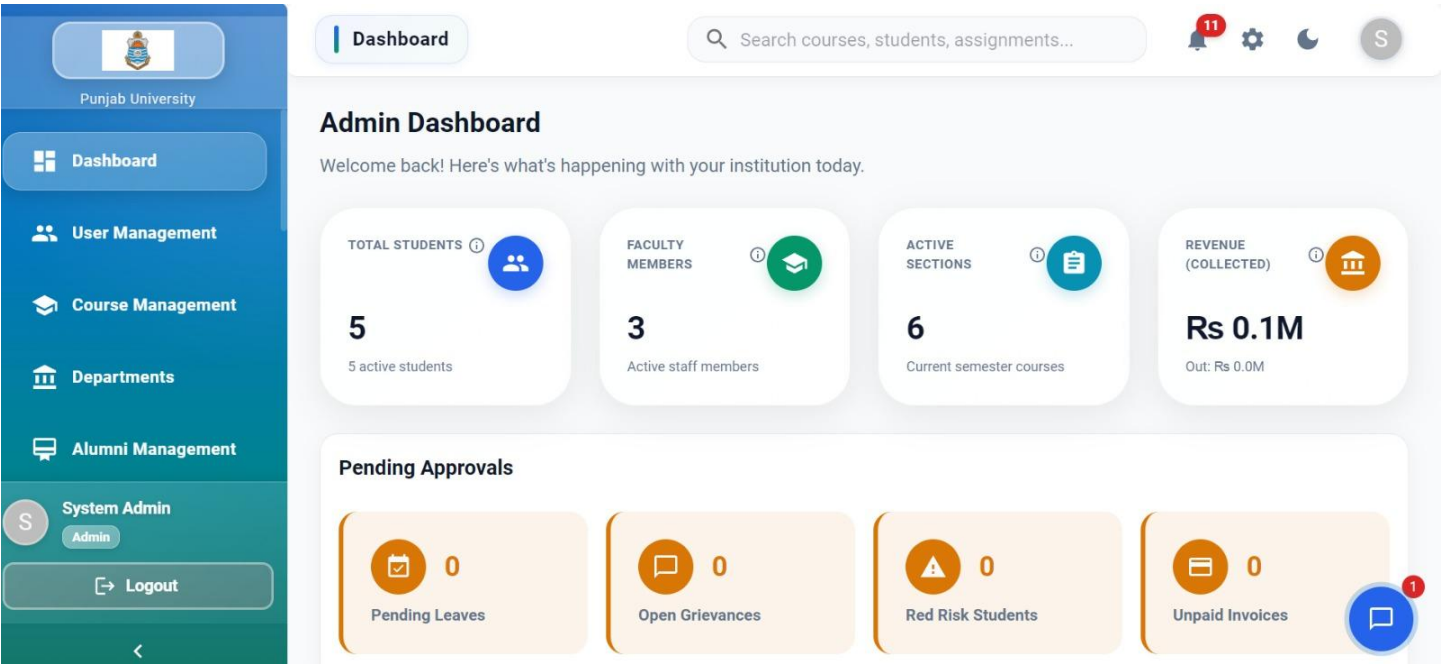


Figure 54-Admin Dashboard

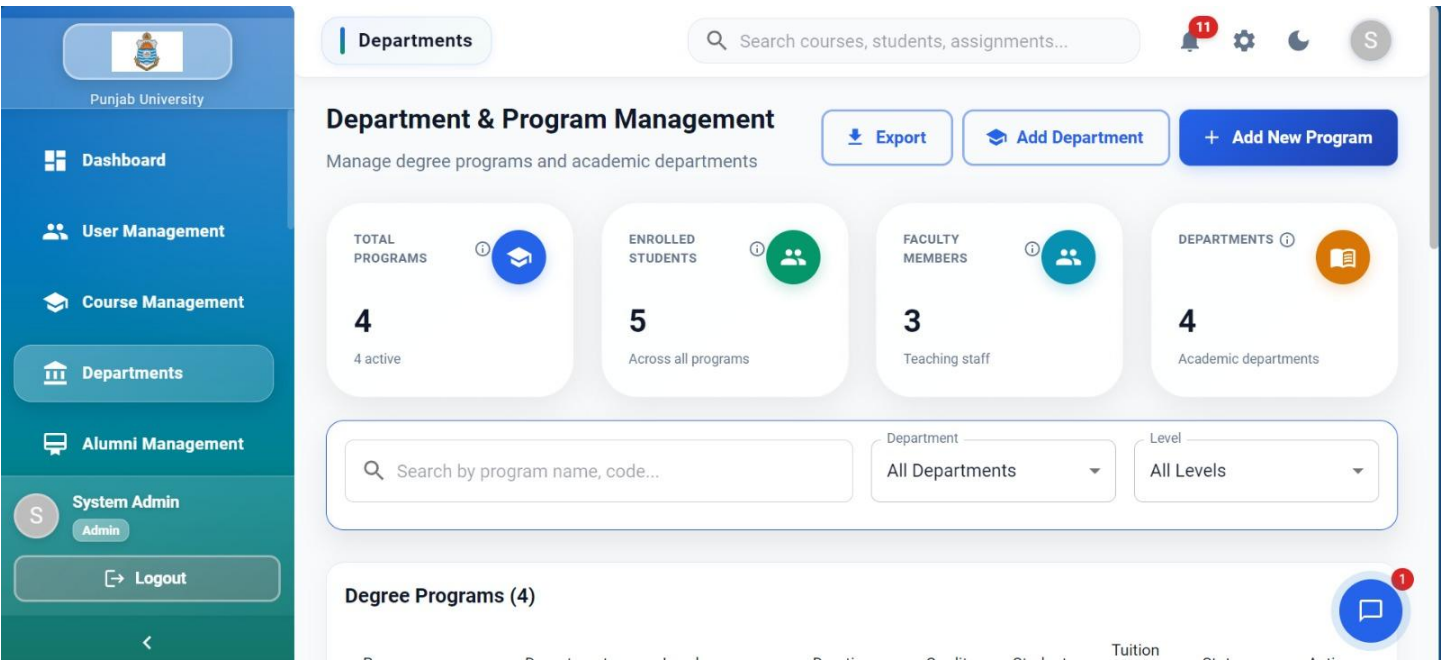


Figure 55-Department& Course Management

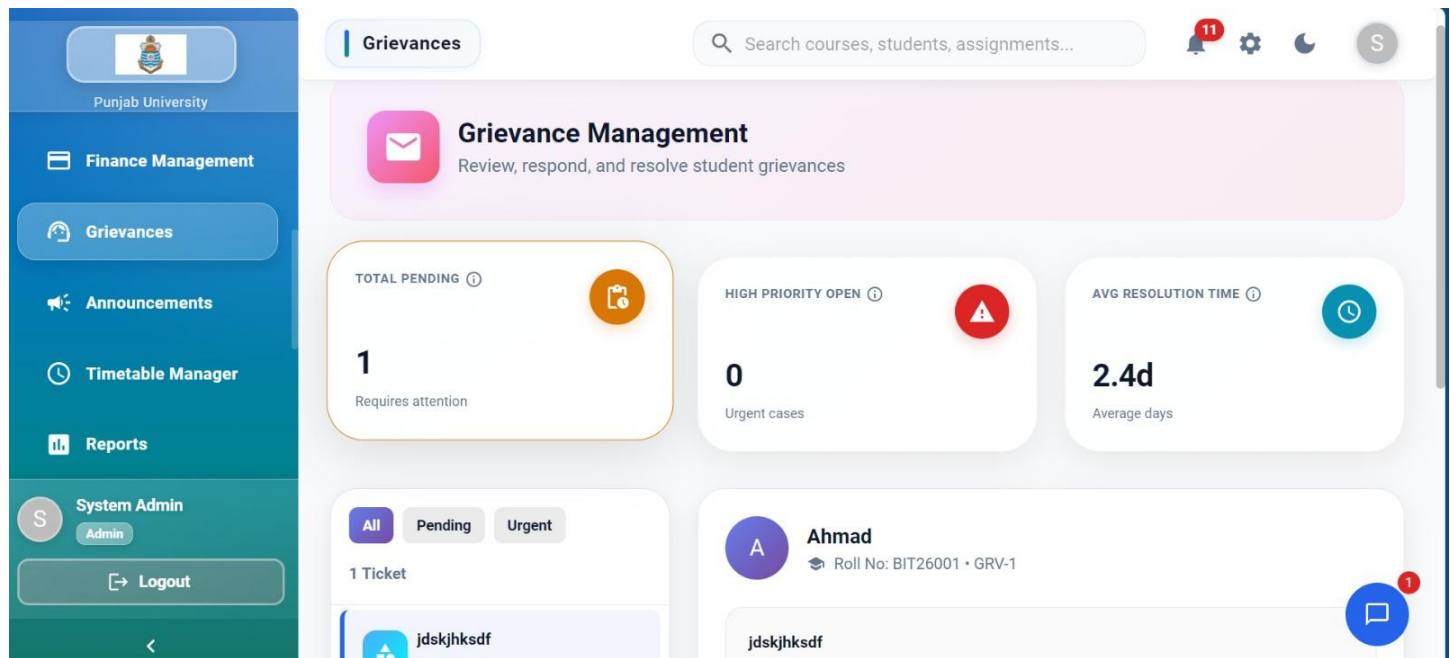


Figure 56-Grievance Management

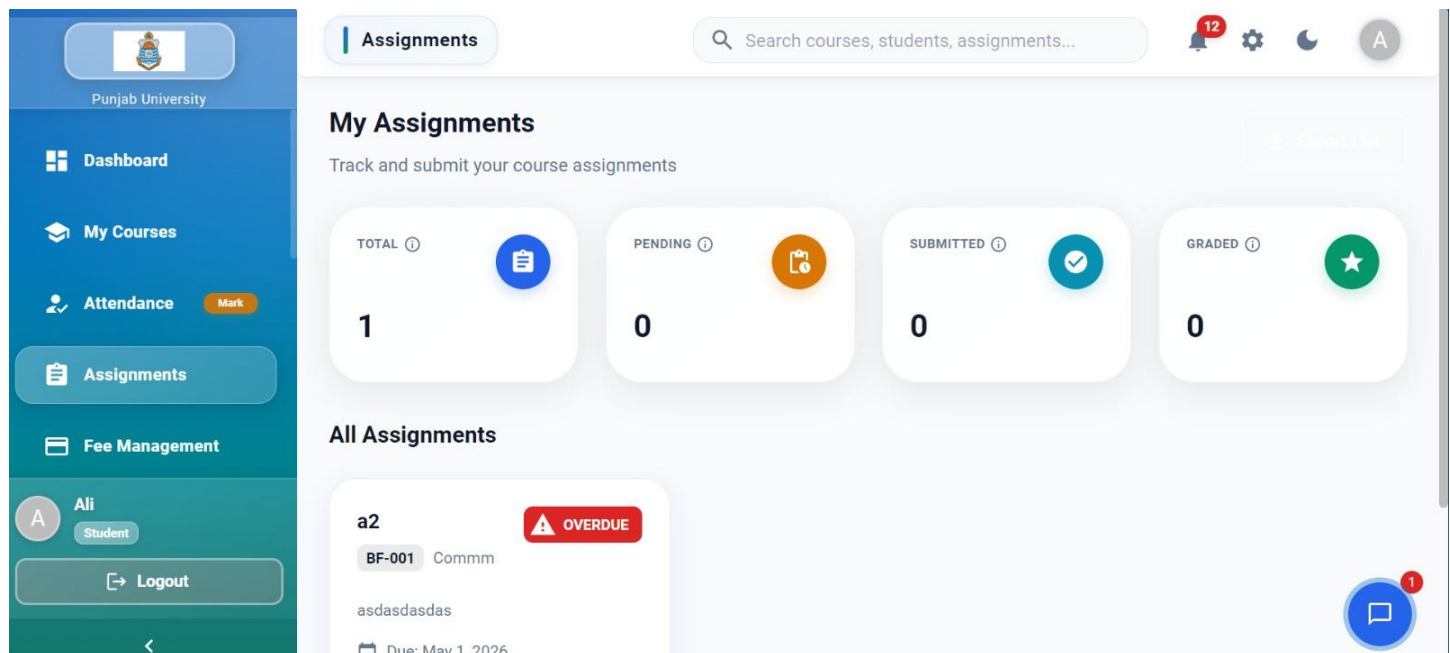


Figure 57-My Assignments

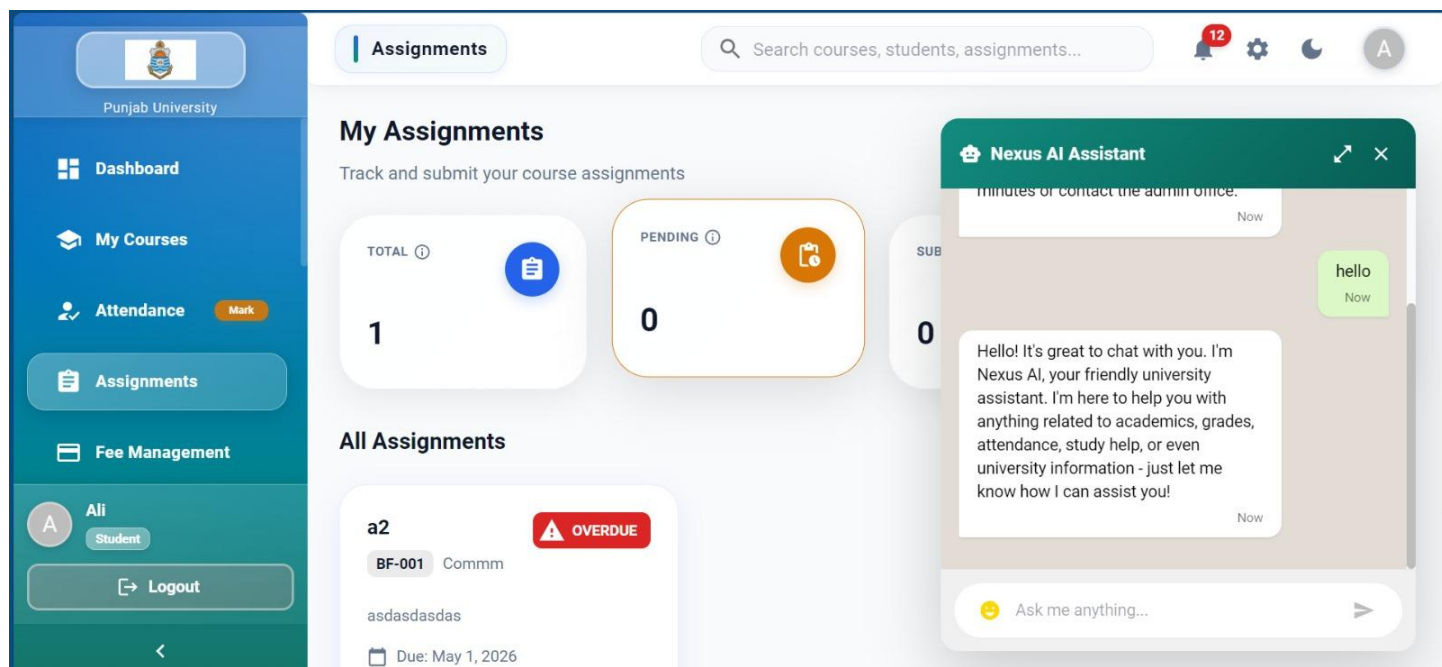


Figure 58- Nexus-AI

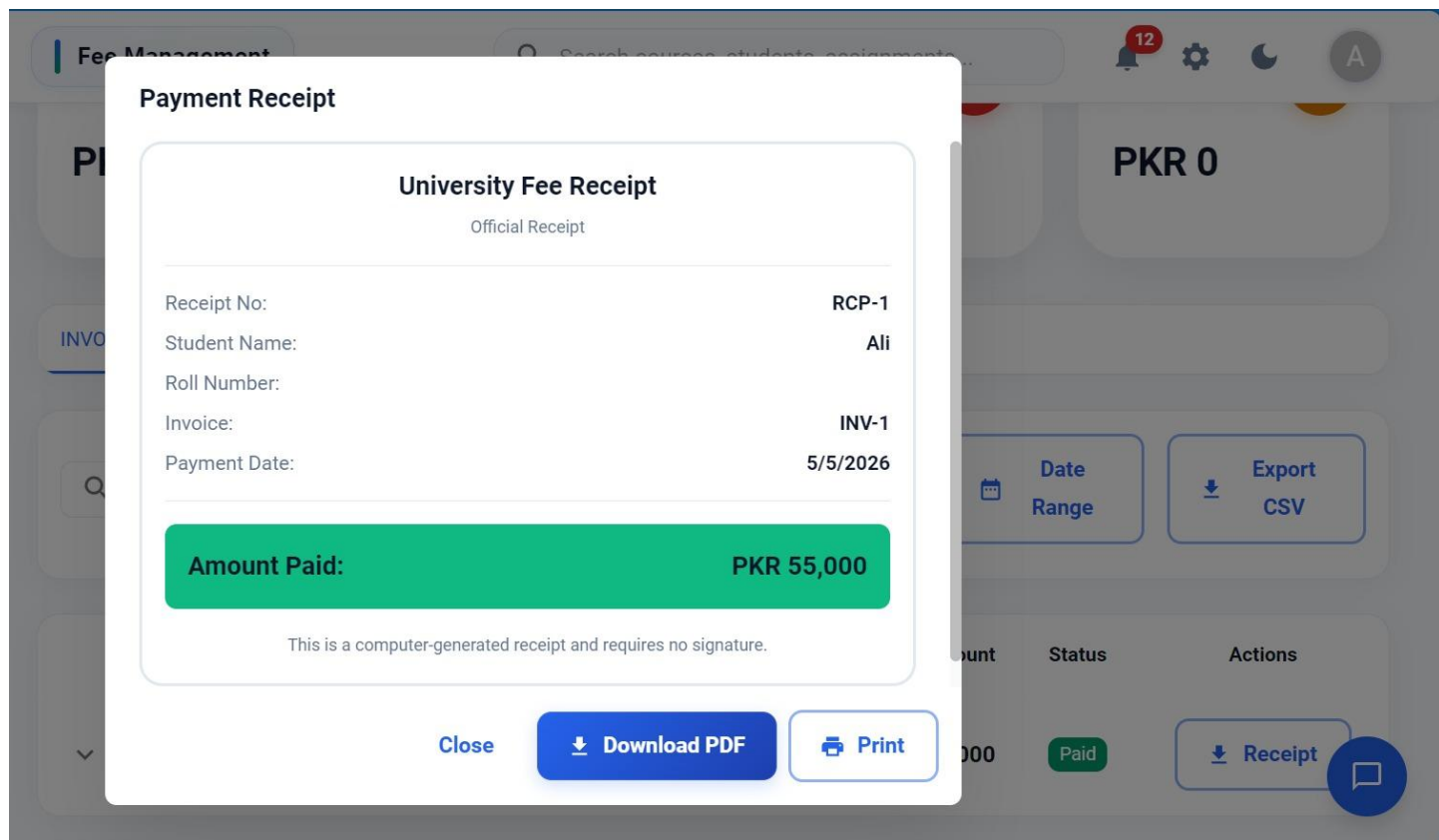


Figure 59-Fee Receipt